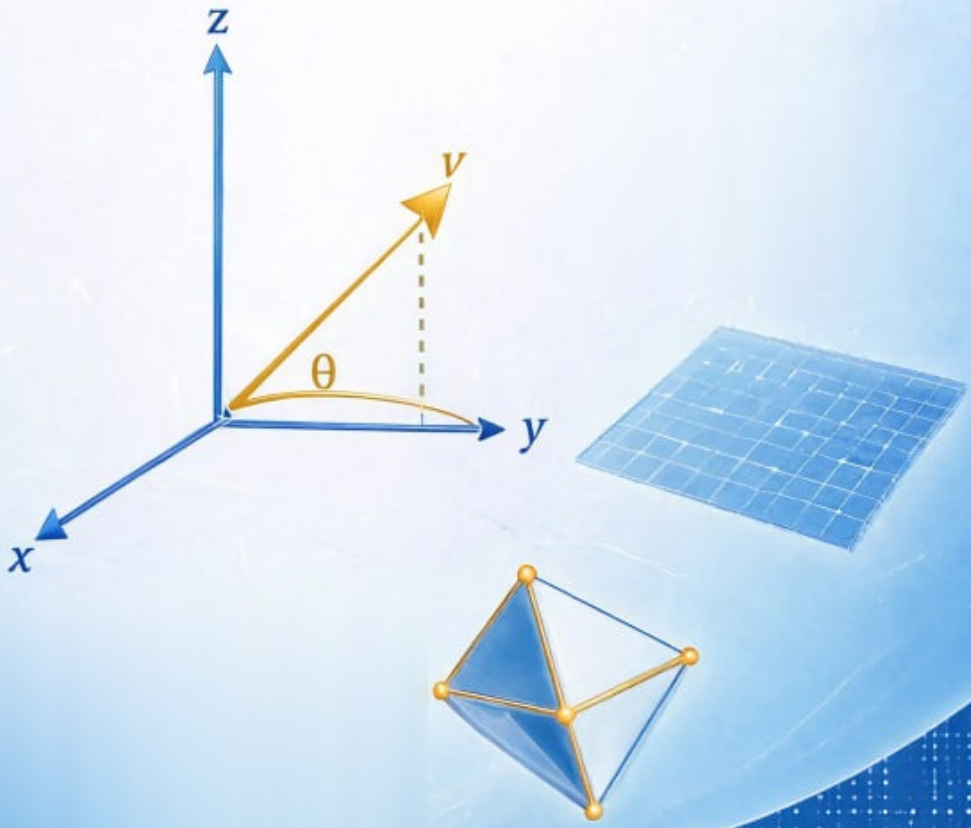
LINEAR ALGEBRA

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LINEAR ALGEBRA

Study Guide

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The study guide covers the 24 topics of the course “Linear Algebra”: matrices and determinants, systems of linear equations, vectors and analytic geometry, linear and Euclidean spaces, linear operators, quadratic forms and iterative methods for linear systems. Each topic contains the necessary theory, examples solved in full detail, applications in information security and short computations in Python, Maple and Mathcad. More than two thousand exercises for independent work are included, all with answers. The guide is intended for university students of information security, computer engineering and other engineering programmes.

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Introduction

Linear algebra is one of the most widely used branches of modern mathematics. Matrices, determinants, systems of linear equations, vector spaces, linear operators and quadratic forms are the working language of engineering computation, computer graphics and machine learning, and they are equally basic to **information security**. The same methods appear in the classical Hill cipher and in modern lattice-based (post-quantum) cryptography, in error-detecting and error-correcting codes, and in the analysis of network traffic and the detection of anomalies.

This study guide accompanies the course “Linear Algebra” and covers the 24 topics of its syllabus. The topics are grouped into five chapters. Chapter 1, *Matrices and Determinants* (Sections 1.1–1.6), deals with operations on matrices, determinants of the second, third and n-th order, minors and cofactors, the rank of a matrix and the inverse matrix. Chapter 2, *Systems of Linear Algebraic Equations, Vectors and Lines in the Plane* (Sections 2.1–2.6), presents Cramer’s rule, the matrix method and the Gauss and Gauss–Jordan methods, then turns to vectors, their scalar, vector and triple products, and the equations of a straight line in the plane. Chapter 3, *Lines and Planes in Space; Linear and Euclidean Spaces* (Sections 3.1–3.5), studies lines and planes in space, bases and subspaces of a linear space, the norm in a Euclidean space and the construction of orthonormal bases. Chapter 4, *Linear Operators* (Sections 4.1–4.3), treats linear operators and their matrices, eigenvalues and eigenvectors, and diagonalization. Chapter 5, *Quadratic Forms, Second-Order Curves and Iterative Methods* (Sections 5.1–5.4), reduces quadratic forms and second-order curves to canonical form and closes with two approximate methods for linear systems, simple (Jacobi) iteration and the Gauss–Seidel method.

All sections are built in the same way. The main definitions and theorems come first, followed by worked examples solved in full detail. A separate subsection shows how the material of the section is used in **information security**, so that the reader sees why an engineer in this field needs it. Each section also contains short examples of the same computations in Python, Maple and Mathcad. The section ends with a **collection of exercises**; every

exercise set has 30 variants, which allows a teacher to give each student in a group an individual assignment, and the sets are supplemented by applied problems. The answers to all exercises are collected at the end of the book in “Answers to the Exercises”, arranged by section and exercise set.

We recommend reading the theory and the worked examples carefully before attempting the exercises, and comparing one’s own results with the answers only after the work is done. Checking the computations on a computer, in Python, MATLAB, Maple or Mathcad, is also useful.

The guide is written for students of information security, computer engineering, software engineering and other engineering programmes at universities, and for anyone who studies linear algebra independently.

Chapter 1

Matrices and Determinants

1.1 The concept of a matrix

Definition 1.1.1. A rectangular array of numbers (or, more generally, of elements of a field) arranged in m rows and n columns is called a **matrix**. It is written as

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij}), \quad i = \overline{1,m}, j = \overline{1,n}.$$

The number a_{ij}, standing at the intersection of the ith row and the jth column, is called an **entry** (element) of the matrix. The **size** of the matrix is written $m \times n$ (read "m by n").

1.1.1 The value of a polynomial at a matrix

Definition 1.1.2. Let A be a square matrix of order n and let $f(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_m x^m$ be a polynomial. The **nonnegative integer powers** of A are defined by

$$A^0 = E, \qquad A^1 = A, \qquad A^k = \underbrace{A \cdot A \cdots A}_{k \text{ times}}.$$

The **value of the polynomial** $f(x)$ at the matrix A is then the matrix

$$f(A) = c_0 E + c_1 A + c_2 A^2 + \cdots + c_m A^m.$$

Note that the constant term c_0 must be multiplied by the identity matrix: c_0 is replaced by c_0E.

Example 1.1.1. Let $f(x) = 3x - 2x^2 + 7$ and $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$. We first compute A^2:

$$A^2 = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = E.$$

Hence

$$f(A) = 3A - 2A^2 + 7E =$$

$$3\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} - 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 7\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 6 \\ 0 & 2 \end{pmatrix}.$$

1.1.2 Types of matrices

- A **square matrix** has as many rows as columns ($m = n$); the number n is then called the **order** of the matrix.
- A **row matrix** consists of a single row (its size is $1 \times n$): $A = (a_1; a_2; \cdots; a_n)$.
- A **column matrix (column vector)** consists of a single column (its size is $m \times 1$): $A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}$.
- The **zero matrix** O is the matrix all of whose entries are zero.
- A **diagonal matrix** is a square matrix whose entries off the main diagonal are all zero: $a_{ij} = 0$ for $i \neq j$.
- The **identity matrix** E (also denoted I) is the diagonal matrix with ones on the main diagonal and zeros elsewhere.

- An **upper (lower) triangular matrix** is a square matrix whose entries below (above) the main diagonal are all zero.
- A **symmetric matrix** is a square matrix satisfying $A = A^T$, that is, $a_{ij} = a_{ji}$ for all i, j.
- The **transpose** A^T of a matrix A is obtained by interchanging the rows and columns of A: if $A = (a_{ij})_{m\times n}$, then $A^T = (a_{ji})_{n\times m}$. In full,

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

The transpose has the following properties: 1) $(A^T)^T = A$;
2) $(\lambda A)^T = \lambda A^T$;
3) $(A + B)^T = A^T + B^T$;
4) $(AB)^T = B^T A^T$.

Example 1.1.2. The matrix $A = \begin{pmatrix} 2 & -1 & 0 \\ 3 & 4 & 5 \end{pmatrix}$ has size 2×3, with $a_{12} = -1$ and $a_{23} = 5$. Its transpose is the 3×2 matrix

$$A^T = \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 0 & 5 \end{pmatrix}.$$

Two matrices $A = (a_{ij})$ and $B = (b_{ij})$ are **equal** if they have the same size and their corresponding entries coincide: $a_{ij} = b_{ij}$ for all i, j.

1.1.3 Arithmetic operations on matrices

Addition and subtraction. The **sum** of two matrices $A = (a_{ij})$ and $B = (b_{ij})$ of the same size $m\times n$ is the matrix $C = A+B = (a_{ij}+b_{ij})$ obtained by adding corresponding entries. In the same way, the **difference** is $A - B = (a_{ij} - b_{ij})$. Only matrices of the same size can be added or subtracted.

Example 1.1.3. For $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix}$ we have

$$A + B = \begin{pmatrix} -1 & 2 \\ 4 & 3 \end{pmatrix}, \qquad A - B = \begin{pmatrix} 3 & 2 \\ 2 & -5 \end{pmatrix}.$$

Multiplication by a number. The product of a matrix $A = (a_{ij})$ and a number (scalar) k is the matrix $kA = (k \cdot a_{ij})$ obtained by multiplying every entry of A by k.

Example 1.1.4. For $A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}$ and $k = -2$,

$$kA = -2A = \begin{pmatrix} -2 & 4 & -6 \\ 0 & -8 & 2 \end{pmatrix}.$$

Matrix multiplication. Let $A = (a_{ik})$ be an $m \times p$ matrix and $B = (b_{kj})$ a $p \times n$ matrix, so that the number of columns of A equals the number of rows of B. The **product** of A and B is the $m \times n$ matrix

$$C = AB = (c_{ij}), \qquad c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}.$$

In words, the (i, j) entry of C is obtained by multiplying the entries of the ith row of A by the corresponding entries of the jth column of B and adding the results.

Example 1.1.5. For the 2×2 matrices $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$,

$$AB = \begin{pmatrix} 1 \cdot 3 + 2 \cdot 2 & 1 \cdot 1 + 2 \cdot 4 \\ 0 \cdot 3 + (-1) \cdot 2 & 0 \cdot 1 + (-1) \cdot 4 \end{pmatrix} =$$

$$= \begin{pmatrix} 7 & 9 \\ -2 & -4 \end{pmatrix}.$$

Example 1.1.6. Matrices of different sizes can also be multiplied, provided the inner dimensions agree. Take the 2×3 matrix $A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}$ and the 3×2 matrix $B = \begin{pmatrix} 3 & 1 \\ 0 & -1 \\ 2 & 4 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 1 \cdot 3 + 0 \cdot 0 + 2 \cdot 2 & 1 \cdot 1 + 0 \cdot (-1) + 2 \cdot 4 \\ -1 \cdot 3 + 3 \cdot 0 + 1 \cdot 2 & -1 \cdot 1 + 3 \cdot (-1) + 1 \cdot 4 \end{pmatrix} =$$

$$= \begin{pmatrix} 7 & 9 \\ -1 & 0 \end{pmatrix},$$

a 2×2 matrix.

Example 1.1.7. In general $AB \neq BA$: matrix multiplication is *not commutative*. For the matrices of the previous example, BA is a 3×3 matrix, so it cannot even be compared with the 2×2 matrix AB.

Transposition.
Transposing twice gives back the original matrix: $(A^T)^T = A$.

Addition of matrices and multiplication of a matrix by a number are called the **linear operations**. Let A, B, C be matrices of the same size $m \times n$, let O be the zero matrix of that size, and let k, l be arbitrary numbers (scalars). The following properties hold.

Properties of addition.

1) $A + B = B + A$ (commutativity)

2) $(A + B) + C = A + (B + C)$ (associativity)

3) $A + O = O + A = A$ (existence of a zero matrix)

4) For every matrix A there is an **additive inverse** $-A = (-1) \cdot A$ such that $A + (-A) = O$.

Properties of multiplication by a number.

1) $k(A + B) = kA + kB$ (distributivity over matrix addition)

2) $(k + l)A = kA + lA$ (distributivity over addition of scalars)

3) $k(lA) = (kl)A = l(kA)$ (associativity)

4) $1 \cdot A = A, \quad 0 \cdot A = O$

5) $(-1) \cdot A = -A$

Properties of multiplication and transposition.

1) $(AB)C = A(BC)$ (associativity of multiplication)

2) $A(B + C) = AB + AC, \quad (A + B)C = AC + BC$ (distributivity)

3) $k(AB) = (kA)B = A(kB)$

4) $AE = EA = A$ (E is the identity matrix of suitable order)

5) $(A + B)^T = A^T + B^T$

6) $(kA)^T = kA^T$

7) $(AB)^T = B^T A^T$ (the order of the factors is reversed)

8) $AB \neq BA$ (in general, multiplication is not commutative)

Example 1.1.8. Let $A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 3 \\ -1 & 2 \end{pmatrix}$, $k = 2$ and $l = -1$. Then

$$kA + lB = 2\begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 3 \\ -1 & 2 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & -2 \\ 4 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 2 & -5 \\ 5 & -2 \end{pmatrix}.$$

With the same matrices, the reader may verify the distributive law $k(A+B) = kA + kB$ for $k = 2$.

Matrix algebra is the mathematical basis of much of modern information security: cryptography, coding theory, network security and digital signal processing all rely on it. We describe a few of these applications.

1.1.4 Matrices in cryptography: the Hill cipher

The **Hill cipher**, one of the classical symmetric encryption algorithms, is built entirely on matrix operations. Each letter is assigned a number (for instance, $A{=}0, B{=}1, \ldots, Z{=}25$), and the plaintext is split into blocks of n letters; each block is written as an $n \times 1$ column vector p. The secret key is an invertible $n \times n$ **key matrix** K, and a block is encrypted by the rule

$$c = Kp \pmod{26}.$$

Decryption uses $p = K^{-1}c \pmod{26}$, where K^{-1} is the inverse of K modulo 26. Such an inverse exists exactly when $\det K$ is coprime to 26, that is, $\gcd(\det K, 26) = 1$; a nonzero determinant alone is not enough.

Example 1.1.9. Take the key matrix $K = \begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix}$ ($\det K = 11$, which is coprime to 26) and the plaintext block $p = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$ (the letters “H”, “E”). The encrypted block is

$$c = Kp = \begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} = \begin{pmatrix} 29 \\ 63 \end{pmatrix} \equiv \begin{pmatrix} 3 \\ 11 \end{pmatrix} \pmod{26},$$

that is, “D”, “L”. The receiver computes $K^{-1} \pmod{26}$ and recovers the plaintext as $p = K^{-1}c$. An attacker, on the other hand, has to find K (or its inverse), so the size of the key matrix and the way it is chosen directly affect the strength of the cipher.

Coding theory: error-detecting and error-correcting codes. Matrices are also used to detect and correct errors that occur when data are transmitted, for example over a network. A **generator matrix** G turns a message vector m into a codeword $c = mG$. A **parity-check matrix** H tests whether a received word c is a codeword: this is the case when $Hc^T = 0$. If $Hc^T \neq 0$, the vector Hc^T, called the **syndrome**, tells us where the error has occurred (this is how Hamming codes work). Codes of this kind detect, and sometimes automatically correct, damage that arises during transmission or storage of data, and so protect the integrity of information, which is one of the aims of information security.

Linear algebra in modern encryption schemes.

Part of modern public-key cryptography, notably **lattice-based cryptography** (which includes schemes believed to resist quantum computers), works with operations on large matrices and vectors. The secret key is a vector, and the public key is obtained by multiplying this vector by a large random matrix and adding some "noise". Recovering the unknown vector from such a perturbed matrix–vector equation is computationally very hard, and the security of the scheme rests on this difficulty.

Image protection and steganography. A digital image is a matrix of pixel values (a colour image is described by three matrices, for the R, G and B channels).

- In **digital watermarking** and **steganography**, hidden data are embedded in selected entries of the pixel matrix (or of a linear transform of it). This is used to protect copyright and to transmit information covertly.
- Methods used in image compression and encryption, such as the **discrete cosine transform** (DCT) and the **singular value decomposition** (SVD), are based directly on linear algebra operations on matrices.

Network security analysis. The connections in a computer network can be represented as a **graph**, which in turn is described by its **adjacency matrix**. Computations with this matrix make it possible to identify vulnerable nodes (those with many links, which are attractive targets), to model possible attack paths such as the spread of a virus, and, in intrusion detection systems (IDS), to represent traffic features as a matrix and detect suspicious activity with machine learning algorithms.

Matrix theory is thus much more than the study of tables of numbers: it provides the mathematical foundation for encryption algorithms, error-correcting

codes, digital signatures and authentication, image protection and network analysis. A specialist in information security therefore needs a working command of matrix operations: addition, multiplication, inversion, computing determinants and so on.

1.1.5 Computing with Python, Maple and Mathcad

The operations of this section are available in every computer algebra system. As an illustration we compute, for $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$, the product AB, the transpose A^T and the value $f(A)$ of $f(x) = 3x - 2x^2 + 7$ found in the examples above. In Python (NumPy) the operator `@` denotes matrix multiplication, whereas `*` multiplies entrywise:

Python

```
import numpy as np
A = np.array([[1, 2], [0, -1]])
B = np.array([[3, 1], [2, 4]])
E = np.eye(2, dtype=int)
print(A @ B)                    # matrix product AB
print(A.T)                      # transpose
print(3*A - 2*(A @ A) + 7*E)  # f(A), f(x)=3x-2x^2+7
```

Output

```
[[ 7  9]
 [-2 -4]]
[[ 1  0]
 [ 2 -1]]
[[8 6]
 [0 2]]
```

In Maple the dot denotes the matrix product, and `A^2` is the matrix power:

Maple

```
with(LinearAlgebra):
A := Matrix([[1, 2], [0, -1]]):
B := Matrix([[3, 1], [2, 4]]):
A . B;
Transpose(A);
3*A - 2*A^2 + 7*IdentityMatrix(2);
```

In Mathcad Prime the matrices are entered with the Matrix button (Ctrl+M), the transpose is the operator `A^T` (Ctrl+1), and the identity matrix is given by `identity(2)`. Writing `+ 7` instead of `+ 7*identity(2)` would add 7 to every entry, which is not the value of the polynomial.

Mathcad

```
A := | 1   2 |        B := | 3  1 |
     | 0  -1 |             | 2  4 |

A*B = |  7   9 |      A^T = | 1   0 |
      | -2  -4 |            | 2  -1 |

3*A - 2*A^2 + 7*identity(2) = | 8  6 |
                              | 0  2 |
```

1.1.6 Exercises on matrix algebra

1. Given a polynomial $f(x)$ and a square matrix A, find $f(A)$.

1. $f(x) = 3x - 2x^2 + 7$ and $A = \begin{pmatrix} -4 & -3 \\ 4 & -3 \end{pmatrix}$.
2. $f(x) = 3x - x^2 + 1$ and $A = \begin{pmatrix} -4 & -3 \\ 2 & 2 \end{pmatrix}$.
3. $f(x) = x - 2x^2 - 3$ and $A = \begin{pmatrix} 4 & 2 \\ -4 & -3 \end{pmatrix}$.
4. $f(x) = 2x - x^2 + 7$ and $A = \begin{pmatrix} -4 & -1 \\ -4 & 4 \end{pmatrix}$.
5. $f(x) = 2x - 3x^2 + 8$ and $A = \begin{pmatrix} -2 & 4 \\ -3 & 0 \end{pmatrix}$.
6. $f(x) = 5x - 2x^2 - 2$ and $A = \begin{pmatrix} -1 & 1 \\ -3 & 4 \end{pmatrix}$.
7. $f(x) = x - x^2 + 1$ and $A = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$.
8. $f(x) = 4x - 4x^2 + 6$ and $A = \begin{pmatrix} 0 & -1 \\ -2 & -1 \end{pmatrix}$.
9. $f(x) = x - 3x^2 + 10$ and $A = \begin{pmatrix} 1 & 3 \\ 0 & -3 \end{pmatrix}$.
10. $f(x) = x - 4x^2$ and $A = \begin{pmatrix} 1 & -2 \\ 3 & 2 \end{pmatrix}$.
11. $f(x) = x - x^2 + 5$ and $A = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$.
12. $f(x) = x - x^2 + 3$ and $A = \begin{pmatrix} 3 & -3 \\ -4 & 0 \end{pmatrix}$.
13. $f(x) = 5x - 4x^2 + 4$ and $A = \begin{pmatrix} 2 & 1 \\ -4 & 3 \end{pmatrix}$.
14. $f(x) = 3x - 2x^2 - 2$ and $A = \begin{pmatrix} 3 & -4 \\ -1 & 0 \end{pmatrix}$.

15. $f(x) = 2x - 2x^2 + 7$ and $A = \begin{pmatrix} 2 & 3 \\ -3 & -2 \end{pmatrix}$.

16. $f(x) = 4x - 4x^2 + 3$ and $A = \begin{pmatrix} -2 & 2 \\ 4 & 0 \end{pmatrix}$.

17. $f(x) = 4x - 3x^2 + 7$ and $A = \begin{pmatrix} -1 & -2 \\ -3 & -2 \end{pmatrix}$.

18. $f(x) = 2x - 2x^2 + 2$ and $A = \begin{pmatrix} -4 & 3 \\ -2 & 0 \end{pmatrix}$.

19. $f(x) = 3x - x^2 - 1$ and $A = \begin{pmatrix} 2 & 4 \\ 1 & 1 \end{pmatrix}$.

20. $f(x) = 2x - x^2 + 9$ and $A = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$.

21. $f(x) = 4x - x^2 + 10$ and $A = \begin{pmatrix} 2 & -4 \\ -1 & -3 \end{pmatrix}$.

22. $f(x) = 2x - 4x^2$ and $A = \begin{pmatrix} -3 & 1 \\ -4 & -3 \end{pmatrix}$.

23. $f(x) = x - 2x^2 - 2$ and $A = \begin{pmatrix} 1 & -4 \\ -3 & -1 \end{pmatrix}$.

24. $f(x) = 5x - 4x^2 - 1$ and $A = \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$.

25. $f(x) = x - x^2 + 10$ and $A = \begin{pmatrix} 3 & 3 \\ 3 & 0 \end{pmatrix}$.

26. $f(x) = x - 2x^2 - 2$ and $A = \begin{pmatrix} 1 & 0 \\ 3 & -2 \end{pmatrix}$.

27. $f(x) = 5x - x^2 + 1$ and $A = \begin{pmatrix} 4 & 1 \\ -2 & 4 \end{pmatrix}$.

28. $f(x) = x - 3x^2 - 3$ and $A = \begin{pmatrix} 0 & 4 \\ 1 & -2 \end{pmatrix}$.

29. $f(x) = 3x - 2x^2 + 5$ and $A = \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix}$.

30. $f(x) = 2x - 2x^2 + 10$ and $A = \begin{pmatrix} 1 & -4 \\ -4 & 0 \end{pmatrix}$.

2. *Show that $A + A^T$ is a symmetric matrix.*

Example 1.1.10. Show that $A + A^T$ is symmetric for $A = \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix}$.

Solution. Here $A^T = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}$, so

$$A + A^T = \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix} + \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 10 \end{pmatrix} = B.$$

Since $B^T = \begin{pmatrix} 2 & 1 \\ 1 & 10 \end{pmatrix} = B$, the matrix $A + A^T$ is symmetric.

In general, for any square matrix A, $(A+A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$.

1. $A = \begin{pmatrix} -2 & -3 & 5 \\ 3 & -1 & 1 \\ 6 & 5 & -1 \end{pmatrix}$.

2. $A = \begin{pmatrix} -5 & -3 \\ -5 & -3 \end{pmatrix}$.

3. $A = \begin{pmatrix} -3 & -1 & -3 \\ 1 & 3 & 3 \\ -6 & 1 & 4 \end{pmatrix}$.

4. $A = \begin{pmatrix} 6 & 4 \\ -5 & 4 \end{pmatrix}$.

5. $A = \begin{pmatrix} 0 & 6 \\ 5 & 6 \end{pmatrix}$.

6. $A = \begin{pmatrix} 1 & -4 \\ 0 & 6 \end{pmatrix}$.

7. $A = \begin{pmatrix} -5 & 6 \\ 5 & 0 \end{pmatrix}$.

8. $A = \begin{pmatrix} 0 & 5 & -5 \\ 5 & -4 & -4 \\ -4 & -6 & -4 \end{pmatrix}$.

9. $A = \begin{pmatrix} 1 & 6 & 4 \\ -4 & 3 & 3 \\ 1 & 4 & -1 \end{pmatrix}$.

10. $A = \begin{pmatrix} 2 & 2 \\ -4 & -6 \end{pmatrix}$.

11. $A = \begin{pmatrix} 6 & 5 \\ 4 & -5 \end{pmatrix}$.

12. $A = \begin{pmatrix} 5 & -4 & 0 \\ -3 & -3 & -6 \\ -2 & -3 & -2 \end{pmatrix}$.

13. $A = \begin{pmatrix} -3 & 6 & 3 \\ -1 & -2 & 2 \\ 0 & -4 & -6 \end{pmatrix}$.

14. $A = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}$.

15. $A = \begin{pmatrix} 2 & -4 & 2 \\ -4 & 2 & 2 \\ -6 & 1 & 6 \end{pmatrix}$.

16. $A = \begin{pmatrix} 3 & -6 \\ 6 & 6 \end{pmatrix}$.

17. $A = \begin{pmatrix} -4 & -4 \\ 1 & 3 \end{pmatrix}$.

18. $A = \begin{pmatrix} 2 & -6 \\ -1 & 4 \end{pmatrix}$.

19. $A = \begin{pmatrix} 2 & 2 & 1 \\ 6 & 6 & -5 \\ 2 & -6 & -3 \end{pmatrix}$.

20. $A = \begin{pmatrix} -2 & -6 \\ 6 & -5 \end{pmatrix}$.

21. $A = \begin{pmatrix} 1 & 2 & -6 \\ 6 & -5 & 1 \\ -1 & 3 & 2 \end{pmatrix}$.

22. $A = \begin{pmatrix} 2 & -3 & 5 \\ -2 & 1 & 2 \\ 2 & 6 & 1 \end{pmatrix}$.

23. $A = \begin{pmatrix} -3 & 5 & 2 \\ -2 & 2 & -3 \\ 1 & -4 & 0 \end{pmatrix}$.

24. $A = \begin{pmatrix} 0 & 1 \\ -1 & -5 \end{pmatrix}$.

25. $A = \begin{pmatrix} 0 & -5 \\ -3 & 4 \end{pmatrix}$.

26. $A = \begin{pmatrix} 6 & -5 \\ 6 & -4 \end{pmatrix}$.

27. $A = \begin{pmatrix} -4 & -2 \\ -4 & 1 \end{pmatrix}$.

28. $A = \begin{pmatrix} 5 & -5 \\ 0 & 1 \end{pmatrix}$.

29. $A = \begin{pmatrix} 4 & -3 \\ -4 & 5 \end{pmatrix}$.

30. $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & -3 & -1 \\ -1 & -5 & 5 \end{pmatrix}$.

3. Express the vector Z as a linear combination of the vectors X and Y.

Example 1.1.11. Let $X = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $Y = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$ and $Z = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix}$. Write Z in the form $Z = \alpha X + \beta Y$.

Solution. The vector equation $\alpha \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix}$ is equivalent to the system

$$\begin{cases} \alpha + 3\beta = 5, \\ 2\alpha = 4, \\ -\alpha + 4\beta = 2. \end{cases}$$

The second equation gives $\alpha = 2$, and then the first gives $2 + 3\beta = 5$, so $\beta = 1$. These values also satisfy the third equation: $-2 + 4 = 2$. Hence $Z = 2X + Y$.

1. $X = \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}$, $Y = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$ and $Z = \begin{pmatrix} 6 \\ 24 \\ 4 \end{pmatrix}$.

2. $X = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$, $Y = \begin{pmatrix} -1 \\ 3 \\ -4 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 \\ -12 \\ -8 \end{pmatrix}$.

3. $X = \begin{pmatrix} -4 \\ -4 \\ -1 \end{pmatrix}$, $Y = \begin{pmatrix} -1 \\ -5 \\ -3 \end{pmatrix}$ and $Z = \begin{pmatrix} 6 \\ -2 \\ -4 \end{pmatrix}$.

4. $X = \begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}$, $Y = \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix}$ and $Z = \begin{pmatrix} 12 \\ 0 \\ 6 \end{pmatrix}$.

5. $X = \begin{pmatrix} -4 \\ -1 \\ -5 \end{pmatrix}$, $Y = \begin{pmatrix} -3 \\ 1 \\ -4 \end{pmatrix}$ and $Z = \begin{pmatrix} 25 \\ 1 \\ 32 \end{pmatrix}$.

6. $X = \begin{pmatrix} -1 \\ -4 \\ 4 \end{pmatrix}$, $Y = \begin{pmatrix} -2 \\ -4 \\ -1 \end{pmatrix}$ and $Z = \begin{pmatrix} -3 \\ 0 \\ -15 \end{pmatrix}$.

7. $X = \begin{pmatrix} -5 \\ 0 \\ 3 \end{pmatrix}$, $Y = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$ and $Z = \begin{pmatrix} 6 \\ 4 \\ -22 \end{pmatrix}$.

8. $X = \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix}$, $Y = \begin{pmatrix} -3 \\ -1 \\ -5 \end{pmatrix}$ and $Z = \begin{pmatrix} -3 \\ 5 \\ 13 \end{pmatrix}$.

9. $X = \begin{pmatrix} -1 \\ 5 \\ -1 \end{pmatrix}$, $Y = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$ and $Z = \begin{pmatrix} 9 \\ 7 \\ -7 \end{pmatrix}$.

10. $X = \begin{pmatrix} 5 \\ -3 \\ -1 \end{pmatrix}$, $Y = \begin{pmatrix} 0 \\ -5 \\ -1 \end{pmatrix}$ and $Z = \begin{pmatrix} -20 \\ 32 \\ 8 \end{pmatrix}$.

11. $X = \begin{pmatrix} -5 \\ 3 \\ 3 \end{pmatrix}$, $Y = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$ and $Z = \begin{pmatrix} -1 \\ 6 \\ 3 \end{pmatrix}$.

12. $X = \begin{pmatrix} -4 \\ 5 \\ 5 \end{pmatrix}$, $Y = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$ and $Z = \begin{pmatrix} -14 \\ 30 \\ 24 \end{pmatrix}$.

13. $X = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$, $Y = \begin{pmatrix} -2 \\ 0 \\ -2 \end{pmatrix}$ and $Z = \begin{pmatrix} -10 \\ 2 \\ 0 \end{pmatrix}$.

14. $X = \begin{pmatrix} 0 \\ -5 \\ -3 \end{pmatrix}$, $Y = \begin{pmatrix} -5 \\ -4 \\ 5 \end{pmatrix}$ and $Z = \begin{pmatrix} 10 \\ -2 \\ -16 \end{pmatrix}$.

15. $X = \begin{pmatrix} -5 \\ -4 \\ 5 \end{pmatrix}$, $Y = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 \\ -8 \\ -25 \end{pmatrix}$.

16. $X = \begin{pmatrix} 2 \\ -3 \\ -3 \end{pmatrix}$, $Y = \begin{pmatrix} -1 \\ 2 \\ -5 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 \\ -1 \\ -8 \end{pmatrix}$.

17. $X = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$, $Y = \begin{pmatrix} -5 \\ -1 \\ -2 \end{pmatrix}$ and $Z = \begin{pmatrix} 13 \\ 2 \\ 2 \end{pmatrix}$.

18. $X = \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix}$, $Y = \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix}$ and $Z = \begin{pmatrix} -16 \\ -2 \\ 5 \end{pmatrix}$.

19. $X = \begin{pmatrix} -2 \\ 3 \\ -5 \end{pmatrix}, Y = \begin{pmatrix} -4 \\ -1 \\ -4 \end{pmatrix}$ and $Z = \begin{pmatrix} -4 \\ -8 \\ 2 \end{pmatrix}$.

20. $X = \begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix}, Y = \begin{pmatrix} -5 \\ -1 \\ -1 \end{pmatrix}$ and $Z = \begin{pmatrix} 11 \\ 8 \\ 2 \end{pmatrix}$.

21. $X = \begin{pmatrix} 4 \\ 3 \\ -3 \end{pmatrix}, Y = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix}$ and $Z = \begin{pmatrix} 19 \\ 15 \\ 0 \end{pmatrix}$.

22. $X = \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}, Y = \begin{pmatrix} 5 \\ -3 \\ -5 \end{pmatrix}$ and $Z = \begin{pmatrix} -2 \\ -10 \\ 6 \end{pmatrix}$.

23. $X = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}, Y = \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix}$ and $Z = \begin{pmatrix} -12 \\ -9 \\ 12 \end{pmatrix}$.

24. $X = \begin{pmatrix} -5 \\ -5 \\ -3 \end{pmatrix}, Y = \begin{pmatrix} 5 \\ 0 \\ -4 \end{pmatrix}$ and $Z = \begin{pmatrix} 5 \\ -10 \\ -18 \end{pmatrix}$.

25. $X = \begin{pmatrix} 3 \\ -5 \\ 5 \end{pmatrix}, Y = \begin{pmatrix} -5 \\ 5 \\ 3 \end{pmatrix}$ and $Z = \begin{pmatrix} -18 \\ 20 \\ 4 \end{pmatrix}$.

26. $X = \begin{pmatrix} -1 \\ -5 \\ 2 \end{pmatrix}, Y = \begin{pmatrix} -4 \\ 3 \\ 3 \end{pmatrix}$ and $Z = \begin{pmatrix} -13 \\ 27 \\ 6 \end{pmatrix}$.

27. $X = \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix}, Y = \begin{pmatrix} -4 \\ -1 \\ -2 \end{pmatrix}$ and $Z = \begin{pmatrix} 8 \\ -1 \\ 3 \end{pmatrix}$.

28. $X = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix}, Y = \begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix}$ and $Z = \begin{pmatrix} -21 \\ -4 \\ -10 \end{pmatrix}$.

29. $X = \begin{pmatrix} -4 \\ 4 \\ -3 \end{pmatrix}$, $Y = \begin{pmatrix} 0 \\ -1 \\ 5 \end{pmatrix}$ and $Z = \begin{pmatrix} 8 \\ -4 \\ -14 \end{pmatrix}$.

30. $X = \begin{pmatrix} 2 \\ -5 \\ 2 \end{pmatrix}$, $Y = \begin{pmatrix} -1 \\ 5 \\ -4 \end{pmatrix}$ and $Z = \begin{pmatrix} -5 \\ 20 \\ -14 \end{pmatrix}$.

Example 1.1.12. Find a and b from $a \begin{pmatrix} 2 \\ -1 \end{pmatrix} + b \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 13 \\ 6 \end{pmatrix}$.

Solution. Comparing the entries of $\begin{pmatrix} 2a + 3b \\ -a + 4b \end{pmatrix} = \begin{pmatrix} 13 \\ 6 \end{pmatrix}$, we obtain the system

$$\begin{cases} 2a + 3b = 13, \\ -a + 4b = 6. \end{cases}$$

From the second equation $a = 4b - 6$. Substituting this into the first gives $2(4b - 6) + 3b = 13$, i.e. $11b = 25$, so $b = \frac{25}{11}$ and $a = 4 \cdot \frac{25}{11} - 6 = \frac{100-66}{11} = \frac{34}{11}$. Thus $a = \frac{34}{11}$, $b = \frac{25}{11}$.

4. Find the unknown numbers a and b from the vector equation.

1. $a \begin{pmatrix} -1 \\ 3 \end{pmatrix} + b \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -18 \\ 45 \end{pmatrix}$.
2. $a \begin{pmatrix} -4 \\ 3 \end{pmatrix} + b \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} -30 \\ 0 \end{pmatrix}$.
3. $a \begin{pmatrix} -5 \\ -1 \end{pmatrix} + b \begin{pmatrix} 2 \\ -4 \end{pmatrix} = \begin{pmatrix} -33 \\ -33 \end{pmatrix}$.
4. $a \begin{pmatrix} 1 \\ -2 \end{pmatrix} + b \begin{pmatrix} -2 \\ -4 \end{pmatrix} = \begin{pmatrix} -5 \\ -22 \end{pmatrix}$.
5. $a \begin{pmatrix} 3 \\ -1 \end{pmatrix} + b \begin{pmatrix} 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 18 \\ -15 \end{pmatrix}$.
6. $a \begin{pmatrix} 0 \\ -2 \end{pmatrix} + b \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -12 \end{pmatrix}$.
7. $a \begin{pmatrix} -3 \\ -5 \end{pmatrix} + b \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} -11 \\ -5 \end{pmatrix}$.
8. $a \begin{pmatrix} -1 \\ -3 \end{pmatrix} + b \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -24 \end{pmatrix}$.
9. $a \begin{pmatrix} -4 \\ 0 \end{pmatrix} + b \begin{pmatrix} -5 \\ 1 \end{pmatrix} = \begin{pmatrix} -68 \\ 8 \end{pmatrix}$.
10. $a \begin{pmatrix} -4 \\ -2 \end{pmatrix} + b \begin{pmatrix} -5 \\ -1 \end{pmatrix} = \begin{pmatrix} -59 \\ -19 \end{pmatrix}$.
11. $a \begin{pmatrix} -4 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -5 \\ 31 \end{pmatrix}$.
12. $a \begin{pmatrix} 1 \\ -1 \end{pmatrix} + b \begin{pmatrix} -5 \\ -1 \end{pmatrix} = \begin{pmatrix} -7 \\ -5 \end{pmatrix}$.

13. $a\begin{pmatrix}5\\-1\end{pmatrix}+b\begin{pmatrix}5\\-3\end{pmatrix}=\begin{pmatrix}55\\-23\end{pmatrix}$.

14. $a\begin{pmatrix}1\\3\end{pmatrix}+b\begin{pmatrix}0\\-2\end{pmatrix}=\begin{pmatrix}7\\5\end{pmatrix}$.

15. $a\begin{pmatrix}-5\\5\end{pmatrix}+b\begin{pmatrix}1\\3\end{pmatrix}=\begin{pmatrix}-22\\34\end{pmatrix}$.

16. $a\begin{pmatrix}-5\\1\end{pmatrix}+b\begin{pmatrix}2\\4\end{pmatrix}=\begin{pmatrix}-8\\28\end{pmatrix}$.

17. $a\begin{pmatrix}2\\-5\end{pmatrix}+b\begin{pmatrix}3\\-3\end{pmatrix}=\begin{pmatrix}35\\-47\end{pmatrix}$.

18. $a\begin{pmatrix}1\\0\end{pmatrix}+b\begin{pmatrix}-1\\-1\end{pmatrix}=\begin{pmatrix}0\\-6\end{pmatrix}$.

19. $a\begin{pmatrix}1\\5\end{pmatrix}+b\begin{pmatrix}-2\\-1\end{pmatrix}=\begin{pmatrix}-7\\37\end{pmatrix}$.

20. $a\begin{pmatrix}-4\\-3\end{pmatrix}+b\begin{pmatrix}5\\-3\end{pmatrix}=\begin{pmatrix}13\\-24\end{pmatrix}$.

21. $a\begin{pmatrix}3\\2\end{pmatrix}+b\begin{pmatrix}3\\-2\end{pmatrix}=\begin{pmatrix}48\\4\end{pmatrix}$.

22. $a\begin{pmatrix}2\\1\end{pmatrix}+b\begin{pmatrix}-3\\3\end{pmatrix}=\begin{pmatrix}-5\\20\end{pmatrix}$.

23. $a\begin{pmatrix}-4\\-3\end{pmatrix}+b\begin{pmatrix}0\\3\end{pmatrix}=\begin{pmatrix}-12\\12\end{pmatrix}$.

24. $a\begin{pmatrix}-2\\0\end{pmatrix}+b\begin{pmatrix}-1\\4\end{pmatrix}=\begin{pmatrix}-12\\8\end{pmatrix}$.

25. $a\begin{pmatrix}1\\1\end{pmatrix}+b\begin{pmatrix}1\\3\end{pmatrix}=\begin{pmatrix}13\\29\end{pmatrix}$.

26. $a\begin{pmatrix}-1\\0\end{pmatrix}+b\begin{pmatrix}-5\\2\end{pmatrix}=\begin{pmatrix}-41\\14\end{pmatrix}$.

27. $a\begin{pmatrix}-3\\5\end{pmatrix}+b\begin{pmatrix}3\\3\end{pmatrix}=\begin{pmatrix}-6\\34\end{pmatrix}$.

28. $a\begin{pmatrix}-1\\-2\end{pmatrix}+b\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}-1\\-10\end{pmatrix}$.

29. $a\begin{pmatrix}-1\\-5\end{pmatrix}+b\begin{pmatrix}-3\\-5\end{pmatrix}=\begin{pmatrix}-35\\-85\end{pmatrix}$.

30. $a\begin{pmatrix}4\\2\end{pmatrix}+b\begin{pmatrix}-5\\-4\end{pmatrix}=\begin{pmatrix}-13\\-20\end{pmatrix}$.

Example 1.1.13. A matrix has **12** entries. What sizes can it have?

Solution. The size $m \times n$ must satisfy $mn = 12$, so we list all ordered pairs of positive divisors of 12 whose product is 12:

$$1 \times 12, \quad 2 \times 6, \quad 3 \times 4, \quad 4 \times 3, \quad 6 \times 2, \quad 12 \times 1.$$

There are therefore six possible sizes.

5. A matrix has a entries. What sizes can it have?

1. $a = 72$
2. $a = 36$
3. $a = 30$
4. $a = 42$
5. $a = 64$
6. $a = 28$
7. $a = 88$
8. $a = 44$

9. $a = 48$
10. $a = 75$
11. $a = 54$
12. $a = 40$
13. $a = 66$
14. $a = 63$
15. $a = 45$
16. $a = 12$
17. $a = 16$
18. $a = 60$
19. $a = 18$
20. $a = 96$
21. $a = 84$
22. $a = 70$
23. $a = 56$
24. $a = 80$
25. $a = 24$
26. $a = 90$
27. $a = 20$
28. $a = 81$
29. $a = 32$
30. $a = 50$

Example 1.1.14. Let $A = \begin{pmatrix} 3 & -1 & 2 \\ 0 & 4 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 & -1 \\ -3 & 0 & 5 \end{pmatrix}$, $\lambda = 2$ and $\mu = -1$. Find $\lambda A + \mu B$.

Solution. We have $2A = \begin{pmatrix} 6 & -2 & 4 \\ 0 & 8 & -6 \end{pmatrix}$ and $(-1)B = \begin{pmatrix} -1 & -2 & 1 \\ 3 & 0 & -5 \end{pmatrix}$, hence

$$\lambda A + \mu B = 2A - B = \begin{pmatrix} 6-1 & -2-2 & 4+1 \\ 0+3 & 8+0 & -6-5 \end{pmatrix} = \begin{pmatrix} 5 & -4 & 5 \\ 3 & 8 & -11 \end{pmatrix}.$$

6. Given matrices A, B and numbers λ, μ, find $\lambda A + \mu B$.

1. $A = \begin{pmatrix} 3 & 4 \\ -2 & 1 \\ -1 & -2 \end{pmatrix}, B = \begin{pmatrix} 4 & -5 \\ -5 & 3 \\ -1 & 2 \end{pmatrix}, \lambda = 1, \mu = -1.$

2. $A = \begin{pmatrix} 3 & -2 & 3 \\ -2 & -5 & 1 \\ 5 & -1 & -5 \end{pmatrix}, B = \begin{pmatrix} -5 & -2 & 2 \\ 5 & 5 & 1 \\ -4 & -1 & -2 \end{pmatrix}, \lambda = 2, \mu = 1.$

3. $A = \begin{pmatrix} 2 & -5 & 0 \\ 1 & 0 & 5 \end{pmatrix}, B = \begin{pmatrix} 1 & -2 & -5 \\ -1 & 3 & -4 \end{pmatrix}, \lambda = -1, \mu = 3.$

4. $A = \begin{pmatrix} -1 & -2 & -2 \\ 2 & -2 & -1 \end{pmatrix}, B = \begin{pmatrix} -1 & -4 & 4 \\ 2 & 4 & -3 \end{pmatrix}, \lambda = -1, \mu = 3.$

5. $A = \begin{pmatrix} 5 & -5 & 4 \\ -3 & 1 & -5 \\ -2 & -5 & 4 \end{pmatrix}, B = \begin{pmatrix} -3 & 1 & -5 \\ -5 & -3 & 1 \\ 2 & 0 & -4 \end{pmatrix}, \lambda = -3, \mu = -2.$

6. $A = \begin{pmatrix} -2 & -3 \\ 5 & 3 \\ 2 & -5 \end{pmatrix}, B = \begin{pmatrix} -1 & 5 \\ 1 & 0 \\ 0 & 2 \end{pmatrix}, \lambda = -2, \mu = -3.$

7. $A = \begin{pmatrix} -4 & -1 \\ -4 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & -4 \\ 3 & -2 \end{pmatrix}, \lambda = 2, \mu = 1.$

8. $A = \begin{pmatrix} 1 & -4 \\ -5 & 2 \\ -2 & 0 \end{pmatrix}, B = \begin{pmatrix} 3 & 2 \\ -2 & 0 \\ 0 & 2 \end{pmatrix}, \lambda = -4, \mu = 2.$

9. $A = \begin{pmatrix} 5 & 1 & -5 \\ 1 & -5 & 2 \end{pmatrix}, B = \begin{pmatrix} -4 & -5 & -1 \\ -2 & -4 & 4 \end{pmatrix}, \lambda = 1, \mu = 1.$

10. $A = \begin{pmatrix} 0 & 4 \\ -5 & -1 \\ 0 & -1 \end{pmatrix}, B = \begin{pmatrix} -1 & -5 \\ 4 & 5 \\ -4 & -5 \end{pmatrix}, \lambda = -1, \mu = -3.$

11. $A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & -3 \\ 2 & -3 & -5 \end{pmatrix}, B = \begin{pmatrix} -1 & -3 & 4 \\ -2 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix}, \lambda = -3, \mu = 4.$

12. $A = \begin{pmatrix} 1 & -3 & -2 \\ 1 & -4 & 5 \end{pmatrix}, B = \begin{pmatrix} -5 & 2 & 3 \\ 3 & 0 & -3 \end{pmatrix}, \lambda = 2, \mu = -3.$

13. $A = \begin{pmatrix} -1 & 4 \\ -4 & -2 \end{pmatrix}, B = \begin{pmatrix} -4 & 1 \\ 2 & 2 \end{pmatrix}, \lambda = -2, \mu = -1.$

14. $A = \begin{pmatrix} 1 & 2 & 4 \\ 5 & -2 & 3 \end{pmatrix}, B = \begin{pmatrix} 5 & -4 & -1 \\ -1 & -1 & 4 \end{pmatrix}, \lambda = 1, \mu = -1.$

15. $A = \begin{pmatrix} -2 & -3 & -2 \\ -2 & -3 & -1 \\ 4 & -2 & 0 \end{pmatrix}, B = \begin{pmatrix} -4 & 1 & -1 \\ -2 & 3 & 3 \\ -2 & 5 & -4 \end{pmatrix}, \lambda = 3, \mu = -4.$

16. $A = \begin{pmatrix} -5 & 2 \\ -2 & 2 \end{pmatrix}, B = \begin{pmatrix} 0 & -5 \\ -1 & -2 \end{pmatrix}, \lambda = -3, \mu = -4.$

17. $A = \begin{pmatrix} 4 & 4 & -2 \\ -4 & 0 & 3 \end{pmatrix}, B = \begin{pmatrix} -3 & 2 & 4 \\ -1 & 5 & -5 \end{pmatrix}, \lambda = -3, \mu = 1.$

18. $A = \begin{pmatrix} -5 & 0 & 0 \\ -3 & -5 & -2 \end{pmatrix}, B = \begin{pmatrix} -1 & -5 & 4 \\ 5 & -2 & -5 \end{pmatrix}, \lambda = 1, \mu = 2.$

19. $A = \begin{pmatrix} -3 & 4 \\ -1 & -4 \\ -2 & -5 \end{pmatrix}, B = \begin{pmatrix} 2 & 3 \\ 2 & -4 \\ 1 & -4 \end{pmatrix}, \lambda = 2, \mu = 4.$

20. $A = \begin{pmatrix} 5 & 3 & -4 \\ 5 & -3 & 1 \end{pmatrix}, B = \begin{pmatrix} -1 & 1 & -1 \\ 5 & -1 & 1 \end{pmatrix}, \lambda = -4, \mu = 1.$

21. $A = \begin{pmatrix} 1 & -5 & 0 \\ 5 & -2 & 1 \\ 1 & -2 & -5 \end{pmatrix}, B = \begin{pmatrix} 1 & -3 & 1 \\ -4 & -4 & 1 \\ 4 & 0 & 2 \end{pmatrix}, \lambda = -2, \mu = -2.$

22. $A = \begin{pmatrix} -5 & 3 \\ -3 & 5 \end{pmatrix}, B = \begin{pmatrix} 1 & -4 \\ 4 & 4 \end{pmatrix}, \lambda = 1, \mu = 4.$

23. $A = \begin{pmatrix} -3 & 0 & -1 \\ -3 & 3 & -3 \end{pmatrix}, B = \begin{pmatrix} -4 & -4 & 1 \\ 2 & -2 & -1 \end{pmatrix}, \lambda = -2, \mu = -4.$

24. $A = \begin{pmatrix} 0 & -5 & 4 \\ 5 & 1 & -4 \\ 4 & -3 & 5 \end{pmatrix}, B = \begin{pmatrix} -2 & 4 & 1 \\ 4 & -2 & 2 \\ -3 & 4 & -2 \end{pmatrix}, \lambda = -4, \mu = 2.$

25. $A = \begin{pmatrix} 1 & 0 & -4 \\ -3 & -2 & -2 \end{pmatrix}, B = \begin{pmatrix} -5 & 3 & 5 \\ -5 & 5 & 0 \end{pmatrix}, \lambda = -3, \mu = 2.$

26. $A = \begin{pmatrix} 3 & 5 & -1 \\ 5 & 1 & -1 \\ 4 & -2 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 5 & 0 \\ 2 & 3 & 2 \\ -3 & -5 & -5 \end{pmatrix}, \lambda = 3, \mu = 3.$

27. $A = \begin{pmatrix} 2 & 4 & 2 \\ -3 & 2 & 1 \end{pmatrix}, B = \begin{pmatrix} -4 & -4 & -3 \\ 0 & 1 & 0 \end{pmatrix}, \lambda = -3, \mu = 3.$

28. $A = \begin{pmatrix} -5 & 5 \\ -3 & -4 \end{pmatrix}, B = \begin{pmatrix} 0 & 3 \\ -4 & -5 \end{pmatrix}, \lambda = 4, \mu = 2.$

29. $A = \begin{pmatrix} -5 & -4 & 4 \\ -4 & -2 & -3 \end{pmatrix}, B = \begin{pmatrix} 2 & -1 & -3 \\ 5 & -2 & -4 \end{pmatrix}, \lambda = 1, \mu = -2.$

30. $A = \begin{pmatrix} 4 & -1 \\ 2 & -3 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -2 \\ 4 & -1 \\ 4 & 3 \end{pmatrix}$, $\lambda = -1, \mu = 1$.

7. Applied problems.

1. **(Hill cipher: encryption.)** Given the key matrix $K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$ and the plaintext block $p = \begin{pmatrix} 9 \\ 17 \end{pmatrix}$, find the encrypted block $c = Kp \pmod{26}$.

2. **(Hill cipher: encryption.)** Compute $c = Kp \pmod{26}$ for $K = \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$ and $p = \begin{pmatrix} 4 \\ 19 \end{pmatrix}$.

3. **(Hill cipher: encryption.)** Find the encrypted block $c = Kp \pmod{26}$ for $K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$ and $p = \begin{pmatrix} 2 \\ 21 \end{pmatrix}$.

4. **(Hill cipher: encryption.)** Find $c = Kp \pmod{26}$ for $K = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$ and $p = \begin{pmatrix} 13 \\ 4 \end{pmatrix}$.

5. **(Validity of a key.)** For the matrix $K = \begin{pmatrix} 4 & 9 \\ 6 & 13 \end{pmatrix}$ compute the determinant and $\gcd(\det K, 26)$. Decide, with justification, whether K can be used as a key for the Hill cipher.

6. **(Validity of a key.)** Carry out the same test as in Problem 5 for the matrix $K = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$ and state your conclusion.

7. **(Coding theory: encoding.)** A generator matrix of the binary $(7, 4)$ Hamming code is

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Find the codeword $c = mG \pmod 2$ for the message vector $m = (1;0;1;1)$.

8. **(Coding theory: error detection.)** In another standard form of the binary $(7,4)$ Hamming code, in which the jth column of the parity-check matrix is the binary representation of j (least significant bit in the first row), the parity-check matrix is

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

The word $c = (1;0;1;1;1;0;1)$ has been received. Compute the syndrome $s = Hc^T \pmod 2$ and use it to decide whether the received word contains an error and, if so, in which bit.

9. **(Network security: the adjacency matrix.)**

A network with five nodes $S_1, \ldots, S_5$ has the adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

Compute A^2 and use it to find the number of paths of length 2 between each pair of nodes. Determine also which nodes have the most connections (and may therefore be the most vulnerable points of the network).

10. **(A matrix polynomial.)** For the key matrix $K = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ compute $f(K) = K^2 - 3K + E$, and use the result to show that $K^{-1} = 3E - K$.

1.2 The concept of a determinant. Determinants of the second and third order

Definition 1.2.1. With every square matrix of order n we associate a single number called its **determinant**. The determinant of a matrix A is denoted by $\det A$, $|A|$ or Δ:

$$\det A = |A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

Determinants are defined **only for square matrices**.

1.2.1 Second-order determinants

Definition 1.2.2. The determinant of a 2×2 matrix is given by

$$\det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} \cdot a_{22} - a_{12} \cdot a_{21}.$$

In words, we subtract the product of the entries on the **secondary diagonal** from the product of the entries on the **main diagonal**.

Example 1.2.1. $\begin{vmatrix} 3 & -2 \\ 5 & 4 \end{vmatrix} = 3 \cdot 4 - (-2) \cdot 5 = 12 + 10 = 22.$

Example 1.2.2. $\begin{vmatrix} -1 & 7 \\ 2 & -3 \end{vmatrix} = (-1) \cdot (-3) - 7 \cdot 2 = 3 - 14 = -11.$

Example 1.2.3. $\begin{vmatrix} 0 & 5 \\ -4 & 1 \end{vmatrix} = 0 \cdot 1 - 5 \cdot (-4) = 0 + 20 = 20.$

1.2.2 Third-order determinants

Definition 1.2.3. Two methods for evaluating a 3×3 determinant are in common use: **Sarrus' rule** and **expansion along the first row** (the Laplace expansion).

Sarrus' rule (the rule of diagonals). The scheme behind Sarrus' rule for a 3×3 determinant is shown in Figure 1.

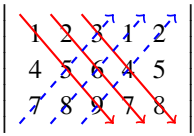


Figure 1.

In practice one copies the first two columns to the right of the matrix. In the resulting 3×5 array the products along the three diagonals running down from left to right are taken with the sign "+" and **added**, while the products along the three diagonals running up from left to right are taken with the sign "–" and **subtracted**. The same products can be read off the triangle scheme of Figure 2.

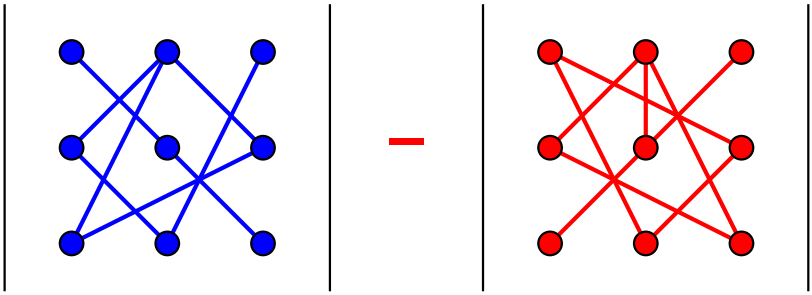


Figure 2.

Either scheme leads to the formula

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \times a_{22} \times a_{33} + a_{12} \times a_{23} \times a_{31} + a_{13} \times a_{21} \times a_{32} -$$

$$- a_{13} \times a_{22} \times a_{31} - a_{12} \times a_{21} \times a_{33} - a_{11} \times a_{23} \times a_{32}.$$

Example 1.2.4. We evaluate by Sarrus' rule

$$\begin{vmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{vmatrix}.$$

The diagonals with the sign "+" give $2\cdot(-1)\cdot 1+1\cdot 2\cdot 4+3\cdot 0\cdot 3 = -2+8+0 = 6$, and those with the sign "−" give $3\cdot(-1)\cdot 4+1\cdot 0\cdot 1+2\cdot 2\cdot 3 = -12+0+12 = 0$. Hence

$$\det A = 6 - 0 = 6.$$

1.2.3 Basic properties of determinants

The following properties hold for determinants of any order.

1) **Transposition:** $\det A^T = \det A$; interchanging the roles of rows and columns does not change the value of a determinant.

2) **Change of sign:** interchanging two rows (or two columns) changes the sign of the determinant.

3) **Zero row (column):** if all entries of some row (or column) are zero, the determinant equals zero.

4) **Common factor:** if all entries of one row (or column) are multiplied by a number k, the determinant is multiplied by k. In other words, a common factor of a row can be taken outside the determinant sign. Since kA has

every one of its n rows multiplied by k,

$$\det(kA) = k^n \det A \quad \text{(for a matrix of order } n\text{)}.$$

5) **Equal rows:** if two rows (or columns) of a determinant are equal, the determinant equals zero.

6) **Proportional rows:** if one row is proportional to another, the determinant equals zero.

7) **Adding a multiple of a row:** adding to one row a multiple k of another row does not change the value of the determinant.

8) **Determinant of a product:** $\det(AB) = \det A \cdot \det B$.

Example 1.2.5. (Property 4).

$$\begin{vmatrix} 6 & 3 \\ 4 & 2 \end{vmatrix} = 3 \cdot \begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 3 \cdot (2 \cdot 2 - 1 \cdot 4) = 3 \cdot 0 = 0.$$

Here we took the common factor 3 out of the first row. In the new determinant the second row is twice the first (Property 6), so the determinant is zero.

Example 1.2.6. (Property 7). We use the properties to evaluate

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \overset{S_2 - 4S_1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{vmatrix} \overset{S_3 - 7S_1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{vmatrix}.$$

Here S_i denotes the ith row. The second and third rows are now proportional ($S_3 = 2S_2$), and therefore $\det A = 0$.

1.2.4 Determinants in information security

Determinants appear naturally in information security and cryptography.

Recovering a key from a linear system. If the system of equations written for the unknown entries of a key has a nonzero determinant, the key is determined uniquely. This is exactly what makes linear ciphers vulnerable to known-plaintext attacks.

The Hill cipher. A key matrix K is admissible only if its determinant satisfies $\gcd(\det K, 26) = 1$. When this condition fails, the inverse matrix $K^{-1} \pmod{26}$ does not exist and decryption is impossible. Two cases are typical. If $\det K \equiv 0 \pmod{26}$, the matrix is **singular** modulo 26 and cannot serve as a key at all. If $\gcd(\det K, 26) \neq 1$, for instance when $\det K$ is even or divisible by 13, the number $\det K$ has no inverse modulo 26. Choosing a key therefore requires evaluating determinants quickly, and for small keys this rests directly on the formulas for determinants of order two and three.

Example 1.2.7. For $K = \begin{pmatrix} 3 & 5 \\ 2 & 7 \end{pmatrix}$ we have $\det K = 3 \cdot 7 - 5 \cdot 2 = 21 - 10 = 11$ and $\gcd(11, 26) = 1$, so K is an admissible Hill cipher key. For $K' = \begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}$ we get $\det K' = 12 - 12 = 0$, and this matrix is **not admissible** as a key.

Example 1.2.8. Compute $\det K$ for $K = \begin{pmatrix} 5 & 3 \\ 2 & 7 \end{pmatrix}$ and decide whether K can be used as a Hill cipher key.

Solution. $\det K = 5 \cdot 7 - 3 \cdot 2 = 35 - 6 = 29$. By the Euclidean algorithm, $\gcd(29, 26) = \gcd(26, 3) = \gcd(3, 2) = \gcd(2, 1) = 1$. The condition $\gcd(\det K, 26) = 1$ holds, so K is an admissible Hill cipher key.

Checking that an inverse exists. Most cryptographic systems need to "undo" a key: to decrypt a message, to verify a signature, and so on. This requires an inverse matrix, which exists only when $\det A \neq 0$. Computing the determinant is therefore the first check in the design of such a system.

Determinants in error-correcting codes. In coding theory the rows of the generator and parity-check matrices must be linearly independent. To verify this one computes the determinants of suitable square submatrices (minors). If

the required minors are nonzero, the code is correctly constructed and is able to detect errors.

Adjacency matrices in network security. In the analysis of network topology, the determinant and other spectral characteristics (eigenvalues) of the adjacency matrix are used to assess how well connected and how resilient a network is. For example, by **Kirchhoff's matrix-tree theorem** the number of spanning trees of a network equals any cofactor of its Laplacian matrix, that is, a certain determinant. This number measures how many backup routes the network has.

1.2.5 Computing with Python, Maple and Mathcad

Determinants of numerical matrices are easily checked by computer. The examples below evaluate the third-order determinant from the Sarrus example and test the Hill cipher key $K = \begin{pmatrix} 5 & 3 \\ 2 & 7 \end{pmatrix}$. In Python we use SymPy, which works with exact integers:

Python

```
from math import gcd
from sympy import Matrix
A = Matrix([[2, 1, 3], [0, -1, 2], [4, 3, 1]])
print(A.det())
K = Matrix([[5, 3], [2, 7]])
d = K.det()
print(d, gcd(int(d), 26))
```

Output

```
6
29 1
```

In Maple the same computation uses the `LinearAlgebra` package:

Maple

```
with(LinearAlgebra):
A := Matrix([[2, 1, 3], [0, -1, 2], [4, 3, 1]]):
Determinant(A);
                                6
K := Matrix([[5, 3], [2, 7]]):
igcd(Determinant(K), 26);
                                1
```

In Mathcad Prime a matrix is entered through the matrix template (Ctrl+M), and the determinant is obtained with the operator $|A|$ from the Matrices/Vectors toolbar (the function `gcd` is built in):

Mathcad

```
     | 2   1  3 |
A := | 0  -1  2 |        |A| = 6
     | 4   3  1 |

K := | 5  3 |            |K| = 29
     | 2  7 |
gcd(|K|, 26) = 1
```

1.2.6 Exercises on determinants

1. Evaluate the second-order determinants.

1. $\begin{vmatrix} 3 & -2 \\ 5 & 4 \end{vmatrix}$
2. $\begin{vmatrix} -1 & 7 \\ 2 & -3 \end{vmatrix}$
3. $\begin{vmatrix} 0 & 5 \\ -4 & 1 \end{vmatrix}$
4. $\begin{vmatrix} 6 & -3 \\ -2 & 8 \end{vmatrix}$
5. $\begin{vmatrix} -5 & 1 \\ 3 & -7 \end{vmatrix}$
6. $\begin{vmatrix} 2 & 4 \\ -1 & 6 \end{vmatrix}$
7. $\begin{vmatrix} 7 & 0 \\ -3 & 5 \end{vmatrix}$
8. $\begin{vmatrix} -4 & 2 \\ 9 & -1 \end{vmatrix}$
9. $\begin{vmatrix} 1 & -8 \\ 3 & 2 \end{vmatrix}$
10. $\begin{vmatrix} -6 & 3 \\ 4 & -5 \end{vmatrix}$
11. $\begin{vmatrix} 8 & -1 \\ 2 & 7 \end{vmatrix}$
12. $\begin{vmatrix} -3 & 9 \\ 1 & -4 \end{vmatrix}$
13. $\begin{vmatrix} 5 & 2 \\ -6 & 3 \end{vmatrix}$
14. $\begin{vmatrix} -2 & -5 \\ 4 & 1 \end{vmatrix}$
15. $\begin{vmatrix} 4 & 7 \\ -3 & -2 \end{vmatrix}$
16. $\begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$
17. $\begin{vmatrix} -7 & 4 \\ 5 & 3 \end{vmatrix}$
18. $\begin{vmatrix} 9 & -6 \\ 2 & 8 \end{vmatrix}$
19. $\begin{vmatrix} 3 & 5 \\ -8 & -4 \end{vmatrix}$
20. $\begin{vmatrix} -4 & -3 \\ 7 & 6 \end{vmatrix}$
21. $\begin{vmatrix} 2 & -9 \\ 1 & 5 \end{vmatrix}$
22. $\begin{vmatrix} 6 & 1 \\ -5 & -8 \end{vmatrix}$
23. $\begin{vmatrix} -3 & 4 \\ 8 & -2 \end{vmatrix}$
24. $\begin{vmatrix} 10 & -7 \\ 3 & 1 \end{vmatrix}$
25. $\begin{vmatrix} -8 & 5 \\ 6 & -9 \end{vmatrix}$
26. $\begin{vmatrix} 5 & -4 \\ -2 & 3 \end{vmatrix}$
27. $\begin{vmatrix} -1 & 6 \\ 7 & -5 \end{vmatrix}$
28. $\begin{vmatrix} 4 & 3 \\ -9 & 2 \end{vmatrix}$
29. $\begin{vmatrix} -6 & -2 \\ 1 & 8 \end{vmatrix}$
30. $\begin{vmatrix} 7 & -5 \\ -4 & 3 \end{vmatrix}$

2. Evaluate the third-order determinants by Sarrus' rule.

1. $\begin{vmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{vmatrix}$
2. $\begin{vmatrix} 1 & -3 & 2 \\ 5 & 0 & -1 \\ -2 & 4 & 3 \end{vmatrix}$
3. $\begin{vmatrix} 3 & 2 & -1 \\ 1 & 4 & 0 \\ -2 & 5 & 3 \end{vmatrix}$
4. $\begin{vmatrix} -1 & 2 & 3 \\ 4 & 0 & -2 \\ 1 & 5 & 1 \end{vmatrix}$
5. $\begin{vmatrix} 5 & -1 & 2 \\ 3 & 4 & -3 \\ 0 & 2 & 1 \end{vmatrix}$
6. $\begin{vmatrix} 2 & 0 & -4 \\ -1 & 3 & 5 \\ 6 & -2 & 1 \end{vmatrix}$
7. $\begin{vmatrix} -3 & 1 & 2 \\ 0 & 5 & -4 \\ 4 & -1 & 3 \end{vmatrix}$
8. $\begin{vmatrix} 1 & 2 & -3 \\ -4 & 0 & 1 \\ 3 & 5 & 2 \end{vmatrix}$
9. $\begin{vmatrix} 4 & -2 & 1 \\ 2 & 3 & -5 \\ -1 & 0 & 6 \end{vmatrix}$
10. $\begin{vmatrix} -2 & 5 & 0 \\ 1 & -3 & 4 \\ 3 & 2 & -1 \end{vmatrix}$
11. $\begin{vmatrix} 0 & 3 & -2 \\ 5 & 1 & 4 \\ -3 & -4 & 2 \end{vmatrix}$
12. $\begin{vmatrix} 3 & -4 & 1 \\ -2 & 0 & 5 \\ 1 & 3 & -3 \end{vmatrix}$
13. $\begin{vmatrix} 1 & 0 & 5 \\ -3 & 2 & -1 \\ 4 & -4 & 3 \end{vmatrix}$
14. $\begin{vmatrix} -5 & 2 & 1 \\ 3 & -1 & 0 \\ 2 & 4 & -3 \end{vmatrix}$
15. $\begin{vmatrix} 2 & -1 & 4 \\ 0 & 3 & -2 \\ 5 & 1 & -1 \end{vmatrix}$
16. $\begin{vmatrix} 4 & 1 & -3 \\ -1 & -2 & 5 \\ 3 & 0 & 2 \end{vmatrix}$
17. $\begin{vmatrix} -2 & 3 & 1 \\ 4 & -5 & 2 \\ 0 & 1 & -4 \end{vmatrix}$
18. $\begin{vmatrix} 1 & -2 & 0 \\ 3 & 4 & -1 \\ -5 & 2 & 3 \end{vmatrix}$
19. $\begin{vmatrix} 6 & 0 & -3 \\ -1 & 2 & 5 \\ 4 & -3 & 1 \end{vmatrix}$
20. $\begin{vmatrix} -3 & 4 & 2 \\ 1 & -1 & -5 \\ 5 & 3 & 0 \end{vmatrix}$
21. $\begin{vmatrix} 0 & -5 & 3 \\ 2 & 1 & -4 \\ -1 & 6 & 2 \end{vmatrix}$
22. $\begin{vmatrix} 5 & 3 & -2 \\ -4 & 0 & 1 \\ 2 & -1 & 4 \end{vmatrix}$
23. $\begin{vmatrix} -1 & 4 & -3 \\ 0 & 2 & 5 \\ 3 & -1 & 1 \end{vmatrix}$
24. $\begin{vmatrix} 2 & -3 & 5 \\ 4 & 1 & 0 \\ -1 & -2 & 3 \end{vmatrix}$
25. $\begin{vmatrix} 3 & 0 & 1 \\ -2 & 5 & -4 \\ 4 & 1 & 2 \end{vmatrix}$
26. $\begin{vmatrix} -4 & 1 & 3 \\ 2 & -3 & 0 \\ 5 & 4 & -2 \end{vmatrix}$
27. $\begin{vmatrix} 1 & 5 & -1 \\ -3 & 0 & 4 \\ 2 & -2 & 3 \end{vmatrix}$
28. $\begin{vmatrix} 0 & -1 & 4 \\ 3 & 2 & -5 \\ -4 & 1 & 0 \end{vmatrix}$
29. $\begin{vmatrix} -2 & 0 & 3 \\ 1 & -4 & 5 \\ 4 & 2 & -1 \end{vmatrix}$
30. $\begin{vmatrix} 3 & -2 & 0 \\ -1 & 5 & 4 \\ 2 & 0 & -3 \end{vmatrix}$

3. Evaluate the third-order determinants by expansion along the first row.

1. $\begin{vmatrix} 1 & 3 & -2 \\ 4 & 0 & 5 \\ -1 & 2 & 3 \end{vmatrix}$

2. $\begin{vmatrix} -2 & 1 & 4 \\ 3 & -5 & 0 \\ 1 & 2 & -3 \end{vmatrix}$

3. $\begin{vmatrix} 5 & -3 & 1 \\ 0 & 2 & -4 \\ -1 & 3 & 5 \end{vmatrix}$

4. $\begin{vmatrix} 3 & 0 & -1 \\ -2 & 4 & 3 \\ 5 & -1 & 2 \end{vmatrix}$

5. $\begin{vmatrix} -4 & 2 & 0 \\ 1 & -3 & 5 \\ 3 & 4 & -2 \end{vmatrix}$

6. $\begin{vmatrix} 0 & 5 & -3 \\ -2 & 1 & 4 \\ 4 & -1 & 2 \end{vmatrix}$

7. $\begin{vmatrix} 2 & -4 & 3 \\ 5 & 0 & -1 \\ -3 & 2 & 4 \end{vmatrix}$

8. $\begin{vmatrix} -1 & 3 & 5 \\ 4 & -2 & 0 \\ 0 & 1 & -3 \end{vmatrix}$

9. $\begin{vmatrix} 6 & 1 & -2 \\ -3 & 0 & 4 \\ 2 & 5 & 1 \end{vmatrix}$

10. $\begin{vmatrix} 1 & -1 & 0 \\ 0 & 3 & -5 \\ -2 & 4 & 1 \end{vmatrix}$

11. $\begin{vmatrix} -3 & 0 & 4 \\ 2 & 5 & -1 \\ 1 & -2 & 3 \end{vmatrix}$

12. $\begin{vmatrix} 4 & -2 & 5 \\ -1 & 3 & 0 \\ 2 & 0 & -4 \end{vmatrix}$

13. $\begin{vmatrix} 0 & 4 & 1 \\ 5 & -3 & 2 \\ -2 & 1 & -5 \end{vmatrix}$

14. $\begin{vmatrix} -5 & 1 & 0 \\ 3 & -2 & 4 \\ 0 & 5 & -1 \end{vmatrix}$

15. $\begin{vmatrix} 2 & 0 & -3 \\ -4 & 1 & 5 \\ 3 & -2 & 0 \end{vmatrix}$

16. $\begin{vmatrix} 1 & -5 & 2 \\ 0 & 4 & -3 \\ -3 & 1 & 0 \end{vmatrix}$

17. $\begin{vmatrix} -2 & 4 & -1 \\ 5 & 0 & 3 \\ 1 & -3 & 2 \end{vmatrix}$

18. $\begin{vmatrix} 3 & -1 & 4 \\ 0 & 2 & -5 \\ -4 & 3 & 1 \end{vmatrix}$

19. $\begin{vmatrix} -1 & 0 & 3 \\ 2 & -4 & 1 \\ 5 & 3 & -2 \end{vmatrix}$

20. $\begin{vmatrix} 4 & 3 & 0 \\ -2 & -1 & 5 \\ 0 & 4 & -3 \end{vmatrix}$

21. $\begin{vmatrix} 0 & -2 & 5 \\ 3 & 1 & -4 \\ -1 & 0 & 2 \end{vmatrix}$

22. $\begin{vmatrix} 5 & 1 & -4 \\ -3 & 0 & 2 \\ 4 & -5 & 1 \end{vmatrix}$

23. $\begin{vmatrix} -4 & 3 & 0 \\ 1 & -1 & -2 \\ 0 & 5 & 4 \end{vmatrix}$

24. $\begin{vmatrix} 2 & 0 & 1 \\ -5 & 3 & -4 \\ 1 & -1 & 5 \end{vmatrix}$

25. $\begin{vmatrix} -1 & -3 & 2 \\ 4 & 0 & -5 \\ 3 & 2 & 1 \end{vmatrix}$

26. $\begin{vmatrix} 3 & 2 & -5 \\ 0 & -4 & 1 \\ -2 & 3 & 0 \end{vmatrix}$

27. $\begin{vmatrix} -3 & 5 & 1 \\ 2 & 0 & -4 \\ 0 & -1 & 3 \end{vmatrix}$

28. $\begin{vmatrix} 1 & -4 & 0 \\ 0 & 5 & -2 \\ 3 & -1 & 4 \end{vmatrix}$

29. $\begin{vmatrix} -5 & 0 & 2 \\ 4 & -3 & 1 \\ -1 & 2 & 0 \end{vmatrix}$

30. $\begin{vmatrix} 0 & 1 & -4 \\ -2 & 3 & 5 \\ 4 & 0 & -1 \end{vmatrix}$

4. Use the properties of determinants to evaluate the following determinants.

1. $\begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix}$

2. $\begin{vmatrix} -5 & 10 \\ 1 & -2 \end{vmatrix}$

3. $\begin{vmatrix} 3 & 0 & -6 \\ 1 & 2 & 4 \\ -2 & 1 & 5 \end{vmatrix}$

4. $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

5. $\begin{vmatrix} 5 & -5 \\ -3 & 3 \end{vmatrix}$

6. $\begin{vmatrix} 2 & 1 & -3 \\ 0 & 0 & 0 \\ 4 & -2 & 5 \end{vmatrix}$

7. $\begin{vmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ 4 & -1 & 3 \end{vmatrix}$

8. $\begin{vmatrix} 6 & -3 & 9 \\ 2 & 4 & -1 \\ 0 & 5 & 3 \end{vmatrix}$

9. $\begin{vmatrix} -4 & 8 \\ 7 & -14 \end{vmatrix}$

10. $\begin{vmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 1 & -3 & 5 \end{vmatrix}$

11. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{vmatrix}$

12. $\begin{vmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \\ 4 & 0 & -2 \end{vmatrix}$

13. $\begin{vmatrix} -2 & 6 \\ 1 & -3 \end{vmatrix}$

14. $\begin{vmatrix} 0 & 0 & 0 \\ 3 & -1 & 5 \\ 2 & 4 & -7 \end{vmatrix}$

15. $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ -1 & 3 & 5 \end{vmatrix}$

16. $\begin{vmatrix} 10 & 5 \\ -4 & -2 \end{vmatrix}$

17. $\begin{vmatrix} 3 & -1 & 2 \\ 0 & 4 & -5 \\ 6 & -2 & 4 \end{vmatrix}$

18. $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{vmatrix}$

19. $\begin{vmatrix} 7 & 0 \\ 0 & -3 \end{vmatrix}$

20. $\begin{vmatrix} 4 & 2 & -6 \\ 0 & 3 & 1 \\ -2 & -1 & 3 \end{vmatrix}$

21. $\begin{vmatrix} 5 & -10 & 15 \\ 1 & 2 & -3 \\ 0 & 4 & 1 \end{vmatrix}$

22. $\begin{vmatrix} -3 & 0 & 0 \\ 1 & 2 & 0 \\ -4 & 5 & 7 \end{vmatrix}$

23. $\begin{vmatrix} 1 & -3 & 5 \\ -2 & 6 & -10 \\ 3 & 1 & 4 \end{vmatrix}$

24. $\begin{vmatrix} 8 & 4 \\ -6 & -3 \end{vmatrix}$

25. $\begin{vmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ 3 & 0 & 4 \end{vmatrix}$

26. $\begin{vmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & -1 & 4 \end{vmatrix}$

27. $\begin{vmatrix} -1 & 5 \\ 2 & -10 \end{vmatrix}$

28. $\begin{vmatrix} 2 & 3 & -1 \\ 4 & 6 & -2 \\ 1 & 5 & 3 \end{vmatrix}$

29. $\begin{vmatrix} 0 & 3 & 0 \\ -1 & 0 & 2 \\ 0 & 5 & 0 \end{vmatrix}$

30. $\begin{vmatrix} 12 & -8 & 4 \\ 3 & 1 & -2 \\ 0 & 5 & 6 \end{vmatrix}$

5. Compute $\det K$*, check the condition* $\gcd(\det K, 26) = 1$*, and decide whether the matrix* K *can be used as a Hill cipher key.*

1. $K = \begin{pmatrix} 3 & 5 \\ 2 & 7 \end{pmatrix}$
2. $K = \begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}$
3. $K = \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$
4. $K = \begin{pmatrix} 7 & 2 \\ 1 & 3 \end{pmatrix}$
5. $K = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$
6. $K = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$
7. $K = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$
8. $K = \begin{pmatrix} 6 & 1 \\ 5 & 4 \end{pmatrix}$
9. $K = \begin{pmatrix} 8 & 3 \\ 7 & 9 \end{pmatrix}$
10. $K = \begin{pmatrix} 10 & 13 \\ 3 & 4 \end{pmatrix}$
11. $K = \begin{pmatrix} 11 & 5 \\ 2 & 9 \end{pmatrix}$
12. $K = \begin{pmatrix} 4 & 9 \\ 6 & 13 \end{pmatrix}$
13. $K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$
14. $K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$
15. $K = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
16. $K = \begin{pmatrix} 15 & 7 \\ 4 & 2 \end{pmatrix}$
17. $K = \begin{pmatrix} 5 & 3 \\ 7 & 9 \end{pmatrix}$
18. $K = \begin{pmatrix} 12 & 5 \\ 1 & 3 \end{pmatrix}$
19. $K = \begin{pmatrix} 6 & 4 \\ 3 & 2 \end{pmatrix}$
20. $K = \begin{pmatrix} 25 & 1 \\ 3 & 7 \end{pmatrix}$
21. $K = \begin{pmatrix} 2 & 11 \\ 1 & 6 \end{pmatrix}$
22. $K = \begin{pmatrix} 13 & 0 \\ 5 & 2 \end{pmatrix}$
23. $K = \begin{pmatrix} 9 & 7 \\ 4 & 5 \end{pmatrix}$
24. $K = \begin{pmatrix} 8 & 6 \\ 3 & 5 \end{pmatrix}$
25. $K = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$
26. $K = \begin{pmatrix} 10 & 9 \\ 7 & 6 \end{pmatrix}$
27. $K = \begin{pmatrix} 4 & 5 \\ 3 & 4 \end{pmatrix}$
28. $K = \begin{pmatrix} 14 & 3 \\ 5 & 1 \end{pmatrix}$
29. $K = \begin{pmatrix} 2 & 7 \\ 5 & 3 \end{pmatrix}$
30. $K = \begin{pmatrix} 11 & 8 \\ 3 & 7 \end{pmatrix}$

1.3 Minors and cofactors. Evaluation of determinants of order n

Definition 1.3.1. Let a_{ij} be an entry of a determinant of order n. The determinant of order $n-1$ obtained by deleting the ith row and the jth

column, which contain a_{ij}, is called the **minor** of a_{ij} and is denoted by M_{ij}.

Example 1.3.1. For the matrix $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{pmatrix}$ we find the minor of the entry $a_{12} = 1$. Deleting the first row and the second column, we obtain

$$M_{12} = \begin{vmatrix} 0 & 2 \\ 4 & 1 \end{vmatrix} = 0 \cdot 1 - 2 \cdot 4 = -8.$$

Example 1.3.2. In the same matrix the minor of $a_{23} = 2$ is

$$M_{23} = \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} = 2 \cdot 3 - 1 \cdot 4 = 2.$$

Example 1.3.3. The minor of $a_{31} = 4$ is

$$M_{31} = \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} = 1 \cdot 2 - 3 \cdot (-1) = 2 + 3 = 5.$$

Cofactors.

Definition 1.3.2. The **cofactor** (algebraic complement) of the entry a_{ij} is the number

$$A_{ij} = (-1)^{i+j} \cdot M_{ij},$$

where M_{ij} is the minor of a_{ij}; the factor $(-1)^{i+j}$ determines the sign.

Example 1.3.4. For the matrix A above,

$$A_{12} = (-1)^{1+2} \cdot M_{12} = (-1)^3 \cdot (-8) = 8,$$
$$A_{23} = (-1)^{2+3} \cdot M_{23} = (-1)^5 \cdot 2 = -2,$$
$$A_{31} = (-1)^{3+1} \cdot M_{31} = (-1)^4 \cdot 5 = 5.$$

Theorem 1.3.1 (Laplace expansion)**.** *A determinant of order n equals the sum of the products of the entries of any of its rows (or columns) by their cofactors. Expansion along the ith row reads*

$$\det A = \sum_{j=1}^{n} a_{ij} \cdot A_{ij} = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in},$$

and expansion along the jth column reads

$$\det A = \sum_{i=1}^{n} a_{ij} \cdot A_{ij} = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}.$$

The theorem shows that every row and every column give the same value, so in practice we expand along the row or column containing **the most zeros**. A closely related fact is often used: if we multiply the entries of one row by the cofactors of a *different* row and add, the result is zero,

$$\sum_{j=1}^{n} a_{ij} \cdot A_{kj} = 0, \quad i \neq k.$$

Indeed, this sum is the expansion along the kth row of the determinant in which the kth row has been replaced by the ith one, and that determinant has two equal rows.

Example 1.3.5. We expand the determinant along the second column:

$$\begin{vmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{vmatrix} = a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32}$$

$$= 1 \cdot (-1)^{1+2} \begin{vmatrix} 0 & 2 \\ 4 & 1 \end{vmatrix} + (-1) \cdot (-1)^{2+2} \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix} + 3 \cdot (-1)^{3+2} \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix}$$

$$= 1 \cdot (-1)(0 - 8) + (-1) \cdot 1 \cdot (2 - 12) + 3 \cdot (-1)(4 - 0)$$

$$= 8 + 10 - 12 = 6,$$

which agrees with the value found by Sarrus' rule in the previous section.

1.3.1 Methods for evaluating determinants of order n

Repeated Laplace expansion. Expanding a determinant of order n along a row (column) reduces the problem to determinants of order $n - 1$. Repeating the procedure, we eventually arrive at determinants of order two.

Example 1.3.6. Consider the fourth-order determinant

$$D = \begin{vmatrix} 1 & 0 & 2 & -1 \\ 3 & 0 & 0 & 5 \\ 2 & 1 & 4 & -3 \\ 1 & 0 & 0 & 0 \end{vmatrix}.$$

The fourth row contains three zeros, so we expand along it:

$$D = 1 \cdot (-1)^{4+1} \begin{vmatrix} 0 & 2 & -1 \\ 0 & 0 & 5 \\ 1 & 4 & -3 \end{vmatrix} = - \begin{vmatrix} 0 & 2 & -1 \\ 0 & 0 & 5 \\ 1 & 4 & -3 \end{vmatrix}.$$

The resulting 3×3 determinant is expanded along its first column:

$$\begin{vmatrix} 0 & 2 & -1 \\ 0 & 0 & 5 \\ 1 & 4 & -3 \end{vmatrix} = 1 \cdot (-1)^{3+1} \begin{vmatrix} 2 & -1 \\ 0 & 5 \end{vmatrix} = 1 \cdot (10 - 0) = 10.$$

Hence $D = -10$.

Reduction by elementary operations. The most efficient approach is to use the properties of determinants to create zeros: adding suitable multiples of one row (column) to the others, we leave a single nonzero entry in some column (row) and then expand along it. The order of the determinant drops by one, and we repeat the process until a 2×2 determinant remains.

Example 1.3.7. Evaluate

$$D = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & 5 & 0 & -1 \\ -1 & 0 & 3 & 2 \\ 3 & 1 & -2 & 4 \end{vmatrix}.$$

We clear the first column by the row operations $S_2 - 2S_1$, $S_3 + S_1$, $S_4 - 3S_1$:

$$D = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & -7 \\ 0 & 2 & 2 & 5 \\ 0 & -5 & 1 & -5 \end{vmatrix}.$$

Expanding along the first column gives $D = 1 \cdot (-1)^{1+1} \cdot D_3$, where

$$D_3 = \begin{vmatrix} 1 & 2 & -7 \\ 2 & 2 & 5 \\ -5 & 1 & -5 \end{vmatrix}.$$

Clearing the first column of D_3 by $S_2 - 2S_1$ and $S_3 + 5S_1$, we get

$$D_3 = \begin{vmatrix} 1 & 2 & -7 \\ 0 & -2 & 19 \\ 0 & 11 & -40 \end{vmatrix} = 1 \cdot \begin{vmatrix} -2 & 19 \\ 11 & -40 \end{vmatrix} = (-2)(-40) - 19 \cdot 11 = 80 - 209 = -129.$$

Therefore $D = -129$.

Example 1.3.8. Evaluate the fifth-order determinant

$$D = \begin{vmatrix} 1 & 0 & 0 & 2 & 0 \\ 3 & 1 & 0 & -1 & 0 \\ 0 & 2 & 1 & 4 & 0 \\ -1 & 0 & 3 & 0 & 0 \\ 2 & 1 & 0 & 3 & 5 \end{vmatrix}.$$

The fifth column has only one nonzero entry, and expanding along it we obtain

$$D = 5 \cdot (-1)^{5+5} \begin{vmatrix} 1 & 0 & 0 & 2 \\ 3 & 1 & 0 & -1 \\ 0 & 2 & 1 & 4 \\ -1 & 0 & 3 & 0 \end{vmatrix} = 5 \cdot D_4.$$

Next we expand D_4 along its first row:

$$D_4 = 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & 4 \\ 0 & 3 & 0 \end{vmatrix} + 2 \cdot (-1)^{1+4} \begin{vmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ -1 & 0 & 3 \end{vmatrix}.$$

The first of these third-order determinants is expanded along its third row,

$$\begin{vmatrix} 1 & 0 & -1 \\ 2 & 1 & 4 \\ 0 & 3 & 0 \end{vmatrix} = 3 \cdot (-1)^{3+2} \begin{vmatrix} 1 & -1 \\ 2 & 4 \end{vmatrix} = -3 \cdot (4 + 2) = -18,$$

and the second along its first row,

$$\begin{vmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ -1 & 0 & 3 \end{vmatrix} = 3(6-0) - 1(0+1) + 0 = 18 - 1 = 17.$$

Thus $D_4 = -18 - 2 \cdot 17 = -52$ and $D = 5 \cdot (-52) = -260$.

1.3.2 The determinant of a triangular matrix

Theorem 1.3.2. *The determinant of a triangular (upper or lower) matrix equals the product of its* ***main diagonal*** *entries:*

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} = a_{11} \cdot a_{22} \cdot \ldots \cdot a_{nn}.$$

To see this, expand repeatedly along the first column, where only the diagonal entry is nonzero. The theorem underlies the method of elementary operations: we bring the determinant to triangular form and multiply the diagonal entries.

Example 1.3.9.

$$\begin{vmatrix} 3 & 5 & -1 & 2 \\ 0 & -2 & 4 & 1 \\ 0 & 0 & 7 & -3 \\ 0 & 0 & 0 & 4 \end{vmatrix} = 3 \cdot (-2) \cdot 7 \cdot 4 = -168.$$

1.3.3 Minors and cofactors in information security

The inverse matrix and decryption. Decryption in the Hill cipher requires the inverse of the key matrix. The formula for the inverse is built directly from **cofactors**:

$$A^{-1} = \frac{1}{\det A} \cdot \widetilde{A}^{T},$$

where $\widetilde{A} = (A_{ij})$ is the matrix of cofactors. Each A_{ij} is computed from the minor M_{ij}, so minors and cofactors lie at the heart of the mechanism by which a cryptosystem "reverses" its key.

Minors in coding theory. For error-correcting codes (Hamming codes, Reed–Solomon codes and others) to work, the rows of the generator matrix G and of the parity-check matrix H must be linearly independent. This is verified by computing various **minors** of the matrix: if at least one minor of order k is nonzero, the rank of the matrix is at least k.

Secret sharing. In **Shamir's secret sharing scheme** a secret key is distributed among n participants in such a way that any k of them can recover it, while fewer cannot. Recovery is based on Lagrange interpolation, which in turn is tied to Vandermonde determinants; the scheme works precisely because these determinants are nonzero for distinct interpolation nodes.

Digital signatures and authentication. Some digital signature schemes, in particular matrix-based ones, verify a signature by computing determinants and minors of certain matrices; the signature is accepted if the determinant takes the expected value.

1.3.4 Computing with Python, Maple and Mathcad

We return to the matrix A from the first examples of this section and compute the minor M_{12}, the cofactor A_{12} and the whole matrix of cofactors, and then the fourth-order determinant evaluated above by elementary operations. Note that SymPy numbers rows and columns from 0, so M_{12} is `A.minor(0, 1)`.

Python

```
from sympy import Matrix
A = Matrix([[2, 1, 3], [0, -1, 2], [4, 3, 1]])
print(A.minor(0, 1), A.cofactor(0, 1))
print(A.cofactor_matrix())
D = Matrix([[1, 2, -1, 3], [2, 5, 0, -1],
            [-1, 0, 3, 2], [3, 1, -2, 4]])
print(D.det())
```

Output

```
-8 8
Matrix([[-7, 8, 4], [8, -10, -2], [5, -4, -2]])
-129
```

In Maple, `Minor(A, i, j)` returns the minor M_{ij} (numbering starts at 1); the cofactor is obtained by attaching the sign:

Maple

```
with(LinearAlgebra):
A := Matrix([[2, 1, 3], [0, -1, 2], [4, 3, 1]]):
Minor(A, 1, 2);
                                -8
(-1)^(1+2)*Minor(A, 1, 2);
                                 8
D4 := Matrix([[1, 2, -1, 3], [2, 5, 0, -1],
              [-1, 0, 3, 2], [3, 1, -2, 4]]):
Determinant(D4);
                               -129
```

Mathcad Prime has no separate minor function, but a submatrix is easy to form: `submatrix(A, ir, jr, ic, jc)` extracts rows `ir..jr` and columns `ic..jc`, and `augment` glues columns together. With the default `ORIGIN = 0`, the minor M_{12} is the determinant of rows 1–2 and columns 0 and 2:

Mathcad

```
     | 2   1  3 |
A := | 0  -1  2 |
     | 4   3  1 |
B := submatrix(A, 1, 2, 0, 2)
M12 := |augment(B<0>, B<2>)|      M12 = -8
A12 := (-1)^(1+2) * M12           A12 = 8

     |  1  2  -1   3 |
D := |  2  5   0  -1 |            |D| = -129
     | -1  0   3   2 |
     |  3  1  -2   4 |
```

Here `B<0>` denotes the column operator (column 0 of B).

1.3.5 Exercises on minors and cofactors

1. For the given matrix, find the minors M_{ij} and the cofactors A_{ij} of the indicated entries.

1. $A = \begin{pmatrix} 3 & -1 & 2 \\ 0 & 4 & -5 \\ 1 & 2 & 3 \end{pmatrix}$ $M_{11}, A_{11}, M_{23}, A_{23}.$

2. $A = \begin{pmatrix} -2 & 1 & 3 \\ 4 & 0 & -1 \\ 5 & 2 & -4 \end{pmatrix}$ $M_{12}, A_{12}, M_{31}, A_{31}.$

3. $A = \begin{pmatrix} 1 & 5 & -3 \\ -2 & 0 & 4 \\ 3 & 1 & 2 \end{pmatrix}$ $M_{22}, A_{22}, M_{13}, A_{13}.$

4. $A = \begin{pmatrix} 4 & -3 & 0 \\ 2 & 1 & -5 \\ -1 & 6 & 3 \end{pmatrix}$ $M_{32}, A_{32}, M_{11}, A_{11}.$

5. $A = \begin{pmatrix} 0 & 2 & -1 \\ 3 & -4 & 5 \\ 1 & 0 & -2 \end{pmatrix}$ $M_{21}, A_{21}, M_{33}, A_{33}.$

6. $A = \begin{pmatrix} -5 & 1 & 4 \\ 0 & 3 & -2 \\ 2 & -1 & 0 \end{pmatrix}$ $M_{13}, A_{13}, M_{22}, A_{22}.$

7. $A = \begin{pmatrix} 2 & 0 & -3 \\ -1 & 5 & 1 \\ 4 & -2 & 3 \end{pmatrix}$ $M_{31}, A_{31}, M_{12}, A_{12}.$

8. $A = \begin{pmatrix} 1 & -4 & 2 \\ 3 & 0 & -5 \\ -2 & 1 & 4 \end{pmatrix}$ $M_{23}, A_{23}, M_{11}, A_{11}.$

9. $A = \begin{pmatrix} -3 & 2 & 0 \\ 5 & -1 & 4 \\ 0 & 3 & -2 \end{pmatrix}$ $M_{32}, A_{32}, M_{21}, A_{21}.$

10. $A = \begin{pmatrix} 6 & -2 & 1 \\ 0 & 4 & -3 \\ 3 & 1 & 5 \end{pmatrix}$ $M_{11}, A_{11}, M_{33}, A_{33}.$

11. $A = \begin{pmatrix} -1 & 3 & 5 \\ 2 & 0 & -4 \\ 4 & -5 & 1 \end{pmatrix}$ $M_{12}, A_{12}, M_{31}, A_{31}.$

12. $A = \begin{pmatrix} 3 & -5 & 0 \\ -1 & 2 & 4 \\ 5 & 0 & -3 \end{pmatrix}$ $M_{22}, A_{22}, M_{13}, A_{13}.$

13. $A = \begin{pmatrix} 0 & 4 & -2 \\ 1 & -3 & 5 \\ -4 & 2 & 0 \end{pmatrix}$ $M_{21}, A_{21}, M_{32}, A_{32}.$

14. $A = \begin{pmatrix} 2 & 1 & -4 \\ -3 & 5 & 0 \\ 0 & -2 & 3 \end{pmatrix}$ $M_{13}, A_{13}, M_{23}, A_{23}.$

15. $A = \begin{pmatrix} -4 & 0 & 3 \\ 2 & -1 & 5 \\ 1 & 4 & -2 \end{pmatrix}$ $M_{11}, A_{11}, M_{32}, A_{32}.$

16. $A = \begin{pmatrix} 5 & -3 & 1 \\ 0 & 2 & -4 \\ -2 & 1 & 0 \end{pmatrix}$ $M_{31}, A_{31}, M_{12}, A_{12}.$

17. $A = \begin{pmatrix} 1 & 0 & -5 \\ -4 & 3 & 2 \\ 3 & -1 & 4 \end{pmatrix}$ $M_{22}, A_{22}, M_{33}, A_{33}.$

18. $A = \begin{pmatrix} -2 & 4 & 0 \\ 1 & -5 & 3 \\ 0 & 2 & -1 \end{pmatrix}$ $M_{13}, A_{13}, M_{21}, A_{21}.$

19. $A = \begin{pmatrix} 3 & 1 & -2 \\ -5 & 0 & 4 \\ 2 & -3 & 1 \end{pmatrix}$ $M_{23}, A_{23}, M_{11}, A_{11}.$

20. $A = \begin{pmatrix} 0 & -2 & 3 \\ 4 & 1 & -5 \\ -1 & 5 & 2 \end{pmatrix}$ $M_{32}, A_{32}, M_{12}, A_{12}.$

21. $A = \begin{pmatrix} -1 & 5 & 0 \\ 3 & -2 & 4 \\ 2 & 0 & -3 \end{pmatrix}$ $M_{11}, A_{11}, M_{23}, A_{23}$.

22. $A = \begin{pmatrix} 4 & 0 & -1 \\ -2 & 3 & 5 \\ 1 & -4 & 2 \end{pmatrix}$ $M_{21}, A_{21}, M_{33}, A_{33}$.

23. $A = \begin{pmatrix} -3 & 1 & 4 \\ 0 & -5 & 2 \\ 5 & 3 & 0 \end{pmatrix}$ $M_{31}, A_{31}, M_{22}, A_{22}$.

24. $A = \begin{pmatrix} 2 & -4 & 1 \\ 5 & 0 & -3 \\ -1 & 2 & 4 \end{pmatrix}$ $M_{12}, A_{12}, M_{33}, A_{33}$.

25. $A = \begin{pmatrix} 1 & 3 & -5 \\ -4 & 0 & 2 \\ 0 & -1 & 3 \end{pmatrix}$ $M_{13}, A_{13}, M_{21}, A_{21}$.

26. $A = \begin{pmatrix} -5 & 2 & 0 \\ 1 & -3 & 4 \\ 3 & 0 & -1 \end{pmatrix}$ $M_{22}, A_{22}, M_{31}, A_{31}$.

27. $A = \begin{pmatrix} 0 & -1 & 3 \\ 2 & 4 & 0 \\ -3 & 5 & -2 \end{pmatrix}$ $M_{32}, A_{32}, M_{11}, A_{11}$.

28. $A = \begin{pmatrix} 3 & 0 & -4 \\ -1 & 2 & 5 \\ 4 & -3 & 1 \end{pmatrix}$ $M_{23}, A_{23}, M_{12}, A_{12}$.

29. $A = \begin{pmatrix} -2 & 5 & 1 \\ 0 & -4 & 3 \\ 1 & 2 & 0 \end{pmatrix}$ $M_{11}, A_{11}, M_{13}, A_{13}$.

30. $A = \begin{pmatrix} 4 & -1 & 0 \\ -3 & 0 & 2 \\ 2 & 5 & -4 \end{pmatrix}$ $M_{21}, A_{21}, M_{32}, A_{32}$.

2. Evaluate the determinant by expanding along the indicated row or column.

1. $\begin{vmatrix} 1 & 0 & 3 \\ 2 & -1 & 4 \\ 0 & 5 & -2 \end{vmatrix}$, row 1.

2. $\begin{vmatrix} -2 & 3 & 0 \\ 1 & 0 & -4 \\ 5 & 2 & 1 \end{vmatrix}$, column 2.

3. $\begin{vmatrix} 4 & -1 & 2 \\ 0 & 3 & 0 \\ -2 & 5 & 1 \end{vmatrix}$, row 2.

4. $\begin{vmatrix} 0 & 2 & -3 \\ 4 & 0 & 1 \\ -1 & 5 & 0 \end{vmatrix}$, column 1.

5. $\begin{vmatrix} 3 & 0 & -1 \\ 0 & 4 & 2 \\ 5 & -3 & 0 \end{vmatrix}$, column 3.

6. $\begin{vmatrix} -1 & 2 & 5 \\ 3 & 0 & 0 \\ 4 & -2 & 1 \end{vmatrix}$, row 2.

7. $\begin{vmatrix} 2 & -3 & 0 \\ 0 & 1 & 4 \\ -5 & 0 & 3 \end{vmatrix}$, row 3.

8. $\begin{vmatrix} 0 & 5 & -1 \\ 2 & 0 & 3 \\ -4 & 1 & 0 \end{vmatrix}$, row 1.

9. $\begin{vmatrix} 1 & -2 & 0 \\ 0 & 3 & -5 \\ 4 & 0 & 2 \end{vmatrix}$, column 3.

10. $\begin{vmatrix} -3 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & -5 & 3 \end{vmatrix}$, row 2.

11. $\begin{vmatrix} 5 & 0 & 0 \\ -1 & 2 & -3 \\ 3 & 4 & 1 \end{vmatrix}$, row 1.

12. $\begin{vmatrix} 0 & -4 & 1 \\ 3 & 2 & 0 \\ -1 & 0 & 5 \end{vmatrix}$, column 2.

13. $\begin{vmatrix} 2 & 1 & 0 \\ -3 & 0 & 4 \\ 0 & 5 & -2 \end{vmatrix}$, column 1.

14. $\begin{vmatrix} 0 & 0 & 3 \\ -2 & 4 & 1 \\ 5 & -1 & 0 \end{vmatrix}$, row 1.

15. $\begin{vmatrix} -1 & 3 & 0 \\ 0 & 2 & -4 \\ 4 & 0 & 5 \end{vmatrix}$, column 3.

16. $\begin{vmatrix} 3 & -2 & 1 \\ 0 & 0 & -5 \\ 2 & 4 & 0 \end{vmatrix}$, row 2.

17. $\begin{vmatrix} 0 & 1 & -3 \\ 5 & 0 & 2 \\ -2 & 4 & 0 \end{vmatrix}$, column 1.

18. $\begin{vmatrix} -4 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & -5 & 4 \end{vmatrix}$, row 3.

19. $\begin{vmatrix} 1 & 0 & -2 \\ 0 & -3 & 5 \\ 3 & 0 & 4 \end{vmatrix}$, column 2.

20. $\begin{vmatrix} 2 & 5 & 0 \\ 0 & -1 & 3 \\ -4 & 0 & 2 \end{vmatrix}$, column 3.

21. $\begin{vmatrix} 0 & -2 & 4 \\ 3 & 1 & 0 \\ -5 & 0 & 2 \end{vmatrix}$, row 2.

22. $\begin{vmatrix} -1 & 0 & 5 \\ 4 & 2 & 0 \\ 0 & -3 & 1 \end{vmatrix}$, row 1.

23. $\begin{vmatrix} 3 & 0 & -4 \\ 0 & 5 & 1 \\ 2 & 0 & 3 \end{vmatrix}$, column 2.

24. $\begin{vmatrix} 0 & 4 & -1 \\ -2 & 0 & 3 \\ 5 & 1 & 0 \end{vmatrix}$, row 3.

25. $\begin{vmatrix} 1 & -5 & 0 \\ 0 & 2 & 3 \\ -3 & 0 & 4 \end{vmatrix}$, column 1.

26. $\begin{vmatrix} -2 & 0 & 3 \\ 4 & -1 & 0 \\ 0 & 5 & -2 \end{vmatrix}$, column 3.

27. $\begin{vmatrix} 5 & 1 & 0 \\ 0 & -3 & 2 \\ -4 & 0 & 4 \end{vmatrix}$, row 3.

28. $\begin{vmatrix} 0 & 2 & -5 \\ 1 & 0 & 4 \\ -3 & 3 & 0 \end{vmatrix}$, column 2.

29. $\begin{vmatrix} -1 & 0 & 0 \\ 3 & -2 & 5 \\ 2 & 4 & -3 \end{vmatrix}$, row 1.

30. $\begin{vmatrix} 4 & -3 & 0 \\ 0 & 1 & -2 \\ 2 & 0 & 5 \end{vmatrix}$, row 2.

3. Evaluate the fourth-order determinants.

1. $\begin{vmatrix} 1 & 0 & 2 & -1 \\ 3 & 0 & 0 & 5 \\ 2 & 1 & 4 & -3 \\ 1 & 0 & 0 & 0 \end{vmatrix}$

3. $\begin{vmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 0 & -2 \\ 4 & 0 & 1 & 5 \\ -1 & 0 & 2 & 0 \end{vmatrix}$

5. $\begin{vmatrix} 0 & 1 & -2 & 3 \\ 1 & 0 & 4 & 0 \\ -3 & 2 & 0 & 1 \\ 0 & 0 & 5 & -1 \end{vmatrix}$

2. $\begin{vmatrix} 2 & -1 & 0 & 3 \\ 0 & 4 & 1 & 0 \\ -1 & 0 & 3 & 2 \\ 0 & 5 & 0 & -4 \end{vmatrix}$

4. $\begin{vmatrix} -2 & 0 & 1 & 0 \\ 3 & -1 & 0 & 4 \\ 0 & 2 & -3 & 0 \\ 1 & 0 & 0 & 5 \end{vmatrix}$

6. $\begin{vmatrix} 3 & 0 & 0 & -2 \\ -1 & 2 & 0 & 4 \\ 0 & 5 & 1 & 0 \\ 4 & 0 & -3 & 0 \end{vmatrix}$

7. $\begin{vmatrix} 1 & -3 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 4 & 0 & 3 & -2 \\ 0 & 1 & 0 & 5 \end{vmatrix}$

8. $\begin{vmatrix} -1 & 0 & 4 & 0 \\ 2 & 3 & 0 & -1 \\ 0 & 0 & 5 & 2 \\ 3 & -2 & 0 & 0 \end{vmatrix}$

9. $\begin{vmatrix} 2 & 1 & 0 & -3 \\ 0 & -2 & 3 & 0 \\ 1 & 0 & 0 & 4 \\ 0 & 4 & -1 & 0 \end{vmatrix}$

10. $\begin{vmatrix} 0 & 0 & 2 & 1 \\ -3 & 1 & 0 & 0 \\ 0 & 4 & -1 & 3 \\ 5 & 0 & 0 & -2 \end{vmatrix}$

11. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix}$

12. $\begin{vmatrix} -1 & 0 & 0 & 3 \\ 2 & -4 & 0 & 0 \\ 0 & 1 & 5 & 0 \\ 3 & 0 & -2 & 1 \end{vmatrix}$

13. $\begin{vmatrix} 4 & 0 & -1 & 0 \\ 0 & 3 & 0 & 2 \\ -2 & 0 & 5 & 0 \\ 0 & 1 & 0 & -3 \end{vmatrix}$

14. $\begin{vmatrix} 1 & -1 & 2 & 0 \\ 3 & 0 & 0 & -4 \\ 0 & 2 & 1 & 3 \\ -2 & 0 & 0 & 5 \end{vmatrix}$

15. $\begin{vmatrix} 0 & 3 & 0 & -1 \\ 2 & 0 & -4 & 0 \\ 0 & 1 & 0 & 3 \\ -5 & 0 & 2 & 0 \end{vmatrix}$

16. $\begin{vmatrix} 2 & 0 & 0 & 1 \\ 0 & -3 & 0 & 0 \\ 4 & 0 & 1 & -2 \\ 0 & 2 & 0 & 3 \end{vmatrix}$

17. $\begin{vmatrix} 1 & 0 & -2 & 3 \\ 0 & 2 & 0 & 0 \\ -1 & 0 & 4 & -1 \\ 3 & 0 & 0 & 2 \end{vmatrix}$

18. $\begin{vmatrix} -3 & 1 & 0 & 0 \\ 0 & 0 & 2 & -4 \\ 5 & -1 & 0 & 0 \\ 0 & 0 & 3 & 1 \end{vmatrix}$

19. $\begin{vmatrix} 0 & 2 & -1 & 0 \\ 1 & 0 & 3 & -2 \\ 0 & -4 & 0 & 5 \\ 3 & 0 & 1 & 0 \end{vmatrix}$

20. $\begin{vmatrix} 1 & 0 & 0 & 0 \\ -2 & 3 & 0 & 0 \\ 4 & -1 & 2 & 0 \\ 0 & 5 & -3 & 1 \end{vmatrix}$

21. $\begin{vmatrix} 2 & -1 & 0 & 4 \\ 0 & 3 & -2 & 0 \\ 1 & 0 & 0 & -3 \\ 0 & 0 & 5 & 1 \end{vmatrix}$

22. $\begin{vmatrix} 0 & 0 & 1 & -2 \\ 0 & 4 & 0 & 3 \\ -1 & 0 & 0 & 5 \\ 2 & -3 & 0 & 0 \end{vmatrix}$

23. $\begin{vmatrix} 5 & 0 & -1 & 0 \\ 0 & 2 & 0 & -3 \\ 3 & 0 & 4 & 0 \\ 0 & -1 & 0 & 2 \end{vmatrix}$

24. $\begin{vmatrix} -2 & 3 & 0 & 0 \\ 1 & 0 & -4 & 0 \\ 0 & 2 & 0 & 5 \\ 0 & 0 & 1 & -3 \end{vmatrix}$

25. $\begin{vmatrix} 1 & 0 & 3 & 0 \\ 0 & -2 & 0 & 4 \\ 2 & 0 & -1 & 0 \\ 0 & 3 & 0 & 5 \end{vmatrix}$

26. $\begin{vmatrix} 0 & 1 & 0 & -3 \\ 4 & 0 & 2 & 0 \\ 0 & -5 & 0 & 1 \\ 3 & 0 & -1 & 0 \end{vmatrix}$

27. $\begin{vmatrix} 3 & 0 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & -1 & 3 & 0 \\ 5 & 0 & 0 & 2 \end{vmatrix}$

28. $\begin{vmatrix} -1 & 2 & 0 & 0 \\ 0 & 0 & 3 & -5 \\ 4 & -1 & 0 & 0 \\ 0 & 0 & 2 & 1 \end{vmatrix}$

29. $\begin{vmatrix} 2 & 0 & -3 & 0 \\ 0 & 1 & 0 & 4 \\ -1 & 0 & 2 & 0 \\ 0 & -3 & 0 & 5 \end{vmatrix}$

30. $\begin{vmatrix} 0 & -2 & 0 & 1 \\ 3 & 0 & 5 & 0 \\ 0 & 4 & 0 & -2 \\ 1 & 0 & -3 & 0 \end{vmatrix}$

4. Using cofactors, compute the sum $\sum_{j=1}^{3} a_{ij}A_{kj}$ *in the cases* $i = k$ *and* $i \neq k$.

Example 1.3.10. For $A = \begin{pmatrix} 2 & -1 & 3 \\ 4 & 0 & -5 \\ 1 & 3 & 2 \end{pmatrix}$ compute 1) $\sum_{j=1}^{3} a_{1j}A_{1j}$ (the case $i = 1,\ k = 1$); 2) $\sum_{j=1}^{3} a_{1j}A_{2j}$ (the case $i = 1,\ k = 2$).

Solution. The cofactors of the first row are

$$A_{11} = \begin{vmatrix} 0 & -5 \\ 3 & 2 \end{vmatrix} = 15, \quad A_{12} = -\begin{vmatrix} 4 & -5 \\ 1 & 2 \end{vmatrix} = -13, \quad A_{13} = \begin{vmatrix} 4 & 0 \\ 1 & 3 \end{vmatrix} = 12.$$

In the first case ($i = 1,\ k = 1$) we obtain

$$\sum_{j=1}^{3} a_{1j}A_{1j} = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} = 2 \cdot 15 + (-1)(-13) + 3 \cdot 12 =$$

$$= 30 + 13 + 36 = 79 = \det A.$$

For the second case we need the cofactors of the second row:

$$A_{21} = -\begin{vmatrix} -1 & 3 \\ 3 & 2 \end{vmatrix} = 11, \quad A_{22} = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1, \quad A_{23} = -\begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} = -7.$$

Then ($i = 1,\ k = 2$)

$$\sum_{j=1}^{3} a_{1j}A_{2j} = a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 2 \cdot 11 + (-1) \cdot 1 + 3 \cdot (-7) =$$

$$= 22 - 1 - 21 = 0.$$

This illustrates the general rule: the entries of a row multiplied by their own cofactors ($i = k$) add up to $\det A$, whereas multiplied by the cofactors of another row ($i \neq k$) they always add up to 0.

1. $A = \begin{pmatrix} 2 & -1 & 3 \\ 4 & 0 & -5 \\ 1 & 3 & 2 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 2$.

2. $A = \begin{pmatrix} -3 & 2 & 1 \\ 0 & 4 & -2 \\ 5 & -1 & 3 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 3$.

3. $A = \begin{pmatrix} 1 & 0 & -4 \\ 3 & -2 & 5 \\ -1 & 4 & 0 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 1$.

4. $A = \begin{pmatrix} 0 & 5 & -1 \\ 2 & -3 & 4 \\ -4 & 1 & 2 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 3$.

5. $A = \begin{pmatrix} -2 & 1 & 0 \\ 3 & 0 & -5 \\ 1 & 4 & 3 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 1$.

6. $A = \begin{pmatrix} 4 & -3 & 2 \\ 0 & 1 & -4 \\ -2 & 5 & 0 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 2$.

7. $A = \begin{pmatrix} 1 & -2 & 5 \\ -3 & 4 & 0 \\ 2 & 0 & -1 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 3$.

8. $A = \begin{pmatrix} -5 & 0 & 3 \\ 1 & -4 & 2 \\ 0 & 3 & -2 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 3$.

9. $A = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 5 & 0 \\ 4 & -3 & 1 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 1$.

10. $A = \begin{pmatrix} 0 & -4 & 2 \\ 5 & 1 & -3 \\ -2 & 0 & 4 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 2$.

11. $A = \begin{pmatrix} -1 & 3 & 0 \\ 2 & -5 & 4 \\ 0 & 1 & -3 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 1$.

12. $A = \begin{pmatrix} 5 & 0 & -3 \\ -2 & 4 & 1 \\ 3 & -1 & 0 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 2$.

13. $A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & -2 \\ -5 & 2 & 1 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 3$.

14. $A = \begin{pmatrix} -4 & 5 & 0 \\ 1 & 0 & -3 \\ 3 & -2 & 4 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 3$.

15. $A = \begin{pmatrix} 0 & 2 & -5 \\ -3 & 1 & 4 \\ 4 & 0 & -1 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 1$.

16. $A = \begin{pmatrix} 1 & -3 & 2 \\ 4 & 0 & -1 \\ -2 & 5 & 3 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 2$.

17. $A = \begin{pmatrix} -2 & 0 & 5 \\ 3 & -4 & 1 \\ 0 & 2 & -3 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 1$.

18. $A = \begin{pmatrix} 3 & 4 & 0 \\ -1 & -2 & 5 \\ 2 & 0 & -4 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 2$.

19. $A = \begin{pmatrix} 0 & -5 & 1 \\ 2 & 3 & -4 \\ -1 & 0 & 2 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 3$.

20. $A = \begin{pmatrix} -3 & 1 & 0 \\ 0 & -2 & 3 \\ 5 & 4 & -1 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 3$.

21. $A = \begin{pmatrix} 4 & 0 & -2 \\ -1 & 3 & 5 \\ 0 & -4 & 1 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 1$.

22. $A = \begin{pmatrix} 1 & 2 & -3 \\ -4 & 0 & 5 \\ 3 & -1 & 0 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 2$.

23. $A = \begin{pmatrix} -5 & 3 & 0 \\ 2 & -1 & 4 \\ 0 & 5 & -2 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 1$.

24. $A = \begin{pmatrix} 0 & -3 & 4 \\ 1 & 5 & 0 \\ -2 & 0 & 3 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 2$.

25. $A = \begin{pmatrix} 2 & 0 & 1 \\ -3 & -4 & 0 \\ 5 & 2 & -5 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 3$.

26. $A = \begin{pmatrix} -1 & 4 & -2 \\ 0 & 0 & 3 \\ 3 & -5 & 1 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 3$.

27. $A = \begin{pmatrix} 3 & -2 & 0 \\ -4 & 1 & 5 \\ 0 & 3 & -1 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 1$.

28. $A = \begin{pmatrix} 0 & 1 & -4 \\ 5 & -3 & 2 \\ -1 & 0 & 3 \end{pmatrix}$; $i = 1, k = 1$; $i = 1, k = 2$.

29. $A = \begin{pmatrix} -2 & 0 & 3 \\ 1 & 5 & -4 \\ 4 & -1 & 0 \end{pmatrix}$; $i = 2, k = 2$; $i = 2, k = 1$.

30. $A = \begin{pmatrix} 5 & -4 & 1 \\ 0 & 2 & -3 \\ -3 & 0 & 4 \end{pmatrix}$; $i = 3, k = 3$; $i = 3, k = 2$.

1.4 Computing the rank of a matrix. Singular and nonsingular matrices

Definition 1.4.1. The **rank** of an $m \times n$ matrix A is the largest order of a nonzero minor of A. It is denoted by rank A, rg A or $r(A)$.

Equivalently, the rank of a matrix is the maximal number of linearly independent rows (or, which gives the same number, of linearly independent columns). We shall use the following basic properties.

- $0 \le \operatorname{rg} A \le \min(m, n)$.
- $\operatorname{rg} A = 0$ if and only if $A = O$ is the zero matrix.
- $\operatorname{rg} A = \operatorname{rg} A^T$: the row rank equals the column rank.
- Elementary transformations do not change the rank of a matrix.

1.4.1 Computing the rank by means of minors

Definition 1.4.2. A matrix A has rank r if

1) A has at least one **nonzero** minor of order r, and
2) every minor of order $r+1$ is zero (or there are no minors of that order).

A nonzero minor of the highest order is called a **basis minor**.

Example 1.4.1. Let us find the rank of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & -1 \end{pmatrix}$. The only minor of order three is the determinant itself:

$$\det A = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= 1(-4-6) - 2(-2-0) + 3(2-0) = -10 + 4 + 6 = 0.$$

Hence $\operatorname{rg} A < 3$. Among the minors of order two we find

$$M = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 - 2 \cdot 0 = 1 \neq 0,$$

so $\operatorname{rg} A = 2$.

Note. Since A in this example is a 3×3 matrix, its only minor of order three is $\det A$, and the equality $\det A = 0$ already shows that $\operatorname{rg} A < 3$.

To prove that $\operatorname{rg} A \geq 2$, a *single* nonzero minor of order two is enough; the remaining minors of order two need not be computed, because they cannot change the rank.

If the first minor of order two we try turns out to be zero, we try another one. If all minors of order two vanish and the matrix is not the zero matrix, then $\operatorname{rg} A = 1$.

For larger matrices checking every minor quickly becomes impractical (a 4×4 matrix already has 16 minors of order three). In such cases one uses Gaussian elimination or the method of bordering minors.

Example 1.4.2. Find the rank of $B = \begin{pmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \end{pmatrix}$. Since B is a 2×3 matrix, $\operatorname{rg} B \leq 2$. We check all minors of order two:

$$\begin{vmatrix} 2 & -1 \\ 4 & -2 \end{vmatrix} = -4 + 4 = 0, \quad \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0,$$

$$\begin{vmatrix} -1 & 3 \\ -2 & 6 \end{vmatrix} = -6 + 6 = 0.$$

All of them vanish, while the minor of order one $a_{11} = 2$ is nonzero. Therefore $\operatorname{rg} B = 1$. This is no surprise: the second row is twice the first, $S_2 = 2S_1$ (here and below S_i denotes the ith row).

1.4.2 Computing the rank by elementary transformations (Gaussian elimination)

Definition 1.4.3. Using elementary row operations, we reduce the matrix to **row echelon form**. The rank of the matrix equals the **number of nonzero rows** of the resulting echelon matrix.

The **elementary row operations** are:

1) interchanging two rows, $S_i \leftrightarrow S_j$;

2) multiplying a row by a nonzero number, $S_i \to k \cdot S_i$, $k \neq 0$;

3) adding to a row a multiple of another row, $S_i \to S_i + k \cdot S_j$.

A matrix is in **row echelon form** if the first nonzero entry of each row (its *leading entry*, or pivot) lies strictly to the right of the leading entry of the row above it, and all zero rows stand at the bottom.

Example 1.4.3. Find the rank of $A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 5 & 0 & -1 \\ -1 & 0 & 3 & 2 \\ 3 & 1 & -2 & 4 \end{pmatrix}$.

Solution. We apply Gaussian elimination. Clearing the first column with the first row and then the second column with the second row, we get

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 5 & 0 & -1 \\ -1 & 0 & 3 & 2 \\ 3 & 1 & -2 & 4 \end{pmatrix} \xrightarrow[S_4-3S_1]{\substack{S_2-2S_1 \\ S_3+S_1}} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & -7 \\ 0 & 2 & 2 & 5 \\ 0 & -5 & 1 & -5 \end{pmatrix}$$

$$\xrightarrow[S_4+5S_2]{S_3-2S_2} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & -7 \\ 0 & 0 & -2 & 19 \\ 0 & 0 & 11 & -40 \end{pmatrix} \xrightarrow{S_4+\frac{11}{2}S_3} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 2 & -7 \\ 0 & 0 & -2 & 19 \\ 0 & 0 & 0 & \frac{129}{2} \end{pmatrix}.$$

The echelon form has four nonzero rows, so $\operatorname{rg} A = 4$ (indeed, $\det A = -129 \neq 0$). ■

Example 1.4.4. Find the rank of $B = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 4 & 0 \\ 3 & 1 & -1 & 2 \\ 1 & 3 & -5 & 2 \end{pmatrix}$.

Solution.

$$\begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 4 & 0 \\ 3 & 1 & -1 & 2 \\ 1 & 3 & -5 & 2 \end{pmatrix} \xrightarrow[S_4 - S_1]{\substack{S_2 - 2S_1 \\ S_3 - 3S_1}} \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 4 & -7 & 2 \\ 0 & 4 & -7 & 2 \end{pmatrix} \xrightarrow{S_4 - S_3}$$

$$\begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{S_2 \leftrightarrow S_3} \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Only two rows remain nonzero, hence $\operatorname{rg} B = 2$. ■

Example 1.4.5. Find the rank of $C = \begin{pmatrix} 2 & 1 & -3 & 5 \\ 4 & 2 & -6 & 10 \\ 1 & 0 & 1 & -2 \end{pmatrix}$.

Solution. It is convenient to bring a row with leading entry 1 to the top:

$$\begin{pmatrix} 2 & 1 & -3 & 5 \\ 4 & 2 & -6 & 10 \\ 1 & 0 & 1 & -2 \end{pmatrix} \xrightarrow{S_1 \leftrightarrow S_3} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 4 & 2 & -6 & 10 \\ 2 & 1 & -3 & 5 \end{pmatrix}$$

$$\xrightarrow[S_3-2S_1]{S_2-4S_1} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 2 & -10 & 18 \\ 0 & 1 & -5 & 9 \end{pmatrix} \xrightarrow{S_2-2S_3} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -5 & 9 \end{pmatrix} \xrightarrow{S_2 \leftrightarrow S_3}$$

$$\begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & -5 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are two nonzero rows, so $\operatorname{rg} C = 2$. ■

Singular and nonsingular matrices.

Definition 1.4.4. A square matrix A of order n is called

- **nonsingular** (nondegenerate) if $\det A \neq 0$, that is, $\operatorname{rg} A = n$;
- **singular** (degenerate) if $\det A = 0$, that is, $\operatorname{rg} A < n$.

A nonsingular matrix A has the following properties:

1) it has an inverse A^{-1}, and the inverse is unique;
2) its rows (and its columns) are linearly independent;
3) the system $Ax = b$ has a unique solution for every right-hand side b;
4) $\operatorname{rg} A = n$ (full rank).

For a singular matrix, on the contrary:

1) there is no inverse matrix;
2) its rows (columns) are linearly dependent;
3) the system $Ax = b$ is either inconsistent or has infinitely many solutions;
4) $\operatorname{rg} A < n$.

Example 1.4.6. For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ we have $\det A = 1 \cdot 4 - 2 \cdot 3 = -2 \neq 0$, so A is **nonsingular** and $\operatorname{rg} A = 2$. For $B = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$ we have $\det B = 2 \cdot 3 - 6 \cdot 1 = 0$, so B is **singular** and $\operatorname{rg} B = 1$: the rows are proportional ($S_1 = 2S_2$), and so are the columns (the second column is three times the first).

Example 1.4.7. Consider $C = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & -3 & 1 \end{pmatrix}$. Expanding along the first row,

$$\det C = 1(3 + 3) - 0 + 2(3 - 6) = 6 - 6 = 0.$$

Thus C is **singular** and $\operatorname{rg} C < 3$. The minor of order two in the upper left corner is

$$\begin{vmatrix} 1 & 0 \\ -1 & 3 \end{vmatrix} = 3 \neq 0,$$

hence $\operatorname{rg} C = 2$.

Rank and systems of linear equations.

Theorem 1.4.1 (Kronecker–Capelli)**.** *A system of linear equations $Ax = b$ is consistent if and only if*

$$\operatorname{rg} A = \operatorname{rg}(A \mid b),$$

that is, the rank of the coefficient matrix A equals the rank of the augmented matrix $(A \mid b)$. Moreover, if n is the number of unknowns, then

- *for $\operatorname{rg} A = \operatorname{rg}(A \mid b) = n$ the system has a **unique solution**;*
- *for $\operatorname{rg} A = \operatorname{rg}(A \mid b) = r < n$ it has **infinitely many solutions**, depending on $n - r$ free unknowns;*

- *for* $\operatorname{rg} A \neq \operatorname{rg}(A \mid b)$ *it has* ***no solutions*** *(the system is inconsistent).*

In the literature this result is also known as the Rouché–Capelli theorem.

Example 1.4.8. Is the system

$$\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 + 4x_2 - 2x_3 = 5 \end{cases}$$

consistent? Its coefficient matrix and augmented matrix are

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{pmatrix}, \quad (A \mid b) = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 5 \end{pmatrix}.$$

The operation $S_2 - 2S_1$ gives $(A \mid b) \to \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & -1 \end{pmatrix}$. Thus $\operatorname{rg} A = 1$, whereas $\operatorname{rg}(A \mid b) = 2$ because the entry -1 survives in the last column. The ranks differ, so the system **has no solutions**.

Example 1.4.9. Apply the Kronecker–Capelli theorem to the system
$\begin{cases} x_1 + 2x_2 - x_3 = 3, \\ 2x_1 + 4x_2 - 2x_3 = 6, \\ x_1 + x_2 + x_3 = 1. \end{cases}$

Solution. Reducing the augmented matrix, we obtain $(A \mid b) = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow[S_3 - S_1]{S_2 - 2S_1} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -2 \end{pmatrix}$. Hence $\operatorname{rg} A = \operatorname{rg}(A \mid b) = 2$. The ranks are equal, but $r = 2 < n = 3$, so the system has infinitely many solutions with $n - r = 1$ free unknown. ■

1.4.3 Rank and singularity in information security

Nonsingular keys in cryptography. In the Hill cipher, as in other matrix-based ciphers, the key matrix must be **nonsingular**. If the key K were singular ($\det K = 0$, $\operatorname{rg} K < n$), the inverse K^{-1} would not exist and decryption would be impossible. Worse, different plaintext blocks could be mapped to the same ciphertext, which defeats the purpose of encryption. For this reason the first thing one checks when choosing a key is the condition $\det K \neq 0$ (for the Hill cipher, invertibility of $\det K$ modulo 26).

Rank in coding theory. In error-correcting codes (Hamming codes, Reed–Solomon codes and others) the rank of the generator matrix G determines how much information the code carries. A $k \times n$ generator matrix must have rank k, that is, linearly independent rows; if $\operatorname{rg} G < k$, two different messages receive the same codeword and cannot be told apart. Likewise, the parity-check matrix H must have rank $n - k$.

Rank analysis of networks. The rank of the Laplacian matrix of a network graph reflects its **connectivity**. For a connected network with n nodes the Laplacian has rank $n - 1$; a smaller rank means that the network splits into several disconnected components, which is an important warning sign in a security audit. The rank of the Laplacian and its null space also help to locate weak points of the network.

Lattice-based cryptography. In modern post-quantum schemes such as LWE (Learning With Errors), rank and nonsingularity are again central. The public key contains a large random matrix, and its having full rank is one of the requirements for the security of the scheme. An attacker trying to recover the secret key exploits linear dependences among the columns of this matrix, so the closer the matrix is to full rank, the harder the attack.

1.4.4 Computing with Python, Maple and Mathcad

Let us check the rank of the matrix B from the second example on Gaussian elimination and the consistency test of the system $x_1 + 2x_2 - x_3 = 3$, $2x_1 + 4x_2 - 2x_3 = 5$. In Python we use SymPy, which works in exact rational arithmetic; the method `rref` also returns the reduced row echelon form.

Python

```
from sympy import Matrix
B = Matrix([[1, -1, 2, 0], [2, -2, 4, 0],
            [3, 1, -1, 2], [1, 3, -5, 2]])
print("rank B =", B.rank())
R, pivots = B.rref()
```

```
for row in R.tolist():
    print(row)
A = Matrix([[1, 2, -1], [2, 4, -2]])
b = Matrix([3, 5])
print(A.rank(), A.row_join(b).rank())
```

Output

```
rank B = 2
[1, 0, 1/4, 1/2]
[0, 1, -7/4, 1/2]
[0, 0, 0, 0]
[0, 0, 0, 0]
1 2
```

The last line shows $\operatorname{rg} A = 1 \neq \operatorname{rg}(A \mid b) = 2$, so the system is inconsistent. In Maple the same computation uses the `LinearAlgebra` package:

Maple

```
with(LinearAlgebra):
B := Matrix([[1,-1,2,0], [2,-2,4,0],
             [3,1,-1,2], [1,3,-5,2]]):
Rank(B);                    # 2
ReducedRowEchelonForm(B);
A := Matrix([[1,2,-1], [2,4,-2]]):
b := Vector([3,5]):
Rank(A), Rank(<A|b>);       # 1, 2
```

In Mathcad Prime the matrix is entered through the matrix template; the built-in functions `rank`, `rref` and `augment` do the rest (Mathcad displays the fractions as decimals).

Mathcad

```
B := | 1  -1   2   0 |
     | 2  -2   4   0 |
     | 3   1  -1   2 |
     | 1   3  -5   2 |
rank(B) = 2
          | 1  0   0.25  0.5 |
rref(B) = | 0  1  -1.75  0.5 |
          | 0  0   0     0   |
          | 0  0   0     0   |
A := | 1  2  -1 |      b := | 3 |
     | 2  4  -2 |           | 5 |
rank(A) = 1     rank(augment(A, b)) = 2
```

1.4.5 Exercises on the rank of a matrix

1. Find the rank of each matrix by means of minors.

1. $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & -1 \end{pmatrix}$

2. $\begin{pmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ 1 & 0 & -1 \end{pmatrix}$

3. $\begin{pmatrix} 3 & 1 & -2 \\ 6 & 2 & -4 \\ -3 & -1 & 2 \end{pmatrix}$

4. $\begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & -3 & 1 \end{pmatrix}$

5. $\begin{pmatrix} 0 & 1 & -3 \\ 2 & -1 & 5 \\ 4 & -2 & 10 \end{pmatrix}$

6. $\begin{pmatrix} 5 & -1 & 2 \\ 3 & 4 & 0 \\ -2 & 9 & -2 \end{pmatrix}$

7. $\begin{pmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ 3 & -6 & 9 \end{pmatrix}$

8. $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & -1 \\ 4 & 5 & -1 \end{pmatrix}$

9. $\begin{pmatrix} -1 & 3 & 2 \\ 2 & -6 & -4 \\ 0 & 0 & 0 \end{pmatrix}$

10. $\begin{pmatrix} 4 & -2 & 1 \\ 0 & 3 & -5 \\ 8 & -4 & 2 \end{pmatrix}$

11. $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$

12. $\begin{pmatrix} 3 & 0 & -1 \\ 0 & 2 & 4 \\ -6 & 0 & 2 \end{pmatrix}$

13. $\begin{pmatrix} 1 & 2 & -1 \\ 3 & 1 & 0 \\ 5 & 5 & -2 \end{pmatrix}$

14. $\begin{pmatrix} -2 & 4 & 1 \\ 1 & -2 & 3 \\ 0 & 0 & 7 \end{pmatrix}$

15. $\begin{pmatrix} 0 & 0 & 5 \\ 0 & 3 & -1 \\ 2 & 0 & 0 \end{pmatrix}$

16. $\begin{pmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ -1 & -3 & -5 \end{pmatrix}$

17. $\begin{pmatrix} 4 & -1 & 2 \\ -8 & 2 & -4 \\ 2 & 3 & -1 \end{pmatrix}$

18. $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -3 \\ 2 & -1 & 0 \end{pmatrix}$

19. $\begin{pmatrix} 3 & 6 & -3 \\ 1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$

20. $\begin{pmatrix} -1 & 0 & 4 \\ 2 & -3 & 1 \\ 1 & -3 & 5 \end{pmatrix}$

21. $\begin{pmatrix} 5 & 10 & -5 \\ 1 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

22. $\begin{pmatrix} 2 & -1 & 0 \\ -4 & 2 & 0 \\ 1 & 3 & -5 \end{pmatrix}$

23. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

24. $\begin{pmatrix} 3 & -2 & 1 \\ 0 & 0 & 0 \\ -6 & 4 & -2 \end{pmatrix}$

25. $\begin{pmatrix} 1 & -1 & 2 \\ 2 & 0 & 1 \\ 3 & -1 & 3 \end{pmatrix}$

26. $\begin{pmatrix} -3 & 1 & 5 \\ 6 & -2 & -10 \\ 0 & 4 & -1 \end{pmatrix}$

27. $\begin{pmatrix} 2 & 4 & -2 \\ -1 & -2 & 1 \\ 3 & 6 & -3 \end{pmatrix}$

28. $\begin{pmatrix} 0 & 3 & 1 \\ -2 & 0 & 4 \\ 2 & -3 & 3 \end{pmatrix}$ 29. $\begin{pmatrix} 1 & -3 & 0 \\ -2 & 6 & 0 \\ 4 & -12 & 0 \end{pmatrix}$ 30. $\begin{pmatrix} 7 & 0 & -1 \\ 0 & 5 & 2 \\ -14 & 0 & 2 \end{pmatrix}$

2. Find the rank of each matrix by Gaussian elimination (elementary row operations).

1. $\begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 4 & 0 \\ 3 & 1 & -1 & 2 \end{pmatrix}$

2. $\begin{pmatrix} 2 & 1 & -3 & 5 \\ 4 & 2 & -6 & 10 \\ 1 & 0 & 1 & -2 \end{pmatrix}$

3. $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{pmatrix}$

4. $\begin{pmatrix} 1 & 0 & -1 & 2 \\ 3 & 1 & 0 & -1 \\ 2 & 1 & 1 & -3 \end{pmatrix}$

5. $\begin{pmatrix} 0 & 1 & 2 & -3 \\ 1 & 0 & -1 & 4 \\ 2 & 1 & 0 & 5 \end{pmatrix}$

6. $\begin{pmatrix} 1 & 3 & -2 \\ 4 & 12 & -8 \\ -2 & -6 & 4 \end{pmatrix}$

7. $\begin{pmatrix} 3 & -1 & 2 & 0 \\ 6 & -2 & 4 & 0 \\ 1 & 1 & -1 & 3 \\ 4 & 0 & 1 & 3 \end{pmatrix}$

8. $\begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \\ -1 & 1 & 5 \\ 2 & 7 & 2 \end{pmatrix}$

9. $\begin{pmatrix} 2 & 0 & 1 & -3 \\ 4 & 0 & 2 & -6 \\ 1 & -1 & 0 & 2 \\ 3 & -1 & 1 & -1 \end{pmatrix}$

10. $\begin{pmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ 5 & -10 & 15 \end{pmatrix}$

11. $\begin{pmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 1 & 1 & 1 & 2 \end{pmatrix}$

12. $\begin{pmatrix} 1 & 1 & -1 & 0 \\ 2 & -1 & 0 & 3 \\ 3 & 0 & -1 & 3 \end{pmatrix}$

13. $\begin{pmatrix} 5 & -2 & 1 \\ -10 & 4 & -2 \\ 15 & -6 & 3 \end{pmatrix}$

14. $\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 1 & 1 & 1 & 4 \\ 2 & 1 & 3 & 5 \end{pmatrix}$

15. $\begin{pmatrix} -1 & 3 & 0 & 2 \\ 2 & -6 & 0 & -4 \\ 0 & 0 & 1 & 5 \end{pmatrix}$

16. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

17. $\begin{pmatrix} 2 & -1 & 0 & 4 \\ 0 & 3 & -2 & 1 \\ 4 & -2 & 0 & 8 \\ 2 & 2 & -2 & 5 \end{pmatrix}$

18. $\begin{pmatrix} 1 & -1 & 2 & -3 \\ 2 & -2 & 4 & -6 \\ -1 & 1 & -2 & 3 \end{pmatrix}$

19. $\begin{pmatrix} 0 & 2 & -1 \\ 3 & 0 & 4 \\ 6 & -4 & 9 \\ 3 & 2 & 3 \end{pmatrix}$

20. $\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$

21. $\begin{pmatrix} 3 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 6 & 2 & -4 & 0 \end{pmatrix}$

22. $\begin{pmatrix} 1 & -3 & 2 \\ -2 & 6 & -4 \\ 4 & -12 & 8 \\ 0 & 0 & 0 \end{pmatrix}$

23. $\begin{pmatrix} 2 & 1 & 0 & -1 \\ 0 & -2 & 3 & 4 \\ 2 & -1 & 3 & 3 \end{pmatrix}$

24. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{pmatrix}$

25. $\begin{pmatrix} 0 & 1 & -2 & 3 \\ 0 & 2 & -4 & 6 \\ 1 & 0 & 1 & -1 \\ 2 & -1 & 4 & -5 \end{pmatrix}$

26. $\begin{pmatrix} 4 & -2 & 6 \\ 2 & -1 & 3 \\ -6 & 3 & -9 \end{pmatrix}$

27. $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$

28. $\begin{pmatrix} 3 & -1 & 2 & 1 \\ 1 & 0 & -1 & 2 \\ 5 & -1 & 1 & 3 \\ 4 & -1 & 1 & 3 \end{pmatrix}$

29. $\begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & 2 \\ 0 & 2 & 2 \\ 2 & 4 & 0 \end{pmatrix}$

30. $\begin{pmatrix} 0 & 0 & 0 \\ 1 & -2 & 3 \\ -2 & 4 & -6 \\ 3 & -6 & 9 \end{pmatrix}$

3. Determine whether each matrix is singular or nonsingular (compute $\det A$*).*

1. $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

2. $\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$

3. $\begin{pmatrix} 5 & -1 \\ 3 & 2 \end{pmatrix}$

4. $\begin{pmatrix} -3 & 9 \\ 1 & -3 \end{pmatrix}$

5. $\begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}$

6. $\begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & -3 & 1 \end{pmatrix}$

7. $\begin{pmatrix} 3 & -1 & 2 \\ 0 & 5 & -4 \\ 6 & -2 & 4 \end{pmatrix}$

8. $\begin{pmatrix} 2 & 1 & -3 \\ 4 & 2 & -6 \\ 1 & 0 & 1 \end{pmatrix}$

9. $\begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \\ 2 & 0 & 5 \end{pmatrix}$

10. $\begin{pmatrix} -1 & 3 & 0 \\ 2 & -1 & 4 \\ 5 & 5 & 4 \end{pmatrix}$

11. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

12. $\begin{pmatrix} 7 & 14 \\ 2 & 4 \end{pmatrix}$

13. $\begin{pmatrix} -2 & 0 & 5 \\ 1 & 3 & -4 \\ 0 & 6 & -3 \end{pmatrix}$

14. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{pmatrix}$

15. $\begin{pmatrix} 3 & 6 & 9 \\ 1 & 2 & 3 \\ -2 & -4 & -6 \end{pmatrix}$

16. $\begin{pmatrix} 8 & -4 \\ -6 & 3 \end{pmatrix}$

17. $\begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

18. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{pmatrix}$

19. $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

20. $\begin{pmatrix} 2 & -3 & 1 \\ 4 & -6 & 2 \\ -1 & 0 & 3 \end{pmatrix}$

21. $\begin{pmatrix} 10 & 5 \\ -2 & -1 \end{pmatrix}$

22. $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$

23. $\begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & -4 \\ 4 & -1 & -2 \end{pmatrix}$

24. $\begin{pmatrix} 6 & 2 \\ 9 & 3 \end{pmatrix}$

25. $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 3 & 0 \\ 4 & 0 & -2 \end{pmatrix}$

26. $\begin{pmatrix} -5 & 1 & 0 \\ 3 & -2 & 4 \\ -7 & 3 & -4 \end{pmatrix}$

27. $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

28. $\begin{pmatrix} 3 & 1 & -2 \\ -6 & -2 & 4 \\ 9 & 3 & -6 \end{pmatrix}$

29. $\begin{pmatrix} 4 & -3 \\ -8 & 6 \end{pmatrix}$

30. $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ -1 & 0 & 6 \end{pmatrix}$

4. Using the Kronecker–Capelli theorem, find the ranks of the coefficient matrix and of the augmented matrix, and decide whether the system is consistent.

1. $\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 + 4x_2 - 2x_3 = 6 \\ x_1 + x_2 + x_3 = 1 \end{cases}$

2. $\begin{cases} x_1 - x_2 + 2x_3 = 1 \\ 2x_1 - 2x_2 + 4x_3 = 3 \end{cases}$

3. $\begin{cases} x_1 + x_2 + x_3 = 1 \\ 2x_1 + 3x_2 - x_3 = 4 \\ 3x_1 + 4x_2 = 5 \end{cases}$

4. $\begin{cases} 2x_1 - x_2 = 3 \\ -4x_1 + 2x_2 = -6 \end{cases}$

5. $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 4x_1 + 5x_2 + 6x_3 = 2 \\ 7x_1 + 8x_2 + 9x_3 = 3 \end{cases}$

6. $\begin{cases} x_1 - 3x_2 = 5 \\ 2x_1 - 6x_2 = 7 \end{cases}$

7. $\begin{cases} x_1 + x_2 - x_3 + x_4 = 2 \\ 2x_1 + 2x_2 - 2x_3 + 2x_4 = 4 \\ x_1 - x_2 + x_3 = 1 \end{cases}$

8. $\begin{cases} 3x_1 + x_2 - 2x_3 = 1 \\ 6x_1 + 2x_2 - 4x_3 = 5 \end{cases}$

9. $\begin{cases} x_1 + 2x_2 = 5 \\ 3x_1 - x_2 = 1 \\ 5x_1 + 3x_2 = 11 \end{cases}$

10. $\begin{cases} x_1 - x_2 + x_3 = 0 \\ 2x_1 + x_2 - x_3 = 0 \\ x_1 + 2x_2 - 2x_3 = 0 \end{cases}$

11. $\begin{cases} x_1 + x_2 = 3 \\ 2x_1 + 2x_2 = 6 \\ 3x_1 + 3x_2 = 9 \end{cases}$

12. $\begin{cases} 2x_1 - x_2 + 3x_3 = 4 \\ 4x_1 - 2x_2 + 6x_3 = 7 \end{cases}$

13. $\begin{cases} x_1 + 3x_2 - 2x_3 = 0 \\ 2x_1 + 6x_2 - 4x_3 = 0 \end{cases}$

14. $\begin{cases} x_1 - x_2 + x_3 - x_4 = 1 \\ 2x_1 + x_2 - x_3 + 2x_4 = 3 \\ x_1 + 2x_2 - 2x_3 + 3x_4 = 2 \end{cases}$

15. $\begin{cases} x_1 + 2x_2 + x_3 = 4 \\ 2x_1 + 4x_2 + 2x_3 = 8 \\ -x_1 - 2x_2 - x_3 = -4 \end{cases}$

16. $\begin{cases} 5x_1 - x_2 = 10 \\ -10x_1 + 2x_2 = -15 \end{cases}$

17. $\begin{cases} x_1 + x_2 + x_3 = 6 \\ x_1 - x_2 + x_3 = 2 \\ x_1 + x_2 - x_3 = 4 \end{cases}$

18. $\begin{cases} x_1 + 2x_2 - 3x_3 = 1 \\ 2x_1 + 4x_2 - 6x_3 = 2 \\ -x_1 - 2x_2 + 3x_3 = -1 \end{cases}$

19. $\begin{cases} x_1 - 2x_2 = 0 \\ 3x_1 + x_2 = 0 \\ 2x_1 + 3x_2 = 0 \end{cases}$

20. $\begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ 2x_1 - x_2 + 3x_3 = 5 \\ x_1 + 2x_2 - x_3 + 4x_4 = -5 \end{cases}$

21. $\begin{cases} 2x_1 + 3x_2 = 7 \\ 4x_1 + 6x_2 = 14 \end{cases}$

22. $\begin{cases} x_1 - x_2 + 2x_3 = 3 \\ 3x_1 - 3x_2 + 6x_3 = 9 \\ -2x_1 + 2x_2 - 4x_3 = -6 \end{cases}$

23. $\begin{cases} x_1 + 2x_2 - x_3 = 1 \\ 3x_1 + x_2 + 2x_3 = 4 \\ 5x_1 + 5x_2 = 6 \end{cases}$

24. $\begin{cases} x_1 - x_2 = 2 \\ 2x_1 + 3x_2 = 1 \\ 4x_1 + x_2 = 5 \end{cases}$

25. $\begin{cases} x_1 + x_2 + x_3 = 1 \\ 2x_1 + 2x_2 + 2x_3 = 2 \\ 3x_1 + 3x_2 + 3x_3 = 3 \end{cases}$

26. $\begin{cases} 3x_1 - 2x_2 + x_3 = 4 \\ 6x_1 - 4x_2 + 2x_3 = 1 \end{cases}$

27. $\begin{cases} x_1 + x_2 = 1 \\ x_1 - x_2 = 3 \\ 2x_1 + x_2 = 4 \end{cases}$

28. $\begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ 2x_1 + 4x_2 + 6x_3 = 0 \end{cases}$

29. $\begin{cases} x_1 - x_2 + x_3 = 2 \\ 2x_1 - 2x_2 + 2x_3 = 4 \\ 3x_1 - 3x_2 + 3x_3 = 7 \end{cases}$

30. $\begin{cases} x_1 + x_2 - x_3 = 3 \\ 2x_1 - x_2 + x_3 = 1 \\ 3x_1 + x_2 - x_3 = 5 \end{cases}$

1.5 The inverse matrix

Definition 1.5.1. Let A be a square matrix of order n. A matrix A^{-1} such that

$$A \cdot A^{-1} = A^{-1} \cdot A = E,$$

where E is the identity matrix of order n, is called the **inverse** of A.

Not every square matrix has an inverse. The answer to the question of existence is given by the determinant.

Theorem 1.5.1. *A matrix A has an inverse if and only if* $\det A \neq 0$*, that is, if and only if A is* ***nonsingular****. In this case the inverse is unique.*

The inverse via cofactors

Definition 1.5.2. If A is an $n \times n$ matrix with $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} \cdot \widetilde{A}^T,$$

where $\widetilde{A} = (A_{ij})$ is the matrix of cofactors (algebraic complements) of A. Its transpose $\widetilde{A}^T$ is called the **adjugate** (or adjoint) matrix and is denoted by adj A.

For 2×2 and 3×3 matrices the computation is organized as follows.

1. Compute $\det A$. If $\det A = 0$, the inverse does not exist.
2. For every entry a_{ij} find the cofactor $A_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the corresponding minor.
3. Form the cofactor matrix $\widetilde{A} = (A_{ij})$ and transpose it to obtain $\operatorname{adj} A = \widetilde{A}^T$.
4. Divide by the determinant: $A^{-1} = \dfrac{1}{\det A} \cdot \widetilde{A}^T$.

It is always a good idea to finish by checking that $A \cdot A^{-1} = E$.

The inverse of a 2×2 matrix. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\det A = ad - bc \neq 0$, then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

In words: interchange the diagonal entries, change the signs of the off-diagonal ones and divide by the determinant.

Example 1.5.1. Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$.

Solution. Here $\det A = 3 \cdot 2 - 1 \cdot 5 = 6 - 5 = 1 \neq 0$, so the inverse exists and

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}.$$

As a check,

$$A \cdot A^{-1} = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 6-5 & -3+3 \\ 10-10 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = E.$$

■

Example 1.5.2. For $A = \begin{pmatrix} 4 & -2 \\ 6 & -3 \end{pmatrix}$ we find $\det A = 4 \cdot (-3) - (-2) \cdot 6 = -12 + 12 = 0$. The determinant vanishes, so A has **no inverse**.

The inverse of a 3×3 matrix.

Example 1.5.3. Find the inverse of $A = \begin{pmatrix} 2 & 5 & 7 \\ 6 & 3 & 4 \\ 5 & -2 & -3 \end{pmatrix}$.

Solution. We first expand the determinant along the first row:

$$\begin{aligned} \det A &= 2\big(3 \cdot (-3) - 4 \cdot (-2)\big) - 5\big(6 \cdot (-3) - 4 \cdot 5\big) \\ &\quad + 7\big(6 \cdot (-2) - 3 \cdot 5\big) \\ &= 2 \cdot (-1) - 5 \cdot (-38) + 7 \cdot (-27) \\ &= -2 + 190 - 189 = -1 \neq 0. \end{aligned}$$

Next we compute the nine cofactors:

$$A_{11} = +\begin{vmatrix} 3 & 4 \\ -2 & -3 \end{vmatrix} = -9 + 8 = -1,$$

$$A_{12} = -\begin{vmatrix} 6 & 4 \\ 5 & -3 \end{vmatrix} = -(-18 - 20) = 38,$$

$$A_{13} = +\begin{vmatrix} 6 & 3 \\ 5 & -2 \end{vmatrix} = -12 - 15 = -27,$$

$$A_{21} = -\begin{vmatrix} 5 & 7 \\ -2 & -3 \end{vmatrix} = -(-15 + 14) = 1,$$

$$A_{22} = +\begin{vmatrix} 2 & 7 \\ 5 & -3 \end{vmatrix} = -6 - 35 = -41,$$

$$A_{23} = -\begin{vmatrix} 2 & 5 \\ 5 & -2 \end{vmatrix} = -(-4 - 25) = 29,$$

$$A_{31} = +\begin{vmatrix} 5 & 7 \\ 3 & 4 \end{vmatrix} = 20 - 21 = -1,$$

$$A_{32} = -\begin{vmatrix} 2 & 7 \\ 6 & 4 \end{vmatrix} = -(8 - 42) = 34,$$

$$A_{33} = +\begin{vmatrix} 2 & 5 \\ 6 & 3 \end{vmatrix} = 6 - 30 = -24.$$

Writing the cofactors of the ith row of A into the ith column, we obtain the adjugate matrix

$$\operatorname{adj} A = \widetilde{A}^T = \begin{pmatrix} -1 & 1 & -1 \\ 38 & -41 & 34 \\ -27 & 29 & -24 \end{pmatrix},$$

and finally

$$A^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & 1 & -1 \\ 38 & -41 & 34 \\ -27 & 29 & -24 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ -38 & 41 & -34 \\ 27 & -29 & 24 \end{pmatrix}.$$

The reader is encouraged to verify that $A \cdot A^{-1} = E$; the computation is also repeated by computer at the end of this section. ■

Properties of the inverse

Let A and B be nonsingular $n \times n$ matrices and let $k \neq 0$ be a scalar. Then

1) $(A^{-1})^{-1} = A$;

2) $(AB)^{-1} = B^{-1}A^{-1}$ (the order of the factors is reversed);

3) $(A^T)^{-1} = (A^{-1})^T$ (inversion commutes with transposition);

4) $(kA)^{-1} = \dfrac{1}{k}A^{-1}$;

5) $\det(A^{-1}) = \dfrac{1}{\det A}$;

6) the inverse A^{-1} is unique.

Example 1.5.4. Let us illustrate property 2 with $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$. By the 2×2 formula, $A^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$. Further,

$$AB = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix},$$

$$(AB)^{-1} = \frac{1}{7-6}\begin{pmatrix} 1 & -2 \\ -3 & 7 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -3 & 7 \end{pmatrix},$$

and indeed

$$B^{-1}A^{-1} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -3 & 7 \end{pmatrix}.$$

Matrix equations. The inverse matrix allows us to solve matrix equations of the form $AX = B$, $XA = B$ and $AXB = C$ with nonsingular A (and B in the last case).

Definition 1.5.3.
- $AX = B \implies X = A^{-1}B$ (multiply by A^{-1} on the left).
- $XA = B \implies X = BA^{-1}$ (multiply by A^{-1} on the right).
- $AXB = C \implies X = A^{-1}CB^{-1}$.

Since matrix multiplication is not commutative, the side on which we multiply matters: in general $A^{-1}B \neq BA^{-1}$.

Example 1.5.5. Solve the equation $AX = B$, where $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$.

Solution. The inverse of A was found in the first example of this section: $A^{-1} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$. Hence

$$X = A^{-1}B = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2+1 & 0-2 \\ -5-3 & 0+6 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ -8 & 6 \end{pmatrix}.$$

Substituting back, we get

$$AX = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -8 & 6 \end{pmatrix} = \begin{pmatrix} 9-8 & -6+6 \\ 15-16 & -10+12 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} = B.$$

■

Example 1.5.6. Solve the equation $XA = B$, where $A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix}$.

Solution. Since $\det A = 5 - 6 = -1 \neq 0$, the matrix A is invertible and $A^{-1} = \frac{1}{-1}\begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$. Now we multiply by A^{-1} on the right:

$$X = BA^{-1} = \begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} -20 - 3 & 8 + 1 \\ -10 + 9 & 4 - 3 \end{pmatrix} = \begin{pmatrix} -23 & 9 \\ -1 & 1 \end{pmatrix}.$$

■

1.5.1 The inverse matrix in information security

Decryption in the Hill cipher. In the Hill cipher a plaintext block p is encrypted by the rule $c = Kp \pmod{26}$. To decrypt, we need the inverse key:

$$p = K^{-1}c \pmod{26}.$$

Here K^{-1} must be computed modulo 26. We find $\det K \pmod{26}$ and check that $\gcd(\det K, 26) = 1$; then we find the multiplicative inverse $(\det K)^{-1} \pmod{26}$ and put

$$K^{-1} \equiv (\det K)^{-1} \cdot \operatorname{adj} K \pmod{26}.$$

Example 1.5.7. Take the key matrix $K = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$. Then $\det K = 5 - 6 = -1 \equiv 25 \pmod{26}$. Since $25 \cdot 25 = 625 = 24 \cdot 26 + 1$, we have $25^{-1} \equiv 25 \pmod{26}$. Moreover, $\operatorname{adj} K = \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix} \equiv \begin{pmatrix} 5 & 24 \\ 23 & 1 \end{pmatrix} \pmod{26}$, and

therefore

$$K^{-1} \equiv 25 \cdot \begin{pmatrix} 5 & 24 \\ 23 & 1 \end{pmatrix} = \begin{pmatrix} 125 & 600 \\ 575 & 25 \end{pmatrix} \equiv \begin{pmatrix} 21 & 2 \\ 3 & 25 \end{pmatrix} \pmod{26}.$$

If the ciphertext is $c = \begin{pmatrix} 10 \\ 3 \end{pmatrix}$, then $p = K^{-1}c = \begin{pmatrix} 21 & 2 \\ 3 & 25 \end{pmatrix} \begin{pmatrix} 10 \\ 3 \end{pmatrix} = \begin{pmatrix} 216 \\ 105 \end{pmatrix} \equiv \begin{pmatrix} 8 \\ 1 \end{pmatrix} \pmod{26}$. With the encoding $A = 0$, $B = 1$, $\ldots$, $Z = 25$ these numbers are the letters "I" and "B".

Digital signatures and authentication. Some challenge–response protocols use matrix transformations: the server sends a matrix A, and the user proves his or her identity by returning its inverse A^{-1} (or the result of applying it). How fast the inverse can be computed largely determines the efficiency of such protocols.

Decoding error-correcting codes. In coding theory, a received word is decoded with the help of the inverse of a suitable part of the generator matrix. For instance, to extract the information bits from a codeword of a systematic code one applies the inverse of the information part of G.

Transformations in computer graphics. Visualization systems, including those that display encrypted 3D data, describe rotations, scalings and translations by matrices. To undo a transformation, one applies the inverse matrix.

1.5.2 Computing with Python, Maple and Mathcad

We repeat by computer the inversion of the 3×3 matrix $A = \begin{pmatrix} 2 & 5 & 7 \\ 6 & 3 & 4 \\ 5 & -2 & -3 \end{pmatrix}$ and the solution of the matrix equation $XA = B$ from the last example (there the matrices are called `P` and `Q`).

Python

```
from sympy import Matrix, eye
A = Matrix([[2, 5, 7], [6, 3, 4], [5, -2, -3]])
print("det A =", A.det())
Ainv = A.inv()
for row in Ainv.tolist():
    print(row)
print(A * Ainv == eye(3))
```

```
P = Matrix([[1, 2], [3, 5]])
Q = Matrix([[4, -1], [2, 3]])
print(Q * P.inv())      # solution of X*P = Q
```

Output

```
det A = -1
[1, -1, 1]
[-38, 41, -34]
[27, -29, 24]
True
Matrix([[-23, 9], [-1, 1]])
```

The method `adjugate()` returns $\operatorname{adj} A$ if one wants to follow the cofactor method literally. In Maple:

Maple

```
with(LinearAlgebra):
A := Matrix([[2,5,7], [6,3,4], [5,-2,-3]]):
Determinant(A);             # -1
Adjoint(A);                 # adjugate matrix
Ainv := MatrixInverse(A);
A . Ainv;                   # identity matrix
P := Matrix([[1,2], [3,5]]):
Q := Matrix([[4,-1], [2,3]]):
X := Q . MatrixInverse(P);  # [[-23,9],[-1,1]]
```

In Mathcad Prime the inverse is obtained with the power -1 and the determinant with the operator $|A|$; the identity matrix is `identity(3)`.

Mathcad

```
A := | 2   5   7 |
     | 6   3   4 |
     | 5  -2  -3 |
|A| = -1
          |   1   -1    1 |
A^-1 =    | -38   41  -34 |
          |  27  -29   24 |
           | 1  0  0 |
A*A^-1 =   | 0  1  0 |
           | 0  0  1 |
P := | 1  2 |     Q := | 4  -1 |
     | 3  5 |          | 2   3 |
X := Q*P^-1       X = | -23  9 |
                      |  -1  1 |
```

1.5.3 Exercises on the inverse matrix

1. Find the inverse of each 2×2 *matrix, or show that it does not exist.*

1. $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$
2. $\begin{pmatrix} 4 & -2 \\ 6 & -3 \end{pmatrix}$
3. $\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$
4. $\begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$
5. $\begin{pmatrix} 5 & 0 \\ 0 & -3 \end{pmatrix}$
6. $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$
7. $\begin{pmatrix} -1 & 3 \\ 2 & -5 \end{pmatrix}$
8. $\begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix}$
9. $\begin{pmatrix} 6 & -4 \\ -9 & 6 \end{pmatrix}$
10. $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$
11. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
12. $\begin{pmatrix} 3 & 6 \\ 2 & 4 \end{pmatrix}$
13. $\begin{pmatrix} -4 & 1 \\ 2 & 3 \end{pmatrix}$
14. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
15. $\begin{pmatrix} 2 & -3 \\ -4 & 6 \end{pmatrix}$
16. $\begin{pmatrix} 5 & 2 \\ -1 & 3 \end{pmatrix}$
17. $\begin{pmatrix} -3 & -1 \\ 6 & 2 \end{pmatrix}$
18. $\begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix}$
19. $\begin{pmatrix} 1 & 4 \\ 2 & 7 \end{pmatrix}$
20. $\begin{pmatrix} -2 & 0 \\ 0 & -5 \end{pmatrix}$
21. $\begin{pmatrix} 4 & 3 \\ -8 & -6 \end{pmatrix}$
22. $\begin{pmatrix} 9 & -1 \\ 4 & 1 \end{pmatrix}$
23. $\begin{pmatrix} -5 & 2 \\ 10 & -4 \end{pmatrix}$
24. $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$
25. $\begin{pmatrix} 3 & -5 \\ 1 & -2 \end{pmatrix}$
26. $\begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$
27. $\begin{pmatrix} -1 & -4 \\ 3 & 11 \end{pmatrix}$
28. $\begin{pmatrix} 6 & 1 \\ -2 & 3 \end{pmatrix}$
29. $\begin{pmatrix} 10 & 5 \\ -2 & -1 \end{pmatrix}$
30. $\begin{pmatrix} 7 & -3 \\ 2 & 1 \end{pmatrix}$

2. Find the inverse of each 3×3 *matrix by the cofactor method.*

1. $\begin{pmatrix} 2 & 5 & 7 \\ 6 & 3 & 4 \\ 5 & -2 & -3 \end{pmatrix}$
2. $\begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & -1 \\ 2 & 1 & 0 \end{pmatrix}$
3. $\begin{pmatrix} 3 & 0 & 1 \\ -1 & 2 & -4 \\ 0 & 5 & 3 \end{pmatrix}$
4. $\begin{pmatrix} -2 & 1 & 3 \\ 4 & 0 & -1 \\ 1 & -2 & 5 \end{pmatrix}$
5. $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -3 \\ -1 & 0 & 4 \end{pmatrix}$
6. $\begin{pmatrix} 4 & -1 & 2 \\ 3 & 5 & 0 \\ -2 & 0 & 1 \end{pmatrix}$

7. $\begin{pmatrix} 0 & 1 & -2 \\ 3 & 0 & 4 \\ -1 & 2 & 0 \end{pmatrix}$

8. $\begin{pmatrix} 1 & 0 & 3 \\ -2 & 1 & 0 \\ 4 & -3 & 2 \end{pmatrix}$

9. $\begin{pmatrix} -3 & 2 & 1 \\ 0 & -4 & 5 \\ 2 & 0 & -1 \end{pmatrix}$

10. $\begin{pmatrix} 5 & -2 & 0 \\ 1 & 3 & -1 \\ 0 & 4 & 2 \end{pmatrix}$

11. $\begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & -2 \\ 1 & 0 & 5 \end{pmatrix}$

12. $\begin{pmatrix} -1 & 0 & 2 \\ 3 & -5 & 1 \\ 2 & 4 & 0 \end{pmatrix}$

13. $\begin{pmatrix} 1 & 3 & -1 \\ -2 & 0 & 4 \\ 0 & 5 & 2 \end{pmatrix}$

14. $\begin{pmatrix} 4 & 0 & -3 \\ 2 & 1 & 5 \\ -1 & -2 & 0 \end{pmatrix}$

15. $\begin{pmatrix} 0 & -3 & 1 \\ 5 & 2 & 0 \\ -4 & 0 & 3 \end{pmatrix}$

16. $\begin{pmatrix} 3 & 1 & 0 \\ -2 & 4 & -1 \\ 0 & -3 & 5 \end{pmatrix}$

17. $\begin{pmatrix} -1 & 2 & -3 \\ 4 & 0 & 1 \\ 0 & -5 & 2 \end{pmatrix}$

18. $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 3 & 4 \\ -3 & 1 & 0 \end{pmatrix}$

19. $\begin{pmatrix} 1 & -4 & 0 \\ 3 & 2 & -5 \\ -2 & 0 & 1 \end{pmatrix}$

20. $\begin{pmatrix} 0 & 5 & -2 \\ -1 & 0 & 3 \\ 4 & -3 & 0 \end{pmatrix}$

21. $\begin{pmatrix} -2 & 0 & 4 \\ 1 & -3 & 0 \\ 0 & 5 & -1 \end{pmatrix}$

22. $\begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & -4 \\ 5 & 0 & 2 \end{pmatrix}$

23. $\begin{pmatrix} 1 & 0 & -5 \\ -3 & 4 & 0 \\ 2 & -1 & 3 \end{pmatrix}$

24. $\begin{pmatrix} 0 & -1 & 3 \\ 2 & 5 & 0 \\ -4 & 0 & 1 \end{pmatrix}$

25. $\begin{pmatrix} 4 & 3 & -1 \\ 0 & -2 & 5 \\ 1 & 0 & 0 \end{pmatrix}$

26. $\begin{pmatrix} -3 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & -1 & 5 \end{pmatrix}$

27. $\begin{pmatrix} 2 & -5 & 1 \\ 0 & 3 & 0 \\ -1 & 0 & 4 \end{pmatrix}$

28. $\begin{pmatrix} 0 & 2 & -4 \\ -3 & 0 & 1 \\ 5 & -1 & 0 \end{pmatrix}$

29. $\begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 3 \\ 3 & 0 & -1 \end{pmatrix}$

30. $\begin{pmatrix} -4 & 3 & 0 \\ 2 & 0 & -5 \\ 0 & 1 & 3 \end{pmatrix}$

3. Solve the matrix equation $AX = B$.

1. $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

2. $A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 1 \\ 2 & -3 \end{pmatrix}$.

3. $A = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$.

4. $A = \begin{pmatrix} 5 & 2 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$.

5. $A = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

6. $A = \begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 4 \\ 2 & -3 \end{pmatrix}$.

7. $A = \begin{pmatrix} -4 & 1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$.

8. $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$.

9. $A = \begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

10. $A = \begin{pmatrix} 9 & -1 \\ 4 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 5 \\ 0 & -2 \end{pmatrix}$.

11. $A = \begin{pmatrix} 3 & -5 \\ 1 & -2 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

12. $A = \begin{pmatrix} 6 & 1 \\ -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 \\ 0 & 5 \end{pmatrix}$.

13. $A = \begin{pmatrix} 1 & 4 \\ 2 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

14. $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 5 & -3 \\ 1 & 2 \end{pmatrix}$.

15. $A = \begin{pmatrix} -1 & 3 \\ 2 & -5 \end{pmatrix}$, $B = \begin{pmatrix} 4 \\ -7 \end{pmatrix}$.

16. $A = \begin{pmatrix} 4 & 3 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$.

17. $A = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

18. $A = \begin{pmatrix} 5 & 0 \\ 0 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 10 & -5 \\ 6 & -9 \end{pmatrix}$.

19. $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 7 \\ -3 \end{pmatrix}$.

20. $A = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 4 & -6 \\ 9 & 3 \end{pmatrix}$.

21. $A = \begin{pmatrix} 7 & -3 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

22. $A = \begin{pmatrix} -3 & -1 \\ 6 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

23. $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$.

24. $A = \begin{pmatrix} 1 & -2 \\ 3 & -5 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

25. $A = \begin{pmatrix} -1 & -4 \\ 3 & 11 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 2 \\ -1 & 5 \end{pmatrix}$.

26. $A = \begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$.

27. $A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 1 \\ -2 & 4 \end{pmatrix}$.

28. $A = \begin{pmatrix} -5 & 2 \\ 10 & -4 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

29. $A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 4 \\ 2 & -5 \end{pmatrix}$.

30. $A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 \\ -2 & 1 \end{pmatrix}$.

4. Solve the matrix equation $XA = B$.

1. $A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix}$.

2. $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 1 \\ 0 & 3 \end{pmatrix}$.

3. $A = \begin{pmatrix} -1 & 3 \\ 2 & -5 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ -3 & 2 \end{pmatrix}$.

4. $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 4 & 2 \end{pmatrix}$.

5. $A = \begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 1 \\ -2 & 0 \end{pmatrix}$.

6. $A = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & -4 \\ 1 & 0 \end{pmatrix}$.

7. $A = \begin{pmatrix} 5 & 2 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$.

8. $A = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ -2 & 4 \end{pmatrix}$.

9. $A = \begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 4 & 3 \end{pmatrix}$.

10. $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -3 \\ 5 & 1 \end{pmatrix}$.

11. $A = \begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 6 & 0 \\ -1 & 5 \end{pmatrix}$.

12. $A = \begin{pmatrix} 9 & -1 \\ 4 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -2 \\ 1 & 4 \end{pmatrix}$.

13. $A = \begin{pmatrix} 3 & -5 \\ 1 & -2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 3 \\ -1 & 2 \end{pmatrix}$.

14. $A = \begin{pmatrix} 6 & 1 \\ -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -3 & 1 \\ 2 & 0 \end{pmatrix}$.

15. $A = \begin{pmatrix} 1 & 4 \\ 2 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 3 & -1 \\ 0 & 2 \end{pmatrix}$.

16. $A = \begin{pmatrix} -4 & 1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 5 & 2 \end{pmatrix}$.

17. $A = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 0 \\ -2 & 3 \end{pmatrix}$.

18. $A = \begin{pmatrix} 5 & 0 \\ 0 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 10 & 0 \\ 0 & 9 \end{pmatrix}$.

19. $A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}$.

20. $A = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 6 & 4 \\ -3 & 6 \end{pmatrix}$.

21. $A = \begin{pmatrix} 7 & -3 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 2 & 5 \end{pmatrix}$.

22. $A = \begin{pmatrix} 1 & -2 \\ 3 & -5 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ -3 & 1 \end{pmatrix}$.

23. $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 3 \\ 4 & -2 \end{pmatrix}$.

24. $A = \begin{pmatrix} -1 & -4 \\ 3 & 11 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 0 \\ -1 & 3 \end{pmatrix}$.

25. $A = \begin{pmatrix} 4 & 3 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ -2 & 4 \end{pmatrix}$.

26. $A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 4 & -3 \\ 2 & 1 \end{pmatrix}$.

27. $A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 3 \\ -1 & 2 \end{pmatrix}$.

28. $A = \begin{pmatrix} -3 & 2 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 6 & -4 \\ 3 & 1 \end{pmatrix}$.

29. $A = \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ -2 & 0 \end{pmatrix}$.

30. $A = \begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -3 \\ 0 & 5 \end{pmatrix}$.

1.6 Computing the inverse matrix by elementary row operations

Definition 1.6.1. The method. To find the inverse of a square matrix A of order n, we write the identity matrix E of order n to the right of A and obtain the $n \times 2n$ **augmented matrix** $(A \mid E)$. Elementary row operations are then applied to the whole augmented matrix until its left half becomes the identity matrix. At that moment the right half is A^{-1}:

$$(A \mid E) \xrightarrow{\text{elementary row operations}} (E \mid A^{-1}).$$

This procedure is known as the **Gauss–Jordan method**.

Recall the three **elementary row operations**:

1) interchanging two rows: $S_i \leftrightarrow S_j$;

2) multiplying a row by a nonzero number: $S_i \to k \cdot S_i$, $k \neq 0$;

3) adding to a row a multiple of another row: $S_i \to S_i + k \cdot S_j$.

If at some stage the left half cannot be reduced to the identity matrix (for instance, a zero appears on the main diagonal and no row below it can replace it), then $\det A = 0$ and A has **no inverse**.

1.6.1 Inverting 2×2 and 3×3 matrices by row operations

Matrices of order 2

Example 1.6.1. Find the inverse of $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ by elementary row operations. **Solution.** We form the augmented matrix

$$(A \mid E) = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array}\right).$$

Dividing the first row by 2 ($S_1 \to \frac{1}{2}S_1$) makes the leading entry equal to 1:

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 5 & 3 & 0 & 1 \end{array}\right).$$

Next we clear the entry 5 below it by $S_2 \to S_2 - 5S_1$:

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{5}{2} & 1 \end{array}\right).$$

Multiplying the second row by 2 gives a unit pivot in the second column,

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & -5 & 2 \end{array}\right),$$

and the operation $S_1 \to S_1 - \frac{1}{2}S_2$ removes the entry $\frac{1}{2}$ above that pivot:

$$\left(\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -5 & 2 \end{array}\right).$$

The left half is now E, hence

$$A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

Check: $A \cdot A^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.\checkmark$

Example 1.6.2. Let $A = \begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}$. One row operation is enough to see what happens:

$$(A \mid E) = \left(\begin{array}{cc|cc} 4 & 6 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right) \xrightarrow{S_2 - \frac{1}{2}S_1} \left(\begin{array}{cc|cc} 4 & 6 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{array}\right).$$

The second row of the left half has become zero, so no pivot can be created in the second column. Therefore $\det A = 0$ and the inverse **does not exist**.

Matrices of order 3

Example 1.6.3. Find the inverse of $A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{pmatrix}$. **Solution.** We start from

$$(A \mid E) = \left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 2 & 3 & 0 & 0 & 0 & 1 \end{array}\right).$$

The first column already has $a_{11} = 1$ and $a_{21} = 0$, so only a_{31} has to be eliminated; the operation $S_3 \to S_3 - 2S_1$ gives

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 0 & 5 & -2 & -2 & 0 & 1 \end{array}\right).$$

We now divide the second row by 2,

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 5 & -2 & -2 & 0 & 1 \end{array}\right),$$

and use the new pivot to clear the second column: $S_1 \to S_1 + S_2$ and $S_3 \to S_3 - 5S_2$. This yields

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & 1 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & -2 & -\frac{5}{2} & 1 \end{array}\right).$$

Multiplying the third row by 2 makes $a_{33} = 1$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & 1 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -4 & -5 & 2 \end{array}\right).$$

Finally, $S_1 \to S_1 - \frac{1}{2}S_3$ and $S_2 \to S_2 + \frac{1}{2}S_3$ remove the entries above the last pivot:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 3 & -1 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & -4 & -5 & 2 \end{array}\right).$$

Consequently,

$$A^{-1} = \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix}.$$

Check:

$$A \cdot A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{pmatrix} \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3+2-4 & 3+2-5 & -1-1+2 \\ 0-4+4 & 0-4+5 & 0+2-2 \\ 6-6+0 & 6-6+0 & -2+3+0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \checkmark$$

Example 1.6.4. Let $A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 3 & 2 & 1 \end{pmatrix}$. Here it is convenient to begin by interchanging the first two rows, so that the first pivot equals 1:

$$\left(\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{S_1 \leftrightarrow S_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 2 & -1 & 0 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[S_3-3S_1]{S_2-2S_1} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 1 & -2 & 0 \\ 0 & 2 & 4 & 0 & -3 & 1 \end{array}\right)$$

$$\xrightarrow{S_2 \to -S_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & -1 & 2 & 0 \\ 0 & 2 & 4 & 0 & -3 & 1 \end{array}\right)$$

$$\xrightarrow{S_3-2S_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & -1 & 2 & 0 \\ 0 & 0 & 8 & 2 & -7 & 1 \end{array}\right) \xrightarrow{S_3 \to \frac{1}{8} S_3}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & -1 & 2 & 0 \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{7}{8} & \frac{1}{8} \end{array}\right)$$

$$\xrightarrow[S_2+2S_3]{S_1+S_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{4} & -\frac{7}{8} & \frac{1}{8} \end{array}\right).$$

Taking out the common factor $\frac{1}{8}$, we obtain

$$A^{-1} = \frac{1}{8} \begin{pmatrix} 2 & 1 & 1 \\ -4 & 2 & 2 \\ 2 & -7 & 1 \end{pmatrix}.$$

Comparison with the cofactor method.

Cofactor (adjugate) method	Row-operation method
Based on the formula $A^{-1} = \frac{1}{\det A}\operatorname{adj} A$	Based on an algorithm (Gauss–Jordan elimination)
Convenient for 2×2 and 3×3 matrices	Much more efficient for matrices of higher order
Each cofactor A_{ij} has to be computed separately	All operations are carried out in a single array
About $O(n!\cdot n)$ operations if determinants are expanded by cofactors	$O(n^3)$ operations
Frequently used in theoretical arguments	Preferred in hand computation and in computer software

1.6.2 Applications of the method in information security

The inverse matrix in AES. The **MixColumns** step of the AES (Advanced Encryption Standard) algorithm multiplies each column of the state by a fixed 4×4 matrix over the field $\mathrm{GF}(2^8)$. Decryption requires the inverse of this matrix, and Gauss–Jordan elimination carried out in $\mathrm{GF}(2^8)$ arithmetic is a natural way to obtain it.

The LWE cryptosystem. In the post-quantum LWE (Learning With Errors) scheme the public key has the form $b = As + e$. Without the noise vector e, the secret s could be recovered by Gauss–Jordan elimination in $O(n^3)$ operations. It is precisely the small added noise that makes elimination useless and keeps the scheme secure.

Inverting the key matrix of the Hill cipher. In practice the key matrix of a Hill cipher may be of order 3 or larger. For such matrices the cofactor method involves a great deal of computation, whereas Gauss–Jordan elimination adapts easily to arithmetic modulo 26 and inverts large key matrices efficiently. The modular version differs from the ordinary one in three respects.

- All operations are performed modulo 26.
- Division is replaced by multiplication by a modular inverse; the inverse $a^{-1} \pmod{26}$ exists only when $\gcd(a, 26) = 1$.
- If a candidate pivot a_{ii} satisfies $\gcd(a_{ii}, 26) \neq 1$, rows are interchanged (or another suitable combination of rows is used) to obtain an invertible pivot.

Efficient computer implementation. Software libraries for information security, encryption libraries among them, compute inverse matrices by Gauss–Jordan

elimination or one of its variants. The method needs $O(n^3)$ arithmetic operations, which is practical even for large matrices; it works in place, on a single array, so it is economical in memory; and with **partial pivoting** it is numerically stable.

Lattice-based cryptography and large matrices. Post-quantum schemes such as LWE and NTRU work with very large matrices, with hundreds or thousands of rows and columns. To invert such a matrix, or merely to decide whether it is invertible, one uses algorithms based on Gaussian elimination; in practice there is no real alternative.

Systematic encoding in coding theory. A generator matrix of a linear code is brought to the systematic form $G = (I_k \mid P)$ by elementary row operations. In essence this means reducing the left block of the matrix to the identity, that is, a form of Gauss–Jordan elimination.

1.6.3 Computing with Python, Maple and Mathcad

Any computer algebra system can carry out the reduction $(A \mid E) \to (E \mid A^{-1})$ directly. We repeat the third example above, with the same 3×3 matrix A. In SymPy the method `rref` returns the reduced row echelon form of the augmented matrix, and its last three columns give A^{-1}:

Python

```
from sympy import Matrix, eye
A = Matrix([[1, -1, 1], [0, 2, -1], [2, 3, 0]])
R, piv = A.row_join(eye(3)).rref()
Ainv = R[:, 3:]        # right half of (E | A^-1)
print(Ainv)
print(A * Ainv == eye(3))
```

Output

```
Matrix([[3, 3, -1], [-2, -2, 1], [-4, -5, 2]])
True
```

In Maple the augmented matrix is written as `<A|E>`, and the command `ReducedRowEchelonForm` performs the Gauss–Jordan reduction; the result can be compared with `MatrixInverse`:

Maple

```
with(LinearAlgebra):
A := Matrix([[1, -1, 1], [0, 2, -1], [2, 3, 0]]):
R := ReducedRowEchelonForm(<A | IdentityMatrix(3)>);
R[1..3, 4..6];
MatrixInverse(A);
```

In Mathcad Prime the functions `augment`, `identity`, `rref` and `submatrix` do the same job (with the default `ORIGIN = 0` the rows and columns are numbered from zero); the operator A^{-1} gives the inverse at once:

Mathcad

```
     | 1  -1   1 |
A := | 0   2  -1 |
     | 2   3   0 |
R := rref(augment(A, identity(3)))

                              |  3   3  -1 |
submatrix(R, 0, 2, 3, 5) =    | -2  -2   1 |
                              | -4  -5   2 |
        |  3   3  -1 |
A^-1 =  | -2  -2   1 |
        | -4  -5   2 |
```

1.6.4 Exercises on inverting matrices by row operations

1. Find the inverses of the following 2×2 *matrices by the Gauss–Jordan method.*

1. $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$
2. $\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$
3. $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$
4. $\begin{pmatrix} 4 & 2 \\ 7 & 4 \end{pmatrix}$
5. $\begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$
6. $\begin{pmatrix} 6 & 1 \\ -2 & 3 \end{pmatrix}$
7. $\begin{pmatrix} 1 & -4 \\ -2 & 7 \end{pmatrix}$
8. $\begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}$
9. $\begin{pmatrix} -3 & 2 \\ 4 & -3 \end{pmatrix}$
10. $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$
11. $\begin{pmatrix} 7 & -2 \\ 3 & -1 \end{pmatrix}$
12. $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$
13. $\begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix}$
14. $\begin{pmatrix} -2 & -1 \\ 5 & 3 \end{pmatrix}$
15. $\begin{pmatrix} 4 & -3 \\ 1 & -1 \end{pmatrix}$
16. $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$
17. $\begin{pmatrix} 9 & -1 \\ 4 & 1 \end{pmatrix}$
18. $\begin{pmatrix} -1 & -3 \\ 2 & 5 \end{pmatrix}$
19. $\begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}$
20. $\begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}$
21. $\begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$

22. $\begin{pmatrix} -4 & 1 \\ 3 & -1 \end{pmatrix}$

23. $\begin{pmatrix} 1 & -2 \\ 3 & -5 \end{pmatrix}$

24. $\begin{pmatrix} 6 & -5 \\ -1 & 1 \end{pmatrix}$

25. $\begin{pmatrix} 3 & 2 \\ -4 & -3 \end{pmatrix}$

26. $\begin{pmatrix} 0 & -2 \\ 3 & 1 \end{pmatrix}$

27. $\begin{pmatrix} 7 & 3 \\ 2 & 1 \end{pmatrix}$

28. $\begin{pmatrix} -3 & -1 \\ 7 & 2 \end{pmatrix}$

29. $\begin{pmatrix} 1 & 0 \\ 0 & -5 \end{pmatrix}$

30. $\begin{pmatrix} 2 & -3 \\ 1 & -1 \end{pmatrix}$

2. Find the inverses of the following 3×3 *matrices by the Gauss–Jordan method.*

1. $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{pmatrix}$

2. $\begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 3 & 2 & 1 \end{pmatrix}$

3. $\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ -1 & 0 & 2 \end{pmatrix}$

4. $\begin{pmatrix} 3 & 0 & 1 \\ -1 & 2 & 0 \\ 0 & -4 & 3 \end{pmatrix}$

5. $\begin{pmatrix} -2 & 1 & 3 \\ 4 & 0 & -1 \\ 1 & -2 & 5 \end{pmatrix}$

6. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 1 \\ 2 & -1 & 0 \end{pmatrix}$

7. $\begin{pmatrix} 0 & 1 & -3 \\ 2 & 0 & 4 \\ -1 & 2 & 1 \end{pmatrix}$

8. $\begin{pmatrix} 4 & -1 & 2 \\ 3 & 5 & 0 \\ -2 & 0 & 1 \end{pmatrix}$

9. $\begin{pmatrix} 1 & 0 & 3 \\ -2 & 1 & 0 \\ 4 & -3 & 2 \end{pmatrix}$

10. $\begin{pmatrix} -3 & 2 & 1 \\ 0 & -4 & 5 \\ 2 & 0 & -1 \end{pmatrix}$

11. $\begin{pmatrix} 5 & -2 & 0 \\ 1 & 3 & -1 \\ 0 & 4 & 2 \end{pmatrix}$

12. $\begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & -2 \\ 1 & 0 & 5 \end{pmatrix}$

13. $\begin{pmatrix} -1 & 0 & 2 \\ 3 & -5 & 1 \\ 2 & 4 & 0 \end{pmatrix}$

14. $\begin{pmatrix} 1 & 3 & -1 \\ -2 & 0 & 4 \\ 0 & 5 & 2 \end{pmatrix}$

15. $\begin{pmatrix} 4 & 0 & -3 \\ 2 & 1 & 5 \\ -1 & -2 & 0 \end{pmatrix}$

16. $\begin{pmatrix} 0 & -3 & 1 \\ 5 & 2 & 0 \\ -4 & 0 & 3 \end{pmatrix}$

17. $\begin{pmatrix} 3 & 1 & 0 \\ -2 & 4 & -1 \\ 0 & -3 & 5 \end{pmatrix}$

18. $\begin{pmatrix} -1 & 2 & -3 \\ 4 & 0 & 1 \\ 0 & -5 & 2 \end{pmatrix}$

19. $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 3 & 4 \\ -3 & 1 & 0 \end{pmatrix}$

20. $\begin{pmatrix} 1 & -4 & 0 \\ 3 & 2 & -5 \\ -2 & 0 & 1 \end{pmatrix}$

21. $\begin{pmatrix} 0 & 5 & -2 \\ -1 & 0 & 3 \\ 4 & -3 & 0 \end{pmatrix}$

22. $\begin{pmatrix} -2 & 0 & 4 \\ 1 & -3 & 0 \\ 0 & 5 & -1 \end{pmatrix}$ 25. $\begin{pmatrix} 0 & -1 & 3 \\ 2 & 5 & 0 \\ -4 & 0 & 1 \end{pmatrix}$ 28. $\begin{pmatrix} 2 & -5 & 1 \\ 0 & 3 & 0 \\ -1 & 0 & 4 \end{pmatrix}$

23. $\begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & -4 \\ 5 & 0 & 2 \end{pmatrix}$ 26. $\begin{pmatrix} 4 & 3 & -1 \\ 0 & -2 & 5 \\ 1 & 0 & 0 \end{pmatrix}$ 29. $\begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 3 \\ 3 & 0 & -1 \end{pmatrix}$

24. $\begin{pmatrix} 1 & 0 & -5 \\ -3 & 4 & 0 \\ 2 & -1 & 3 \end{pmatrix}$ 27. $\begin{pmatrix} -3 & 0 & 2 \\ 1 & 4 & 0 \\ 0 & -1 & 5 \end{pmatrix}$ 30. $\begin{pmatrix} -4 & 3 & 0 \\ 2 & 0 & -5 \\ 0 & 1 & 3 \end{pmatrix}$

3. Solve the matrix equation $AX = B$ by first finding A^{-1} with the Gauss–Jordan method.

1. $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$.
2. $A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 5 & -1 \\ 2 & 3 \end{pmatrix}$.
3. $A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.
4. $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ -1 & 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.
5. $A = \begin{pmatrix} 5 & 1 \\ -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 4 \\ -1 & 2 \end{pmatrix}$.
6. $A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 1 \\ 2 & -1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$.
7. $A = \begin{pmatrix} -1 & 4 \\ 2 & -7 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$.
8. $A = \begin{pmatrix} 3 & 0 & 1 \\ -1 & 2 & 0 \\ 0 & -4 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix}$.
9. $A = \begin{pmatrix} 4 & -3 \\ 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$.
10. $A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 3 & 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 5 \\ -2 \end{pmatrix}$.
11. $A = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 7 \\ -2 \end{pmatrix}$.
12. $A = \begin{pmatrix} -2 & 1 & 3 \\ 4 & 0 & -1 \\ 1 & -2 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$.
13. $A = \begin{pmatrix} 6 & -5 \\ -1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 1 \\ 0 & -2 \end{pmatrix}$.
14. $A = \begin{pmatrix} 4 & -1 & 2 \\ 3 & 5 & 0 \\ -2 & 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix}$.
15. $A = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$.
16. $A = \begin{pmatrix} 1 & 0 & 3 \\ -2 & 1 & 0 \\ 4 & -3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix}$.
17. $A = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 3 \\ 1 & 0 \end{pmatrix}$.

18. $A = \begin{pmatrix} -3 & 2 & 1 \\ 0 & -4 & 5 \\ 2 & 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix}$.

19. $A = \begin{pmatrix} 9 & -1 \\ 4 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$.

20. $A = \begin{pmatrix} 5 & -2 & 0 \\ 1 & 3 & -1 \\ 0 & 4 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}$.

21. $A = \begin{pmatrix} -1 & -3 \\ 2 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 2 & 4 \end{pmatrix}$.

22. $A = \begin{pmatrix} 0 & 1 & -3 \\ 2 & 0 & 4 \\ -1 & 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix}$.

23. $A = \begin{pmatrix} 7 & 3 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

24. $A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & -2 \\ 1 & 0 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$.

25. $A = \begin{pmatrix} -3 & -1 \\ 7 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 \\ -7 & 1 \end{pmatrix}$.

26. $A = \begin{pmatrix} -1 & 0 & 2 \\ 3 & -5 & 1 \\ 2 & 4 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -4 \\ 0 \\ 7 \end{pmatrix}$.

27. $A = \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix}$.

28. $A = \begin{pmatrix} 1 & 3 & -1 \\ -2 & 0 & 4 \\ 0 & 5 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$.

29. $A = \begin{pmatrix} 2 & -3 \\ 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$.

30. $A = \begin{pmatrix} 3 & -2 & 0 \\ 0 & 1 & -4 \\ 5 & 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 6 \\ 2 \end{pmatrix}$.

4. Applied problems (10 problems).

1. **(Hill cipher: inverse modulo 26.)** Using the Gauss–Jordan method, find the inverse modulo 26 of the key matrix $K = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$. Then decrypt the ciphertext $c = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$.

2. **(Key matrix of order 3.)** Compute the determinant of the key matrix $K = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix}$ and decide whether K is invertible modulo 26. If it is, find K^{-1} by elementary row operations.

3. **(A matrix equation.)** Solve the equation $AX + B = C$ for X, where $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix}$, $C = \begin{pmatrix} 5 & -3 \\ 0 & 7 \end{pmatrix}$. (Compute A^{-1} by the Gauss–Jordan method.)

4. **(Verification.)** For $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ -1 & 0 & 2 \end{pmatrix}$ find A^{-1} by the Gauss–Jordan method and verify that $A \cdot A^{-1} = E$.

5. **(Uniqueness of the inverse.)** Find the inverse of $A = \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix}$ (a) by the formula for 2×2 matrices and (b) by the Gauss–Jordan method. Show that the two answers coincide.

6. **(Coding theory: systematic form.)** Apply elementary row operations to the generator matrix

$$G = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

so as to reduce its left 3×3 block to the identity, and write down the systematic generator matrix $G' = (I_3 \mid P)$ (all arithmetic modulo 2).

7. **(Network security.)** The Laplacian matrix of a network with five nodes is

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & 0 & -1 & -1 \\ -1 & 0 & 2 & -1 & 0 \\ 0 & -1 & -1 & 3 & -1 \\ 0 & -1 & 0 & -1 & 2 \end{pmatrix}.$$

Show that $\det L = 0$, i.e. L is singular, and explain why L therefore has no inverse.

8. **(A matrix with a parameter.)** For which values of a does the matrix $A = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{pmatrix}$ have an inverse? For $a = 1$ find A^{-1} by the Gauss–Jordan method.

9. **(Inverse of a product.)** For $A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 0 & -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ find A^{-1} and B^{-1} by the Gauss–Jordan method. Then verify that $(AB)^{-1} = B^{-1}A^{-1}$.

10. **(Computer security: a binary transformation.)** Consider the matrix over the binary field $\mathrm{GF}(2)$

$$M = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix},$$

of the kind used in the linear layers of AES-type ciphers. Compute its determinant modulo 2 and decide whether M is invertible.

Chapter 2

Systems of Linear Algebraic Equations, Vectors and Lines in the Plane

2.1 Cramer's rule for systems of linear algebraic equations

Definition 2.1.1. Consider a system of n linear equations in n unknowns

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \dots\dots\dots\dots\dots\dots\dots\dots\dots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n. \end{cases}$$

The determinant $\Delta = \det A$ of its coefficient matrix is called the **determinant of the system**. The determinant obtained from Δ by replacing its jth column with the column of constant terms $b = (b_1; \dots; b_n)^T$ is denoted by Δ_j; we call it the jth **auxiliary determinant**.

Theorem 2.1.1 (Cramer's rule)**.** *If the determinant of the system is nonzero ($\Delta \neq 0$), the system has a unique solution, given by*

$$x_j = \frac{\Delta_j}{\Delta}, \qquad j = 1, 2, \dots, n. \tag{2.1.1}$$

In practice one first computes Δ. If $\Delta = 0$, Cramer's rule does not apply: the system is then either inconsistent or has infinitely many solutions. If $\Delta \neq 0$, one computes the auxiliary determinants $\Delta_1, \Delta_2, \dots, \Delta_n$, finds the unknowns

from the formulas $x_j = \Delta_j/\Delta$, and checks the result by substituting it into the system.

2.1.1 Cramer's rule for 2×2 systems

For the system $\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$ the three determinants are

$$\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}, \quad \Delta_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}.$$

Example 2.1.1. Solve the system $\begin{cases} 3x_1 + 2x_2 = 7, \\ x_1 - x_2 = -1 \end{cases}$ by Cramer's rule.

Solution.

$$\Delta = \begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} = -3 - 2 = -5 \neq 0, \quad \Delta_1 = \begin{vmatrix} 7 & 2 \\ -1 & -1 \end{vmatrix} = -7 + 2 = -5,$$

$$\Delta_2 = \begin{vmatrix} 3 & 7 \\ 1 & -1 \end{vmatrix} = -3 - 7 = -10.$$

Hence $x_1 = \dfrac{-5}{-5} = 1$ and $x_2 = \dfrac{-10}{-5} = 2$.

Check: $3 \cdot 1 + 2 \cdot 2 = 7$ ✓; $1 - 2 = -1$ ✓.

Example 2.1.2. Solve the system $\begin{cases} 2x_1 - 5x_2 = 4, \\ 3x_1 + x_2 = -2. \end{cases}$

Solution. We have $\Delta = 2 \cdot 1 - (-5) \cdot 3 = 17$, $\Delta_1 = 4 \cdot 1 - (-5)(-2) = -6$ and $\Delta_2 = 2 \cdot (-2) - 4 \cdot 3 = -16$, so

$$x_1 = -\frac{6}{17}, \qquad x_2 = -\frac{16}{17}.$$

2.1.2 Cramer's rule for 3×3 systems

Example 2.1.3. Solve the system $\begin{cases} x_1 + x_2 + x_3 = 6, \\ 2x_1 - x_2 + x_3 = 3, \\ x_1 + 2x_2 - x_3 = 2 \end{cases}$ by Cramer's rule.

Solution. Expanding the determinant of the system along the first row, we find

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 1(1-2) - 1(-2-1) + 1(4+1) = -1 + 3 + 5 = 7 \neq 0.$$

The auxiliary determinants are

$$\Delta_1 = \begin{vmatrix} 6 & 1 & 1 \\ 3 & -1 & 1 \\ 2 & 2 & -1 \end{vmatrix} = 7, \qquad \Delta_2 = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 14,$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 6 \\ 2 & -1 & 3 \\ 1 & 2 & 2 \end{vmatrix} = 21.$$

Therefore $x_1 = \dfrac{7}{7} = 1$, $x_2 = \dfrac{14}{7} = 2$, $x_3 = \dfrac{21}{7} = 3$.

Check: $1 + 2 + 3 = 6$, $2 - 2 + 3 = 3$, $1 + 4 - 3 = 2$ ✓.

2.1.3 The case $\Delta = 0$

Theorem 2.1.2. *If $\Delta = 0$ and at least one of the auxiliary determinants is nonzero, the system is inconsistent. If $\Delta = 0$ and all $\Delta_j = 0$, the system is either inconsistent or has infinitely many solutions; in this case it has to be investigated by Gaussian elimination.*

Example 2.1.4. For the system $\begin{cases} x_1 + 2x_2 = 3, \\ 2x_1 + 4x_2 = 7 \end{cases}$ we have $\Delta = 4 - 4 = 0$ and $\Delta_1 = 3 \cdot 4 - 2 \cdot 7 = -2 \neq 0$. The system is therefore **inconsistent**: its equations contradict each other.

Example 2.1.5. For the system $\begin{cases} x_1 - 3x_2 = 2, \\ -2x_1 + 6x_2 = -4 \end{cases}$ we have $\Delta = 6-6 = 0$, $\Delta_1 = 12 - 12 = 0$ and $\Delta_2 = -4 + 4 = 0$. The second equation is the first one multiplied by -2, so the system has **infinitely many solutions**: $x_1 = 2 + 3t$, $x_2 = t$, $t \in \mathbb{R}$.

Remark 2.1.1. For n unknowns Cramer's rule requires $n + 1$ determinants of order n. It is therefore convenient mainly for $n = 2$ or 3 and in theoretical work, for example when a system depends on parameters. For large systems Gaussian elimination is far cheaper.

2.1.4 Applications of Cramer's rule in information security

Recovering a block of the Hill cipher. In the Hill cipher a plaintext block p is encrypted as $c = Kp$. If the key matrix K is known, recovering p amounts to solving the linear system $Kp = c$. For small keys (2×2 or 3×3) this can be done directly by Cramer's rule, and the condition $\Delta = \det K \neq 0$ is a necessary condition for the key to be admissible.

Example 2.1.6. Suppose that a block encrypted with the key $K = \begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix}$ (without reduction modulo 26) is $c = \begin{pmatrix} 29 \\ 63 \end{pmatrix}$. We solve $Kp = c$:

$$\Delta = 21 - 10 = 11, \quad \Delta_1 = \begin{vmatrix} 29 & 2 \\ 63 & 7 \end{vmatrix} = 203 - 126 = 77,$$

$$\Delta_2 = \begin{vmatrix} 3 & 29 \\ 5 & 63 \end{vmatrix} = 189 - 145 = 44.$$

Thus $p_1 = 77/11 = 7$ and $p_2 = 44/11 = 4$, i.e. $p = (7;4)$, which corresponds to the letters "H" and "E" ($A = 0, \ldots, Z = 25$).

Key recovery in a known-plaintext attack. If an attacker knows several plaintext–ciphertext pairs, the unknown entries of the key satisfy a system of linear equations. When the determinant of this system is nonzero, Cramer's rule recovers the key uniquely. Because the Hill cipher is **linear**, it offers no resistance to a known-plaintext attack.

Systems with parameters and the key space. For a family of keys depending on a parameter, the roots of the equation $\Delta(k) = 0$ give the inadmissible keys. For instance, for $K = \begin{pmatrix} 1 & k \\ k & 4 \end{pmatrix}$ we get $\Delta = 4 - k^2$, so the values $k = \pm 2$ cannot be used.

2.1.5 Computing with Python, Maple and Mathcad

We solve the 3×3 system of the third example once more. In Python it is enough to replace the columns of A by b in turn and divide the determinants:

Python

```
from sympy import Matrix
A = Matrix([[1, 1, 1], [2, -1, 1], [1, 2, -1]])
b = Matrix([6, 3, 2])
D = A.det()
x = []
for j in range(3):
    Aj = A.copy()
    Aj[:, j] = b              # j-th column replaced by b
    x.append(Aj.det() / D)
print(D, x)
```

Output

```
7 [1, 2, 3]
```

Maple does the same with `Determinant` and `Column`; the command `LinearSolve` serves as a check:

Maple

```
with(LinearAlgebra):
A := Matrix([[1, 1, 1], [2, -1, 1], [1, 2, -1]]):
b := Vector([6, 3, 2]):
D0 := Determinant(A);
D1 := Determinant(<b | Column(A, 2) | Column(A, 3)>);
D2 := Determinant(<Column(A, 1) | b | Column(A, 3)>);
```

```
D3 := Determinant(<Column(A, 1) | Column(A, 2) | b>);
D1/D0, D2/D0, D3/D0;
LinearSolve(A, b);
```

In Mathcad Prime the auxiliary matrices are built from the columns of A (the operator $A^{\langle j \rangle}$, typed with Ctrl+Shift+C) and the function `augment`; $|A|$ is the determinant, and `lsolve(A, b)` gives the solution directly (with `ORIGIN = 0` the columns are numbered from zero):

Mathcad

```
     | 1   1   1 |          | 6 |
A := | 2  -1   1 |    b := | 3 |
     | 1   2  -1 |          | 2 |
A1 := augment(b, A<1>, A<2>)
A2 := augment(A<0>, b, A<2>)
A3 := augment(A<0>, A<1>, b)
|A| = 7   |A1| = 7   |A2| = 14   |A3| = 21
x1 := |A1|/|A|   x2 := |A2|/|A|   x3 := |A3|/|A|
x1 = 1           x2 = 2           x3 = 3
                 | 1 |
lsolve(A, b) =   | 2 |
                 | 3 |
```

2.1.6 Exercises on Cramer's rule

1. Solve the following 2×2 systems by Cramer's rule.

1. $\begin{cases} -x_1 - 4x_2 = 26 \\ 4x_2 = -20 \end{cases}$

2. $\begin{cases} 2x_1 - 5x_2 = -22 \\ -x_1 + 3x_2 = 12 \end{cases}$

3. $\begin{cases} 2x_1 = -10 \\ -6x_1 + 3x_2 = 21 \end{cases}$

4. $\begin{cases} -2x_1 = 10 \\ -4x_1 + 2x_2 = 26 \end{cases}$

5. $\begin{cases} -3x_1 - x_2 = -10 \\ -5x_1 + 2x_2 = -35 \end{cases}$

6. $\begin{cases} 3x_1 - 6x_2 = -21 \\ 3x_1 - 3x_2 = -9 \end{cases}$

7. $\begin{cases} 2x_1 = 2 \\ 6x_1 - x_2 = 3 \end{cases}$

8. $\begin{cases} x_1 - x_2 = 10 \\ -2x_1 - 3x_2 = 0 \end{cases}$

9. $\begin{cases} 5x_1 + 6x_2 = 3 \\ -3x_1 - 5x_2 = 1 \end{cases}$

10. $\begin{cases} 2x_1 + x_2 = 0 \\ -x_1 + 5x_2 = -11 \end{cases}$

11. $\begin{cases} 3x_1 + 6x_2 = 9 \\ -x_1 - x_2 = -4 \end{cases}$

12. $\begin{cases} 4x_1 + 3x_2 = 7 \\ 4x_1 + x_2 = -3 \end{cases}$

13. $\begin{cases} 4x_2 = -4 \\ -x_1 - 6x_2 = 5 \end{cases}$

14. $\begin{cases} -4x_1 + 3x_2 = 15 \\ -5x_1 + x_2 = 27 \end{cases}$

15. $\begin{cases} -3x_1 = 15 \\ x_2 = -4 \end{cases}$

16. $\begin{cases} x_1 = -4 \\ 2x_1 - 2x_2 = -8 \end{cases}$

17. $\begin{cases} 2x_1 - 2x_2 = -10 \\ 5x_1 = -5 \end{cases}$

18. $\begin{cases} -3x_2 = 12 \\ -4x_1 - 5x_2 = 36 \end{cases}$

19. $\begin{cases} x_1 + 3x_2 = -20 \\ -4x_1 - 2x_2 = 20 \end{cases}$

20. $\begin{cases} -4x_1 = -12 \\ 2x_1 - x_2 = 3 \end{cases}$

21. $\begin{cases} 3x_1 + 4x_2 = -14 \\ 4x_1 + 5x_2 = -19 \end{cases}$

22. $\begin{cases} 6x_1 + 4x_2 = 0 \\ 6x_1 + 2x_2 = 0 \end{cases}$

23. $\begin{cases} 4x_1 = -20 \\ -6x_1 - 3x_2 = 39 \end{cases}$

24. $\begin{cases} 3x_1 - 4x_2 = -15 \\ 2x_1 - 5x_2 = -17 \end{cases}$

25. $\begin{cases} -x_1 + 3x_2 = -10 \\ -x_1 + x_2 = 0 \end{cases}$

26. $\begin{cases} -2x_1 - 5x_2 = -5 \\ -4x_1 - 5x_2 = -15 \end{cases}$

27. $\begin{cases} 2x_1 - x_2 = 10 \\ -4x_1 + 5x_2 = -38 \end{cases}$

28. $\begin{cases} -x_1 - 4x_2 = -5 \\ -x_1 + 6x_2 = 15 \end{cases}$

29. $\begin{cases} -3x_1 - 3x_2 = -12 \\ 2x_1 + x_2 = 3 \end{cases}$

30. $\begin{cases} x_1 - 2x_2 = 5 \\ -3x_1 + 5x_2 = -14 \end{cases}$

2. Solve the following 3×3 *systems by Cramer's rule.*

1. $\begin{cases} 3x_1 + x_2 + x_3 = 5 \\ -3x_1 - x_2 - 3x_3 = -11 \\ -x_1 + 3x_2 - x_3 = -7 \end{cases}$

2. $\begin{cases} -4x_1 + 3x_2 + x_3 = -8 \\ -3x_1 - 3x_2 + 2x_3 = -15 \\ -x_1 + 3x_2 - 2x_3 = 7 \end{cases}$

3. $\begin{cases} 2x_1 + 3x_2 + 2x_3 = 6 \\ -3x_1 - 2x_2 - 2x_3 = -11 \\ -2x_1 - 4x_2 - 2x_3 = -4 \end{cases}$

4. $\begin{cases} -x_1 + x_2 = -2 \\ 4x_1 + 2x_2 - 2x_3 = 12 \\ -4x_1 + x_2 + 3x_3 = -2 \end{cases}$

5. $\begin{cases} 4x_1 + 4x_2 + 3x_3 = -16 \\ -3x_1 + 4x_2 - 4x_3 = -16 \\ -x_1 - x_2 = 7 \end{cases}$

6. $\begin{cases} 4x_1 - x_2 + 3x_3 = -5 \\ -2x_1 + 2x_2 - 3x_3 = -2 \\ 2x_1 + 3x_2 + x_3 = -7 \end{cases}$

7. $\begin{cases} -3x_1 - x_2 = -8 \\ -3x_1 - 2x_2 + x_3 = -10 \\ -2x_1 - 2x_3 = 0 \end{cases}$

8. $\begin{cases} 2x_1 + 3x_2 - 2x_3 = -1 \\ -x_1 - 2x_2 + 2x_3 = 2 \\ 4x_1 + 2x_2 + x_3 = 7 \end{cases}$

9. $\begin{cases} x_1 - 3x_2 + x_3 = 3 \\ -4x_1 + x_2 + 4x_3 = 9 \\ 3x_1 + 3x_2 - 4x_3 = -7 \end{cases}$

10. $\begin{cases} 4x_1 + 3x_2 + x_3 = -15 \\ -3x_1 - 4x_3 = 12 \\ -2x_1 + 2x_2 - 3x_3 = 1 \end{cases}$

11. $\begin{cases} -3x_2 - x_3 = -6 \\ -3x_1 - 3x_3 = -12 \\ 3x_1 - 4x_2 + x_3 = 4 \end{cases}$

12. $\begin{cases} -2x_1 - 4x_2 + 4x_3 = 2 \\ -x_1 - 3x_2 - 2x_3 = 1 \\ -4x_2 - 2x_3 = 0 \end{cases}$

13. $\begin{cases} -4x_2 - 4x_3 = 0 \\ -4x_1 + 4x_2 + 4x_3 = 4 \\ -x_1 + 4x_2 + 3x_3 = 4 \end{cases}$

14. $\begin{cases} 2x_1 + 3x_2 + 4x_3 = 0 \\ 2x_1 + 4x_2 = -10 \\ -x_1 - x_2 + x_3 = 5 \end{cases}$

15. $\begin{cases} -3x_1 + 2x_2 + 4x_3 = 18 \\ -x_2 = 0 \\ -4x_1 + 3x_2 - 2x_3 = 2 \end{cases}$

16. $\begin{cases} -4x_1 + x_3 = 2 \\ x_1 + 4x_2 + x_3 = 1 \\ -x_1 - 4x_2 = -3 \end{cases}$

17. $\begin{cases} -4x_1 + x_2 + 2x_3 = -26 \\ -3x_1 + 3x_2 = -24 \\ 4x_1 - x_2 - x_3 = 23 \end{cases}$

18. $\begin{cases} -x_1 - 3x_2 + 4x_3 = 30 \\ -2x_1 + 2x_2 + x_3 = 0 \\ 3x_1 - 2x_2 = 2 \end{cases}$

19. $\begin{cases} 2x_1 + 4x_2 - 2x_3 = -18 \\ 4x_1 + 4x_2 - 4x_3 = -28 \\ -x_1 - 3x_2 - 4x_3 = 6 \end{cases}$

20. $\begin{cases} -3x_1 + 2x_2 + 3x_3 = 1 \\ 4x_1 - 4x_2 - 4x_3 = 4 \\ 4x_1 - x_2 + 3x_3 = 13 \end{cases}$

21. $\begin{cases} -x_2 - x_3 = 5 \\ -x_1 + 3x_2 + 3x_3 = -15 \\ 2x_1 - 3x_2 + 3x_3 = 9 \end{cases}$

22. $\begin{cases} -3x_1 - 2x_2 + x_3 = -12 \\ -2x_3 = 6 \\ -4x_1 + 3x_2 - 4x_3 = 0 \end{cases}$

23. $\begin{cases} 4x_1 - x_2 = 13 \\ -3x_1 + 3x_2 - 4x_3 = -3 \\ 3x_2 - 3x_3 = 9 \end{cases}$

24. $\begin{cases} 2x_1 - x_2 - x_3 = -8 \\ -3x_1 - 3x_2 - 2x_3 = -6 \\ 4x_1 + x_3 = -8 \end{cases}$

25. $\begin{cases} -3x_1 + x_2 - x_3 = -8 \\ 3x_1 + 3x_2 + 2x_3 = 22 \\ -4x_1 - 2x_2 - 4x_3 = -26 \end{cases}$

26. $\begin{cases} -2x_2 + 2x_3 = 2 \\ x_1 + 2x_2 + x_3 = 5 \\ -3x_1 + x_2 - 4x_3 = -10 \end{cases}$

27. $\begin{cases} -3x_1 - x_2 - 4x_3 = 0 \\ x_3 = 2 \\ -3x_1 + 2x_2 + 2x_3 = 15 \end{cases}$

28. $\begin{cases} 2x_1 + 4x_2 + x_3 = 1 \\ -x_1 + x_2 + 2x_3 = -11 \\ -4x_1 + 2x_2 + 4x_3 = -30 \end{cases}$

29. $\begin{cases} -4x_1 + 2x_2 + 3x_3 = 20 \\ -2x_1 + 3x_3 = 10 \\ -4x_1 + 4x_2 - 2x_3 = 16 \end{cases}$

30. $\begin{cases} 4x_1 - x_2 + 3x_3 = -7 \\ x_1 + 3x_2 + 2x_3 = -14 \\ -2x_1 + 4x_2 - x_3 = -8 \end{cases}$

3. Applied problems (10 problems).

1. **(Hill cipher.)** A plaintext block p was encrypted as $c = Kp$ (without reduction modulo 26) with the key matrix $K = \begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix}$. Given the ciphertext $c = \begin{pmatrix} 29 \\ 63 \end{pmatrix}$, solve the system $Kp = c$ by Cramer's rule and find the plaintext p.

2. **(Hill cipher.)** A plaintext block p was encrypted as $c = Kp$ (without reduction modulo 26) with the key matrix $K = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$. Given the ciphertext

$c = \begin{pmatrix} 37 \\ 31 \end{pmatrix}$, solve the system $Kp = c$ by Cramer's rule and find the plaintext p.

3. **(Hill cipher.)** A plaintext block p was encrypted as $c = Kp$ (without reduction modulo 26) with the key matrix $K = \begin{pmatrix} 5 & 1 \\ 3 & 2 \end{pmatrix}$. Given the ciphertext $c = \begin{pmatrix} 59 \\ 62 \end{pmatrix}$, solve the system $Kp = c$ by Cramer's rule and find the plaintext p.

4. **(A parameter.)** For which values of k does the system $\begin{cases} x_1 + kx_2 = 3 \\ kx_1 + 4x_2 = 6 \end{cases}$ have a unique solution? Find the solution for $k = 1$.

5. **(A parameter.)** For which values of a can the system $\begin{cases} ax_1 + 2x_2 = 1 \\ 3x_1 + (a+1)x_2 = 2 \end{cases}$ not be solved by Cramer's rule?

6. **(Determinants.)** For the system $\begin{cases} 2x_1 - x_2 + x_3 = 5 \\ x_1 + 3x_2 - 2x_3 = -6 \\ 3x_1 + x_2 + 4x_3 = 10 \end{cases}$ compute the determinants $\Delta, \Delta_1, \Delta_2, \Delta_3$ and find the solution.

7. **(Network traffic.)** The numbers x_1, x_2, x_3 of packets (in thousands) that passed through three servers satisfy the system $\begin{cases} x_1 + x_2 + x_3 = 12 \\ 2x_1 - x_2 + x_3 = 9 \\ x_1 + 2x_2 - x_3 = 5 \end{cases}$. Find them by Cramer's rule.

8. **(A fractional solution.)** Solve the system $\begin{cases} 3x_1 + 2x_2 = 7 \\ 4x_1 - 5x_2 = 3 \end{cases}$ by Cramer's rule.

9. **(The case $\Delta = 0$.)** Investigate the systems $\begin{cases} 2x_1 - 4x_2 = 6 \\ -x_1 + 2x_2 = -3 \end{cases}$ and $\begin{cases} 2x_1 - 4x_2 = 6 \\ -x_1 + 2x_2 = 1 \end{cases}$.

10. **(Key recovery.)** The key $K = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of a Hill cipher is unknown, but it is known that $K\binom{1}{0} = \binom{3}{5}$ and $K\binom{1}{1} = \binom{5}{12}$. Find a, b, c, d using Cramer's rule.

2.2 The matrix method for systems of linear equations

Definition 2.2.1. Let $Ax = b$ be a system of n linear equations in n unknowns. If the coefficient matrix A is nonsingular ($\det A \neq 0$), the system has a unique solution, given by

$$x = A^{-1} \cdot b,$$

where A^{-1} is the inverse of A and b is the column vector of constant terms.

The method follows directly from the definition. We write down the coefficient matrix A and the column b and compute $\det A$; if $\det A = 0$, the matrix method cannot be applied. Otherwise we find A^{-1}, either from the cofactors or by Gauss–Jordan elimination, and form the product $x = A^{-1}b$. Substituting the vector x back into the original equations gives a quick check of the arithmetic.

We stress that the matrix method applies only to square systems (n equations in n unknowns) with $\det A \neq 0$, that is, to systems with exactly one solution.

2.2.1 Solving 2×2 and 3×3 systems by the matrix method

The 2×2 case.

Example 2.2.1. $\begin{cases} 3x_1 + x_2 = 5, \\ 5x_1 + 2x_2 = 7. \end{cases}$

Solution. Here $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ and $b = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$. Since $\det A = 3 \cdot 2 - 1 \cdot 5 = 1 \neq 0$, the system has a unique solution. For a 2×2 matrix the inverse is obtained by swapping the diagonal entries, changing the signs of the off-diagonal ones and dividing by the determinant:

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}.$$

Multiplying A^{-1} by b, we obtain

$$x = A^{-1}b = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}\begin{pmatrix} 5 \\ 7 \end{pmatrix} = \begin{pmatrix} 2\cdot 5 + (-1)\cdot 7 \\ (-5)\cdot 5 + 3\cdot 7 \end{pmatrix} = \begin{pmatrix} 10 - 7 \\ -25 + 21 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}.$$

Answer: $x_1 = 3, x_2 = -4$.
These values satisfy both equations: $3\cdot 3 + (-4) = 5$ and $5\cdot 3 + 2\cdot(-4) = 7$.

Example 2.2.2. $\begin{cases} -2x_1 + 3x_2 = 1, \\ x_1 - x_2 = 4. \end{cases}$

Solution. We have $A = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\det A = (-2)(-1) - 3\cdot 1 = -1 \neq 0$. Hence

$$A^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & -3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix},$$

and

$$x = \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 + 12 \\ 1 + 8 \end{pmatrix} = \begin{pmatrix} 13 \\ 9 \end{pmatrix}.$$

Answer: $x_1 = 13, x_2 = 9$.
Check: $-2\cdot 13 + 3\cdot 9 = -26 + 27 = 1$ and $13 - 9 = 4$.

The 3×3 case.

Example 2.2.3. $\begin{cases} x_1 - x_2 + x_3 = 0, \\ 2x_2 - x_3 = 1, \\ 2x_1 + 3x_2 = 5. \end{cases}$

Solution. The coefficient matrix and the column of constants are $A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{pmatrix}$ and $b = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}$. Expanding $\det A$ along the first row, we get

$$\det A = 1 \begin{vmatrix} 2 & -1 \\ 3 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} + 1 \begin{vmatrix} 0 & 2 \\ 2 & 3 \end{vmatrix} = 1 \cdot 3 + 1 \cdot 2 + 1 \cdot (-4) = 1,$$

so A is nonsingular. Next we compute the cofactors:

$$A_{11} = \begin{vmatrix} 2 & -1 \\ 3 & 0 \end{vmatrix} = 3, \qquad A_{12} = -\begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} = -2, \qquad A_{13} = \begin{vmatrix} 0 & 2 \\ 2 & 3 \end{vmatrix} = -4,$$

$$A_{21} = -\begin{vmatrix} -1 & 1 \\ 3 & 0 \end{vmatrix} = 3, \qquad A_{22} = \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = -2, \qquad A_{23} = -\begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = -5,$$

$$A_{31} = \begin{vmatrix} -1 & 1 \\ 2 & -1 \end{vmatrix} = -1, \qquad A_{32} = -\begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix} = 1, \qquad A_{33} = \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} = 2.$$

Placing the cofactors of the ith row of A in the ith column, we obtain the adjugate matrix, and therefore

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix}.$$

It remains to multiply by b:

$$x = A^{-1}b = \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 0+3-5 \\ 0-2+5 \\ 0-5+10 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}.$$

Answer: $x_1 = -2, x_2 = 3, x_3 = 5$.

Substitution confirms the result: $(-2) - 3 + 5 = 0$, $2 \cdot 3 - 5 = 1$, $2 \cdot (-2) + 3 \cdot 3 = 5$.

Example 2.2.4. $\begin{cases} 2x_1 - x_2 = 1, \\ x_1 - x_3 = -1, \\ 3x_1 + 2x_2 + x_3 = 4. \end{cases}$

Solution. Here $A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 3 & 2 & 1 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$, and expansion along the first row gives $\det A = 2(0+2) + 1 \cdot (1+3) + 0 = 8 \neq 0$. The cofactors are

$$\begin{aligned} A_{11} &= 2, & A_{12} &= -4, & A_{13} &= 2, \\ A_{21} &= 1, & A_{22} &= 2, & A_{23} &= -7, \\ A_{31} &= 1, & A_{32} &= 2, & A_{33} &= 1, \end{aligned}$$

so that

$$A^{-1} = \frac{1}{8}\begin{pmatrix} 2 & 1 & 1 \\ -4 & 2 & 2 \\ 2 & -7 & 1 \end{pmatrix}$$

and

$$x = \frac{1}{8}\begin{pmatrix} 2 & 1 & 1 \\ -4 & 2 & 2 \\ 2 & -7 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 2-1+4 \\ -4-2+8 \\ 2+7+4 \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 5 \\ 2 \\ 13 \end{pmatrix} = \begin{pmatrix} 5/8 \\ 1/4 \\ 13/8 \end{pmatrix}.$$

For instance, the first equation holds: $2 \cdot \frac{5}{8} - \frac{1}{4} = \frac{10}{8} - \frac{2}{8} = 1$.

Comparison of the matrix method with Cramer's rule.

Matrix method ($x = A^{-1}b$)	**Cramer's rule** ($x_j = \Delta_j/\Delta$)
Requires the inverse matrix	Requires only determinants
Once A^{-1} is known, each new b costs one multiplication	The determinants must be recomputed for every b
A^{-1} has n^2 entries	$n+1$ determinants of order n
Computing A^{-1} is expensive for large n	The work grows very fast for $n > 3$
Also suits matrix equations such as $AXB = C$	Only for systems of the form $Ax = b$

Thus, when several systems with the same matrix A and different right-hand sides have to be solved, the matrix method is preferable, since A^{-1} is found only once. For a single small system Cramer's rule may be simpler.

2.2.2 Applications of the matrix method in information security

Encryption and decryption with the Hill cipher. The Hill cipher is a direct application of the matrix method. A plaintext vector p is encrypted by multiplying it by the key matrix K, which gives the ciphertext $c = Kp$. To decrypt, one multiplies the ciphertext by the inverse matrix: $p = K^{-1}c$, which is exactly the formula $x = A^{-1}b$. Since every ciphertext block is decrypted with the same matrix K^{-1}, the cipher illustrates well the main advantage of the method: the inverse is computed once and then used for many multiplications.

Example 2.2.5. Let the key matrix be $K = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ and let three ciphertext blocks $c_1 = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$, $c_2 = \begin{pmatrix} 10 \\ 18 \end{pmatrix}$, $c_3 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ be given.

Having found $K^{-1} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$ once, we obtain

$$p_1 = K^{-1}c_1 = \begin{pmatrix} 3 \\ -4 \end{pmatrix}, \quad p_2 = K^{-1}c_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad p_3 = K^{-1}c_3 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}.$$

All three blocks are decrypted with the same matrix K^{-1}.

Linear cryptanalysis. In cryptanalysis (the study of breaking ciphers), when several plaintext–ciphertext pairs are known, finding the key matrix reduces to a matrix equation $AX = B$ with the unknown key X. Its solution is $X = A^{-1}B$.

Decoding in coding theory. For systematic codes the matrix method is used to extract the information part of a received word. If the generator matrix has the form $G = (I_k \mid P)$, the information bits can be read off directly from the received vector c. If G is not in systematic form, one has to apply the inverse of the information part of G.

Digital signal processing. Linear filters used to filter and analyse signals in information security (for example, in intrusion detection systems) are described

by relations of the form $y = Hx$. The signal x is recovered by the formula $x = H^{-1}y$, which is one more instance of the matrix method.

2.2.3 Computing with Python, Maple and Mathcad

Let us solve the system of the third example once more by computer. In Python the SymPy library computes the determinant, the inverse matrix and the product $A^{-1}b$ in exact rational arithmetic.

Python

```
from sympy import Matrix
A = Matrix([[1, -1, 1], [0, 2, -1], [2, 3, 0]])
b = Matrix([0, 1, 5])
print(A.det())
Ainv = A.inv()
print(Ainv)
print((Ainv*b).T)
```

Output

```
1
Matrix([[3, 3, -1], [-2, -2, 1], [-4, -5, 2]])
Matrix([[-2, 3, 5]])
```

In Maple the same computation uses the `LinearAlgebra` package; the command `LinearSolve(A, b)` gives the solution directly and serves as a check.

Maple

```
with(LinearAlgebra):
A := Matrix([[1, -1, 1], [0, 2, -1], [2, 3, 0]]):
b := Vector([0, 1, 5]):
Determinant(A);           # 1
Ainv := MatrixInverse(A);
x := Ainv . b;            # <-2, 3, 5>
LinearSolve(A, b);        # the same vector
```

In Mathcad Prime the matrices are entered through the matrix template, the inverse is obtained with the exponent -1, and the evaluation sign = displays the result; the built-in function `lsolve(A, b)` returns the same vector.

Mathcad

```
     | 1  -1   1 |          | 0 |
A := | 0   2  -1 |     b := | 1 |
     | 2   3   0 |          | 5 |

|A| = 1
```

```
          |  3   3  -1 |              | -2 |
A^-1 =    | -2  -2   1 |   A^-1*b =   |  3 |
          | -4  -5   2 |              |  5 |

                    | -2 |
lsolve(A, b) =      |  3 |
                    |  5 |
```

2.2.4 Exercises on the matrix method

1. Solve the following 2×2 systems by the matrix method.

1. $\begin{cases} 3x_1 + x_2 = 5 \\ 5x_1 + 2x_2 = 7 \end{cases}$

2. $\begin{cases} -2x_1 + 3x_2 = 1 \\ x_1 - x_2 = 4 \end{cases}$

3. $\begin{cases} x_1 + 2x_2 = 8 \\ 3x_1 - x_2 = 3 \end{cases}$

4. $\begin{cases} 4x_1 - x_2 = 6 \\ 2x_1 + 3x_2 = 8 \end{cases}$

5. $\begin{cases} -x_1 + 5x_2 = 9 \\ 2x_1 - 3x_2 = -1 \end{cases}$

6. $\begin{cases} 7x_1 + 2x_2 = 11 \\ 3x_1 + x_2 = 5 \end{cases}$

7. $\begin{cases} x_1 - 4x_2 = -7 \\ -2x_1 + 7x_2 = 11 \end{cases}$

8. $\begin{cases} 5x_1 + 3x_2 = 19 \\ 2x_1 - x_2 = 5 \end{cases}$

9. $\begin{cases} -3x_1 + 2x_2 = 7 \\ x_1 + x_2 = 6 \end{cases}$

10. $\begin{cases} 6x_1 - 5x_2 = 1 \\ -2x_1 + 3x_2 = 5 \end{cases}$

11. $\begin{cases} x_1 + x_2 = 4 \\ 2x_1 - x_2 = 5 \end{cases}$

12. $\begin{cases} 8x_1 + 5x_2 = 13 \\ 3x_1 + 2x_2 = 5 \end{cases}$

13. $\begin{cases} -x_1 + 2x_2 = 3 \\ 4x_1 - 5x_2 = -2 \end{cases}$

14. $\begin{cases} 2x_1 + 7x_2 = 16 \\ x_1 - 3x_2 = -5 \end{cases}$

15. $\begin{cases} 3x_1 - 4x_2 = -1 \\ x_1 + 2x_2 = 7 \end{cases}$

16. $\begin{cases} 9x_1 - x_2 = 8 \\ 4x_1 + x_2 = 5 \end{cases}$

17. $\begin{cases} x_1 + 6x_2 = 13 \\ 2x_1 - x_2 = 0 \end{cases}$

18. $\begin{cases} 5x_1 - x_2 = 4 \\ -3x_1 + 2x_2 = 1 \end{cases}$

19. $\begin{cases} 4x_1 + 3x_2 = 18 \\ -x_1 + 2x_2 = 1 \end{cases}$

20. $\begin{cases} x_1 - x_2 = 0 \\ 3x_1 + 4x_2 = 7 \end{cases}$

21. $\begin{cases} 2x_1 + x_2 = 8 \\ x_1 + 3x_2 = 14 \end{cases}$

22. $\begin{cases} -4x_1 + 3x_2 = 5 \\ 2x_1 + x_2 = 9 \end{cases}$

23. $\begin{cases} x_1 - 5x_2 = -9 \\ 3x_1 + 2x_2 = 4 \end{cases}$

24. $\begin{cases} 7x_1 - x_2 = 6 \\ x_1 + 3x_2 = 8 \end{cases}$

25. $\begin{cases} 3x_1 + 2x_2 = 11 \\ x_1 - 4x_2 = -9 \end{cases}$

26. $\begin{cases} -2x_1 - 3x_2 = -7 \\ x_1 + 5x_2 = 12 \end{cases}$

27. $\begin{cases} 3x_1 - x_2 = 5 \\ -x_1 + 4x_2 = 6 \end{cases}$

28. $\begin{cases} x_1 + 4x_2 = 9 \\ -3x_1 + 2x_2 = 7 \end{cases}$

29. $\begin{cases} 8x_1 - 3x_2 = 5 \\ 2x_1 + x_2 = 7 \end{cases}$

30. $\begin{cases} 2x_1 - 3x_2 = -4 \\ -x_1 + x_2 = 1 \end{cases}$

2. Solve the following 3×3 systems by the matrix method.

1. $\begin{cases} x_1 - x_2 + x_3 = 0 \\ 2x_2 - x_3 = 1 \\ 2x_1 + 3x_2 = 5 \end{cases}$

2. $\begin{cases} 2x_1 - x_2 + 3x_3 = 9 \\ x_1 - x_3 = -2 \\ 3x_1 + 2x_2 + x_3 = 7 \end{cases}$

3. $\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 - x_2 + 3x_3 = 1 \\ x_1 + x_2 + x_3 = 6 \end{cases}$

4. $\begin{cases} -x_1 + x_2 + x_3 = 1 \\ x_1 - x_2 + x_3 = 3 \\ x_1 + x_2 - x_3 = 5 \end{cases}$

5. $\begin{cases} 3x_1 - x_2 + 2x_3 = 5 \\ x_1 + 2x_2 - x_3 = 0 \\ 2x_1 + x_2 + 3x_3 = 11 \end{cases}$

6. $\begin{cases} x_1 + x_2 + x_3 = 6 \\ x_1 - x_2 + 2x_3 = 5 \\ 2x_1 + x_2 - x_3 = 1 \end{cases}$

7. $\begin{cases} 2x_1 + 3x_2 - x_3 = 4 \\ x_1 - x_2 + 2x_3 = 7 \\ 3x_1 + x_2 + x_3 = 10 \end{cases}$

8. $\begin{cases} 4x_1 - x_2 + 2x_3 = 8 \\ x_1 + 3x_2 - x_3 = 5 \\ -2x_1 + x_2 + 3x_3 = 1 \end{cases}$

9. $\begin{cases} x_1 - 2x_2 + 3x_3 = 7 \\ -x_1 + x_2 - x_3 = -3 \\ 2x_1 + 3x_2 - 2x_3 = -1 \end{cases}$

10. $\begin{cases} 3x_1 + 2x_2 + x_3 = 10 \\ x_1 - x_2 + 4x_3 = 12 \\ 2x_1 + x_2 - 3x_3 = -5 \end{cases}$

11. $\begin{cases} x_1 + 3x_2 + 2x_3 = 8 \\ 2x_1 - x_2 + x_3 = 3 \\ -x_1 + 2x_2 - 3x_3 = -7 \end{cases}$

12. $\begin{cases} x_1 - x_2 + x_3 = 2 \\ 2x_1 + x_2 - 3x_3 = -1 \\ x_1 + 2x_2 + x_3 = 7 \end{cases}$

13. $\begin{cases} 5x_1 - 2x_2 + x_3 = 3 \\ x_1 + 3x_2 - 2x_3 = 1 \\ -x_1 + x_2 + 4x_3 = 8 \end{cases}$

14. $\begin{cases} -2x_1 + x_2 + 3x_3 = 4 \\ x_1 + 2x_2 - x_3 = 5 \\ 3x_1 - x_2 + 2x_3 = 6 \end{cases}$

15. $\begin{cases} x_1 + x_2 - 2x_3 = -3 \\ 3x_1 - 2x_2 + x_3 = 4 \\ 2x_1 + x_2 + 3x_3 = 11 \end{cases}$

16. $\begin{cases} 2x_1 - 3x_2 + x_3 = 0 \\ x_1 + x_2 - 2x_3 = -4 \\ -x_1 + 2x_2 + x_3 = 5 \end{cases}$

17. $\begin{cases} x_1 + 4x_2 - x_3 = 7 \\ -2x_1 + x_2 + 3x_3 = 2 \\ x_1 - x_2 + 2x_3 = 3 \end{cases}$

18. $\begin{cases} 3x_1 - x_2 - 2x_3 = -1 \\ x_1 + 2x_2 + x_3 = 8 \\ -x_1 + 3x_2 - x_3 = 5 \end{cases}$

19. $\begin{cases} -x_1 - 2x_2 + x_3 = 1 \\ 2x_1 + x_2 - 3x_3 = 0 \\ -x_1 - x_2 + 2x_3 = 3 \end{cases}$

20. $\begin{cases} 4x_1 + x_2 - x_3 = 6 \\ x_1 - 3x_2 + 2x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 9 \end{cases}$

21. $\begin{cases} x_1 + x_2 + x_3 = 9 \\ 2x_1 + x_2 - x_3 = 2 \\ x_1 - 2x_2 + 3x_3 = 13 \end{cases}$

22. $\begin{cases} 3x_1 + x_2 - 2x_3 = 3 \\ -x_1 + 2x_2 + x_3 = 4 \\ x_1 + x_2 + 3x_3 = 10 \end{cases}$

23. $\begin{cases} 2x_1 - x_2 + x_3 = 4 \\ x_1 + 3x_2 - 2x_3 = -1 \\ -x_1 + x_2 + x_3 = 2 \end{cases}$

24. $\begin{cases} -x_1 + x_2 + 4x_3 = 5 \\ 2x_1 + 3x_2 - x_3 = 7 \\ x_1 - 2x_2 + x_3 = 0 \end{cases}$

25. $\begin{cases} x_1 - 3x_2 + 2x_3 = 1 \\ 3x_1 + x_2 - x_3 = 8 \\ 2x_1 - x_2 + 3x_3 = 7 \end{cases}$

26. $\begin{cases} x_1 + 2x_2 - 3x_3 = -4 \\ -2x_1 + x_2 + x_3 = 3 \\ 3x_1 - x_2 + 2x_3 = 5 \end{cases}$

27. $\begin{cases} 2x_1 + x_2 + 2x_3 = 10 \\ 2x_1 - x_2 + x_3 = 3 \\ -x_1 + 2x_2 - x_3 = 0 \end{cases}$

28. $\begin{cases} x_1 - x_2 - x_3 = -3 \\ -x_1 + 2x_2 + 3x_3 = 10 \\ 2x_1 + x_2 - 2x_3 = -1 \end{cases}$

29. $\begin{cases} 3x_1 + 2x_2 - x_3 = 7 \\ x_1 - x_2 + 3x_3 = 6 \\ -2x_1 + x_2 + x_3 = 1 \end{cases}$

30. $\begin{cases} x_1 + x_2 - x_3 = 4 \\ -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - x_2 + 3x_3 = 5 \end{cases}$

3. Solve the systems $Ax = b$ with the given matrix A for each of the given vectors b, computing A^{-1} only once.

1. $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, (b) $b = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, (c) $b = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$.

2. $A = \begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix}$; (a) $b = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, (b) $b = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$, (c) $b = \begin{pmatrix} -3 \\ 7 \end{pmatrix}$.

3. $A = \begin{pmatrix} 5 & 2 \\ -1 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$, (b) $b = \begin{pmatrix} 11 \\ -2 \end{pmatrix}$, (c) $b = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$.

4. $A = \begin{pmatrix} 3 & -1 \\ 4 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 2 \\ 9 \end{pmatrix}$, (b) $b = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$, (c) $b = \begin{pmatrix} 8 \\ 0 \end{pmatrix}$.

5. $A = \begin{pmatrix} -2 & 1 \\ 1 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 0 \\ 7 \end{pmatrix}$, (b) $b = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$, (c) $b = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$.

6. $A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{pmatrix}$; (a) $b = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}$, (b) $b = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$.

7. $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -3 \\ -1 & 0 & 2 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$, (b) $b = \begin{pmatrix} 3 \\ 0 \\ 5 \end{pmatrix}$.

8. $A = \begin{pmatrix} 3 & 0 & 1 \\ -1 & 2 & 0 \\ 0 & -4 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 4 \\ -1 \\ 7 \end{pmatrix}$, (b) $b = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}$.

9. $A = \begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$, (b) $b = \begin{pmatrix} -3 \\ 11 \end{pmatrix}$, (c) $b = \begin{pmatrix} 1 \\ -5 \end{pmatrix}$.

10. $A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 1 \\ 2 & -1 & 0 \end{pmatrix}$; (a) $b = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix}$, (b) $b = \begin{pmatrix} 0 \\ 4 \\ -3 \end{pmatrix}$.

11. $A = \begin{pmatrix} 7 & -2 \\ 3 & -1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, (b) $b = \begin{pmatrix} -5 \\ 0 \end{pmatrix}$, (c) $b = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$.

12. $A = \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$, (b) $b = \begin{pmatrix} 7 \\ 0 \end{pmatrix}$, (c) $b = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

13. $A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 3 & 2 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$, (b) $b = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$.

14. $A = \begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, (b) $b = \begin{pmatrix} 13 \\ 5 \end{pmatrix}$, (c) $b = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$.

15. $A = \begin{pmatrix} -2 & 1 & 3 \\ 4 & 0 & -1 \\ 1 & -2 & 5 \end{pmatrix}$; (a) $b = \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix}$, (b) $b = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix}$.

16. $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$; (a) $b = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$, (b) $b = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$, (c) $b = \begin{pmatrix} 10 \\ 15 \end{pmatrix}$.

17. $A = \begin{pmatrix} 4 & -1 & 2 \\ 3 & 5 & 0 \\ -2 & 0 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 7 \\ -1 \\ 3 \end{pmatrix}$, (b) $b = \begin{pmatrix} 0 \\ 8 \\ -4 \end{pmatrix}$.

18. $A = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, (b) $b = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$, (c) $b = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

19. $A = \begin{pmatrix} 0 & -3 & 1 \\ 5 & 2 & 0 \\ -4 & 0 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}$, (b) $b = \begin{pmatrix} -3 \\ 0 \\ 7 \end{pmatrix}$.

20. $A = \begin{pmatrix} 6 & 1 \\ -2 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$, (b) $b = \begin{pmatrix} 13 \\ -5 \end{pmatrix}$, (c) $b = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$.

21. $A = \begin{pmatrix} 1 & 3 & -1 \\ -2 & 0 & 4 \\ 0 & 5 & 2 \end{pmatrix}$; (a) $b = \begin{pmatrix} 4 \\ -6 \\ 3 \end{pmatrix}$, (b) $b = \begin{pmatrix} -1 \\ 2 \\ 7 \end{pmatrix}$.

22. $A = \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, (b) $b = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$, (c) $b = \begin{pmatrix} 11 \\ -7 \end{pmatrix}$.

23. $A = \begin{pmatrix} 3 & 1 & 0 \\ -2 & 4 & -1 \\ 0 & -3 & 5 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ 6 \\ -2 \end{pmatrix}$, (b) $b = \begin{pmatrix} -5 \\ 0 \\ 3 \end{pmatrix}$.

24. $A = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, (b) $b = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$, (c) $b = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$.

25. $A = \begin{pmatrix} -1 & 2 & -3 \\ 4 & 0 & 1 \\ 0 & -5 & 2 \end{pmatrix}$; (a) $b = \begin{pmatrix} 0 \\ 9 \\ -5 \end{pmatrix}$, (b) $b = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.

26. $A = \begin{pmatrix} 9 & -1 \\ 4 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 8 \\ 5 \end{pmatrix}$, (b) $b = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$, (c) $b = \begin{pmatrix} 17 \\ 9 \end{pmatrix}$.

27. $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 3 & 4 \\ -3 & 1 & 0 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$, (b) $b = \begin{pmatrix} -3 \\ 5 \\ 0 \end{pmatrix}$.

28. $A = \begin{pmatrix} -3 & 2 \\ 4 & -3 \end{pmatrix}$; (a) $b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, (b) $b = \begin{pmatrix} 7 \\ -8 \end{pmatrix}$, (c) $b = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$.

29. $A = \begin{pmatrix} 1 & -4 & 0 \\ 3 & 2 & -5 \\ -2 & 0 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$, (b) $b = \begin{pmatrix} -1 \\ 6 \\ 0 \end{pmatrix}$.

30. $A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$; (a) $b = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, (b) $b = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$, (c) $b = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$.

2.3 The Gauss and Gauss–Jordan methods for systems of linear equations

Definition 2.3.1. Gaussian elimination is a general method for solving systems of linear equations. Elementary row operations are applied to the augmented matrix of the system in order to bring it to **row echelon form**; the unknowns are then found by **back substitution**.

Accordingly, the method consists of two stages. In the **forward elimination** the augmented matrix is reduced to row echelon form; in the **back substitution** the unknowns are determined one after another, starting from the last equation.

Definition 2.3.2. The **Gauss–Jordan method** is a modification of Gaussian elimination in which the reduction is continued beyond row echelon form until the matrix reaches **reduced row echelon form** (RREF). No back substitution is then needed: the values of the unknowns are read off directly.

For a system of m equations in n unknowns the **augmented matrix** is

$$(A \mid b) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right).$$

Unlike Cramer's rule and the matrix method, Gaussian elimination works also when the number of equations differs from the number of unknowns ($m \neq n$) and when $\det A = 0$. In the row operations below, S_i denotes the ith row of the matrix.

2.3.1 Gaussian elimination: a system with a unique solution

Example 2.3.1. $\begin{cases} 2x_1 + x_2 - x_3 = 1, \\ x_1 - x_2 + 2x_3 = 5, \\ 3x_1 + 2x_2 + x_3 = 8. \end{cases}$

Solution. We begin the forward elimination with the augmented matrix. To have a leading coefficient equal to 1, we first interchange the first two rows:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 1 & -1 & 2 & 5 \\ 3 & 2 & 1 & 8 \end{array}\right) \xrightarrow{S_1 \leftrightarrow S_2} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 2 & 1 & -1 & 1 \\ 3 & 2 & 1 & 8 \end{array}\right).$$

Subtracting suitable multiples of the new first row from the other two rows and then eliminating the second entry of the third row, we obtain

$$\xrightarrow[S_3 - 3S_1]{S_2 - 2S_1} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 3 & -5 & -9 \\ 0 & 5 & -5 & -7 \end{array}\right) \xrightarrow{S_3 - \frac{5}{3}S_2} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 3 & -5 & -9 \\ 0 & 0 & \frac{10}{3} & 8 \end{array}\right).$$

The back substitution starts with the last equation, which gives

$$\frac{10}{3}x_3 = 8 \;\Rightarrow\; x_3 = \frac{24}{10} = \frac{12}{5}.$$

Substituting this value into the second equation, we find

$$3x_2 - 5 \cdot \frac{12}{5} = -9 \;\Rightarrow\; 3x_2 = -9 + 12 = 3 \;\Rightarrow\; x_2 = 1,$$

and the first equation then yields

$$x_1 - 1 + 2 \cdot \frac{12}{5} = 5 \;\Rightarrow\; x_1 = 5 + 1 - \frac{24}{5} = \frac{30 - 24}{5} = \frac{6}{5}.$$

The fractions here are a little inconvenient. In hand computations one often rescales a row, for instance $S_3 \to \frac{3}{10}S_3$, to keep the numbers simple; the result, of course, does not depend on this.

Answer: $x_1 = \frac{6}{5}, x_2 = 1, x_3 = \frac{12}{5}$.

Example 2.3.2. $\begin{cases} x_1 + x_2 + x_3 = 6, \\ x_1 - x_2 + 2x_3 = 5, \\ 2x_1 + x_2 - x_3 = 1. \end{cases}$

Solution. Here the first coefficient is already 1, so the forward elimination proceeds at once:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 2 & 1 & -1 & 1 \end{array}\right) \xrightarrow[S_3-2S_1]{S_2-S_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -1 & -3 & -11 \end{array}\right)$$

$$\xrightarrow{S_3-\frac{1}{2}S_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & -\frac{7}{2} & -\frac{21}{2} \end{array}\right).$$

Back substitution gives successively

$$-\frac{7}{2}x_3 = -\frac{21}{2} \;\Rightarrow\; x_3 = 3, \qquad -2x_2 + 3 = -1 \;\Rightarrow\; x_2 = 2,$$
$$x_1 + 2 + 3 = 6 \;\Rightarrow\; x_1 = 1.$$

Answer: $x_1 = 1, x_2 = 2, x_3 = 3$.
Check: $1 + 2 + 3 = 6$, $1 - 2 + 6 = 5$, $2 + 2 - 3 = 1$.

2.3.2 The Gauss–Jordan method

In the Gauss–Jordan method the matrix is brought to reduced row echelon form: every leading entry equals 1, and all other entries in the column of a leading entry, above and below it, are zero.

Example 2.3.3. Let us solve the system of the previous example by the Gauss–Jordan method, starting from the row echelon form obtained there. We first scale the third row so that its leading entry becomes 1:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & -\frac{7}{2} & -\frac{21}{2} \end{array}\right) \xrightarrow{S_3 \to -\frac{2}{7}S_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

Next we clear the entry above it in the second row and normalize the second row:

$$\xrightarrow{S_2-S_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow{S_2\to-\frac{1}{2}S_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

Finally, subtracting the second and third rows from the first, we arrive at

$$\xrightarrow[S_1-S_3]{S_1-S_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

The solution can now be read off directly: $x_1 = 1, x_2 = 2, x_3 = 3$.

2.3.3 Inconsistent systems and systems with infinitely many solutions

An inconsistent system.

Example 2.3.4. $\begin{cases} x_1 + 2x_2 = 3, \\ 2x_1 + 4x_2 = 5. \end{cases}$

Solution.

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 5 \end{array}\right) \xrightarrow{S_2-2S_1} \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & -1 \end{array}\right).$$

The last row corresponds to the equation $0 \cdot x_1 + 0 \cdot x_2 = -1$, which is a contradiction. Hence the system is **inconsistent**: it has no solution.

A system with infinitely many solutions.

Example 2.3.5. $\begin{cases} x_1 - x_2 + 2x_3 = 3, \\ 2x_1 - 2x_2 + 4x_3 = 6, \\ x_1 + x_2 - x_3 = 1. \end{cases}$

Solution.

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 2 & -2 & 4 & 6 \\ 1 & 1 & -1 & 1 \end{array}\right) \xrightarrow[S_3-S_1]{S_2-2S_1} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & -3 & -2 \end{array}\right) \xrightarrow{S_2\leftrightarrow S_3} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 2 & -3 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

The ranks of the coefficient matrix and of the augmented matrix coincide, $\operatorname{rank} A = \operatorname{rank}(A \mid b) = 2$, but they are less than the number of unknowns, which is 3. So the system is consistent and has one free unknown; we take $x_3 = t$ as a parameter. The second row gives $2x_2 - 3t = -2$, that is, $x_2 = \frac{3t-2}{2}$, and from the first row

$$x_1 = 3 + x_2 - 2t = 3 + \frac{3t-2}{2} - 2t = 2 - \frac{t}{2}.$$

Answer: $x_1 = 2 - \frac{t}{2}$, $x_2 = \frac{3t-2}{2}$, $x_3 = t$, where $t \in \mathbb{R}$ is arbitrary.

Example 2.3.6. $\begin{cases} x_1 + 2x_2 - x_3 + x_4 = 2, \\ 2x_1 + 4x_2 - 2x_3 + 3x_4 = 5, \\ x_1 + 2x_2 - x_3 + 2x_4 = 3. \end{cases}$

Solution.

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 2 & 4 & -2 & 3 & 5 \\ 1 & 2 & -1 & 2 & 3 \end{array}\right) \xrightarrow[S_3-S_1]{S_2-2S_1} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow{S_3-S_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Here $\operatorname{rank} A = \operatorname{rank}(A \mid b) = 2$ and there are four unknowns, so $4 - 2 = 2$ of them are free. We choose $x_2 = s$ and $x_3 = t$. The second row gives $x_4 = 1$, and then the first row gives $x_1 = 2 - 2s + t - 1 = 1 - 2s + t$.

Answer: $x_1 = 1 - 2s + t$, $x_2 = s$, $x_3 = t$, $x_4 = 1$, where $s, t \in \mathbb{R}$.

Comparison of the Gauss and Gauss–Jordan methods.

Gaussian elimination	Gauss–Jordan method
Reduction to row echelon form followed by back substitution	Reduction to RREF; the answer is read off directly
Needs fewer operations	Needs more operations, but the procedure is uniform
Back substitution requires solving the equations one by one	No further solving is needed
Also yields the determinant and the rank	Also yields the inverse matrix: $(A \mid E) \to (E \mid A^{-1})$

2.3.4 Applications of Gaussian elimination in information security

Linear equations in cryptographic algorithms. The security analysis of many modern ciphers (AES, linearized versions of GOST, and others) uses the method of **linear cryptanalysis**. One writes down linear approximations of the encryption process, which leads to a large system of linear equations. Gaussian elimination is the main tool for solving such systems, because they may contain thousands of equations, they need not have a unique solution ($\det A = 0$ is possible), and they usually have to be solved over the binary field $\mathrm{GF}(2)$.

Gaussian elimination over $\mathrm{GF}(2)$. In information security one often works over the field $\mathrm{GF}(2) = \{0, 1\}$, in which addition is the XOR operation ($\oplus$) and multiplication is the AND operation. Gaussian elimination works over this field without change. Adding one row to another is a bitwise XOR, and no scaling of rows is ever needed, since the only nonzero element is 1 and $1^{-1} = 1$. This version of the method is widely used in cryptanalysis, in coding theory and in the analysis of hash functions.

Example 2.3.7. Solve the following system over $\mathrm{GF}(2)$:

$$\begin{cases} x_1 \oplus x_2 \oplus x_3 = 1, \\ x_1 \oplus x_3 = 0, \\ x_2 \oplus x_3 = 1. \end{cases}$$

Solution.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array}\right) \xrightarrow{S_2 \oplus S_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right) \xrightarrow{S_3 \oplus S_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right).$$

Back substitution gives $x_3 = 0$, $x_2 = 1$ and $x_1 = 1 \oplus x_2 \oplus x_3 = 1 \oplus 1 \oplus 0 = 0$.

Answer: $x_1 = 0, x_2 = 1, x_3 = 0$.

Syndrome decoding in coding theory. To detect and correct errors in a received word one has to solve the syndrome equation $Hc^T = s$. For large codes this is done by Gaussian elimination.

Lattice-based cryptography. In LWE and other lattice-based schemes, an attack that recovers the secret key reduces to solving a system of the form $Ax \approx b \pmod q$. Gaussian elimination and its modular version are standard tools in the security analysis of such schemes.

2.3.5 Computing with Python, Maple and Mathcad

The reduced row echelon form of an augmented matrix is computed by a single command in each of the three systems. We take the system of the second example. In SymPy the method `rref()` returns the reduced matrix together with the indices of the pivot columns.

Python

```
from sympy import Matrix
Ab = Matrix([[1, 1, 1, 6],
             [1, -1, 2, 5],
             [2, 1, -1, 1]])
R, pivots = Ab.rref()
print(R)
print(pivots)
```

Output

```
Matrix([[1, 0, 0, 1], [0, 1, 0, 2], [0, 0, 1, 3]])
(0, 1, 2)
```

The last column of R contains the solution $x_1 = 1, x_2 = 2, x_3 = 3$.

In Maple, `GaussianElimination` produces a row echelon form (the result of the forward elimination), while `ReducedRowEchelonForm` carries out the full Gauss–Jordan reduction.

Maple

```
with(LinearAlgebra):
A := Matrix([[1, 1, 1], [1, -1, 2], [2, 1, -1]]):
b := Vector([6, 5, 1]):
GaussianElimination(<A | b>);
ReducedRowEchelonForm(<A | b>);
# [[1, 0, 0, 1], [0, 1, 0, 2], [0, 0, 1, 3]]
```

In Mathcad Prime the augmented matrix is formed with `augment(A, b)` and reduced with the built-in function `rref`.

Mathcad

```
     | 1   1   1 |          | 6 |
A := | 1  -1   2 |     b := | 5 |
     | 2   1  -1 |          | 1 |

                       | 1  0  0  1 |
rref(augment(A, b)) =  | 0  1  0  2 |
                       | 0  0  1  3 |
```

2.3.6 Exercises on the Gauss and Gauss–Jordan methods

1. Solve the following 2×2 systems by Gaussian elimination.

1. $\begin{cases} x_1 + 3x_2 = 7 \\ 2x_1 - x_2 = 0 \end{cases}$

2. $\begin{cases} 3x_1 - 2x_2 = 4 \\ x_1 + x_2 = 3 \end{cases}$

3. $\begin{cases} -x_1 + 4x_2 = 5 \\ 2x_1 + x_2 = 7 \end{cases}$

4. $\begin{cases} 5x_1 - x_2 = 9 \\ -2x_1 + 3x_2 = 4 \end{cases}$

5. $\begin{cases} x_1 + 2x_2 = 8 \\ 3x_1 - x_2 = 3 \end{cases}$

6. $\begin{cases} 4x_1 + 3x_2 = 14 \\ x_1 - 2x_2 = -1 \end{cases}$

7. $\begin{cases} -2x_1 + x_2 = 1 \\ x_1 + 5x_2 = 13 \end{cases}$

8. $\begin{cases} 7x_1 - 3x_2 = 5 \\ x_1 + x_2 = 3 \end{cases}$

9. $\begin{cases} 2x_1 + x_2 = 5 \\ -x_1 + 4x_2 = 7 \end{cases}$

10. $\begin{cases} x_1 - x_2 = 2 \\ 3x_1 + 2x_2 = 11 \end{cases}$

11. $\begin{cases} 6x_1 + x_2 = 13 \\ -x_1 + 3x_2 = 4 \end{cases}$

12. $\begin{cases} -3x_1 + 2x_2 = 1 \\ x_1 - x_2 = 0 \end{cases}$

13. $\begin{cases} x_1 + x_2 = 5 \\ 4x_1 - 3x_2 = 6 \end{cases}$

14. $\begin{cases} 2x_1 - 5x_2 = -6 \\ x_1 + x_2 = 3 \end{cases}$

15. $\begin{cases} 8x_1 + x_2 = 17 \\ 3x_1 - 2x_2 = 0 \end{cases}$

16. $\begin{cases} -x_1 + 2x_2 = 3 \\ 5x_1 - x_2 = 8 \end{cases}$

17. $\begin{cases} 3x_1 + 4x_2 = 19 \\ x_1 - x_2 = 1 \end{cases}$

18. $\begin{cases} x_1 + 6x_2 = 14 \\ 2x_1 - x_2 = 1 \end{cases}$

19. $\begin{cases} 4x_1 - x_2 = 7 \\ -x_1 + 3x_2 = 2 \end{cases}$

20. $\begin{cases} 2x_1 + 3x_2 = 12 \\ x_1 - 2x_2 = -1 \end{cases}$

21. $\begin{cases} 5x_1 + 2x_2 = 17 \\ -x_1 + x_2 = 1 \end{cases}$

22. $\begin{cases} x_1 - 4x_2 = -3 \\ 3x_1 + x_2 = 10 \end{cases}$

23. $\begin{cases} -2x_1 + x_2 = -3 \\ x_1 + 2x_2 = 4 \end{cases}$

24. $\begin{cases} x_1 + x_2 = 0 \\ 2x_1 - 3x_2 = 5 \end{cases}$

25. $\begin{cases} 9x_1 - x_2 = 8 \\ x_1 + 2x_2 = 5 \end{cases}$

26. $\begin{cases} -x_1 + 5x_2 = 9 \\ 3x_1 + x_2 = 7 \end{cases}$

27. $\begin{cases} 2x_1 - x_2 = 3 \\ -x_1 + 3x_2 = 4 \end{cases}$

28. $\begin{cases} x_1 + 7x_2 = 15 \\ 3x_1 - x_2 = 5 \end{cases}$

29. $\begin{cases} 4x_1 + x_2 = 10 \\ x_1 - 3x_2 = -5 \end{cases}$

30. $\begin{cases} 6x_1 - 5x_2 = 2 \\ x_1 + x_2 = 3 \end{cases}$

2. Solve the following 3×3 *systems by Gaussian elimination (forward elimination and back substitution).*

1. $\begin{cases} x_1 + x_2 + x_3 = 6 \\ x_1 - x_2 + 2x_3 = 5 \\ 2x_1 + x_2 - x_3 = 1 \end{cases}$

2. $\begin{cases} 2x_1 - x_2 + 3x_3 = 9 \\ x_1 + 2x_2 - x_3 = 0 \\ 3x_1 + x_2 + 2x_3 = 11 \end{cases}$

3. $\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 - x_2 + 3x_3 = 1 \\ x_1 + x_2 + x_3 = 6 \end{cases}$

4. $\begin{cases} -x_1 + x_2 + x_3 = 1 \\ x_1 - x_2 + x_3 = 3 \\ x_1 + x_2 - x_3 = 5 \end{cases}$

5. $\begin{cases} 3x_1 - x_2 + 2x_3 = 5 \\ x_1 + 2x_2 - x_3 = 0 \\ 2x_1 + x_2 + 3x_3 = 11 \end{cases}$

6. $\begin{cases} x_1 + 3x_2 + 2x_3 = 8 \\ 2x_1 - x_2 + x_3 = 3 \\ -x_1 + 2x_2 - 3x_3 = -7 \end{cases}$

7. $\begin{cases} 4x_1 - x_2 + 2x_3 = 8 \\ x_1 + 3x_2 - x_3 = 5 \\ -2x_1 + x_2 + 3x_3 = 1 \end{cases}$

8. $\begin{cases} x_1 - 2x_2 + 3x_3 = 7 \\ -x_1 + x_2 - x_3 = -3 \\ 2x_1 + 3x_2 - 2x_3 = -1 \end{cases}$

9. $\begin{cases} 2x_1 + 3x_2 - x_3 = 4 \\ x_1 - x_2 + 2x_3 = 7 \\ 3x_1 + x_2 + x_3 = 10 \end{cases}$

10. $\begin{cases} x_1 - x_2 + x_3 = 2 \\ 2x_1 + x_2 - 3x_3 = -1 \\ x_1 + 2x_2 + x_3 = 7 \end{cases}$

11. $\begin{cases} 5x_1 - 2x_2 + x_3 = 3 \\ x_1 + 3x_2 - 2x_3 = 1 \\ -x_1 + x_2 + 4x_3 = 8 \end{cases}$

12. $\begin{cases} x_1 + x_2 - 2x_3 = -3 \\ 3x_1 - 2x_2 + x_3 = 4 \\ 2x_1 + x_2 + 3x_3 = 11 \end{cases}$

13. $\begin{cases} -2x_1 + x_2 + 3x_3 = 4 \\ x_1 + 2x_2 - x_3 = 5 \\ 3x_1 - x_2 + 2x_3 = 6 \end{cases}$

14. $\begin{cases} 2x_1 - 3x_2 + x_3 = 0 \\ x_1 + x_2 - 2x_3 = -4 \\ -x_1 + 2x_2 + x_3 = 5 \end{cases}$

15. $\begin{cases} x_1 + 4x_2 - x_3 = 7 \\ -2x_1 + x_2 + 3x_3 = 2 \\ x_1 - x_2 + 2x_3 = 3 \end{cases}$

16. $\begin{cases} 3x_1 - x_2 - 2x_3 = -1 \\ x_1 + 2x_2 + x_3 = 8 \\ -x_1 + 3x_2 - x_3 = 5 \end{cases}$

17. $\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ 2x_1 + x_2 - 3x_3 = 0 \\ -x_1 - x_2 + 2x_3 = 3 \end{cases}$

18. $\begin{cases} 4x_1 + x_2 - x_3 = 6 \\ x_1 - 3x_2 + 2x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 9 \end{cases}$

19. $\begin{cases} -x_1 + 2x_2 - 3x_3 = -8 \\ 2x_1 - x_2 + x_3 = 5 \\ x_1 + x_2 + 2x_3 = 7 \end{cases}$

20. $\begin{cases} x_1 + x_2 + x_3 = 9 \\ 2x_1 + x_2 - x_3 = 2 \\ x_1 - 2x_2 + 3x_3 = 13 \end{cases}$

21. $\begin{cases} 3x_1 + x_2 - 2x_3 = 3 \\ -x_1 + 2x_2 + x_3 = 4 \\ x_1 + x_2 + 3x_3 = 10 \end{cases}$

22. $\begin{cases} 2x_1 - x_2 + x_3 = 4 \\ x_1 + 3x_2 - 2x_3 = -1 \\ -x_1 + x_2 + x_3 = 2 \end{cases}$

23. $\begin{cases} x_1 - 3x_2 + 2x_3 = 1 \\ 3x_1 + x_2 - x_3 = 8 \\ 2x_1 - x_2 + 3x_3 = 7 \end{cases}$

24. $\begin{cases} -x_1 + x_2 + 4x_3 = 5 \\ 2x_1 + 3x_2 - x_3 = 7 \\ x_1 - 2x_2 + x_3 = 0 \end{cases}$

25. $\begin{cases} x_1 + 2x_2 - 3x_3 = -4 \\ -2x_1 + x_2 + x_3 = 3 \\ 3x_1 - x_2 + 2x_3 = 5 \end{cases}$

26. $\begin{cases} 2x_1 + x_2 + 2x_3 = 10 \\ x_1 - x_2 + x_3 = 3 \\ -x_1 + 2x_2 - x_3 = 0 \end{cases}$

27. $\begin{cases} x_1 - x_2 - x_3 = -3 \\ -x_1 + 2x_2 + 3x_3 = 10 \\ 2x_1 + x_2 - 2x_3 = -1 \end{cases}$

28. $\begin{cases} 3x_1 + 2x_2 - x_3 = 7 \\ x_1 - x_2 + 3x_3 = 6 \\ -2x_1 + x_2 + x_3 = 1 \end{cases}$

29. $\begin{cases} x_1 + x_2 - x_3 = 4 \\ -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - x_2 + 3x_3 = 5 \end{cases}$

30. $\begin{cases} 2x_1 + x_2 - x_3 = 3 \\ x_1 - x_2 + 2x_3 = 4 \\ -x_1 + 2x_2 + x_3 = 5 \end{cases}$

3. Solve the following systems by the Gauss–Jordan method (reduction to RREF).

1. $\begin{cases} x_1 + 2x_2 - x_3 = 4 \\ 2x_1 - x_2 + 3x_3 = 1 \\ x_1 + x_2 + x_3 = 6 \end{cases}$

2. $\begin{cases} 3x_1 - x_2 = 5 \\ x_1 + 4x_2 = 6 \end{cases}$

3. $\begin{cases} x_1 - x_2 + 2x_3 = 3 \\ 2x_1 + x_2 - x_3 = 5 \\ -x_1 + 3x_2 + x_3 = 4 \end{cases}$

4. $\begin{cases} 2x_1 + x_2 = 7 \\ x_1 - 3x_2 = -4 \end{cases}$

5. $\begin{cases} -x_1 + 2x_2 + x_3 = 0 \\ x_1 + x_2 - 3x_3 = 4 \\ 2x_1 - x_2 + x_3 = 3 \end{cases}$

6. $\begin{cases} 4x_1 - x_2 + 2x_3 = 11 \\ x_1 + 2x_2 - x_3 = 2 \\ -2x_1 + x_2 + 3x_3 = 3 \end{cases}$

7. $\begin{cases} x_1 + x_2 = 3 \\ 2x_1 - 3x_2 = -4 \end{cases}$

8. $\begin{cases} x_1 + x_2 + x_3 = 6 \\ x_1 - x_2 + 2x_3 = 5 \\ 2x_1 + x_2 - x_3 = 1 \end{cases}$

9. $\begin{cases} 3x_1 + 2x_2 - x_3 = 7 \\ x_1 - x_2 + 3x_3 = 6 \\ -2x_1 + x_2 + x_3 = 1 \end{cases}$

10. $\begin{cases} 5x_1 - 2x_2 = 8 \\ x_1 + x_2 = 5 \end{cases}$

11. $\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ 2x_1 + x_2 - 3x_3 = 0 \\ -x_1 - x_2 + 2x_3 = 3 \end{cases}$

12. $\begin{cases} x_1 + 3x_2 = 10 \\ 4x_1 - x_2 = 1 \end{cases}$

13. $\begin{cases} 2x_1 - x_2 + x_3 = 4 \\ x_1 + 3x_2 - 2x_3 = -1 \\ -x_1 + x_2 + x_3 = 2 \end{cases}$

14. $\begin{cases} -2x_1 + x_2 = 3 \\ x_1 + x_2 = 6 \end{cases}$

15. $\begin{cases} x_1 + x_2 - 2x_3 = -3 \\ 3x_1 - 2x_2 + x_3 = 4 \\ 2x_1 + x_2 + 3x_3 = 11 \end{cases}$

16. $\begin{cases} x_1 - x_2 + x_3 = 2 \\ 2x_1 + x_2 - 3x_3 = -1 \\ x_1 + 2x_2 + x_3 = 7 \end{cases}$

17. $\begin{cases} 3x_1 + x_2 = 8 \\ -x_1 + 2x_2 = 1 \end{cases}$

18. $\begin{cases} 4x_1 + x_2 - x_3 = 6 \\ x_1 - 3x_2 + 2x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 9 \end{cases}$

19. $\begin{cases} x_1 + x_2 + x_3 = 9 \\ 2x_1 + x_2 - x_3 = 2 \\ x_1 - 2x_2 + 3x_3 = 13 \end{cases}$

20. $\begin{cases} 2x_1 - x_2 = 3 \\ x_1 + 4x_2 = 9 \end{cases}$

21. $\begin{cases} x_1 + 2x_2 - 3x_3 = -4 \\ -2x_1 + x_2 + x_3 = 3 \\ 3x_1 - x_2 + 2x_3 = 5 \end{cases}$

22. $\begin{cases} -x_1 + 2x_2 - 3x_3 = -8 \\ 2x_1 - x_2 + x_3 = 5 \\ x_1 + x_2 + 2x_3 = 7 \end{cases}$

23. $\begin{cases} x_1 + 5x_2 = 11 \\ 3x_1 - x_2 = 5 \end{cases}$

24. $\begin{cases} 3x_1 - x_2 - 2x_3 = -1 \\ x_1 + 2x_2 + x_3 = 8 \\ -x_1 + 3x_2 - x_3 = 5 \end{cases}$

25. $\begin{cases} x_1 - 3x_2 + 2x_3 = 1 \\ 3x_1 + x_2 - x_3 = 8 \\ 2x_1 - x_2 + 3x_3 = 7 \end{cases}$

26. $\begin{cases} 6x_1 + x_2 = 13 \\ -x_1 + 3x_2 = 4 \end{cases}$

27. $\begin{cases} 2x_1 + x_2 + 2x_3 = 10 \\ x_1 - x_2 + x_3 = 3 \\ -x_1 + 2x_2 - x_3 = 0 \end{cases}$

28. $\begin{cases} -x_1 + x_2 + 4x_3 = 5 \\ 2x_1 + 3x_2 - x_3 = 7 \\ x_1 - 2x_2 + x_3 = 0 \end{cases}$

29. $\begin{cases} 7x_1 - x_2 = 6 \\ x_1 + 3x_2 = 8 \end{cases}$

30. $\begin{cases} x_1 - x_2 - x_3 = -3 \\ -x_1 + 2x_2 + 3x_3 = 10 \\ 2x_1 + x_2 - 2x_3 = -1 \end{cases}$

4. Use Gaussian elimination to decide whether each system is consistent; if it is, find its general solution.

1. $\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ 2x_1 + 4x_2 - 2x_3 = 6 \\ x_1 + x_2 + x_3 = 1 \end{cases}$

2. $\begin{cases} x_1 - x_2 + 2x_3 = 1 \\ 2x_1 - 2x_2 + 4x_3 = 3 \end{cases}$

3. $\begin{cases} x_1 + x_2 + x_3 = 1 \\ 2x_1 + 2x_2 + 2x_3 = 2 \\ 3x_1 + 3x_2 + 3x_3 = 3 \end{cases}$

4. $\begin{cases} x_1 + 2x_2 = 5 \\ 2x_1 + 4x_2 = 10 \\ 3x_1 + 6x_2 = 15 \end{cases}$

5. $\begin{cases} 2x_1 - x_2 + 3x_3 = 4 \\ 4x_1 - 2x_2 + 6x_3 = 7 \end{cases}$

6. $\begin{cases} x_1 - x_2 + x_3 - x_4 = 2 \\ 2x_1 - 2x_2 + 2x_3 - 2x_4 = 4 \\ x_1 + x_2 - x_3 + x_4 = 0 \end{cases}$

7. $\begin{cases} 3x_1 - x_2 = 5 \\ -6x_1 + 2x_2 = -10 \end{cases}$

8. $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 4x_1 + 5x_2 + 6x_3 = 2 \\ 7x_1 + 8x_2 + 9x_3 = 3 \end{cases}$

9. $\begin{cases} x_1 - 2x_2 = 3 \\ -2x_1 + 4x_2 = 1 \end{cases}$

10. $\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 3 \\ 2x_1 + 2x_2 - 2x_3 + 4x_4 = 6 \\ x_1 - x_2 + x_3 = 1 \end{cases}$

11. $\begin{cases} x_1 + 3x_2 - 2x_3 = 0 \\ 2x_1 + 6x_2 - 4x_3 = 0 \end{cases}$

12. $\begin{cases} x_1 + x_2 = 3 \\ 2x_1 + 2x_2 = 6 \\ 3x_1 + 3x_2 = 10 \end{cases}$

13. $\begin{cases} x_1 - x_2 + 2x_3 = 3 \\ 2x_1 - 2x_2 + 4x_3 = 6 \\ x_1 + x_2 - x_3 = 1 \end{cases}$

14. $\begin{cases} 2x_1 + x_2 - x_3 = 4 \\ 4x_1 + 2x_2 - 2x_3 = 8 \\ -2x_1 - x_2 + x_3 = -4 \end{cases}$

15. $\begin{cases} x_1 - x_2 + x_3 = 2 \\ 2x_1 - 2x_2 + 2x_3 = 4 \\ 3x_1 - 3x_2 + 3x_3 = 7 \end{cases}$

16. $\begin{cases} x_1 + 2x_2 + x_3 + x_4 = 5 \\ 2x_1 + 4x_2 + 2x_3 + 2x_4 = 10 \end{cases}$

17. $\begin{cases} x_1 + x_2 = 4 \\ x_1 - x_2 = 2 \\ 2x_1 + x_2 = 7 \end{cases}$

18. $\begin{cases} x_1 + 2x_2 - x_3 = 1 \\ 3x_1 + 6x_2 - 3x_3 = 3 \\ -x_1 - 2x_2 + x_3 = -1 \end{cases}$

19. $\begin{cases} 5x_1 - x_2 = 10 \\ -10x_1 + 2x_2 = -15 \end{cases}$

20. $\begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ 2x_1 - x_2 + 3x_3 = 5 \\ x_1 + 2x_2 - x_3 + 4x_4 = -5 \end{cases}$

21. $\begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ 2x_1 + 4x_2 + 6x_3 = 0 \\ 3x_1 + 6x_2 + 9x_3 = 0 \end{cases}$

22. $\begin{cases} x_1 - x_2 = 2 \\ 2x_1 - 2x_2 = 4 \\ 3x_1 - 3x_2 = 6 \end{cases}$

23. $\begin{cases} 2x_1 + 3x_2 = 7 \\ 4x_1 + 6x_2 = 15 \end{cases}$

24. $\begin{cases} x_1 + x_2 - x_3 = 3 \\ 2x_1 + 2x_2 - 2x_3 = 6 \\ -x_1 - x_2 + x_3 = -3 \end{cases}$

25. $\begin{cases} x_1 + 2x_2 - x_3 + 3x_4 = 4 \\ 2x_1 + 4x_2 - 2x_3 + 6x_4 = 8 \\ x_1 + 2x_2 - x_3 + 4x_4 = 5 \end{cases}$

26. $\begin{cases} x_1 - 3x_2 = 5 \\ -2x_1 + 6x_2 = -10 \\ 3x_1 - 9x_2 = 15 \end{cases}$

27. $\begin{cases} x_1 + x_2 + x_3 = 5 \\ 2x_1 + 2x_2 + 2x_3 = 10 \\ x_1 + x_2 + x_3 = 8 \end{cases}$

28. $\begin{cases} x_1 + 2x_2 = 3 \\ 2x_1 + 4x_2 = 6 \\ -x_1 - 2x_2 = -3 \end{cases}$

29. $\begin{cases} 3x_1 - x_2 + 2x_3 = 4 \\ 6x_1 - 2x_2 + 4x_3 = 9 \end{cases}$

30. $\begin{cases} x_1 + x_2 + x_3 = 1 \\ 2x_1 + 3x_2 - x_3 = 4 \\ 3x_1 + 4x_2 = 5 \end{cases}$

2.4 Arithmetic operations on vectors: multiplication of a vector by a number, addition and subtraction of vectors

Definition 2.4.1. A quantity determined by its length and its direction is called a **vector**. The directed segment with initial point A and terminal point B is denoted by $\overrightarrow{AB}$. Once a Cartesian coordinate system is fixed in space, a vector is determined by its coordinates: $\vec{a} = (a_1; a_2; a_3) = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, where $\vec{i}, \vec{j}, \vec{k}$ are the unit vectors of the coordinate axes.

If $A(x_1; y_1; z_1)$ and $B(x_2; y_2; z_2)$, then

$$\overrightarrow{AB} = (x_2 - x_1;\ y_2 - y_1;\ z_2 - z_1),$$
$$|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

The **length (magnitude)** of $\vec{a}$ is $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$. The vector of length 0 is the **zero vector** $\vec{0}$, and a vector of length 1 is called a **unit vector**.

2.4.1 Addition, subtraction and multiplication by a number

Definition 2.4.2. For vectors $\vec{a} = (a_1; a_2; a_3)$, $\vec{b} = (b_1; b_2; b_3)$ and a number λ,

$$\vec{a} \pm \vec{b} = (a_1 \pm b_1;\ a_2 \pm b_2;\ a_3 \pm b_3), \qquad \lambda\vec{a} = (\lambda a_1;\ \lambda a_2;\ \lambda a_3).$$

Geometrically, the sum is constructed by the **triangle rule** or the **parallelogram rule**. The vector $\lambda\vec{a}$ has length $|\lambda|\,|\vec{a}|$; it points in the same direction as $\vec{a}$ when $\lambda > 0$ and in the opposite direction when $\lambda < 0$.

Properties of the linear operations.

1) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, $\quad (\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$;

2) $\vec{a} + \vec{0} = \vec{a}$, $\quad \vec{a} + (-\vec{a}) = \vec{0}$;

3) $\lambda(\vec{a} + \vec{b}) = \lambda\vec{a} + \lambda\vec{b}$, $\quad (\lambda + \mu)\vec{a} = \lambda\vec{a} + \mu\vec{a}$;

4) $\lambda(\mu\vec{a}) = (\lambda\mu)\vec{a}$, $\quad 1 \cdot \vec{a} = \vec{a}$.

Example 2.4.1. Let $\vec{a} = (2; -1; 3)$ and $\vec{b} = (1; 4; -2)$. Find $\vec{a} + \vec{b}$, $\vec{a} - \vec{b}$, $3\vec{a} - 2\vec{b}$ and $|\vec{a}|$.

Solution. Adding and subtracting coordinatewise gives $\vec{a} + \vec{b} = (3; 3; 1)$ and $\vec{a} - \vec{b} = (1; -5; 5)$. Further,

$$3\vec{a} - 2\vec{b} = (6; -3; 9) - (2; 8; -4) = (4; -11; 13),$$
$$|\vec{a}| = \sqrt{4 + 1 + 9} = \sqrt{14}.$$

Example 2.4.2. Given the points $A(1;2;-1)$ and $B(4;0;5)$, find $\overrightarrow{AB}$, its length and the midpoint of the segment AB.

Solution. We have $\overrightarrow{AB} = (4-1;\ 0-2;\ 5+1) = (3;-2;6)$ and $|\overrightarrow{AB}| = \sqrt{9+4+36} = 7$. The midpoint is $M\left(\frac{1+4}{2}; \frac{2+0}{2}; \frac{-1+5}{2}\right) = M\left(\frac{5}{2}; 1; 2\right)$.

2.4.2 Collinear vectors. Unit vectors

Definition 2.4.3. Vectors lying on one line or on parallel lines are called **collinear**. Vectors $\vec{a}$ and $\vec{b} \neq \vec{0}$ are collinear if and only if $\vec{a} = \lambda\vec{b}$ for some number λ, that is, if and only if their coordinates are proportional:

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}.$$

The unit vector in the direction of $\vec{a} \neq \vec{0}$ is $\vec{a}^{\,0} = \dfrac{\vec{a}}{|\vec{a}|} = (\cos\alpha; \cos\beta; \cos\gamma)$, where α, β, γ are the angles that $\vec{a}$ makes with the coordinate axes (the direction cosines satisfy $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$).

Example 2.4.3. For $\vec{a} = (2;-6;3)$ we have $|\vec{a}| = \sqrt{4+36+9} = 7$, hence $\vec{a}^{\,0} = \left(\frac{2}{7}; -\frac{6}{7}; \frac{3}{7}\right)$. Since $\vec{b} = (-4;12;-6) = -2\vec{a}$, the vectors $\vec{a}$ and $\vec{b}$ are collinear and point in opposite directions.

Example 2.4.4. Find the point M dividing the segment with endpoints $A(1;-2;4)$ and $B(7;4;-2)$ in the ratio $AM : MB = \lambda = 1 : 2$.

Solution. By the section formula $M = \dfrac{A + \lambda B}{1+\lambda}$ with $\lambda = \frac{1}{2}$, $x = \dfrac{1+\frac{7}{2}}{\frac{3}{2}} = 3$, $y = \dfrac{-2+2}{\frac{3}{2}} = 0$, $z = \dfrac{4-1}{\frac{3}{2}} = 2$. Thus $M(3;0;2)$.

2.4.3 Applications of vectors in information security

Feature vectors. In an intrusion detection system (IDS) each network session is described by a feature vector $\vec{x} = (x_1; x_2; \ldots; x_n)$ whose coordinates are the number of packets, the number of bytes transferred, the duration of the session, the number of errors, and so on. The mean vector of normal activity, $\vec{m} = \frac{1}{N}(\vec{x}_1 + \cdots + \vec{x}_N)$, is computed with nothing more than vector addition and multiplication by a number. For a new session the difference vector $\vec{x} - \vec{m}$ is used to detect anomalous behavior.

XOR as addition of binary vectors. Binary vectors are added coordinate-wise modulo 2 (the XOR operation). The one-time pad (Vernam cipher) is built on exactly this operation: $c = p \oplus k$ and $p = c \oplus k$.

Example 2.4.5. For the plaintext $p = (1; 0; 1; 1; 0)$ and the key $k = (0; 1; 1; 0; 1)$ we obtain $c = p \oplus k = (1; 1; 0; 1; 1)$; decryption gives back $c \oplus k = (1; 0; 1; 1; 0) = p$.

Lattices. In lattice-based cryptography a lattice is the set of all integer linear combinations of basis vectors $\vec{b}_1, \ldots, \vec{b}_n$: $L = \{k_1\vec{b}_1 + \cdots + k_n\vec{b}_n : k_i \in \mathbb{Z}\}$. The problem of finding a shortest nonzero vector of a lattice (SVP) underlies the security of ciphers designed to resist quantum computers.

2.4.4 Computing with Python, Maple and Mathcad

The following short programs repeat the first two examples of this topic: linear combinations of vectors, the length of a vector and the midpoint of a segment.

Python

```
import numpy as np
a = np.array([2, -1, 3])
b = np.array([1, 4, -2])
print(a + b, a - b, 3*a - 2*b)
print(np.linalg.norm(a))
A = np.array([1, 2, -1])
B = np.array([4, 0, 5])
print(B - A, np.linalg.norm(B - A), (A + B)/2)
```

Output

```
[3 3 1] [ 1 -5  5] [  4 -11  13]
3.7416573867739413
[ 3 -2  6] 7.0 [2.5 1.  2. ]
```

In Maple, column vectors are entered with angle brackets, and the Euclidean length is returned by `Norm(a, 2)`.

Maple

```
with(LinearAlgebra):
a := <2, -1, 3>:  b := <1, 4, -2>:
a + b;  a - b;  3*a - 2*b;
Norm(a, 2);                  # sqrt(14)
A := <1, 2, -1>:  B := <4, 0, 5>:
B - A;  Norm(B - A, 2);  (A + B)/2;
```

In Mathcad Prime the vectors are typed as one-column matrices; the absolute-value operator $|\cdot|$ applied to a vector gives its length.

Mathcad

```
a := |  2 |      b := |  1 |
     | -1 |           |  4 |
     |  3 |           | -2 |

3*a - 2*b = |   4 |      |a| = 3.742
            | -11 |
            |  13 |
```

2.4.5 Exercises on operations with vectors

1. Vectors $\vec{a}$ and $\vec{b}$ are given. Find the indicated linear combination.

1. $\vec{a} = (4; -5; 1)$, $\vec{b} = (2; 4; -5)$; find $-2\vec{a} + 3\vec{b}$.
2. $\vec{a} = (2; -1; 5)$, $\vec{b} = (-3; -5; 3)$; find $-\vec{a} - 3\vec{b}$.
3. $\vec{a} = (-2; 0; -5)$, $\vec{b} = (1; -3; 4)$; find $-\vec{a} - \vec{b}$.
4. $\vec{a} = (5; -1; 2)$, $\vec{b} = (-3; 5; -1)$; find $2\vec{a} - \vec{b}$.
5. $\vec{a} = (-3; 2; -2)$, $\vec{b} = (2; 4; 1)$; find $-3\vec{a} + \vec{b}$.
6. $\vec{a} = (-5; -2; -3)$, $\vec{b} = (-2; -1; 3)$; find $-\vec{a} + 3\vec{b}$.
7. $\vec{a} = (-2; 0; 5)$, $\vec{b} = (3; 2; 1)$; find $-3\vec{a} + 2\vec{b}$.

8. $\vec{a} = (4; 0; 3)$, $\vec{b} = (-3; -2; 1)$; find $-2\vec{a} - 3\vec{b}$.

9. $\vec{a} = (-5; 2; -1)$, $\vec{b} = (4; 5; -4)$; find $\vec{a} + 3\vec{b}$.

10. $\vec{a} = (-4; -3; 1)$, $\vec{b} = (4; 0; 4)$; find $-2\vec{a} - 3\vec{b}$.

11. $\vec{a} = (-4; 2; -3)$, $\vec{b} = (-2; 0; 1)$; find $-2\vec{a} + 2\vec{b}$.

12. $\vec{a} = (-1; -3; 4)$, $\vec{b} = (3; -3; -4)$; find $-\vec{a} - \vec{b}$.

13. $\vec{a} = (-3; 5; 5)$, $\vec{b} = (-3; -3; 2)$; find $3\vec{a} - \vec{b}$.

14. $\vec{a} = (0; 1; -2)$, $\vec{b} = (-5; 3; -5)$; find $-\vec{a} - \vec{b}$.

15. $\vec{a} = (-2; -4; -1)$, $\vec{b} = (2; 1; 4)$; find $-2\vec{a} + 3\vec{b}$.

16. $\vec{a} = (2; 5; -2)$, $\vec{b} = (3; -1; 3)$; find $\vec{a} - 3\vec{b}$.

17. $\vec{a} = (-3; 2; 5)$, $\vec{b} = (2; 1; -3)$; find $\vec{a} + \vec{b}$.

18. $\vec{a} = (0; 3; 1)$, $\vec{b} = (2; -3; 3)$; find $-3\vec{a} + 3\vec{b}$.

19. $\vec{a} = (-5; -5; 2)$, $\vec{b} = (3; -4; -5)$; find $-\vec{a} - 3\vec{b}$.

20. $\vec{a} = (-3; -4; 4)$, $\vec{b} = (2; 2; -3)$; find $3\vec{a} + 3\vec{b}$.

21. $\vec{a} = (3; 2; 0)$, $\vec{b} = (-1; 2; 1)$; find $-\vec{a} + 2\vec{b}$.

22. $\vec{a} = (3; 3; -3)$, $\vec{b} = (-2; 0; 4)$; find $-3\vec{a} + 2\vec{b}$.

23. $\vec{a} = (-2; 5; 4)$, $\vec{b} = (4; 5; 2)$; find $-3\vec{a} + 3\vec{b}$.

24. $\vec{a} = (-4; -1; 1)$, $\vec{b} = (-2; -1; -4)$; find $-3\vec{a} + \vec{b}$.

25. $\vec{a} = (3; -4; -1)$, $\vec{b} = (-3; 5; 2)$; find $-2\vec{a} - \vec{b}$.

26. $\vec{a} = (4; 1; -2)$, $\vec{b} = (0; -3; -4)$; find $\vec{a} - 2\vec{b}$.

27. $\vec{a} = (1; -2; -4)$, $\vec{b} = (5; -1; 2)$; find $3\vec{a} + 2\vec{b}$.

28. $\vec{a} = (3; 1; 2)$, $\vec{b} = (5; 3; -2)$; find $-\vec{a} + \vec{b}$.

29. $\vec{a} = (0; -4; 5)$, $\vec{b} = (5; -5; -2)$; find $\vec{a} + 3\vec{b}$.

30. $\vec{a} = (1; 5; -2)$, $\vec{b} = (-4; 1; -4)$; find $3\vec{a} - 3\vec{b}$.

2. For the given points A and B, find the vector $\overrightarrow{AB}$, its length and the midpoint M of the segment AB.

1. $A(-1; -2; -1)$, $B(1; 0; 6)$.
2. $A(-3; 4; 5)$, $B(-2; 1; 5)$.
3. $A(-2; 2; 1)$, $B(2; -2; 0)$.
4. $A(1; 4; -1)$, $B(-1; 5; -2)$.
5. $A(1; 1; 1)$, $B(3; 1; -5)$.
6. $A(5; 2; 5)$, $B(-1; 6; 5)$.
7. $A(0; 6; -4)$, $B(3; 0; 5)$.
8. $A(3; -5; 3)$, $B(5; -2; -6)$.
9. $A(3; -1; 4)$, $B(-1; -5; 5)$.
10. $A(0; 3; -1)$, $B(-2; 3; -4)$.
11. $A(-3; -4; -2)$, $B(-5; 2; 6)$.
12. $A(-6; -3; 3)$, $B(5; -5; 0)$.
13. $A(-3; 3; 3)$, $B(1; 6; 3)$.
14. $A(-6; 0; -3)$, $B(-1; 3; -3)$.
15. $A(2; 6; 3)$, $B(6; 6; -1)$.
16. $A(-6; 4; 5)$, $B(3; -6; -6)$.
17. $A(3; -1; 0)$, $B(-1; 3; 4)$.
18. $A(-4; 6; -4)$, $B(-5; 3; 6)$.
19. $A(-1; -4; -6)$, $B(5; 2; 2)$.
20. $A(6; -1; -5)$, $B(0; 1; 3)$.
21. $A(-1; -6; -4)$, $B(1; 5; 1)$.
22. $A(3; 1; 5)$, $B(1; 3; -4)$.

23. $A(-1;\ 0;\ -1)$, $B(2;\ 2;\ 1)$.

24. $A(0;\ -3;\ -5)$, $B(0;\ 4;\ -4)$.

25. $A(-6;\ -4;\ 4)$, $B(4;\ -3;\ 3)$.

26. $A(0;\ 6;\ -1)$, $B(2;\ 0;\ -6)$.

27. $A(3;\ -6;\ -4)$, $B(-2;\ -3;\ -6)$.

28. $A(3;\ -4;\ -5)$, $B(4;\ 3;\ -1)$.

29. $A(-4;\ -2;\ -6)$, $B(0;\ 3;\ 6)$.

30. $A(0;\ -5;\ -1)$, $B(6;\ -3;\ 1)$.

3. Applied problems.

1. **(Unit vector.)** Find the unit vector in the direction of $\vec{a} = (2; -1; 2)$.

2. **(Collinearity.)** Are the vectors $\vec{a} = (2; -4; 6)$ and $\vec{b} = (-3; 6; -9)$ collinear? How are they directed relative to each other?

3. **(Parameters.)** For which values of m and n are the vectors $\vec{a} = (1; m; 3)$ and $\vec{b} = (2; 4; n)$ collinear?

4. **(Parallelogram.)** The points $A(1; 0; 2)$, $B(3; 1; -1)$, $C(4; -2; 0)$ are vertices of a parallelogram $ABCD$. Find the vertex D.

5. **(Centroid.)** Find the centroid of the triangle with vertices $A(2; -1; 3)$, $B(4; 3; -1)$, $C(0; 1; 1)$.

6. **(Direction cosines.)** Find the direction cosines of the vector $\vec{a} = (1; 2; -2)$.

7. **(Feature vectors.)** The login activity of two users over four time slots is given by the vectors $\vec{u} = (5; 2; 0; 3)$ and $\vec{v} = (1; 4; 2; 3)$. Find $\vec{u} - \vec{v}$ and $\frac{1}{2}(\vec{u} + \vec{v})$.

8. **(Decomposition.)** Express the vector $\vec{c} = (7; -1)$ as a linear combination of $\vec{a} = (2; 1)$ and $\vec{b} = (1; -1)$.

9. **(Length.)** Given $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\vec{a} \perp \vec{b}$, find $|\vec{a} + \vec{b}|$ and $|\vec{a} - \vec{b}|$.

10. **(Dividing a segment.)** Find the point M that divides the segment with endpoints $A(1; -2; 4)$ and $B(7; 4; -2)$ in the ratio $AM : MB = 1 : 2$.

2.5 Scalar, vector and mixed products of vectors

2.5.1 The scalar product

Definition 2.5.1. The **scalar (dot) product** of vectors $\vec{a}$ and $\vec{b}$ is the number equal to the product of their lengths and the cosine of the angle between them:

$$\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\varphi.$$

In coordinates, $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$.

Properties: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$; $(\lambda\vec{a}) \cdot \vec{b} = \lambda(\vec{a} \cdot \vec{b})$; $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$; $\vec{a} \cdot \vec{a} = |\vec{a}|^2$; $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$.

The angle between two vectors and the projection of one vector onto another are given by

$$\cos\varphi = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|\,|\vec{b}|}, \qquad \mathrm{pr}_{\vec{b}}\vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}.$$

Example 2.5.1. Find the angle between the vectors $\vec{a} = (1; 2; -2)$ and $\vec{b} = (3; 0; 4)$.

Solution. Here $\vec{a} \cdot \vec{b} = 3 + 0 - 8 = -5$, $|\vec{a}| = \sqrt{1 + 4 + 4} = 3$ and $|\vec{b}| = \sqrt{9 + 0 + 16} = 5$. Hence $\cos\varphi = \dfrac{-5}{3 \cdot 5} = -\dfrac{1}{3}$ and $\varphi = \arccos\left(-\frac{1}{3}\right) \approx 109.5°$, an obtuse angle. The projection of $\vec{a}$ onto $\vec{b}$ is $\mathrm{pr}_{\vec{b}}\vec{a} = -\frac{5}{5} = -1$.

2.5.2 The vector product

Definition 2.5.2. The **vector (cross) product** of vectors $\vec{a}$ and $\vec{b}$ is the vector $\vec{c} = \vec{a} \times \vec{b}$ such that 1) $|\vec{c}| = |\vec{a}|\,|\vec{b}| \sin\varphi$; 2) $\vec{c} \perp \vec{a}$ and $\vec{c} \perp \vec{b}$;

3) $\vec{a}, \vec{b}, \vec{c}$ form a right-handed triple. In coordinates,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2;\ a_3b_1 - a_1b_3;\ a_1b_2 - a_2b_1).$$

Properties: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$; $\vec{a} \times \vec{a} = \vec{0}$; $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$. The length $|\vec{a} \times \vec{b}|$ equals the area of the parallelogram spanned by $\vec{a}$ and $\vec{b}$, so the triangle with sides $\vec{a}$ and $\vec{b}$ has area $S_{\triangle} = \frac{1}{2}|\vec{a} \times \vec{b}|$.

Example 2.5.2. For $\vec{a} = (1; 2; -2)$ and $\vec{b} = (3; 0; 4)$ find $\vec{a} \times \vec{b}$ and the area of the parallelogram spanned by these vectors.

Solution. Expanding the determinant along the first row, we get

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -2 \\ 3 & 0 & 4 \end{vmatrix} = (8 - 0)\vec{i} - (4 + 6)\vec{j} + (0 - 6)\vec{k} = (8; -10; -6).$$

Therefore $S = |\vec{a} \times \vec{b}| = \sqrt{64 + 100 + 36} = \sqrt{200} = 10\sqrt{2}$. As a check, $(8; -10; -6) \cdot (1; 2; -2) = 8 - 20 + 12 = 0$, so the result is indeed perpendicular to $\vec{a}$ (and similarly to $\vec{b}$).

2.5.3 The mixed product

Definition 2.5.3. The number $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is called the **mixed (scalar triple) product** of $\vec{a}, \vec{b}, \vec{c}$ and is denoted by $(\vec{a}, \vec{b}, \vec{c})$. In coordinates,

$$(\vec{a}, \vec{b}, \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

Theorem 2.5.1. *The absolute value* $|(\vec{a}, \vec{b}, \vec{c})|$ *equals the volume of the parallelepiped spanned by the three vectors; the volume of the tetrahedron they span is* $V = \frac{1}{6}|(\vec{a}, \vec{b}, \vec{c})|$. *Three vectors are* ***coplanar*** *(parallel to one plane) if and only if* $(\vec{a}, \vec{b}, \vec{c}) = 0$.

Example 2.5.3. Find the volume of the parallelepiped spanned by $\vec{a} = (1; 2; -2)$, $\vec{b} = (3; 0; 4)$ and $\vec{c} = (0; 1; 1)$.

Solution. From the previous example, $\vec{a} \times \vec{b} = (8; -10; -6)$, so $(\vec{a}, \vec{b}, \vec{c}) = 0 - 10 - 6 = -16$ and $V = 16$ cubic units. In particular, the vectors are not coplanar.

2.5.4 Applications of vector products in information security

Cosine similarity. When texts such as e-mail messages or malware signatures are analyzed, a document is represented by the vector of its word frequencies. The similarity of two documents is measured by $\cos\varphi = \dfrac{\vec{u} \cdot \vec{v}}{|\vec{u}|\,|\vec{v}|}$: a value close to 1 means that the documents are very similar, and a value close to 0 that they are almost unrelated. This measure is widely used in spam filters and in plagiarism detection.

Example 2.5.4. For the frequency vectors $\vec{u} = (3; 0; 2; 1)$ and $\vec{v} = (1; 1; 2; 0)$, $\cos\varphi = \dfrac{3 + 0 + 4 + 0}{\sqrt{14}\sqrt{6}} = \dfrac{7}{2\sqrt{21}} \approx 0.76$, so the two documents are fairly similar.

Schemes based on the inner product. In the LWE cryptosystem encryption has the form $b = \langle \vec{a}, \vec{s} \rangle + e \pmod{q}$, where $\langle \vec{a}, \vec{s} \rangle$ is the scalar product of a public vector with the secret key vector and e is a small "noise" term.

Computer graphics and 3D security. The vector product gives the normal of a surface, while the sign of a mixed product tells on which side of a plane a point lies. Both are used in collision tests and in 3D face models for biometrics.

2.5.5 Computing with Python, Maple and Mathcad

We recompute the three examples above: the scalar product and the cosine of the angle, the vector product with the area of the parallelogram, and the mixed product as a 3×3 determinant.

Python

```
import sympy as sp
a = sp.Matrix([1, 2, -2])
b = sp.Matrix([3, 0, 4])
c = sp.Matrix([0, 1, 1])
cosphi = a.dot(b)/(a.norm()*b.norm())
print(a.dot(b), cosphi)
print(a.cross(b).T, a.cross(b).norm())
print(sp.Matrix.hstack(a, b, c).det())
```

Output

```
-5 -1/3
Matrix([[8, -10, -6]]) 10*sqrt(2)
-16
```

The Maple commands `DotProduct`, `CrossProduct` and `Determinant` of the `LinearAlgebra` package do the same.

Maple

```
with(LinearAlgebra):
a := <1, 2, -2>: b := <3, 0, 4>: c := <0, 1, 1>:
DotProduct(a, b);                  # -5
DotProduct(a, b)/(Norm(a,2)*Norm(b,2));
CrossProduct(a, b);                # <8, -10, -6>
Norm(CrossProduct(a, b), 2);       # 10*sqrt(2)
Determinant(<a | b | c>);          # -16
```

In Mathcad Prime the dot product is the ordinary multiplication operator applied to two vectors, the cross product has its own operator (`Ctrl`+8), and `augment` assembles the columns of the determinant.

Mathcad

```
a := |  1 |    b := | 3 |    c := | 0 |
     |  2 |         | 0 |         | 1 |
     | -2 |         | 4 |         | 1 |

a*b = -5            a x b = |   8 |
                            | -10 |
                            |  -6 |
|a x b| = 14.142    |augment(a, b, c)| = -16
```

2.5.6 Exercises on products of vectors

1. Find the scalar product of the vectors and the cosine of the angle between them.

1. $\vec{a} = (2; 3; 2), \vec{b} = (2; 3; 4)$.
2. $\vec{a} = (-2; -3; 3), \vec{b} = (2; 5; 4)$.
3. $\vec{a} = (-3; -4; 2), \vec{b} = (-1; -3; -4)$.
4. $\vec{a} = (3; 5; -5), \vec{b} = (4; 1; 2)$.
5. $\vec{a} = (5; 4; 5), \vec{b} = (-3; 4; -5)$.
6. $\vec{a} = (3; -4; -5), \vec{b} = (-5; -2; -2)$.
7. $\vec{a} = (4; -5; 2), \vec{b} = (0; 2; 4)$.
8. $\vec{a} = (-2; 3; -2), \vec{b} = (5; -1; 2)$.
9. $\vec{a} = (-5; 5; -4), \vec{b} = (2; 5; -1)$.
10. $\vec{a} = (1; 3; -4), \vec{b} = (-1; 0; -2)$.
11. $\vec{a} = (3; -1; -5), \vec{b} = (-4; 4; -4)$.
12. $\vec{a} = (1; -4; -1), \vec{b} = (1; -4; -5)$.
13. $\vec{a} = (5; -5; -2), \vec{b} = (-2; -5; 2)$.
14. $\vec{a} = (1; 1; 1), \vec{b} = (-4; 4; 5)$.
15. $\vec{a} = (-2; 5; -1), \vec{b} = (0; -4; -1)$.
16. $\vec{a} = (0; -5; 1), \vec{b} = (-4; -3; -2)$.
17. $\vec{a} = (-4; -5; -5), \vec{b} = (2; 2; -3)$.
18. $\vec{a} = (5; 3; -2), \vec{b} = (2; 3; -2)$.
19. $\vec{a} = (-3; 1; 5), \vec{b} = (1; -4; 1)$.
20. $\vec{a} = (1; -2; -5), \vec{b} = (-1; 4; -1)$.
21. $\vec{a} = (-5; -2; -3), \vec{b} = (1; 4; 5)$.

22. $\vec{a} = (4; -4; -5)$, $\vec{b} = (-3; -2; 2)$.

23. $\vec{a} = (-1; -5; 4)$, $\vec{b} = (0; -1; 1)$.

24. $\vec{a} = (-4; -4; -4)$, $\vec{b} = (-2; 4; 5)$.

25. $\vec{a} = (-2; -5; 4)$, $\vec{b} = (0; 0; 4)$.

26. $\vec{a} = (2; -3; 4)$, $\vec{b} = (2; 4; -3)$.

27. $\vec{a} = (1; -3; 5)$, $\vec{b} = (-3; -1; -2)$.

28. $\vec{a} = (4; -2; -2)$, $\vec{b} = (-3; 5; 3)$.

29. $\vec{a} = (-2; 5; 1)$, $\vec{b} = (2; 4; -4)$.

30. $\vec{a} = (1; -5; -4)$, $\vec{b} = (-4; -5; 3)$.

2. Find the vector product and the area of the parallelogram spanned by the vectors.

1. $\vec{a} = (0; -1; 2)$, $\vec{b} = (0; 2; 3)$.

2. $\vec{a} = (0; 4; -2)$, $\vec{b} = (-3; -2; -1)$.

3. $\vec{a} = (3; 4; -3)$, $\vec{b} = (0; -1; -1)$.

4. $\vec{a} = (-4; -3; 0)$, $\vec{b} = (2; 3; -1)$.

5. $\vec{a} = (-4; -4; -2)$, $\vec{b} = (0; 1; 4)$.

6. $\vec{a} = (-2; -3; 1)$, $\vec{b} = (-2; 3; 1)$.

7. $\vec{a} = (4; -2; -4)$, $\vec{b} = (-4; 3; 1)$.

8. $\vec{a} = (0; -4; -4)$, $\vec{b} = (-3; 3; -3)$.

9. $\vec{a} = (0; 1; -2)$, $\vec{b} = (-3; -3; 3)$.

10. $\vec{a} = (4; 1; -4)$, $\vec{b} = (-2; 1; 1)$.

11. $\vec{a} = (-3; 3; -3)$, $\vec{b} = (2; -4; 3)$.

12. $\vec{a} = (-4; 2; 2)$, $\vec{b} = (-4; -3; -3)$.

13. $\vec{a} = (-3; -3; 0)$, $\vec{b} = (2; 1; 2)$.

14. $\vec{a} = (3; 3; 3), \vec{b} = (4; -3; 4)$.

15. $\vec{a} = (4; -4; 0), \vec{b} = (-3; 3; -4)$.

16. $\vec{a} = (-1; -3; 3), \vec{b} = (3; 0; -4)$.

17. $\vec{a} = (1; 0; -2), \vec{b} = (-1; 4; -2)$.

18. $\vec{a} = (1; 3; 3), \vec{b} = (-1; 1; 2)$.

19. $\vec{a} = (0; -1; 2), \vec{b} = (-1; -1; 2)$.

20. $\vec{a} = (-1; 1; -1), \vec{b} = (-2; -2; 3)$.

21. $\vec{a} = (1; -4; -3), \vec{b} = (0; -2; -3)$.

22. $\vec{a} = (3; 3; 0), \vec{b} = (-1; 2; 2)$.

23. $\vec{a} = (4; 3; 1), \vec{b} = (3; 1; -3)$.

24. $\vec{a} = (-4; 0; -4), \vec{b} = (0; 1; 0)$.

25. $\vec{a} = (-4; -2; 2), \vec{b} = (3; -1; -4)$.

26. $\vec{a} = (0; -1; -2), \vec{b} = (-4; -3; 3)$.

27. $\vec{a} = (-3; 4; 1), \vec{b} = (-3; -1; -1)$.

28. $\vec{a} = (3; 0; -2), \vec{b} = (-4; 3; 4)$.

29. $\vec{a} = (-4; -2; -1), \vec{b} = (0; 1; 4)$.

30. $\vec{a} = (4; 4; -2), \vec{b} = (2; -1; -3)$.

3. Applied problems.

1. **(Mixed product.)** Let $\vec{a} = (0; 2; 0)$, $\vec{b} = (-2; 0; 0)$, $\vec{c} = (-2; 2; -3)$. Compute $(\vec{a}, \vec{b}, \vec{c})$. Are the vectors coplanar? Find the volume of the parallelepiped they span.

2. **(Mixed product.)** Let $\vec{a} = (0; 1; 1)$, $\vec{b} = (2; 1; 3)$, $\vec{c} = (3; -1; 0)$. Compute $(\vec{a}, \vec{b}, \vec{c})$. Are the vectors coplanar? Find the volume of the parallelepiped they span.

3. **(Mixed product.)** Let $\vec{a} = (-1; 2; -2)$, $\vec{b} = (-3; 2; 3)$, $\vec{c} = (3; 3; 2)$. Compute $(\vec{a}, \vec{b}, \vec{c})$. Are the vectors coplanar? Find the volume of the parallelepiped they span.

4. **(Mixed product.)** Let $\vec{a} = (2; -3; -2)$, $\vec{b} = (-2; 0; -3)$, $\vec{c} = (-1; 1; 3)$. Compute $(\vec{a}, \vec{b}, \vec{c})$. Are the vectors coplanar? Find the volume of the parallelepiped they span.

5. **(Mixed product.)** Let $\vec{a} = (-1; -1; 2)$, $\vec{b} = (3; -3; -1)$, $\vec{c} = (1; -3; -3)$. Compute $(\vec{a}, \vec{b}, \vec{c})$. Are the vectors coplanar? Find the volume of the parallelepiped they span.

6. **(Mixed product.)** Let $\vec{a} = (0; -1; 1)$, $\vec{b} = (0; 3; -1)$, $\vec{c} = (0; -3; -2)$. Compute $(\vec{a}, \vec{b}, \vec{c})$. Are the vectors coplanar? Find the volume of the parallelepiped they span.

7. **(Tetrahedron.)** Find the volume of the tetrahedron with vertices $A(1; 1; 1)$, $B(2; 0; 3)$, $C(0; 2; 1)$, $D(3; 1; 4)$.

8. **(Parameter.)** For which value of λ are the vectors $\vec{a} = (1; 2; \lambda)$, $\vec{b} = (0; 1; 3)$, $\vec{c} = (2; 1; 1)$ coplanar?

9. **(Perpendicularity.)** For which value of m are the vectors $\vec{a} = (m; 3; -2)$ and $\vec{b} = (4; m; 5)$ perpendicular?

10. **(Area of a triangle.)** Find the area of the triangle with vertices $A(1; 0; 0)$, $B(0; 2; 0)$, $C(0; 0; 3)$.

2.6 Equations of a straight line in the plane in various forms. The angle between two lines. The distance from a point to a line

2.6.1 Forms of the equation of a line

Definition 2.6.1. The **general equation** of a straight line in the plane is

$$Ax + By + C = 0 \qquad (A^2 + B^2 \neq 0);$$

the vector $\vec{n} = (A; B)$ is perpendicular to the line (the **normal vector**), and $\vec{s} = (-B; A)$ is parallel to it (a **direction vector**).

Form	**Equation**
Slope-intercept	$y = kx + b,\ k = \tan\alpha$
Point-slope	$y - y_0 = k(x - x_0)$
Through two points	$\dfrac{x - x_1}{x_2 - x_1} = \dfrac{y - y_1}{y_2 - y_1}$
Intercept form	$\dfrac{x}{a} + \dfrac{y}{b} = 1$
Canonical (symmetric)	$\dfrac{x - x_0}{m} = \dfrac{y - y_0}{n},\ \vec{s} = (m; n)$
Normal form	$x\cos\alpha + y\sin\alpha - p = 0$

Example 2.6.1. Find the equation of the line through the points $A(1; 2)$ and $B(3; -2)$ and write it in several forms.

Solution. The equation of a line through two points gives

$$\frac{x-1}{3-1} = \frac{y-2}{-2-2} \implies -4(x-1) = 2(y-2) \implies 2x + y - 4 = 0.$$

In slope-intercept form this is $y = -2x + 4$ (so $k = -2$), and in intercept form $\dfrac{x}{2} + \dfrac{y}{4} = 1$.

2.6.2 The angle between two lines

If the lines are given by $y = k_1x + b_1$ and $y = k_2x + b_2$, then

$$\tan\varphi = \left|\frac{k_2 - k_1}{1 + k_1k_2}\right|.$$

If they are given by general equations $A_1x+B_1y+C_1 = 0$ and $A_2x+B_2y+C_2 = 0$, then

$$\cos\varphi = \frac{|A_1A_2 + B_1B_2|}{\sqrt{A_1^2 + B_1^2}\sqrt{A_2^2 + B_2^2}}.$$

Condition of parallelism: $k_1 = k_2$, or $\dfrac{A_1}{A_2} = \dfrac{B_1}{B_2}$. **Condition of perpendicularity:** $k_1 k_2 = -1$, or $A_1A_2 + B_1B_2 = 0$.

Example 2.6.2. Find the angle between the lines $y = 2x+1$ and $y = -3x+2$.

Solution. With $k_1 = 2$ and $k_2 = -3$ we get $\tan\varphi = \left|\dfrac{-3-2}{1-6}\right| = 1$, hence $\varphi = 45°$.

Example 2.6.3. Find the lines through $M(2;-3)$ that are a) parallel, b) perpendicular to the line $3x - 4y + 7 = 0$.

Solution. a) A parallel line has the same normal vector $(3;-4)$, so its equation is $3(x-2) - 4(y+3) = 0$, that is, $3x - 4y - 18 = 0$. b) For a perpendicular line the vector $(3;-4)$ is a direction vector: $\dfrac{x-2}{3} = \dfrac{y+3}{-4}$, that is, $4x + 3y + 1 = 0$.

2.6.3 The distance from a point to a line

Theorem 2.6.1. *The distance from the point $M_0(x_0;y_0)$ to the line $Ax + By + C = 0$ is*

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

Example 2.6.4. Find the distance from $M(3;4)$ to the line $3x - 4y + 5 = 0$.

Solution. $d = \dfrac{|3\cdot 3 - 4\cdot 4 + 5|}{\sqrt{9+16}} = \dfrac{|-2|}{5} = \dfrac{2}{5}$.

Example 2.6.5. To find the distance between the parallel lines $3x+4y-10=0$ and $3x+4y+15=0$, we take any point of the first line, say $M(2;1)$, and compute its distance to the second one: $d=\dfrac{|6+4+15|}{5}=5$.

2.6.4 Applications of lines in information security

A linear classifier. The simplest model of a spam filter or an intrusion detection system works with two features (x, the number of errors in a session, and y, the request rate) and separates the "normal" and "suspicious" classes by a line $Ax+By+C=0$. For a new observation $(x_0;y_0)$ the sign of Ax_0+By_0+C determines the class, and the distance d to the line measures how confident the decision is. This idea lies at the heart of **support vector machines** (SVM).

Example 2.6.6. Let the separating line be $x+2y-10=0$. For the session $P(2;3)$ we get $2+6-10=-2<0$ (the "normal" class), and for $Q(6;5)$ we get $6+10-10=6>0$ (the "suspicious" class). The distances are $d_P=\frac{2}{\sqrt{5}}$ and $d_Q=\frac{6}{\sqrt{5}}$; since Q is farther from the boundary, it is classified as suspicious with greater confidence.

Shamir's $(2,2)$ secret sharing scheme. The secret number s is taken to be the y-intercept of a line $y=kx+s$, and each of the two participants receives one point of the line. A single point reveals nothing about s, whereas two points determine the line, and hence s, uniquely.

Example 2.6.7. The participants hold the points $(1;17)$ and $(3;27)$. The slope is $k=\frac{27-17}{3-1}=5$, so the line is $y=5x+12$ and the secret is $s=12$.

2.6.5 Computing with Python, Maple and Mathcad

The geometry module of SymPy finds the equation of a line through two points, the distance from a point to a line and the angle between two lines. The code below reproduces the first, third and fifth examples of this topic; the line $3x-4y+5=0$ is entered through two of its points.

Python

```
from sympy import Point, Line, deg
from sympy.abc import x, y
l = Line(Point(1, 2), Point(3, -2))
print(l.equation(x, y))
m = Line(Point(1, 2), Point(5, 5))   # 3x-4y+5=0
print(m.equation(x, y), m.distance(Point(3, 4)))
l1 = Line(Point(0, 1), Point(1, 3))  # y=2x+1
l2 = Line(Point(0, 2), Point(1, -1)) # y=-3x+2
print(deg(l1.smallest_angle_between(l2)))
```

Output

```
4*x + 2*y - 8
-3*x + 4*y - 5 2/5
45
```

The equations $4x + 2y - 8 = 0$ and $-3x + 4y - 5 = 0$ are, of course, multiples of $2x + y - 4 = 0$ and $3x - 4y + 5 = 0$. Maple has a similar `geometry` package.

Maple

```
with(geometry):
point(A, 1, 2): point(B, 3, -2):
line(l, [A, B], [x, y]):
Equation(l);              # equivalent to 2x+y-4=0
line(m, 3*x - 4*y + 5 = 0, [x, y]):
point(M, 3, 4):
distance(M, m);           # 2/5
line(l1, y = 2*x + 1, [x, y]):
line(l2, y = -3*x + 2, [x, y]):
FindAngle(l1, l2);        # Pi/4 (or 3*Pi/4)
```

In Mathcad Prime it is convenient to define the distance formula as a user function and evaluate it.

Mathcad

```
A := 3     B := -4     C := 5
             |A*x0 + B*y0 + C|
d(x0, y0) := -----------------
              sqrt(A^2 + B^2)
d(3, 4) = 0.4

k1 := 2    k2 := -3
phi := atan(|(k2 - k1)/(1 + k1*k2)|)
phi = 45 deg
```

2.6.6 Exercises on lines in the plane

1. Write the general equation of the line through the two given points.

1. $A(1;-2)$, $B(4;2)$
2. $A(4;-1)$, $B(-4;0)$
3. $A(-6;-1)$, $B(1;-2)$
4. $A(4;6)$, $B(1;5)$
5. $A(3;-3)$, $B(2;-6)$
6. $A(4;3)$, $B(-4;1)$
7. $A(-1;-4)$, $B(-1;-3)$
8. $A(-6;3)$, $B(-3;-5)$
9. $A(2;4)$, $B(-1;4)$
10. $A(0;6)$, $B(-5;-6)$
11. $A(-6;4)$, $B(2;-3)$
12. $A(-5;0)$, $B(1;-5)$
13. $A(4;0)$, $B(-4;2)$
14. $A(-1;3)$, $B(2;-4)$
15. $A(5;-6)$, $B(2;-4)$
16. $A(2;-5)$, $B(0;3)$
17. $A(0;4)$, $B(3;1)$
18. $A(1;3)$, $B(0;2)$
19. $A(-6;4)$, $B(5;-5)$
20. $A(-3;6)$, $B(4;-2)$
21. $A(-1;-1)$, $B(5;0)$
22. $A(4;-2)$, $B(-5;-2)$
23. $A(-3;-1)$, $B(-1;-1)$
24. $A(2;3)$, $B(6;6)$
25. $A(2;-4)$, $B(-6;0)$
26. $A(0;-6)$, $B(2;-6)$
27. $A(-3;4)$, $B(0;-6)$
28. $A(0;-3)$, $B(6;3)$
29. $A(-5;6)$, $B(2;-3)$
30. $A(-4;-5)$, $B(5;-2)$

2. Find the distance from the point M to the given line.

1. $M(0;4)$, $-6x-2y+6=0$
2. $M(3;2)$, $4x+5y-8=0$
3. $M(-1;2)$, $-x-3y+1=0$
4. $M(-2;-3)$, $x+5y+6=0$
5. $M(-3;-2)$, $x+2y+2=0$
6. $M(4;-5)$, $6x+5y-2=0$
7. $M(-3;-2)$, $-2x+6y+1=0$
8. $M(-5;-2)$, $6x-6y+7=0$
9. $M(3;3)$, $3x-5y+2=0$
10. $M(5;-1)$, $-4x-6y+1=0$
11. $M(3;3)$, $4x-5y-2=0$
12. $M(-1;5)$, $x+y+2=0$
13. $M(2;-1)$, $x+5y-4=0$
14. $M(0;3)$, $3x-6y-1=0$
15. $M(-2;-1)$, $4x+5y+5=0$
16. $M(5;-5)$, $2x-2y-9=0$
17. $M(3;-1)$, $5x-6y-5=0$
18. $M(1;2)$, $-4x-3y-3=0$
19. $M(4;-1)$, $-6x+2y+9=0$
20. $M(1;4)$, $6x-6y+4=0$
21. $M(4;3)$, $-5x+6y-3=0$
22. $M(4;-3)$, $-x-y+3=0$
23. $M(-5;-5)$, $-6x+5y-5=0$

24. $M(1;-4),\ -5x+2y-3=0$
25. $M(-1;-1),\ 5x-y-3=0$
26. $M(2;4),\ 6x+5y+8=0$
27. $M(3;4),\ 3x-5y-3=0$
28. $M(4;-2),\ 5x+4y+3=0$
29. $M(-4;5),\ -5x-y+1=0$
30. $M(-1;-1),\ -x-4y-7=0$

3. Applied problems.

1. **(Angle.)** Find the angle between the lines $2x-y+3=0$ and $x+2y-4=0$; determine whether they are parallel or perpendicular.
2. **(Angle.)** Find the angle between the lines $3x+4y-1=0$ and $6x+8y+5=0$; determine whether they are parallel or perpendicular.
3. **(Angle.)** Find the angle between the lines $x-3y+2=0$ and $2x+y-1=0$; determine whether they are parallel or perpendicular.
4. **(Angle.)** Find the angle between the lines $x+y=0$ and $x-y+4=0$; determine whether they are parallel or perpendicular.
5. **(Angle.)** Find the angle between the lines $2x+3y-6=0$ and $4x-y+2=0$; determine whether they are parallel or perpendicular.
6. **(Parallel line.)** Find the line through $M(2;-3)$ parallel to $3x-4y+7=0$.
7. **(Perpendicular line.)** Find the line through $M(1;4)$ perpendicular to $2x+y-5=0$.
8. **(Intercept form.)** Write the line $3x-2y+12=0$ in intercept form.
9. **(Distance between parallel lines.)** Find the distance between the lines $3x+4y-10=0$ and $3x+4y+15=0$.
10. **(Altitude of a triangle.)** In the triangle with vertices $A(1;1)$, $B(4;5)$, $C(7;1)$, find the length of the altitude dropped from C to AB.

Chapter 3

Lines and Planes in Space; Linear and Euclidean Spaces

3.1 Lines in space. Equations of a line in space, the angle between lines, the distance between lines

3.1.1 Equations of a line in space

Definition 3.1.1. The line passing through the point $M_0(x_0; y_0; z_0)$ parallel to the vector $\vec{s} = (m; n; p)$ has the **canonical equations**

$$\frac{x - x_0}{m} = \frac{y - y_0}{n} = \frac{z - z_0}{p}$$

and the **parametric equations** $x = x_0 + mt$, $y = y_0 + nt$, $z = z_0 + pt$, $t \in \mathbb{R}$. The vector $\vec{s}$ is called a **direction vector** of the line.

Remark 3.1.1. One of the denominators in the canonical equations may be zero. For instance, $\frac{x-1}{2} = \frac{y+3}{0} = \frac{z}{5}$ is understood to mean that the line lies in the plane $y = -3$.

The line through two points M_1 and M_2 is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}.$$

A line can also be given as the intersection of two planes (**general equations**):

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0, \\ A_2x + B_2y + C_2z + D_2 = 0; \end{cases}$$

in this case one may take $\vec{s} = \vec{n}_1 \times \vec{n}_2$.

Example 3.1.1. Write the canonical and parametric equations of the line through the points $M_1(1;-2;3)$ and $M_2(3;1;-1)$.

Solution. Here $\vec{s} = \overrightarrow{M_1M_2} = (2;3;-4)$, so

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-4}; \qquad x = 1+2t,\ y = -2+3t,\ z = 3-4t.$$

Example 3.1.2. Reduce the line $\begin{cases} x+y-z+1=0, \\ 2x-y+z-4=0 \end{cases}$ to canonical form.

Solution. A direction vector is $\vec{s} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -1 \\ 2 & -1 & 1 \end{vmatrix} = (0;-3;-3) \parallel (0;1;1)$. To find a point on the line we put $z = 0$; then $x + y = -1$ and $2x - y = 4$, whence $x = 1$, $y = -2$. Therefore the line is $\dfrac{x-1}{0} = \dfrac{y+2}{1} = \dfrac{z}{1}$.

3.1.2 The angle between two lines

The angle between two lines with direction vectors $\vec{s}_1$ and $\vec{s}_2$ is given by

$$\cos\varphi = \frac{|\vec{s}_1 \cdot \vec{s}_2|}{|\vec{s}_1|\,|\vec{s}_2|}.$$

The lines are parallel when $\vec{s}_1 \parallel \vec{s}_2$ and perpendicular when $\vec{s}_1 \cdot \vec{s}_2 = 0$.

Example 3.1.3. For the lines $\dfrac{x-1}{1} = \dfrac{y}{2} = \dfrac{z+1}{2}$ and $\dfrac{x}{2} = \dfrac{y-3}{-1} = \dfrac{z}{2}$ we have $\cos\varphi = \dfrac{|2-2+4|}{3\cdot 3} = \dfrac{4}{9}$, so $\varphi \approx 63.6°$.

3.1.3 The distance from a point to a line and between two lines

Theorem 3.1.1. *The distance from a point M_0 to the line passing through M_1 parallel to $\vec{s}$ equals*

$$d = \frac{|\overrightarrow{M_1M_0} \times \vec{s}|}{|\vec{s}|}.$$

The distance between skew lines $(M_1, \vec{s}_1)$ and $(M_2, \vec{s}_2)$ equals

$$d = \frac{|(\overrightarrow{M_1M_2}, \vec{s}_1, \vec{s}_2)|}{|\vec{s}_1 \times \vec{s}_2|}.$$

Example 3.1.4. Find the distance from $M_0(2;3;1)$ to the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z+1}{2}$.

Solution. The line passes through $M_1(1;0;-1)$ and has direction $\vec{s} = (1;2;2)$, while $\overrightarrow{M_1M_0} = (1;3;2)$. Hence $\overrightarrow{M_1M_0} \times \vec{s} = (6-4;\ 2-2;\ 2-3) = (2;0;-1)$ and $d = \frac{\sqrt{5}}{3}$.

Example 3.1.5. Consider the skew lines $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$ (the Ox axis) and $\frac{x}{0} = \frac{y-1}{1} = \frac{z-2}{0}$. Here $\overrightarrow{M_1M_2} = (0;1;2)$, $\vec{s}_1 \times \vec{s}_2 = (1;0;0) \times (0;1;0) = (0;0;1)$ and $(\overrightarrow{M_1M_2}, \vec{s}_1, \vec{s}_2) = 2$; therefore $d = 2$.

3.1.4 Lines in space and information security

Blakley's scheme. In Blakley's secret sharing scheme the secret is encoded as a coordinate of a point in space, and each participant receives a plane (or a line) passing through that point. Two planes meet in a line and three planes in general meet in a single point. Consequently, three participants together can

recover the secret point, whereas any two of them know only a line, that is, infinitely many candidates.

Trajectories and anomaly detection. In tracking systems (for drones or for vehicle monitoring, say) the motion of an object is modelled by the line $\vec{r}(t) = \vec{r}_0 + t\vec{s}$. If the distance d from an observed position to the predicted line exceeds a prescribed threshold, the signal may have been forged (spoofing).

Example 3.1.6. Let the predicted trajectory be $x = t$, $y = 2t$, $z = 2t$, and let the observed position be $P(3;3;6)$. Then $\overrightarrow{OP} \times \vec{s} = (3;3;6) \times (1;2;2) = (-6;0;3)$ and $d = \dfrac{\sqrt{45}}{3} = \sqrt{5} \approx 2.24$. With an admissible threshold of 1, the observation is flagged as suspicious.

3.1.5 Computing with Python, Maple and Mathcad

We recompute the distance from $M_0(2;3;1)$ to the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z+1}{2}$ and the angle between this line and the line with direction vector $(2;-1;2)$. In SymPy the cross product is `cross` and the length of a vector is `norm`:

Python

```
from sympy import Matrix, acos, deg

M0 = Matrix([2, 3, 1])          # the point
M1 = Matrix([1, 0, -1])         # a point of the line
s  = Matrix([1, 2, 2])          # direction vector
w  = (M0 - M1).cross(s)
print("cross =", w.T)
print("d =", w.norm() / s.norm())

s2 = Matrix([2, -1, 2])
c = abs(s.dot(s2)) / (s.norm() * s2.norm())
print("cos =", c, " angle =", deg(acos(c)).evalf(4))
```

Output

```
cross = Matrix([[2, 0, -1]])
d = sqrt(5)/3
cos = 4/9  angle = 63.61
```

In Maple the same computation uses `CrossProduct`, `DotProduct` and the Euclidean norm `Norm(v, 2)`:

Maple

```
with(LinearAlgebra):
M0 := <2, 3, 1>: M1 := <1, 0, -1>: s := <1, 2, 2>:
```

```
w := CrossProduct(M0 - M1, s);
d := Norm(w, 2)/Norm(s, 2);          # sqrt(5)/3
s2 := <2, -1, 2>:
c := abs(DotProduct(s, s2))/(Norm(s, 2)*Norm(s2, 2));
evalf(arccos(c)*180/Pi);             # 63.61220004
```

In Mathcad Prime the vectors are entered as column matrices, the cross product is the vector-product operator (Ctrl+8), and $|v|$ is the magnitude operator:

Mathcad

```
      | 2 |          | 1 |         | 1 |
M0 := | 3 |    M1 := | 0 |    s := | 2 |
      | 1 |          |-1 |         | 2 |


                           |  2 |
w := (M0 - M1) x s     w = |  0 |
                           | -1 |
d := |w| / |s|         d = 0.745
```

3.1.6 Exercises on lines in space

1. Write the canonical equations of the line through the points M_1 and M_2.

1. $M_1(-1; -1; 5)$, $M_2(5; -3; 5)$
2. $M_1(-3; -4; 3)$, $M_2(-2; -1; -5)$
3. $M_1(1; -3; 5)$, $M_2(4; -5; -1)$
4. $M_1(-3; -4; -1)$, $M_2(2; 1; -3)$
5. $M_1(-1; 0; -2)$, $M_2(2; 3; 4)$
6. $M_1(5; -4; 0)$, $M_2(4; 5; -1)$
7. $M_1(2; 3; 4)$, $M_2(-3; 2; 5)$
8. $M_1(2; 3; -3)$, $M_2(-1; -2; -3)$
9. $M_1(3; 0; -1)$, $M_2(0; 2; -1)$
10. $M_1(4; -1; 1)$, $M_2(-3; 4; 2)$
11. $M_1(3; -2; 4)$, $M_2(-2; -2; 0)$
12. $M_1(-3; -4; 1)$, $M_2(5; 5; 2)$
13. $M_1(1; -5; 1)$, $M_2(-5; -2; -3)$
14. $M_1(2; 2; 5)$, $M_2(-1; -3; 1)$
15. $M_1(-1; 0; -2)$, $M_2(5; 0; 3)$
16. $M_1(5; -3; 4)$, $M_2(-2; 2; -2)$
17. $M_1(4; -5; 4)$, $M_2(4; -5; 0)$
18. $M_1(4; 4; 0)$, $M_2(-3; -1; -2)$
19. $M_1(-3; 5; 3)$, $M_2(-4; -2; -3)$
20. $M_1(-4; 1; 5)$, $M_2(4; 3; 3)$
21. $M_1(-5; 2; 0)$, $M_2(5; -4; 3)$
22. $M_1(-5; -3; -1)$, $M_2(-4; 1; -1)$
23. $M_1(-1; 4; 4)$, $M_2(-4; 2; 3)$

24. $M_1(0; -5; -5)$, $M_2(0; 0; -3)$
25. $M_1(2; 3; 3)$, $M_2(3; 2; -2)$
26. $M_1(5; -2; -1)$, $M_2(-5; 1; 3)$
27. $M_1(-3; -3; -1)$, $M_2(1; 5; -2)$
28. $M_1(-1; 1; 4)$, $M_2(-2; 0; -2)$
29. $M_1(4; -1; 1)$, $M_2(-2; 5; 0)$
30. $M_1(1; -5; 0)$, $M_2(1; 2; -4)$

2. Find the angle between the lines.

1. $\frac{x}{-3} = \frac{y-1}{-4} = \frac{z}{-4}$ and $\frac{x+1}{0} = \frac{y+3}{-1} = \frac{z+1}{3}$
2. $\frac{x+1}{4} = \frac{y-3}{2} = \frac{z+2}{-2}$ and $\frac{x+2}{-2} = \frac{y+1}{-4} = \frac{z+1}{-3}$
3. $\frac{x+1}{2} = \frac{y}{0} = \frac{z}{-4}$ and $\frac{x-1}{1} = \frac{y-1}{-2} = \frac{z}{-3}$
4. $\frac{x+1}{-4} = \frac{y+2}{1} = \frac{z-1}{4}$ and $\frac{x-2}{0} = \frac{y+2}{-2} = \frac{z+3}{-4}$
5. $\frac{x+1}{-1} = \frac{y+1}{-2} = \frac{z-3}{-3}$ and $\frac{x-3}{3} = \frac{y+2}{1} = \frac{z+3}{-3}$
6. $\frac{x-1}{-1} = \frac{y+3}{-1} = \frac{z-3}{4}$ and $\frac{x-1}{3} = \frac{y+3}{-1} = \frac{z+3}{2}$
7. $\frac{x+2}{-2} = \frac{y-1}{4} = \frac{z+1}{1}$ and $\frac{x+1}{4} = \frac{y+3}{2} = \frac{z+1}{0}$
8. $\frac{x-2}{-4} = \frac{y+1}{-4} = \frac{z-3}{-2}$ and $\frac{x+3}{-4} = \frac{y+1}{-3} = \frac{z-3}{4}$
9. $\frac{x+1}{2} = \frac{y}{-2} = \frac{z}{0}$ and $\frac{x}{-4} = \frac{y}{-3} = \frac{z+1}{-3}$
10. $\frac{x+3}{-2} = \frac{y-1}{-2} = \frac{z+1}{3}$ and $\frac{x+1}{0} = \frac{y+1}{-4} = \frac{z+2}{4}$
11. $\frac{x+1}{-1} = \frac{y+2}{1} = \frac{z}{2}$ and $\frac{x-1}{-2} = \frac{y-1}{3} = \frac{z}{2}$
12. $\frac{x}{-1} = \frac{y-3}{2} = \frac{z-1}{3}$ and $\frac{x+2}{-1} = \frac{y+2}{-4} = \frac{z+3}{-4}$
13. $\frac{x-2}{-4} = \frac{y-3}{-2} = \frac{z+3}{0}$ and $\frac{x+2}{-3} = \frac{y-2}{3} = \frac{z-2}{4}$
14. $\frac{x-3}{2} = \frac{y-1}{-4} = \frac{z+2}{4}$ and $\frac{x-2}{-4} = \frac{y+3}{-2} = \frac{z-2}{0}$
15. $\frac{x-1}{4} = \frac{y+3}{2} = \frac{z+2}{0}$ and $\frac{x+1}{-1} = \frac{y-1}{3} = \frac{z-1}{-4}$
16. $\frac{x-2}{-4} = \frac{y}{3} = \frac{z+2}{1}$ and $\frac{x-1}{-1} = \frac{y-2}{-4} = \frac{z+3}{4}$
17. $\frac{x-2}{-3} = \frac{y+1}{1} = \frac{z-2}{-2}$ and $\frac{x+2}{3} = \frac{y-2}{1} = \frac{z+1}{-2}$

18. $\frac{x+3}{-3} = \frac{y-1}{0} = \frac{z-2}{-2}$ and $\frac{x+2}{-2} = \frac{y-1}{-2} = \frac{z-2}{-3}$

19. $\frac{x-3}{4} = \frac{y-1}{-4} = \frac{z-1}{3}$ and $\frac{x}{1} = \frac{y+3}{2} = \frac{z-3}{3}$

20. $\frac{x+2}{3} = \frac{y-2}{4} = \frac{z+1}{0}$ and $\frac{x}{1} = \frac{y-1}{2} = \frac{z+3}{-3}$

21. $\frac{x}{3} = \frac{y}{2} = \frac{z-1}{0}$ and $\frac{x-2}{-4} = \frac{y+3}{3} = \frac{z+3}{2}$

22. $\frac{x+1}{0} = \frac{y-3}{-1} = \frac{z-1}{-4}$ and $\frac{x+2}{-2} = \frac{y}{4} = \frac{z-1}{1}$

23. $\frac{x+3}{0} = \frac{y-1}{-4} = \frac{z}{-1}$ and $\frac{x-2}{-3} = \frac{y+2}{-3} = \frac{z+1}{-2}$

24. $\frac{x+2}{-2} = \frac{y}{-4} = \frac{z-1}{-2}$ and $\frac{x+2}{-2} = \frac{y-2}{-4} = \frac{z-3}{-4}$

25. $\frac{x+3}{2} = \frac{y+1}{-3} = \frac{z-1}{3}$ and $\frac{x+3}{4} = \frac{y-1}{-2} = \frac{z-3}{-3}$

26. $\frac{x+1}{4} = \frac{y+2}{-2} = \frac{z+1}{-3}$ and $\frac{x+1}{2} = \frac{y-3}{0} = \frac{z-1}{2}$

27. $\frac{x-3}{4} = \frac{y-2}{2} = \frac{z+3}{0}$ and $\frac{x+2}{0} = \frac{y+3}{-1} = \frac{z-3}{-3}$

28. $\frac{x+3}{-3} = \frac{y}{0} = \frac{z}{-2}$ and $\frac{x-1}{-1} = \frac{y+3}{0} = \frac{z+3}{-3}$

29. $\frac{x+1}{4} = \frac{y-3}{2} = \frac{z+1}{-2}$ and $\frac{x+3}{3} = \frac{y+3}{-1} = \frac{z-2}{3}$

30. $\frac{x}{3} = \frac{y+2}{-3} = \frac{z-2}{-3}$ and $\frac{x+3}{0} = \frac{y-2}{3} = \frac{z-1}{-4}$

3. Further problems (10 problems).

1. **(Point–line distance.)** Find the distance from the point $M_0(-4; 1; 1)$ to the line $\frac{x+2}{-2} = \frac{y+2}{0} = \frac{z+1}{0}$.

2. **(Point–line distance.)** Find the distance from the point $M_0(-2; -2; -2)$ to the line $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z+1}{-3}$.

3. **(Point–line distance.)** Find the distance from the point $M_0(-3; 3; 2)$ to the line $\frac{x-2}{3} = \frac{y}{1} = \frac{z+3}{0}$.

4. **(Point–line distance.)** Find the distance from the point $M_0(-4; 2; 2)$ to the line $\frac{x-1}{-2} = \frac{y+2}{1} = \frac{z-3}{0}$.

5. **(Point–line distance.)** Find the distance from the point $M_0(-1; 2; -2)$ to the line $\frac{x+1}{-1} = \frac{y}{-3} = \frac{z}{1}$.

6. **(Point–line distance.)** Find the distance from the point $M_0(3; 3; -3)$ to the line $\frac{x-3}{1} = \frac{y-1}{-3} = \frac{z-3}{3}$.

7. **(Point–line distance.)** Find the distance from the point $M_0(-2; 4; -4)$ to the line $\frac{x-1}{1} = \frac{y-3}{2} = \frac{z-3}{0}$.

8. **(Parametric equations.)** Write the parametric equations of the line $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z}{4}$ and find the point corresponding to $t = 2$.

9. **(Intersection.)** Do the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{1}$ and $\frac{x-2}{1} = \frac{y-4}{1} = \frac{z-4}{2}$ intersect? If they do, find the point of intersection.

10. **(Skew lines.)** Find the distance between the lines $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$ (the Ox axis) and $\frac{x}{0} = \frac{y-1}{1} = \frac{z-2}{0}$.

3.2 Planes in space. The general equation of a plane. Relative position of lines and planes in space

3.2.1 Equations of a plane

Definition 3.2.1. The **general equation** of a plane in space is

$$Ax + By + Cz + D = 0 \qquad (A^2 + B^2 + C^2 \neq 0),$$

where $\vec{n} = (A; B; C)$ is a **normal vector**, that is, a vector perpendicular to the plane.

Several other forms are often more convenient. The plane through the point $M_0(x_0; y_0; z_0)$ with normal vector $\vec{n}$ is $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$. The plane through three points not on one line is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0,$$

and a plane cutting off segments a, b, c on the coordinate axes has the **intercept form** $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Example 3.2.1. Write an equation of the plane through the points $M_1(1;0;2)$, $M_2(2;1;0)$ and $M_3(0;1;1)$.

Solution. We have $\overrightarrow{M_1M_2} = (1;1;-2)$ and $\overrightarrow{M_1M_3} = (-1;1;-1)$, so a normal vector is

$$\vec{n} = \overrightarrow{M_1M_2} \times \overrightarrow{M_1M_3} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -2 \\ -1 & 1 & -1 \end{vmatrix} = (1;3;2).$$

The plane is $1(x-1)+3(y-0)+2(z-2)=0$, that is, $x+3y+2z-5=0$. As a check, substituting M_2 gives $2+3+0-5=0$ and substituting M_3 gives $0+3+2-5=0$.

3.2.2 The distance from a point to a plane. The angle between planes

Theorem 3.2.1. *The distance from the point $M_0(x_0;y_0;z_0)$ to the plane $Ax+By+Cz+D=0$ is*

$$d = \frac{|Ax_0+By_0+Cz_0+D|}{\sqrt{A^2+B^2+C^2}}.$$

The angle between two planes is found from $\cos\varphi = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}$; the planes are parallel when $\vec{n}_1 \parallel \vec{n}_2$ and perpendicular when $\vec{n}_1 \cdot \vec{n}_2 = 0$.

Example 3.2.2. The distance from $P(2;-1;3)$ to the plane $2x-y+2z-1=0$ is $d = \dfrac{|4+1+6-1|}{\sqrt{4+1+4}} = \dfrac{10}{3}$.

Example 3.2.3. For the planes $x+y-z=0$ and $2x-y+z-3=0$ we find $\vec{n}_1 \cdot \vec{n}_2 = 2-1-1=0$, so the planes are perpendicular.

3.2.3 Relative position of a line and a plane

Let a line $\frac{x-x_0}{m} = \frac{y-y_0}{n} = \frac{z-z_0}{p}$ and a plane $Ax + By + Cz + D = 0$ be given. The angle ψ between them is determined by

$$\sin\psi = \frac{|Am + Bn + Cp|}{\sqrt{A^2 + B^2 + C^2}\sqrt{m^2 + n^2 + p^2}}.$$

The line is parallel to the plane (or lies in it) if and only if $Am + Bn + Cp = 0$, and it is perpendicular to the plane if and only if $\frac{A}{m} = \frac{B}{n} = \frac{C}{p}$. To find the point where the line meets the plane, we substitute the parametric equations of the line into the equation of the plane and solve for t.

Example 3.2.4. Find the point of intersection of the line $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z}{-1}$ and the plane $x + 2y + z - 6 = 0$.

Solution. Substituting $x = 1 + 2t$, $y = -1 + t$, $z = -t$ into the equation of the plane gives $(1 + 2t) + 2(-1 + t) - t - 6 = 0$, i.e. $3t = 7$ and $t = \frac{7}{3}$. The point of intersection is $\left(\frac{17}{3}; \frac{4}{3}; -\frac{7}{3}\right)$.

3.2.4 Planes and information security

Blakley's secret sharing scheme. The secret number s is taken as the first coordinate of a point $P(s; y_0; z_0)$ in space. Each participant receives the equation of a randomly chosen plane through P. Three such planes meet (in general) in exactly one point, which is P, whereas two planes determine only a line. This is a threshold scheme in which all three shares are needed to recover the secret.

Example 3.2.5. Suppose the participants hold the planes $x + y + z = 6$, $x - y + 2z = 5$ and $2x + y - z = 1$. Their common point is the solution $(1; 2; 3)$ of the corresponding system, so the secret is $s = 1$.

Separating hyperplanes. When the feature space is three-dimensional, a linear classifier is given by a plane $Ax + By + Cz + D = 0$, and the formula

for the distance from a point to a plane is used to measure the margin of the decision boundary.

3.2.5 Computing with Python, Maple and Mathcad

The following code finds the plane through $M_1(1;0;2)$, $M_2(2;1;0)$, $M_3(0;1;1)$ and recovers the secret point of the Blakley example by solving the system of three plane equations.

Python

```
from sympy import Matrix, symbols, expand

x, y, z = symbols('x y z')
M1 = Matrix([1, 0, 2])
M2 = Matrix([2, 1, 0])
M3 = Matrix([0, 1, 1])
n = (M2 - M1).cross(M3 - M1)
print("n =", n.T)
print(expand(n.dot(Matrix([x, y, z]) - M1)), "= 0")

# Blakley: intersection of three planes
A = Matrix([[1, 1, 1], [1, -1, 2], [2, 1, -1]])
b = Matrix([6, 5, 1])
print("P =", A.solve(b).T)
```

Output

```
n = Matrix([[1, 3, 2]])
x + 3*y + 2*z - 5 = 0
P = Matrix([[1, 2, 3]])
```

In Maple we use `CrossProduct`, `DotProduct` (with `conjugate=false`, since the second vector is symbolic) and `LinearSolve`:

Maple

```
with(LinearAlgebra):
M1 := <1, 0, 2>: M2 := <2, 1, 0>: M3 := <0, 1, 1>:
n := CrossProduct(M2 - M1, M3 - M1);      # <1, 3, 2>
expand(DotProduct(n, <x, y, z> - M1, conjugate=false));
                                          # x+3*y+2*z-5
A := Matrix([[1, 1, 1], [1, -1, 2], [2, 1, -1]]):
LinearSolve(A, <6, 5, 1>);                # <1, 2, 3>
```

In Mathcad Prime the normal vector is obtained with the vector-product operator and the system is solved with `lsolve`:

Mathcad

```
                                    | 1 |
n := (M2 - M1) x (M3 - M1)      n = | 3 |
```

```
                                          | 2 |
     | 1   1   1 |        | 6 |                  | 1 |
A := | 1  -1   2 |   b := | 5 |    lsolve(A, b) = | 2 |
     | 2   1  -1 |        | 1 |                  | 3 |
```

3.2.6 Exercises on planes in space

1. Write an equation of the plane through the three given points.

1. $M_1(-3; 4; -1)$, $M_2(0; 0; 0)$, $M_3(-3; 3; 0)$
2. $M_1(3; 2; 2)$, $M_2(-3; 0; -1)$, $M_3(1; 1; 0)$
3. $M_1(1; 4; -2)$, $M_2(-2; 4; 0)$, $M_3(-2; -4; -3)$
4. $M_1(-3; 1; -4)$, $M_2(-3; 0; -1)$, $M_3(2; 2; 3)$
5. $M_1(-3; -3; 1)$, $M_2(-2; -3; 3)$, $M_3(4; -1; 0)$
6. $M_1(3; -1; 3)$, $M_2(0; 4; 0)$, $M_3(-3; -3; -3)$
7. $M_1(0; 0; -3)$, $M_2(-4; -2; 2)$, $M_3(-3; 4; -3)$
8. $M_1(2; 3; -2)$, $M_2(4; 2; 3)$, $M_3(0; 3; 3)$
9. $M_1(2; 2; -3)$, $M_2(0; 3; 2)$, $M_3(-1; 3; 3)$
10. $M_1(-3; -2; 3)$, $M_2(0; 3; -4)$, $M_3(2; 2; -4)$
11. $M_1(1; 1; 3)$, $M_2(0; 1; 0)$, $M_3(4; -1; -4)$
12. $M_1(4; 3; 0)$, $M_2(-3; 2; -3)$, $M_3(-4; -1; 0)$
13. $M_1(-4; -4; 4)$, $M_2(0; 4; 1)$, $M_3(0; -3; -4)$
14. $M_1(-2; -1; -4)$, $M_2(-1; 0; 4)$, $M_3(0; -2; -1)$
15. $M_1(-1; 1; 0)$, $M_2(1; -4; 4)$, $M_3(-2; 4; -1)$
16. $M_1(1; 3; 2)$, $M_2(-1; 2; 4)$, $M_3(4; 1; 3)$
17. $M_1(-4; -1; 2)$, $M_2(3; 1; -3)$, $M_3(0; -3; -2)$
18. $M_1(-1; -4; -2)$, $M_2(-1; 0; 3)$, $M_3(3; -2; 4)$

19. $M_1(2; -4; -3)$, $M_2(0; 1; 4)$, $M_3(2; 2; -2)$

20. $M_1(3; -1; -3)$, $M_2(-2; -2; -2)$, $M_3(-4; -1; 1)$

21. $M_1(4; -4; 2)$, $M_2(3; 0; 3)$, $M_3(1; -3; 1)$

22. $M_1(-4; 4; -3)$, $M_2(-3; 2; -3)$, $M_3(-4; -2; -3)$

23. $M_1(-4; 0; -4)$, $M_2(-2; 2; -2)$, $M_3(-4; 3; 3)$

24. $M_1(3; 4; -4)$, $M_2(0; -3; 0)$, $M_3(-1; 0; 4)$

25. $M_1(-4; 4; -1)$, $M_2(-4; 1; -3)$, $M_3(-3; 2; 2)$

26. $M_1(3; 0; 1)$, $M_2(1; -4; 0)$, $M_3(-2; 0; 0)$

27. $M_1(-1; 4; 3)$, $M_2(-4; -3; 3)$, $M_3(1; 2; 2)$

28. $M_1(0; 2; -1)$, $M_2(-4; 2; 3)$, $M_3(-1; 2; 4)$

29. $M_1(-3; -1; -1)$, $M_2(-1; 0; -3)$, $M_3(-1; -3; 3)$

30. $M_1(4; -4; -4)$, $M_2(-1; 0; 2)$, $M_3(3; -1; -3)$

2. Find the distance from the point M to the plane.

1. $M(0; 2; -2)$, $4x - 5y + 5z - 4 = 0$
2. $M(2; -2; 5)$, $-5y + 3z + 2 = 0$
3. $M(-1; -5; -3)$, $-x - y + z + 8 = 0$
4. $M(-1; 2; -1)$, $3x + 4y + z - 5 = 0$
5. $M(3; -5; 5)$, $2x - 6y - 4z - 8 = 0$
6. $M(0; 5; -3)$, $-2x + y - z = 0$
7. $M(3; -4; -3)$, $-4x + 6y - 6z - 3 = 0$
8. $M(4; 2; 4)$, $3x - 3y - 2z + 3 = 0$
9. $M(-2; 3; 5)$, $6x - y - 4z + 2 = 0$
10. $M(-4; -5; -5)$, $-x - y + 2z + 4 = 0$
11. $M(-2; -2; -4)$, $-2x - 2y - 4z = 0$
12. $M(-1; 4; 2)$, $2x + 5y - 3z + 3 = 0$
13. $M(2; 4; -4)$, $-5x + 6z - 6 = 0$
14. $M(4; 3; -2)$, $5x + 2y - 3z - 8 = 0$
15. $M(1; -2; 3)$, $2x + 2y + 6z + 9 = 0$
16. $M(-4; -4; -4)$, $-5x - 4y - z + 6 = 0$

17. $M(-4;\,-3;\,2)$, $2x+3y-3z-2=0$
18. $M(3;\,-4;\,5)$, $2x+y-z+3=0$
19. $M(0;\,-1;\,-3)$, $4x-y-5z+7=0$
20. $M(5;\,0;\,-5)$, $x+y+5z-6=0$
21. $M(-5;\,3;\,1)$, $2x+y-z-9=0$
22. $M(0;\,3;\,-3)$, $-3x-3y+2z-6=0$
23. $M(-4;\,1;\,5)$, $-2x-6z-4=0$
24. $M(1;\,-3;\,3)$, $-x+y-3z+8=0$
25. $M(-1;\,-2;\,-1)$, $2x+6y+5z+4=0$
26. $M(-1;\,-2;\,3)$, $-3x-4y+5z-6=0$
27. $M(-2;\,1;\,-1)$, $4x-6y-3z-5=0$
28. $M(-2;\,5;\,5)$, $-5x-3y-3z-9=0$
29. $M(3;\,-2;\,-2)$, $6x-2y-6z-1=0$
30. $M(-1;\,-3;\,-5)$, $-x-4y-3z-8=0$

3. Further problems (10 problems).

1. **(Two planes.)** Find the angle between the planes $x+3y+3z-1=0$ and $2y+z+2=0$.
2. **(Two planes.)** Find the angle between the planes $3y-2z-3=0$ and $2x-3y+2=0$.
3. **(Two planes.)** Find the angle between the planes $-2x+2y+2z+3=0$ and $2x-y-2z+4=0$.
4. **(Two planes.)** Find the angle between the planes $2x-y-3z-5=0$ and $2y+2=0$.
5. **(Two planes.)** Find the angle between the planes $x-y=0$ and $-x-y+z+3=0$.
6. **(Line and plane.)** Find the point of intersection of the line $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z}{-1}$ and the plane $x+2y+z-6=0$.
7. **(Given a normal.)** Write an equation of the plane through $M(2;-1;3)$ with normal vector $\vec{n}=(1;-2;4)$.
8. **(Parallel plane.)** Write an equation of the plane through $M(1;2;-1)$ parallel to the plane $3x-y+2z-5=0$.

9. **(Intercept form.)** Write the equation of the plane $2x - 3y + 6z - 12 = 0$ in intercept form.

10. **(Distance between parallel planes.)** Find the distance between the planes $x - 2y + 2z - 3 = 0$ and $x - 2y + 2z + 9 = 0$.

3.3 Expansion of a vector of a linear space in a basis. Subspaces of a linear space

Definition 3.3.1. A set V on whose elements addition and multiplication by real numbers are defined is called a **linear (vector) space** if these operations satisfy the following eight axioms ($x, y, z \in V$, $\alpha, \beta \in \mathbb{R}$):

1) $x + y = y + x$; 2) $(x + y) + z = x + (y + z)$;

3) $\exists\, \theta \in V$: $x + \theta = x$; 4) $\forall x\ \exists (-x)$: $x + (-x) = \theta$;

5) $1 \cdot x = x$; 6) $\alpha(\beta x) = (\alpha\beta)x$;

7) $(\alpha + \beta)x = \alpha x + \beta x$; 8) $\alpha(x + y) = \alpha x + \alpha y$.

Familiar examples are the n-dimensional arithmetic space $\mathbb{R}^n$, the space P_n of polynomials of degree at most n, the space $M_{m\times n}$ of $m \times n$ matrices and the space $C[a; b]$ of functions continuous on $[a; b]$.

3.3.1 Linear dependence. Basis and dimension

Definition 3.3.2. Elements $x_1, \ldots, x_k$ are called **linearly dependent** if $\alpha_1 x_1 + \cdots + \alpha_k x_k = \theta$ for some numbers $\alpha_1, \ldots, \alpha_k$ not all equal to zero. Otherwise, that is, if this equality holds only for $\alpha_1 = \cdots = \alpha_k = 0$, they are called **linearly independent**.

Definition 3.3.3. A system $e_1, \ldots, e_n$ of elements of V is called a **basis** if it is linearly independent and every element of V is a linear combination of

$e_1, \ldots, e_n$; the number n is the **dimension** of the space ($\dim V = n$). The numbers $\xi_1, \ldots, \xi_n$ in the expansion $x = \xi_1 e_1 + \cdots + \xi_n e_n$ are called the **coordinates** of x in this basis; they are uniquely determined.

Theorem 3.3.1. *Vectors $\vec{a}_1, \ldots, \vec{a}_n$ of $\mathbb{R}^n$ form a basis if and only if the determinant made up of their coordinates is nonzero. More generally, the rank of a system of k vectors equals the dimension of their linear span.*

Example 3.3.1. Show that the vectors $\vec{e}_1 = (1;1;0)$, $\vec{e}_2 = (0;1;1)$, $\vec{e}_3 = (1;0;1)$ form a basis of $\mathbb{R}^3$ and expand $\vec{x} = (2;3;1)$ in this basis.

Solution. Since $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 1(1-0) - 0 + 1(1-0) = 2 \neq 0$, the vectors form a basis. Writing $\vec{x} = \xi_1\vec{e}_1 + \xi_2\vec{e}_2 + \xi_3\vec{e}_3$ coordinatewise, we obtain

$$\begin{cases} \xi_1 + \xi_3 = 2, \\ \xi_1 + \xi_2 = 3, \\ \xi_2 + \xi_3 = 1. \end{cases}$$

Adding the three equations gives $\xi_1 + \xi_2 + \xi_3 = 3$, from which $\xi_3 = 0$, $\xi_1 = 2$ and $\xi_2 = 1$. Thus $\vec{x} = 2\vec{e}_1 + \vec{e}_2$.

Example 3.3.2. Examine the system $\vec{a} = (1;2;3)$, $\vec{b} = (2;1;0)$, $\vec{c} = (3;3;3)$.

Solution. Here $\vec{c} = \vec{a} + \vec{b}$, i.e. $\vec{a} + \vec{b} - \vec{c} = \vec{0}$, so the system is linearly dependent. Its rank is 2, and the linear span is two-dimensional (a plane through the origin).

3.3.2 Subspaces

Definition 3.3.4. A nonempty subset L of a space V is called a **subspace** if $x + y \in L$ and $\alpha x \in L$ for all $x, y \in L$ and $\alpha \in \mathbb{R}$. The set of all linear combinations of elements $x_1, \ldots, x_k$ is called their **linear span** and is denoted by $L(x_1, \ldots, x_k)$.

Example 3.3.3. The set $L = \{(x_1; x_2; x_3) : x_1 + x_2 + x_3 = 0\}$ is a subspace of $\mathbb{R}^3$: sums and scalar multiples of solutions of a homogeneous equation are again solutions. Since $x_1 = -x_2 - x_3$, we can write $(x_1; x_2; x_3) = x_2(-1; 1; 0) + x_3(-1; 0; 1)$; hence the vectors $(-1; 1; 0)$, $(-1; 0; 1)$ form a basis of L and $\dim L = 2$.

On the other hand, $M = \{(x_1; x_2) : x_1 + x_2 = 1\}$ is not a subspace, because $(0; 0) \notin M$.

Theorem 3.3.2. *The solutions of a homogeneous system $Ax = 0$ form a subspace of $\mathbb{R}^n$ of dimension $n - \operatorname{rank} A$; any basis of this subspace is called a* ***fundamental system of solutions****.*

3.3.3 Linear spaces and information security

Linear codes. A binary (n, k) linear code is a k-dimensional subspace of $\mathbb{F}_2^n$. The rows of the generator matrix G form a basis of this subspace, and every codeword is a linear combination (mod 2) of the basis vectors: $c = mG$. The code therefore contains 2^k codewords.

Example 3.3.4. For $G = \left(\begin{smallmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{smallmatrix}\right)$ the messages are encoded as $00 \to 0000$, $10 \to 1011$, $01 \to 0110$, $11 \to 1101$. These four words form a two-dimensional subspace of $\mathbb{F}_2^4$.

Parity-check matrix. A code can equally well be described as the solution space of the homogeneous system $Hc^T = 0$; then $\dim C = n - \operatorname{rank} H$.

Linear cryptanalysis. The linear part of a cipher is treated as a linear map of a vector space over $\mathbb{F}_2$. If the analysis yields r independent linear relations among the n key bits, the admissible keys form a set of dimension $n - r$ (the solutions of a linear system), and an exhaustive search shrinks from 2^n to 2^{n-r} keys.

3.3.4 Computing with Python, Maple and Mathcad

We check the first two examples: the coordinates of $\vec{x} = (2; 3; 1)$ in the basis $\vec{e}_1, \vec{e}_2, \vec{e}_3$, and the rank of the system $\vec{a}, \vec{b}, \vec{c}$ together with the linear relation between its vectors (a null vector of the matrix whose columns are $\vec{a}, \vec{b}, \vec{c}$).

Python

```
from sympy import Matrix

e1 = Matrix([1, 1, 0])
e2 = Matrix([0, 1, 1])
e3 = Matrix([1, 0, 1])
A = Matrix.hstack(e1, e2, e3)   # basis vectors as columns
print("det =", A.det())
print("coordinates:", A.solve(Matrix([2, 3, 1])).T)

B = Matrix([[1, 2, 3], [2, 1, 0], [3, 3, 3]])  # a, b, c
print("rank =", B.rank())
print("null space of B^T:", B.T.nullspace()[0].T)
```

Output

```
det = 2
coordinates: Matrix([[2, 1, 0]])
rank = 2
null space of B^T: Matrix([[-1, -1, 1]])
```

The last line says that $-\vec{a} - \vec{b} + \vec{c} = \vec{0}$. The same results in Maple:

Maple

```
with(LinearAlgebra):
A := Matrix([[1, 0, 1], [1, 1, 0], [0, 1, 1]]):
Determinant(A);                  # 2
LinearSolve(A, <2, 3, 1>);       # <2, 1, 0>
B := Matrix([[1, 2, 3], [2, 1, 0], [3, 3, 3]]):
Rank(B);                         # 2
NullSpace(Transpose(B));         # {<-1, -1, 1>}
```

In Mathcad Prime the coordinates are given by `lsolve` and the rank by `rank`:

Mathcad

```
     | 1  0  1 |        | 2 |                  | 2 |
A := | 1  1  0 |   x := | 3 |    lsolve(A, x) = | 1 |
```

```
     | 0  1  1 |        | 1 |                  | 0 |
|A| = 2
     | 1  2  3 |
B := | 2  1  0 |        rank(B) = 2
     | 3  3  3 |
```

3.3.5 Exercises on linear spaces and bases

1. Expand the vector $\vec{x}$ in the basis $\vec{e}_1, \vec{e}_2, \vec{e}_3$.

1. $\vec{e}_1 = (2; 2; -2)$, $\vec{e}_2 = (-1; 2; -1)$, $\vec{e}_3 = (-1; 0; 0)$; $\vec{x} = (1; -4; 2)$
2. $\vec{e}_1 = (-1; 0; 2)$, $\vec{e}_2 = (1; 0; 0)$, $\vec{e}_3 = (2; -1; 1)$; $\vec{x} = (-3; -1; 3)$
3. $\vec{e}_1 = (-1; 0; 2)$, $\vec{e}_2 = (0; 0; 2)$, $\vec{e}_3 = (-2; 1; 2)$; $\vec{x} = (-7; 2; 4)$
4. $\vec{e}_1 = (-1; -1; 0)$, $\vec{e}_2 = (-1; 2; 1)$, $\vec{e}_3 = (0; -2; -1)$; $\vec{x} = (0; -8; -3)$
5. $\vec{e}_1 = (1; -1; 2)$, $\vec{e}_2 = (2; -1; 2)$, $\vec{e}_3 = (-2; -1; 1)$; $\vec{x} = (-1; 0; -1)$
6. $\vec{e}_1 = (0; 0; 1)$, $\vec{e}_2 = (-1; 0; 2)$, $\vec{e}_3 = (2; -1; -1)$; $\vec{x} = (5; -4; 0)$
7. $\vec{e}_1 = (-2; 1; 2)$, $\vec{e}_2 = (-1; 1; 0)$, $\vec{e}_3 = (-1; -1; 1)$; $\vec{x} = (-11; 2; 9)$
8. $\vec{e}_1 = (-1; -1; -1)$, $\vec{e}_2 = (1; 2; 2)$, $\vec{e}_3 = (-1; -2; 0)$; $\vec{x} = (0; 0; -2)$
9. $\vec{e}_1 = (2; 0; 2)$, $\vec{e}_2 = (2; 0; 1)$, $\vec{e}_3 = (0; 1; -2)$; $\vec{x} = (4; 2; 1)$
10. $\vec{e}_1 = (2; -1; -2)$, $\vec{e}_2 = (0; -1; 0)$, $\vec{e}_3 = (-1; -2; 0)$; $\vec{x} = (-2; 7; 4)$
11. $\vec{e}_1 = (2; 1; 0)$, $\vec{e}_2 = (-2; -2; 2)$, $\vec{e}_3 = (0; 1; -1)$; $\vec{x} = (-6; -11; 12)$
12. $\vec{e}_1 = (0; 2; 0)$, $\vec{e}_2 = (1; 1; 2)$, $\vec{e}_3 = (0; 1; -1)$; $\vec{x} = (3; 3; 6)$
13. $\vec{e}_1 = (2; 2; -1)$, $\vec{e}_2 = (0; -1; 0)$, $\vec{e}_3 = (2; 2; -2)$; $\vec{x} = (-2; 0; 4)$
14. $\vec{e}_1 = (-1; -1; -1)$, $\vec{e}_2 = (1; -1; 0)$, $\vec{e}_3 = (0; 1; 0)$; $\vec{x} = (3; 3; 3)$
15. $\vec{e}_1 = (1; 0; 2)$, $\vec{e}_2 = (1; 2; 1)$, $\vec{e}_3 = (1; -1; 2)$; $\vec{x} = (-9; 0; -16)$
16. $\vec{e}_1 = (1; -1; -2)$, $\vec{e}_2 = (2; 1; -1)$, $\vec{e}_3 = (0; -2; -1)$; $\vec{x} = (4; 4; -1)$
17. $\vec{e}_1 = (2; 0; -1)$, $\vec{e}_2 = (-1; 2; -1)$, $\vec{e}_3 = (2; -2; 1)$; $\vec{x} = (7; -14; 9)$
18. $\vec{e}_1 = (1; 1; 0)$, $\vec{e}_2 = (-1; 0; 2)$, $\vec{e}_3 = (-1; -1; -2)$; $\vec{x} = (0; 4; 12)$

19. $\vec{e}_1 = (-1; 0; 0), \vec{e}_2 = (-1; 0; 2), \vec{e}_3 = (-1; 1; 2);\ \vec{x} = (-2; 3; 10)$
20. $\vec{e}_1 = (-1; 2; -1), \vec{e}_2 = (-1; -1; 0), \vec{e}_3 = (1; -1; 0);\ \vec{x} = (2; 0; -2)$
21. $\vec{e}_1 = (-1; 2; 0), \vec{e}_2 = (-1; -1; -1), \vec{e}_3 = (1; 2; 2);\ \vec{x} = (-9; -2; -8)$
22. $\vec{e}_1 = (-1; 2; -2), \vec{e}_2 = (1; 2; -1), \vec{e}_3 = (1; 1; -1);\ \vec{x} = (4; 16; -13)$
23. $\vec{e}_1 = (-2; 2; -1), \vec{e}_2 = (1; 1; 1), \vec{e}_3 = (1; 2; 2);\ \vec{x} = (-4; 1; -5)$
24. $\vec{e}_1 = (1; 2; 0), \vec{e}_2 = (0; 1; 0), \vec{e}_3 = (-2; -2; 1);\ \vec{x} = (-6; -7; 4)$
25. $\vec{e}_1 = (-2; -2; 0), \vec{e}_2 = (1; -1; -1), \vec{e}_3 = (0; -1; 0);\ \vec{x} = (7; 5; -1)$
26. $\vec{e}_1 = (-1; -1; 2), \vec{e}_2 = (-2; 2; 2), \vec{e}_3 = (0; 1; 0);\ \vec{x} = (-3; 13; 0)$
27. $\vec{e}_1 = (1; 2; -1), \vec{e}_2 = (1; -2; 0), \vec{e}_3 = (2; -2; -1);\ \vec{x} = (6; -4; -4)$
28. $\vec{e}_1 = (-1; -2; -1), \vec{e}_2 = (0; -1; -1), \vec{e}_3 = (-1; 0; -1);\ \vec{x} = (4; 3; 5)$
29. $\vec{e}_1 = (1; -2; -2), \vec{e}_2 = (1; 0; 0), \vec{e}_3 = (1; -1; 0);\ \vec{x} = (4; 1; 2)$
30. $\vec{e}_1 = (-2; 1; 1), \vec{e}_2 = (-1; 1; 0), \vec{e}_3 = (0; -1; 2);\ \vec{x} = (4; -5; 2)$

2. Determine whether the system of vectors is linearly independent and find the dimension of its linear span.

1. $\vec{a} = (-3; -2; -2), \vec{b} = (1; 1; -1), \vec{c} = (7; 5; 3)$
2. $\vec{a} = (0; 3; -3), \vec{b} = (-3; 0; -3), \vec{c} = (-3; 3; -3)$
3. $\vec{a} = (-1; -2; -3), \vec{b} = (-1; -3; -1), \vec{c} = (0; 2; -4)$
4. $\vec{a} = (1; 0; 2), \vec{b} = (1; 0; 3), \vec{c} = (1; 2; -3)$
5. $\vec{a} = (-2; -1; -3), \vec{b} = (2; 2; 2), \vec{c} = (-4; -3; -5)$
6. $\vec{a} = (2; -1; -1), \vec{b} = (0; -1; 3), \vec{c} = (2; 2; 0)$
7. $\vec{a} = (1; 0; 3), \vec{b} = (2; 0; 2), \vec{c} = (0; 0; -4)$
8. $\vec{a} = (1; -2; 3), \vec{b} = (1; -3; -3), \vec{c} = (-2; 0; 3)$
9. $\vec{a} = (1; 2; 3), \vec{b} = (-3; 3; -1), \vec{c} = (-4; 10; 4)$
10. $\vec{a} = (-2; -1; 0), \vec{b} = (1; 0; 1), \vec{c} = (2; 3; -1)$

11. $\vec{a} = (1; -2; 0)$, $\vec{b} = (-1; 2; 0)$, $\vec{c} = (0; 0; 0)$
12. $\vec{a} = (1; 3; 2)$, $\vec{b} = (1; 1; 3)$, $\vec{c} = (1; 1; -2)$
13. $\vec{a} = (0; -1; -3)$, $\vec{b} = (2; 2; -2)$, $\vec{c} = (4; 5; -1)$
14. $\vec{a} = (2; -2; 2)$, $\vec{b} = (2; -1; -3)$, $\vec{c} = (0; 2; 3)$
15. $\vec{a} = (1; -2; 2)$, $\vec{b} = (-1; -1; 2)$, $\vec{c} = (2; -4; 4)$
16. $\vec{a} = (3; 1; -2)$, $\vec{b} = (3; -1; 1)$, $\vec{c} = (-3; 1; -2)$
17. $\vec{a} = (1; -1; 2)$, $\vec{b} = (-1; 2; 2)$, $\vec{c} = (-1; 1; -2)$
18. $\vec{a} = (1; -3; 1)$, $\vec{b} = (0; 1; 1)$, $\vec{c} = (-2; 3; 1)$
19. $\vec{a} = (1; 2; -1)$, $\vec{b} = (-3; 0; 1)$, $\vec{c} = (-1; -2; 1)$
20. $\vec{a} = (-2; 0; -2)$, $\vec{b} = (-1; 0; -2)$, $\vec{c} = (-1; -2; 2)$
21. $\vec{a} = (-3; 3; -3)$, $\vec{b} = (-3; 2; 1)$, $\vec{c} = (-6; 6; -6)$
22. $\vec{a} = (-2; 0; 1)$, $\vec{b} = (-1; -2; -1)$, $\vec{c} = (-1; 2; 1)$
23. $\vec{a} = (-3; 3; -3)$, $\vec{b} = (0; -3; 3)$, $\vec{c} = (-6; 9; -9)$
24. $\vec{a} = (2; -1; -2)$, $\vec{b} = (2; -2; -1)$, $\vec{c} = (3; -3; 0)$
25. $\vec{a} = (-1; 0; -2)$, $\vec{b} = (3; 0; -3)$, $\vec{c} = (-1; 0; 7)$
26. $\vec{a} = (-3; -2; -3)$, $\vec{b} = (-3; 0; 2)$, $\vec{c} = (0; 3; 2)$
27. $\vec{a} = (1; 0; 3)$, $\vec{b} = (-1; -3; 1)$, $\vec{c} = (1; 0; 3)$
28. $\vec{a} = (0; -3; 3)$, $\vec{b} = (0; -3; 3)$, $\vec{c} = (0; -1; 1)$
29. $\vec{a} = (-1; -1; -2)$, $\vec{b} = (3; 1; -1)$, $\vec{c} = (-2; -2; -4)$
30. $\vec{a} = (-3; -2; -3)$, $\vec{b} = (-2; -1; 3)$, $\vec{c} = (3; -1; 1)$

3. Further problems (10 problems).

1. **(A subspace.)** Show that the set $L = \{(x_1; x_2; x_3) : x_1 + x_2 + x_3 = 0\}$ is a subspace of $\mathbb{R}^3$ and find a basis of L and its dimension.
2. **(Not a subspace.)** Is the set $M = \{(x_1; x_2) : x_1 + x_2 = 1\}$ a subspace of $\mathbb{R}^2$?

3. **(Membership in a span.)** Does the vector $\vec{x} = (4; 1; 5)$ belong to the linear span of $\vec{a} = (1; -1; 2)$ and $\vec{b} = (2; 3; 1)$?

4. **(Membership in a span.)** Does the vector $\vec{x} = (1; 2; 4)$ belong to the linear span of $\vec{a} = (1; 0; 1)$ and $\vec{b} = (0; 1; 1)$?

5. **(A space of polynomials.)** In P_2, expand $p(x) = x^2 + 3x + 2$ in the basis $1,\ x - 1,\ (x - 1)^2$.

6. **(A space of matrices.)** What is the dimension of the space of 2×2 matrices? Write down its standard basis.

7. **(Change of basis.)** Find the coordinates of $\vec{x} = (5; 1)$ in the basis $\vec{e}_1 = (1; 1)$, $\vec{e}_2 = (1; -1)$.

8. **(A binary space.)** Are the vectors $\vec{a} = (1; 1; 0)$, $\vec{b} = (0; 1; 1)$, $\vec{c} = (1; 0; 1)$ of $\mathbb{F}_2^3$ linearly independent (over $\mathbb{F}_2$, i.e. mod 2)?

9. **(A homogeneous system.)** Find a basis and the dimension of the solution space of the system $\begin{cases} x_1 + 2x_2 - x_3 = 0 \\ 2x_1 + 4x_2 - 2x_3 = 0 \end{cases}$.

10. **(A code subspace.)** How many codewords does the binary code with generator matrix $G = \left(\begin{smallmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{smallmatrix}\right)$ have? List them.

3.4 Definition of a Euclidean space. The norm of an element of a Euclidean space

Definition 3.4.1. Let E be a real linear space, and suppose that to every pair of elements $x, y \in E$ a real number (x, y) is assigned so that the following axioms hold. Then (x, y) is called an **inner (scalar) product**, and E is called a **Euclidean space**:

1) $(x, y) = (y, x)$ (symmetry);

2) $(x + y, z) = (x, z) + (y, z)$; $\quad (\lambda x, y) = \lambda(x, y)$ (linearity);

3) $(x, x) \geq 0$, and $(x, x) = 0 \iff x = \theta$ (positivity).

Examples. 1) The standard inner product on $\mathbb{R}^n$ is $(x, y) = x_1y_1 + x_2y_2 + \cdots + x_ny_n$. 2) On $\mathbb{R}^2$ one can take the weighted product $(x, y) = 2x_1y_1 + 3x_2y_2$. 3) On $C[a; b]$ the formula $(f, g) = \int_a^b f(t)g(t)\,dt$ defines an inner product.

3.4.1 Norm, distance and angle

Definition 3.4.2. The **norm (length)** of an element x is $\|x\| = \sqrt{(x, x)}$; the **distance** between two elements is $\rho(x, y) = \|x - y\|$; the angle φ between them is defined by

$$\cos\varphi = \frac{(x, y)}{\|x\|\,\|y\|} \qquad (x, y \neq \theta).$$

If $(x, y) = 0$, the elements x and y are said to be **orthogonal**, and we write $x \perp y$.

Theorem 3.4.1. *In every Euclidean space:*

1) $|(x, y)| \leq \|x\|\,\|y\|$ *(the **Cauchy–Schwarz** or **Cauchy–Bunyakovsky–Schwarz inequality**);*

2) $\|x + y\| \leq \|x\| + \|y\|$ *(the **triangle inequality**);*

3) if $x \perp y$, *then* $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ *(the **Pythagorean theorem**).*

Example 3.4.1. In $\mathbb{R}^4$ let $x = (1; 2; -1; 3)$ and $y = (2; 0; 1; 1)$. Find (x, y), $\|x\|$, $\|y\|$, $\cos\varphi$ and $\rho(x, y)$.

Solution. We have $(x, y) = 2+0-1+3 = 4$, $\|x\| = \sqrt{1 + 4 + 1 + 9} = \sqrt{15}$ and $\|y\| = \sqrt{4 + 0 + 1 + 1} = \sqrt{6}$; hence

$$\cos\varphi = \frac{4}{\sqrt{15}\sqrt{6}} = \frac{4}{3\sqrt{10}}, \qquad \rho(x, y) = \|(-1; 2; -2; 2)\| = \sqrt{13}.$$

The Cauchy–Schwarz inequality is satisfied, as it must be: $4 \leq \sqrt{90} \approx 9.49$.

Example 3.4.2. Are the vectors $x = (1; 2)$ and $y = (3; -1)$ orthogonal with respect to the inner product $(x, y) = 2x_1y_1 + 3x_2y_2$ on $\mathbb{R}^2$?

Solution. Here $(x, y) = 2 \cdot 1 \cdot 3 + 3 \cdot 2 \cdot (-1) = 0$, so the vectors are orthogonal. With the standard inner product, however, $(x, y) = 3 - 2 = 1 \neq 0$. Thus orthogonality depends on the choice of inner product.

Example 3.4.3. In $C[0; 1]$ take $f(t) = t$ and $g(t) = 1$. Then $(f, g) = \int_0^1 t\,dt = \frac{1}{2}$, $\|f\| = \sqrt{\int_0^1 t^2 dt} = \frac{1}{\sqrt{3}}$, $\|g\| = 1$, and $\cos\varphi = \frac{1/2}{1/\sqrt{3}} = \frac{\sqrt{3}}{2}$, so that $\varphi = 30^\circ$.

Definition 3.4.3. To **normalize** an element $x \neq \theta$ means to pass to the unit element $x^0 = \dfrac{x}{\|x\|}$.

3.4.2 Applications of the norm in information security

Hamming distance. In the space of binary words, $d(u, v)$ is the number of positions in which u and v differ; it equals the number of ones in $u \oplus v$. If a code has minimum distance $d_{\min}$, it detects up to $d_{\min} - 1$ errors and corrects up to $\left\lfloor \frac{d_{\min}-1}{2} \right\rfloor$ errors.

Example 3.4.4. For $u = 1011010$ and $v = 1001110$ we get $u \oplus v = 0010100$, hence $d(u, v) = 2$.

Anomaly detection. A user's activity is described by a vector $\vec{x}$; if its Euclidean distance $\|\vec{x} - \vec{m}\|$ from the normal profile $\vec{m}$ exceeds a given threshold, the behaviour is flagged as suspicious. In biometric authentication (face, fingerprint) templates are also stored as vectors, and they are compared by means of a distance or the cosine similarity.

Lattice cryptography. The shortest vector problem (SVP) asks for a nonzero lattice vector of smallest norm. In high dimensions it is extremely

hard computationally, which is why it serves as one of the basic hard problems of post-quantum cryptography.

3.4.3 Computing with Python, Maple and Mathcad

The quantities of the first example above can be obtained in a few lines. In Python we use SymPy, which keeps the square roots exact:

Python

```
from sympy import Matrix, radsimp
x = Matrix([1, 2, -1, 3])
y = Matrix([2, 0, 1, 1])
print(x.dot(y), x.norm(), y.norm())
print(radsimp(x.dot(y)/(x.norm()*y.norm())))
print((x - y).norm())
```

Output

```
4 sqrt(15) sqrt(6)
2*sqrt(10)/15
sqrt(13)
```

Note that $\frac{2\sqrt{10}}{15} = \frac{4}{3\sqrt{10}}$, in agreement with the hand computation. In Maple the same results are given by the `LinearAlgebra` package (`VectorAngle` returns the angle itself):

Maple

```
with(LinearAlgebra):
x := <1, 2, -1, 3>:  y := <2, 0, 1, 1>:
DotProduct(x, y);              # 4
Norm(x, 2);  Norm(y, 2);       # sqrt(15), sqrt(6)
DotProduct(x, y)/(Norm(x, 2)*Norm(y, 2));
Norm(x - y, 2);                # sqrt(13)
VectorAngle(x, y);
```

In Mathcad the vectors are entered as columns; the dot product is the multiplication operator applied to two vectors, and $|x|$ is the length of a vector:

Mathcad

```
     | 1 |        | 2 |
x := | 2 |   y := | 0 |
     |-1 |        | 1 |
     | 3 |        | 1 |
x.y = 4     |x| = 3.873     |y| = 2.449
  x.y
-------- = 0.422     |x - y| = 3.606
|x|.|y|
```

3.4.4 Exercises on Euclidean spaces and norms

1. Find the inner product and the norms of the vectors.

1. $x = (0; 2; 2; -1)$, $y = (1; -2; 2; -5)$
2. $x = (1; 5; -1; -2)$, $y = (5; -2; -5; -1)$
3. $x = (-1; 0; 5; -3)$, $y = (4; -1; -5; -2)$
4. $x = (4; -1; -5; -3)$, $y = (4; 5; 5; -5)$
5. $x = (2; 2; 4; 5)$, $y = (-1; -2; -1; 0)$
6. $x = (-1; 1; -4; 0)$, $y = (2; 1; 3; 5)$
7. $x = (-3; 4; -1; 4)$, $y = (-5; -1; -4; -5)$
8. $x = (3; 0; -2; 2)$, $y = (-3; -1; -1; 0)$
9. $x = (2; 2; -4; -3)$, $y = (2; -5; 2; 2)$
10. $x = (-5; 2; -4; 2)$, $y = (4; -4; 2; 5)$
11. $x = (-5; -3; -2; 1)$, $y = (0; -5; 3; -5)$
12. $x = (5; 5; 1; 4)$, $y = (0; 2; 3; 5)$
13. $x = (5; 5; -4; -2)$, $y = (0; -4; -4; 3)$
14. $x = (-4; -2; -5; 1)$, $y = (5; -5; -4; 5)$
15. $x = (-5; 1; -3; 5)$, $y = (-1; -2; -3; 4)$
16. $x = (-5; -2; 3; -2)$, $y = (-4; -4; 5; -3)$
17. $x = (-1; 1; 1; -5)$, $y = (1; 2; 2; 2)$
18. $x = (-1; -5; -5; -5)$, $y = (-2; 1; 5; -1)$
19. $x = (3; 1; -4; -2)$, $y = (-4; 1; 5; -4)$
20. $x = (-1; -4; 2; 1)$, $y = (-3; 1; -2; -3)$
21. $x = (-3; 1; 1; 3)$, $y = (-2; 1; -3; 0)$

22. $x = (5; 2; -3; 1)$, $y = (-2; 5; 1; -5)$

23. $x = (2; -1; 4; -1)$, $y = (-4; -4; -1; 2)$

24. $x = (-2; 3; 5; 2)$, $y = (-2; 4; 2; 3)$

25. $x = (-3; 2; 3; -4)$, $y = (5; 5; -4; 5)$

26. $x = (0; -5; 2; -4)$, $y = (2; -5; 3; 0)$

27. $x = (3; -3; -4; 0)$, $y = (1; -1; 1; -4)$

28. $x = (-2; -2; 2; -2)$, $y = (-5; 5; -5; -2)$

29. $x = (0; 1; 4; 4)$, $y = (-2; -5; -1; 5)$

30. $x = (-5; 1; 5; -5)$, $y = (-3; 5; 3; -2)$

2. Find the cosine of the angle between the vectors and the distance between them.

1. $x = (-2; 3; 5; 3)$, $y = (5; 5; -1; 3)$

2. $x = (-5; 4; 4; 3)$, $y = (-2; 2; 4; 2)$

3. $x = (0; -3; 2; -3)$, $y = (-4; -2; -5; -4)$

4. $x = (0; -1; 5; 5)$, $y = (-4; -3; -5; -3)$

5. $x = (4; -4; -2; 0)$, $y = (-2; 4; 1; -4)$

6. $x = (-3; -4; 2; -4)$, $y = (-5; 4; -3; -3)$

7. $x = (5; -3; -2; 1)$, $y = (-5; 0; -4; 5)$

8. $x = (1; 1; -1; -4)$, $y = (4; 1; 4; -2)$

9. $x = (0; 2; 0; 1)$, $y = (-5; 0; -2; -4)$

10. $x = (-2; 1; 1; -5)$, $y = (4; -2; 1; 5)$

11. $x = (3; 0; -5; 3)$, $y = (1; 0; -5; 5)$

12. $x = (-3; 1; -2; 0)$, $y = (-3; 1; -5; 1)$

13. $x = (-5; 1; 5; 2)$, $y = (5; -5; -3; -5)$

14. $x = (-3; -2; 5; 4)$, $y = (-1; 4; 5; 4)$

15. $x = (1; -5; -5; -4)$, $y = (3; 3; -5; -3)$

16. $x = (0; 4; 4; -4)$, $y = (4; 3; -1; 2)$

17. $x = (-3; 4; 1; -1)$, $y = (4; 4; 1; 3)$

18. $x = (-4; -3; 4; -5)$, $y = (-1; -1; -1; 5)$

19. $x = (2; 0; -4; 5)$, $y = (-1; 1; -1; 5)$

20. $x = (4; 2; -5; -4)$, $y = (-4; 0; 1; -1)$

21. $x = (2; -1; -2; -1)$, $y = (-5; 2; -5; -2)$

22. $x = (-4; 1; -3; 4)$, $y = (-2; 0; 1; 4)$

23. $x = (-4; -1; -5; 0)$, $y = (-3; 0; 1; 4)$

24. $x = (-1; 0; -2; -3)$, $y = (5; -2; 3; -5)$

25. $x = (3; -4; -1; -5)$, $y = (-5; 3; -4; -1)$

26. $x = (-2; -4; -4; 2)$, $y = (2; 0; -5; -5)$

27. $x = (-4; -1; 5; 1)$, $y = (-4; 0; 1; 4)$

28. $x = (2; 1; -2; 4)$, $y = (-2; -1; -3; -2)$

29. $x = (5; -3; -3; 1)$, $y = (0; -4; -3; -5)$

30. $x = (5; -2; -3; 5)$, $y = (1; -4; -1; -1)$

3. Applied problems (10 problems).

1. **(Orthogonality.)** Are the vectors $x = (1; 2; -1; 3)$ and $y = (2; 1; 1; -1)$ of $\mathbb{R}^4$ orthogonal?

2. **(Normalization.)** Normalize the vector $x = (1; -2; 2; 4)$.

3. **(Cauchy–Schwarz.)** Verify the inequality $|(x, y)| \leq \|x\| \, \|y\|$ for $x = (1; 2; 3)$ and $y = (4; -1; 2)$.

4. **(Triangle inequality.)** Verify the inequality $\|x + y\| \leq \|x\| + \|y\|$ for $x = (3; 0; 4)$ and $y = (0; 5; 12)$.

5. **(A parameter.)** For which value of λ are the vectors $x = (1; \lambda; 2; -1)$ and $y = (3; 2; \lambda; 1)$ orthogonal?

6. **(Weighted inner product.)** On $\mathbb{R}^2$ the inner product $(x, y) = 2x_1y_1 + 3x_2y_2$ is given. Find (x, y), $\|x\|$ and $\|y\|$ for $x = (1; 2)$ and $y = (3; -1)$.

7. **(A function space.)** On $C[0; 1]$ let $(f, g) = \int_0^1 f(t)g(t)\,dt$. Find (f, g) and $\|f\|$ for $f = t$ and $g = 1$.

8. **(Hamming distance.)** Find the Hamming distance between the binary words $u = 1011010$ and $v = 1001110$.

9. **(Cosine similarity.)** The word-frequency vectors of two documents are $u = (3; 0; 2; 1)$ and $v = (1; 1; 2; 0)$. Find $\cos(u, v)$.

10. **(Pythagorean theorem.)** Verify that $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ for $x = (1; 1; 0; 0)$ and $y = (0; 0; 2; 2)$.

3.5 Constructing an orthonormal basis in a Euclidean space

Definition 3.5.1. A basis $e_1, \dots, e_n$ of a Euclidean space is called **orthogonal** if $(e_i, e_j) = 0$ for $i \neq j$. If, in addition, $\|e_i\| = 1$ for every i, the basis is called **orthonormal**; in this case $(e_i, e_j) = \delta_{ij}$, where $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$.

In an orthonormal basis the coordinates of a vector are simply inner products, and the inner product itself takes the standard form:

$$x = \sum_{i=1}^{n} (x, e_i)\, e_i, \qquad (x, y) = \sum_{i=1}^{n} \xi_i \eta_i,$$

where ξ_i and η_i are the coordinates of x and y.

3.5.1 The Gram–Schmidt orthogonalization process

Theorem 3.5.1 (Gram–Schmidt)**.** *From a linearly independent system $a_1, a_2, \ldots, a_k$ an orthogonal system $b_1, \ldots, b_k$ is obtained by the formulas*

$$b_1 = a_1, \qquad b_m = a_m - \sum_{j=1}^{m-1} \frac{(a_m, b_j)}{(b_j, b_j)}\, b_j, \quad m = 2, \ldots, k.$$

Normalizing, $e_m = \dfrac{b_m}{\|b_m\|}$, we arrive at an orthonormal system. Moreover, $L(a_1, \ldots, a_m) = L(e_1, \ldots, e_m)$ for every m.

Example 3.5.1. Orthonormalize the vectors $a_1 = (1; 1; 0)$, $a_2 = (1; 0; 1)$, $a_3 = (0; 1; 1)$.

Solution. We put $b_1 = a_1 = (1; 1; 0)$; then $(b_1, b_1) = 2$. Since $(a_2, b_1) = 1$, the second vector is $b_2 = a_2 - \frac{1}{2} b_1 = \left(\frac{1}{2}; -\frac{1}{2}; 1\right)$, and $(b_2, b_2) = \frac{3}{2}$. For the third vector we need $(a_3, b_1) = 1$ and $(a_3, b_2) = \frac{1}{2}$, which give

$$b_3 = a_3 - \frac{1}{2} b_1 - \frac{1/2}{3/2} b_2 = (0; 1; 1) - \left(\tfrac{1}{2}; \tfrac{1}{2}; 0\right) - \left(\tfrac{1}{6}; -\tfrac{1}{6}; \tfrac{1}{3}\right) = \left(-\tfrac{2}{3}; \tfrac{2}{3}; \tfrac{2}{3}\right).$$

As a check, $(b_1, b_3) = -\frac{2}{3} + \frac{2}{3} = 0$ and $(b_2, b_3) = -\frac{1}{3} - \frac{1}{3} + \frac{2}{3} = 0$. It remains to normalize:

$$e_1 = \frac{1}{\sqrt{2}}(1; 1; 0), \quad e_2 = \frac{1}{\sqrt{6}}(1; -1; 2), \quad e_3 = \frac{1}{\sqrt{3}}(-1; 1; 1).$$

3.5.2 Orthogonal projection

Definition 3.5.2. The **orthogonal projection** of an element x onto the direction of $e \neq \theta$ is $\mathrm{pr}_e x = \dfrac{(x, e)}{(e, e)}\, e$; the element $h = x - \mathrm{pr}_e x$ is called the **orthogonal component** of x (we have $h \perp e$).

Example 3.5.2. The projection of $x = (3;1;2)$ onto $e = (1;1;0)$ is $\mathrm{pr}_e x = \frac{4}{2}(1;1;0) = (2;2;0)$, and $h = (1;-1;2)$. Indeed, $(h,e) = 1 - 1 = 0$.

Definition 3.5.3. A square matrix Q whose columns form an orthonormal system is called an **orthogonal matrix**. Equivalently, $Q^T Q = E$, that is, $Q^{-1} = Q^T$; in this case $\det Q = \pm 1$. An orthogonal matrix preserves lengths of vectors and angles between them.

3.5.3 Applications of orthonormal bases in information security

Orthogonal transforms and compression. The discrete cosine transform (DCT) is used to compress images and audio and to embed digital watermarks. Its basis vectors form an orthonormal system, so the inverse transform amounts to a transposition: $x = Q^T y$. A watermark is hidden in one of the coefficients.

Lattice basis reduction (LLL). Attacks on lattice-based cryptosystems use the LLL algorithm. At each stage it computes the Gram–Schmidt coefficients and brings the basis to a "nearly orthogonal" form. The shorter and the closer to orthogonal the basis that can be found, the easier it is to recover the secret key.

Quantum cryptography (BB84). In this key distribution protocol photon states are prepared in two different orthonormal bases, $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$. A measurement in the wrong basis gives a random result, and this is what allows an eavesdropper to be detected.

3.5.4 Computing with Python, Maple and Mathcad

SymPy has a ready-made function `GramSchmidt`; applied to the vectors of the first example it reproduces b_1, b_2, b_3 and, with the option `orthonormal=True`, the vectors e_1, e_2, e_3:

Python

```
from sympy import Matrix, GramSchmidt
a = [Matrix([1, 1, 0]), Matrix([1, 0, 1]),
     Matrix([0, 1, 1])]
for b in GramSchmidt(a):
    print(b.T)
```

```
for e in GramSchmidt(a, orthonormal=True):
    print(e.T)
```

Output

```
Matrix([[1, 1, 0]])
Matrix([[1/2, -1/2, 1]])
Matrix([[-2/3, 2/3, 2/3]])
Matrix([[sqrt(2)/2, sqrt(2)/2, 0]])
Matrix([[sqrt(6)/6, -sqrt(6)/6, sqrt(6)/3]])
Matrix([[-sqrt(3)/3, sqrt(3)/3, sqrt(3)/3]])
```

Maple's `LinearAlgebra` package has the same procedure; the option `normalized` returns an orthonormal list:

Maple

```
with(LinearAlgebra):
a1 := <1, 1, 0>: a2 := <1, 0, 1>: a3 := <0, 1, 1>:
GramSchmidt([a1, a2, a3]);
GramSchmidt([a1, a2, a3], normalized);
```

Mathcad has no built-in Gram–Schmidt function, but the formulas of the theorem can be typed directly; the dot product and the length $|b|$ are built-in operators:

Mathcad

```
      | 1 |         | 1 |         | 0 |
a1 := | 1 |   a2 := | 0 |   a3 := | 1 |
      | 0 |         | 1 |         | 1 |
b1 := a1
            a2.b1
b2 := a2 - ------- b1
            b1.b1
            a3.b1        a3.b2
b3 := a3 - ------- b1 - ------- b2
            b1.b1        b2.b2
b3 = |-0.667|    e3 := b3/|b3|    e3 = |-0.577|
     | 0.667|                          | 0.577|
     | 0.667|                          | 0.577|
```

3.5.5 Exercises on orthonormal bases

1. Orthogonalize the vectors by the Gram–Schmidt process and then normalize them by the rule $\vec{e}_i = \vec{b}_i / \|\vec{b}_i\|$.

1. $\vec{a}_1 = (1; 0; 3), \vec{a}_2 = (-1; -1; -1)$

2. $\vec{a}_1 = (-2; 3; 2), \vec{a}_2 = (2; 1; 2)$
3. $\vec{a}_1 = (-1; -3; -3), \vec{a}_2 = (-2; 0; 3)$
4. $\vec{a}_1 = (2; 0; -1), \vec{a}_2 = (3; 1; 3)$
5. $\vec{a}_1 = (-1; 2; 3), \vec{a}_2 = (2; 2; 0)$
6. $\vec{a}_1 = (-2; 1; -3), \vec{a}_2 = (-2; 3; 3)$
7. $\vec{a}_1 = (-2; -2; 2), \vec{a}_2 = (1; 1; 3)$
8. $\vec{a}_1 = (2; -2; -1), \vec{a}_2 = (1; -3; 2)$
9. $\vec{a}_1 = (2; -3; -1), \vec{a}_2 = (0; -3; 1)$
10. $\vec{a}_1 = (3; 0; 2), \vec{a}_2 = (1; -2; 2)$
11. $\vec{a}_1 = (3; 0; 3), \vec{a}_2 = (1; -1; -3)$
12. $\vec{a}_1 = (0; -1; 1), \vec{a}_2 = (-3; 3; 2)$
13. $\vec{a}_1 = (-1; -3; 0), \vec{a}_2 = (-1; 1; 3)$
14. $\vec{a}_1 = (-3; 3; 0), \vec{a}_2 = (-2; -3; 1)$
15. $\vec{a}_1 = (-2; -2; 1), \vec{a}_2 = (-1; -3; -1)$
16. $\vec{a}_1 = (-3; 3; -1), \vec{a}_2 = (-2; 2; 1)$
17. $\vec{a}_1 = (3; 1; 1), \vec{a}_2 = (1; -2; -1)$
18. $\vec{a}_1 = (2; -1; -1), \vec{a}_2 = (-1; -2; -1)$
19. $\vec{a}_1 = (-3; -3; 3), \vec{a}_2 = (-1; -3; 0)$
20. $\vec{a}_1 = (2; -1; 0), \vec{a}_2 = (2; -3; 2)$
21. $\vec{a}_1 = (-2; -1; -1), \vec{a}_2 = (-3; 2; 2)$
22. $\vec{a}_1 = (0; -3; 1), \vec{a}_2 = (1; 1; 2)$
23. $\vec{a}_1 = (-2; 0; 0), \vec{a}_2 = (2; 1; 3)$
24. $\vec{a}_1 = (0; -2; -2), \vec{a}_2 = (-1; -2; 2)$
25. $\vec{a}_1 = (-3; -2; -3), \vec{a}_2 = (3; 1; -1)$

26. $\vec{a}_1 = (3; -1; 3), \vec{a}_2 = (-1; -3; -2)$

27. $\vec{a}_1 = (0; -3; 2), \vec{a}_2 = (3; -1; 3)$

28. $\vec{a}_1 = (-1; 2; 0), \vec{a}_2 = (0; -3; 1)$

29. $\vec{a}_1 = (-3; 1; 2), \vec{a}_2 = (-2; -2; 1)$

30. $\vec{a}_1 = (-1; 1; -2), \vec{a}_2 = (3; -1; -3)$

2. Orthogonalize the three vectors by the Gram–Schmidt process.

1. $\vec{a}_1 = (-2; -1; -1), \vec{a}_2 = (1; 1; 0), \vec{a}_3 = (-1; 0; 2)$

2. $\vec{a}_1 = (-2; 2; 0), \vec{a}_2 = (2; 1; 2), \vec{a}_3 = (2; -2; -1)$

3. $\vec{a}_1 = (2; 0; -1), \vec{a}_2 = (-1; 0; 0), \vec{a}_3 = (2; -1; 1)$

4. $\vec{a}_1 = (0; -2; 2), \vec{a}_2 = (1; 2; -2), \vec{a}_3 = (2; -2; -1)$

5. $\vec{a}_1 = (2; 1; 2), \vec{a}_2 = (2; 1; -1), \vec{a}_3 = (-1; 1; 2)$

6. $\vec{a}_1 = (2; -2; -1), \vec{a}_2 = (1; 0; -2), \vec{a}_3 = (1; 0; 1)$

7. $\vec{a}_1 = (0; -2; 2), \vec{a}_2 = (0; 0; -2), \vec{a}_3 = (-1; -2; -2)$

8. $\vec{a}_1 = (-1; -2; 2), \vec{a}_2 = (-2; 1; -2), \vec{a}_3 = (1; -2; -1)$

9. $\vec{a}_1 = (2; 0; 2), \vec{a}_2 = (0; 0; 1), \vec{a}_3 = (0; 2; 1)$

10. $\vec{a}_1 = (-2; -2; 0), \vec{a}_2 = (2; -2; 0), \vec{a}_3 = (0; 0; -2)$

11. $\vec{a}_1 = (0; -1; 1), \vec{a}_2 = (-1; 1; -2), \vec{a}_3 = (2; 0; 0)$

12. $\vec{a}_1 = (-1; 0; 0), \vec{a}_2 = (-2; -1; -1), \vec{a}_3 = (1; 1; -1)$

13. $\vec{a}_1 = (2; 0; 1), \vec{a}_2 = (2; -2; 2), \vec{a}_3 = (-1; 1; 1)$

14. $\vec{a}_1 = (2; 2; 2), \vec{a}_2 = (2; -1; 2), \vec{a}_3 = (-1; -1; 0)$

15. $\vec{a}_1 = (1; -2; -1), \vec{a}_2 = (0; 2; -1), \vec{a}_3 = (1; -1; -1)$

16. $\vec{a}_1 = (1; 2; -1), \vec{a}_2 = (2; 1; 0), \vec{a}_3 = (0; 2; -1)$

17. $\vec{a}_1 = (-2; 2; -1), \vec{a}_2 = (1; 1; -2), \vec{a}_3 = (-2; -1; 0)$

18. $\vec{a}_1 = (1; 2; -2), \vec{a}_2 = (0; -1; 0), \vec{a}_3 = (2; 1; -1)$

19. $\vec{a}_1 = (-1; 0; 0), \vec{a}_2 = (-1; 1; -2), \vec{a}_3 = (2; 1; 0)$

20. $\vec{a}_1 = (1; 2; 1), \vec{a}_2 = (1; 0; -1), \vec{a}_3 = (1; 0; 1)$

21. $\vec{a}_1 = (-1; 0; 1), \vec{a}_2 = (-1; -1; -2), \vec{a}_3 = (-1; -1; 1)$

22. $\vec{a}_1 = (0; -1; 1), \vec{a}_2 = (1; 1; -2), \vec{a}_3 = (-1; -2; 1)$

23. $\vec{a}_1 = (1; 1; 1), \vec{a}_2 = (-2; -2; -1), \vec{a}_3 = (-1; 0; -1)$

24. $\vec{a}_1 = (-1; -2; 1), \vec{a}_2 = (0; -2; -2), \vec{a}_3 = (-1; 0; -2)$

25. $\vec{a}_1 = (0; 2; 2), \vec{a}_2 = (2; 0; 0), \vec{a}_3 = (0; 1; 2)$

26. $\vec{a}_1 = (-2; 0; -2), \vec{a}_2 = (-2; 2; -2), \vec{a}_3 = (2; 2; 0)$

27. $\vec{a}_1 = (0; 1; -2), \vec{a}_2 = (2; -1; -2), \vec{a}_3 = (0; 2; 0)$

28. $\vec{a}_1 = (0; 0; -1), \vec{a}_2 = (1; 1; -2), \vec{a}_3 = (1; 0; 2)$

29. $\vec{a}_1 = (-2; -1; 2), \vec{a}_2 = (0; 1; 2), \vec{a}_3 = (2; -2; -2)$

30. $\vec{a}_1 = (-2; 1; 2), \vec{a}_2 = (1; -1; 1), \vec{a}_3 = (-2; -1; -2)$

3. Applied problems (10 problems).

1. **(Projection.)** Find the orthogonal projection of $\vec{x} = (0; -4; -3)$ onto the direction of $\vec{e} = (3; -3; -2)$ and the corresponding orthogonal component.

2. **(Projection.)** Find the orthogonal projection of $\vec{x} = (3; -1; 0)$ onto the direction of $\vec{e} = (3; 2; -1)$ and the corresponding orthogonal component.

3. **(Projection.)** Find the orthogonal projection of $\vec{x} = (0; 1; 4)$ onto the direction of $\vec{e} = (-3; 0; -3)$ and the corresponding orthogonal component.

4. **(Projection.)** Find the orthogonal projection of $\vec{x} = (2; -4; 3)$ onto the direction of $\vec{e} = (-3; -1; 1)$ and the corresponding orthogonal component.

5. **(Projection.)** Find the orthogonal projection of $\vec{x} = (1; 3; -4)$ onto the direction of $\vec{e} = (3; 0; -3)$ and the corresponding orthogonal component.

6. **(Orthogonal complement.)** Find a unit vector orthogonal to both $\vec{a} = (1; 2; -1)$ and $\vec{b} = (0; 1; 1)$.

7. **(Checking an orthonormal basis.)** Verify that $\vec{e}_1 = \frac{1}{3}(1;2;2)$, $\vec{e}_2 = \frac{1}{3}(2;1;-2)$, $\vec{e}_3 = \frac{1}{3}(2;-2;1)$ form an orthonormal basis.

8. **(Coordinates.)** Using inner products, find the coordinates of $\vec{x} = (3;3;0)$ in the orthonormal basis of Problem 7.

9. **(Orthogonal matrix.)** Verify that $Q^T Q = E$ for $Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$ and find Q^{-1}.

10. **(Basis of a subspace.)** Construct an orthonormal basis of the plane $x_1 + x_2 - x_3 = 0$.

Chapter 4

Linear Operators

4.1 Operations on linear operators. The space of linear operators. The matrix of a linear operator under a change of basis

Definition 4.1.1. A mapping $\mathcal{A}\colon V \to V$ of a linear space V into itself is called a **linear operator** if for all $x, y \in V$ and all $\lambda \in \mathbb{R}$

$$\mathcal{A}(x+y) = \mathcal{A}x + \mathcal{A}y, \qquad \mathcal{A}(\lambda x) = \lambda\,\mathcal{A}x.$$

Definition 4.1.2. Let $e_1, \dots, e_n$ be a basis and $\mathcal{A}e_j = \sum_{i=1}^{n} a_{ij} e_i$. The matrix $A = (a_{ij})$ is called the **matrix of the operator** in this basis: its jth column consists of the coordinates of $\mathcal{A}e_j$. The coordinates of $y = \mathcal{A}x$ are then found from the formula $Y = AX$.

Example 4.1.1. Write down the matrix of the operator $\mathcal{A}(x_1; x_2) = (2x_1 - x_2;\ x_1 + 3x_2)$ in the standard basis and find the image of $x = (1; -2)$.

Solution. Since $\mathcal{A}e_1 = \mathcal{A}(1;0) = (2;1)$ and $\mathcal{A}e_2 = \mathcal{A}(0;1) = (-1;3)$, we get $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$ and $\mathcal{A}x = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$.

4.1.1 Operations on linear operators

Let A and B be the matrices of operators $\mathcal{A}$ and $\mathcal{B}$ in a fixed basis. Then $\mathcal{A}+\mathcal{B}$ has the matrix $A + B$, and $\lambda\mathcal{A}$ has the matrix λA. The product $\mathcal{A}\mathcal{B}$ (first $\mathcal{B}$ is

applied, then $\mathcal{A}$) has the matrix AB. Finally, the inverse operator $\mathcal{A}^{-1}$ exists if and only if $\det A \neq 0$, and its matrix is A^{-1}. Consequently, the set of linear operators on an n-dimensional space is itself a linear space; it is isomorphic to the space of $n \times n$ matrices and has dimension n^2.

Example 4.1.2. Let $\mathcal{R}$ be the rotation of the plane through 90°, with matrix $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and let $\mathcal{S}$ be the reflection in the Ox axis, $S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Then

$$SR = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad RS = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Since $SR \neq RS$, multiplication of operators is not commutative.

4.1.2 The matrix of an operator in a new basis

Theorem 4.1.1. *Let T be the transition matrix from a basis e to a basis f (the columns of T are the coordinates of the vectors f_j in the basis e). Then the matrix of the operator $\mathcal{A}$ in the basis f is*

$$A' = T^{-1}AT.$$

Similar matrices have the same determinant, trace and characteristic polynomial.

Example 4.1.3. In the basis e an operator has the matrix $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. Find its matrix A' in the new basis $f_1 = e_1 + e_2$, $f_2 = e_1$.

Solution. Here $T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $T^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$, so

$$A' = T^{-1}AT = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

In the new basis the matrix of the operator is diagonal: $\mathcal{A}f_1 = 3f_1$ and $\mathcal{A}f_2 = f_2$. As a check, $\det A = \det A' = 3$ and $\operatorname{tr} A = \operatorname{tr} A' = 4$.

Definition 4.1.3. The set $\ker \mathcal{A} = \{x \colon \mathcal{A}x = \theta\}$ is called the **kernel** of the operator, and $\operatorname{Im} \mathcal{A} = \{\mathcal{A}x \colon x \in V\}$ its **image**; they satisfy $\dim \ker \mathcal{A} + \dim \operatorname{Im} \mathcal{A} = \dim V$.

4.1.3 Applications of linear operators in information security

The encryption operator. The Hill cipher is the linear operator $\mathcal{K}(p) = Kp \pmod{26}$ on the space $\mathbb{Z}_{26}^n$. Decryption is possible only if this operator is invertible, i.e. $\gcd(\det K, 26) = 1$. Two successive encryptions $\mathcal{K}_2\mathcal{K}_1$ again form a Hill cipher, with key K_2K_1; for this reason "double encryption" does not make the cipher any stronger.

The linear layers of AES. The *ShiftRows* and *MixColumns* steps of AES are linear operators over $\mathrm{GF}(2^8)$. They provide "diffusion", while the nonlinear *SubBytes* step provides "confusion". A cipher built from linear layers alone is easily broken by linear algebra.

Example 4.1.4. Encrypting first with $K_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and then with $K_2 = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ is the same as encrypting once with the key $K = K_2K_1 = \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$.

4.1.4 Computing with Python, Maple and Mathcad

The change of basis in the example above is a single matrix product. In SymPy:

Python

```
from sympy import Matrix
A = Matrix([[1, 2], [0, 3]])
T = Matrix([[1, 1], [1, 0]])
B = T.inv() * A * T
print(B)
print(A.det(), B.det(), A.trace(), B.trace())
```

Output

```
Matrix([[3, 0], [0, 1]])
3 3 4 4
```

In Maple the matrix product is written with a dot:

Maple

```
with(LinearAlgebra):
A := Matrix([[1, 2], [0, 3]]):
T := Matrix([[1, 1], [1, 0]]):
B := MatrixInverse(T) . A . T;
Determinant(B);  Trace(B);
```

In Mathcad the inverse is entered as the power -1, and `tr(B)` gives the trace:

Mathcad

```
A := | 1  2 |      T := | 1  1 |
     | 0  3 |           | 1  0 |
B := T^-1.A.T        B = | 3  0 |
                         | 0  1 |
|B| = 3      tr(B) = 4
```

4.1.5 Exercises on linear operators

1. The matrices A and B of two operators are given. Find the matrices of the indicated operators.

1. $A = \begin{pmatrix} -2 & -3 \\ 2 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -1 & -2 \\ -2 & 0 \end{pmatrix}$; $A + 3B$, AB, BA

2. $A = \begin{pmatrix} 0 & -2 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ -2 & -1 \end{pmatrix}$; $A + 3B$, AB, BA

3. $A = \begin{pmatrix} -3 & -1 \\ 1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 3 \\ -2 & 3 \end{pmatrix}$; $A - B$, AB, BA

4. $A = \begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix}$; $A + 3B$, AB, BA

5. $A = \begin{pmatrix} 3 & 3 \\ 3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 2 \\ -1 & -1 \end{pmatrix}$; $A - 2B$, AB, BA

6. $A = \begin{pmatrix} -1 & 0 \\ -2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$; $A + 2B$, AB, BA

7. $A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$; $A + 2B$, AB, BA

8. $A = \begin{pmatrix} -3 & -1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -2 \\ 0 & 2 \end{pmatrix}$; $A + 3B$, AB, BA

9. $A = \begin{pmatrix} 3 & -2 \\ -2 & -2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 2 \\ 3 & 0 \end{pmatrix}$; $A - B$, AB, BA

10. $A = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -3 & 2 \\ -2 & 1 \end{pmatrix}$; $A + 2B$, AB, BA

11. $A = \begin{pmatrix} 3 & -1 \\ 3 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 3 \\ 2 & 1 \end{pmatrix}$; $A + B$, AB, BA

12. $A = \begin{pmatrix} -3 & 0 \\ -2 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 0 \\ -3 & -2 \end{pmatrix}$; $A + 3B$, AB, BA

13. $A = \begin{pmatrix} -3 & 1 \\ -1 & -2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 2 \\ 0 & 3 \end{pmatrix}$; $A + 2B$, AB, BA

14. $A = \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -3 \\ -2 & 3 \end{pmatrix}$; $A + 3B$, AB, BA

15. $A = \begin{pmatrix} 3 & -3 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 \\ -2 & 0 \end{pmatrix}$; $A + B$, AB, BA

16. $A = \begin{pmatrix} -2 & -1 & 2 \\ 0 & 2 & 2 \\ -1 & -2 & -1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 2 & 3 \\ 1 & -1 & -2 \\ 0 & 2 & 1 \end{pmatrix}$; $A - 2B$, AB, BA

17. $A = \begin{pmatrix} 1 & 0 & -2 \\ 3 & -2 & 1 \\ 2 & 3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -3 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & -3 \end{pmatrix}$; $A - 2B$, AB, BA

18. $A = \begin{pmatrix} 3 & 1 & 3 \\ -1 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 3 & 3 \\ 3 & -2 & -2 \end{pmatrix}$; $A - B$, AB, BA

19. $A = \begin{pmatrix} 3 & 3 & -2 \\ -1 & 2 & 2 \\ -1 & 0 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 3 & -3 \\ -1 & 3 & 1 \\ 2 & -2 & 3 \end{pmatrix}$; $A + 2B$, AB, BA

20. $A = \begin{pmatrix} -2 & -2 & -1 \\ -3 & -2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -2 & -2 & -1 \\ -3 & 1 & -1 \\ -3 & -1 & -1 \end{pmatrix}$; $A - 2B$, AB, BA

21. $A = \begin{pmatrix} 1 & -1 & -1 \\ 2 & -2 & 0 \\ 2 & 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 2 & -3 \\ -1 & 0 & 3 \end{pmatrix}$; $A + 3B$, AB, BA

22. $A = \begin{pmatrix} -3 & 1 & -1 \\ 0 & -1 & -3 \\ -2 & -1 & -2 \end{pmatrix}$, $B = \begin{pmatrix} -1 & -2 & 2 \\ 2 & 2 & 0 \\ -3 & -3 & 3 \end{pmatrix}$; $A + 3B$, AB, BA

23. $A = \begin{pmatrix} -1 & 3 & 3 \\ 1 & 3 & 0 \\ 1 & 1 & -3 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 2 & -3 \\ 3 & 0 & 3 \\ -1 & -2 & 3 \end{pmatrix}$; $A + 3B$, AB, BA

24. $A = \begin{pmatrix} -3 & -2 & -1 \\ 3 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 0 & 2 \\ -1 & 0 & 3 \end{pmatrix}$; $A - 2B$, AB, BA

25. $A = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 2 & -2 \\ -3 & -3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 3 & -1 \\ 2 & -3 & 1 \\ -3 & -1 & -2 \end{pmatrix}$; $A - B$, AB, BA

26. $A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & -3 \\ -2 & -2 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 & -1 \\ -3 & 1 & 0 \\ 2 & 2 & 1 \end{pmatrix}$; $A - B$, AB, BA

27. $A = \begin{pmatrix} -1 & 0 & -2 \\ 0 & 1 & -2 \\ 1 & -1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -3 & 1 & 1 \\ 2 & 3 & -1 \\ 1 & -3 & 1 \end{pmatrix}$; $A + B$, AB, BA

28. $A = \begin{pmatrix} -1 & 3 & 2 \\ -1 & 2 & 1 \\ -3 & -2 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 3 & 1 \\ 1 & -2 & -3 \\ -2 & -1 & 3 \end{pmatrix}$; $A - B$, AB, BA

29. $A = \begin{pmatrix} -2 & 0 & -3 \\ 3 & 1 & 3 \\ -2 & -3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 & -2 \\ 0 & -3 & 2 \\ 2 & 3 & 0 \end{pmatrix}$; $A + B$, AB, BA

30. $A = \begin{pmatrix} -2 & -2 & 3 \\ 0 & -1 & 2 \\ -1 & 0 & -1 \end{pmatrix}, B = \begin{pmatrix} -2 & -2 & 0 \\ 3 & 2 & -1 \\ 2 & 2 & 2 \end{pmatrix}$; $A + 2B$, AB, BA

2. The matrix A of an operator in a basis e and the transition matrix T are given. Find the matrix $A' = T^{-1}AT$ in the new basis.

1. $A = \begin{pmatrix} 4 & 0 \\ 1 & -1 \end{pmatrix}, T = \begin{pmatrix} 0 & -1 \\ 1 & -2 \end{pmatrix}$

2. $A = \begin{pmatrix} -3 & 1 \\ -2 & 1 \end{pmatrix}, T = \begin{pmatrix} -1 & -2 \\ 0 & -1 \end{pmatrix}$

3. $A = \begin{pmatrix} -1 & -2 \\ -2 & 1 \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

4. $A = \begin{pmatrix} 1 & 4 \\ -3 & 0 \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$

5. $A = \begin{pmatrix} 0 & -1 \\ 3 & -2 \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

6. $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}, T = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}$

7. $A = \begin{pmatrix} 4 & 3 \\ -3 & 3 \end{pmatrix}, T = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$

8. $A = \begin{pmatrix} -2 & 2 \\ 2 & -4 \end{pmatrix}, T = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$

9. $A = \begin{pmatrix} -2 & 4 \\ 1 & 1 \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$

10. $A = \begin{pmatrix} 2 & 4 \\ 2 & -2 \end{pmatrix}, T = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}$

11. $A = \begin{pmatrix} 4 & -1 \\ 3 & 3 \end{pmatrix}, T = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}$

12. $A = \begin{pmatrix} 0 & -4 \\ -4 & -3 \end{pmatrix}, T = \begin{pmatrix} -1 & 0 \\ -2 & 1 \end{pmatrix}$

13. $A = \begin{pmatrix} 0 & 2 \\ -2 & 3 \end{pmatrix}, T = \begin{pmatrix} -1 & 3 \\ 0 & -1 \end{pmatrix}$

14. $A = \begin{pmatrix} -2 & -3 \\ 0 & 2 \end{pmatrix}, T = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

15. $A = \begin{pmatrix} -2 & 3 \\ 4 & 3 \end{pmatrix}, T = \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix}$

16. $A = \begin{pmatrix} -2 & -2 \\ 0 & 2 \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$

17. $A = \begin{pmatrix} 4 & -2 \\ 2 & 2 \end{pmatrix}, T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

18. $A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, T = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$

19. $A = \begin{pmatrix} -1 & -4 \\ -4 & -3 \end{pmatrix}, T = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$

20. $A = \begin{pmatrix} 4 & 4 \\ -1 & 3 \end{pmatrix}, T = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$

21. $A = \begin{pmatrix} 0 & -2 \\ 4 & -1 \end{pmatrix}, T = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$

22. $A = \begin{pmatrix} -3 & -3 \\ -2 & -4 \end{pmatrix}, T = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$

23. $A = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}, T = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}$

24. $A = \begin{pmatrix} 1 & 3 \\ 3 & 4 \end{pmatrix}, T = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}$

25. $A = \begin{pmatrix} 0 & 2 \\ -4 & -1 \end{pmatrix}, T = \begin{pmatrix} -1 & 0 \\ -2 & 1 \end{pmatrix}$

26. $A = \begin{pmatrix} -4 & -1 \\ -2 & 0 \end{pmatrix}, T = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$

27. $A = \begin{pmatrix} 3 & 1 \\ 0 & -2 \end{pmatrix}, T = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

28. $A = \begin{pmatrix} -4 & -3 \\ -3 & -4 \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

29. $A = \begin{pmatrix} -4 & 0 \\ -4 & -1 \end{pmatrix}, T = \begin{pmatrix} -2 & -1 \\ 3 & 2 \end{pmatrix}$

30. $A = \begin{pmatrix} 3 & 3 \\ 2 & 1 \end{pmatrix}, T = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$

3. Applied problems (10 problems).

1. **(Linearity.)** Is the operator $\mathcal{A}(x_1; x_2) = (x_1 + 2x_2;\ 3x_1 - x_2)$ linear? Write down its matrix.

2. **(A nonlinear map.)** Is the operator $\mathcal{A}(x_1; x_2) = (x_1 + 1;\ x_2)$ linear?

3. **(Rotation.)** Write down the matrix of the rotation of the plane through 90° and find the image of $\vec{x} = (3; 1)$.

4. **(Projection.)** Write down the matrix of the projection of $\mathbb{R}^3$ onto the Oxy plane and verify that $\mathcal{A}^2 = \mathcal{A}$.

5. **(Reflection.)** Write down the matrix of the reflection in the Ox axis and verify that $\mathcal{A}^2 = E$.

6. **(Image of a vector.)** Find the image of $\vec{x} = (1; -1; 2)$ under the operator with matrix $A = \left(\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 2 & 0 & 3 \end{smallmatrix}\right)$.

7. **(Kernel.)** Find the kernel and the image of the operator with matrix $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$.

8. **(Composition.)** Find the matrix of the operator that first rotates the plane through 90° and then reflects it in the Ox axis.

9. **(Inverse operator.)** Find the matrix of the inverse of the operator with matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$.

10. **(The Hill cipher as an operator.)** Let $\mathcal{K}(p) = Kp \pmod{26}$ with $K = \left(\begin{smallmatrix} 3 & 2 \\ 5 & 7 \end{smallmatrix}\right)$. Show that $\mathcal{K}$ is invertible on $\mathbb{Z}_{26}^2$.

4.2 Eigenvalues and eigenvectors of a linear operator

Definition 4.2.1. A number λ is called an **eigenvalue** of the operator $\mathcal{A}$ (or of its matrix A) if there is a nonzero vector x such that $\mathcal{A}x = \lambda x$; every such vector x is called an **eigenvector** corresponding to λ.

The equation $Ax = \lambda x$ is equivalent to the homogeneous system $(A-\lambda E)x = 0$. This system has a nonzero solution if and only if its matrix is singular, that is,

$$\det(A - \lambda E) = 0. \tag{4.2.1}$$

Equation (4.2.1) is called the **characteristic equation** of A, and its left-hand side, a polynomial of degree n in λ, is the **characteristic polynomial**.

The computation therefore splits into two stages. We first solve the characteristic equation and obtain the eigenvalues $\lambda_1, \dots, \lambda_n$. Then, for each eigenvalue λ_k, we solve the homogeneous system $(A - \lambda_k E)x = 0$; a fundamental system of its solutions is a maximal set of linearly independent eigenvectors belonging to λ_k.

Properties. The eigenvalues, counted with multiplicity, satisfy $\lambda_1 + \cdots + \lambda_n = \operatorname{tr} A$ and $\lambda_1 \cdots \lambda_n = \det A$. The eigenvalues of a triangular matrix are its diagonal entries. If λ is an eigenvalue of A, then λ^k is an eigenvalue of A^k and, for nonsingular A, $1/\lambda$ is an eigenvalue of A^{-1}, with the same eigenvector. Eigenvectors belonging to distinct eigenvalues are linearly independent. Finally, a real symmetric matrix has only real eigenvalues, and its eigenvectors belonging to distinct eigenvalues are orthogonal.

Example 4.2.1. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Solution. The characteristic equation is

$$\det(A - \lambda E) = \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = \lambda^2 - 7\lambda + 10 = 0,$$

so $\lambda_1 = 2$ and $\lambda_2 = 5$. For $\lambda_1 = 2$ the system $(A - 2E)x = 0$ reduces to the single equation $2x_1 + x_2 = 0$, whence $x^{(1)} = c(1; -2)$. For $\lambda_2 = 5$ we

obtain $-x_1 + x_2 = 0$ (the second equation $2x_1 - 2x_2 = 0$ is proportional to it), so $x^{(2)} = c(1;1)$. In both cases $c \neq 0$.

As a check, $A(1;1)^T = (5;5)^T = 5(1;1)^T$; moreover, $\lambda_1 + \lambda_2 = 7 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = 10 = \det A$.

Example 4.2.2. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & -3 \end{pmatrix}$.

Solution. Expanding along the first row, we get

$$\det(A - \lambda E) = (2-\lambda)\big[(3-\lambda)(-3-\lambda) - 16\big] = (2-\lambda)(\lambda^2 - 25) = 0,$$

so $\lambda_1 = 2$, $\lambda_2 = 5$, $\lambda_3 = -5$. For $\lambda = 2$ the system gives $x_2 = x_3 = 0$ with x_1 arbitrary, hence $x^{(1)} = (1;0;0)$. For $\lambda = 5$ we have $x_1 = 0$ and $-2x_2 + 4x_3 = 0$, hence $x^{(2)} = (0;2;1)$. For $\lambda = -5$ we have $x_1 = 0$ and $8x_2 + 4x_3 = 0$, hence $x^{(3)} = (0;1;-2)$.

Since A is symmetric, the three eigenvectors must be pairwise orthogonal; indeed, $(0;2;1) \cdot (0;1;-2) = 0$, and both vectors are clearly orthogonal to $(1;0;0)$.

Example 4.2.3. For the matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ of the rotation of the plane through 90°, the characteristic equation is $\lambda^2 + 1 = 0$, which has no real roots. This agrees with geometry: a rotation through 90° leaves no nonzero vector on its own line.

4.2.1 Applications of eigenvalues and eigenvectors in information security

PageRank and reputation systems. The importance of the nodes of a network (web sites, users) is computed as the eigenvector of the transition matrix corresponding to the eigenvalue $\lambda = 1$. Trust scores used to detect phishing sites and botnets are obtained in exactly this way.

Example 4.2.4. For the transition matrix $P = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}$ the system $(P - E)x = 0$ reduces to $-0.1x_1 + 0.2x_2 = 0$, that is, $x_1 = 2x_2$. Normalizing so that the components add up to one, we obtain the stationary distribution $x = \left(\frac{2}{3}; \frac{1}{3}\right)$: in the long run the system spends two thirds of the time in state 1.

Spectral analysis and anomaly detection. The eigenvectors of the covariance matrix of network traffic that belong to its largest eigenvalues (the principal components of PCA) span the subspace of "normal" traffic; observations lying far from this subspace are flagged as possible attacks. The eigenvalues of the adjacency matrix of a graph, in turn, describe the connectivity of a network and reveal its weak points.

4.2.2 Computing with Python, Maple and Mathcad

All three systems find eigenvalues and eigenvectors directly. We take the matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ of the first example. In Python, SymPy factors the characteristic polynomial and returns, for each eigenvalue, its multiplicity and a basis of eigenvectors.

Python

```
import sympy as sp
A = sp.Matrix([[4, 1], [2, 3]])
lam = sp.symbols('lambda')
print(sp.factor((A - lam*sp.eye(2)).det()))
for val, mult, vecs in A.eigenvects():
    print(val, mult, list(vecs[0]))
```

Output

```
(lambda - 5)*(lambda - 2)
2 1 [-1/2, 1]
5 1 [1, 1]
```

The eigenvector $(-\frac{1}{2}; 1)$ is a multiple of $(1; -2)$, as it should be.

In Maple the same computation uses the `LinearAlgebra` package; `Eigenvectors` returns the column of eigenvalues and a matrix whose columns are the corresponding eigenvectors.

Maple

```
with(LinearAlgebra):
A := Matrix([[4, 1], [2, 3]]):
CharacteristicPolynomial(A, lambda);
                 lambda^2 - 7 lambda + 10
Eigenvectors(A);
                 [2]  [-1/2  1]
                 [5], [ 1    1]
```

Mathcad Prime has the built-in functions `eigenvals(A)`, `eigenvecs(A)` and `eigenvec(A, z)`; the eigenvectors are returned normalized to unit length.

Mathcad

```
A := | 4  1 |
     | 2  3 |
eigenvals(A) = | 2 |
               | 5 |
eigenvec(A, 2) = | -0.447 |     eigenvec(A, 5) = | 0.707 |
                 |  0.894 |                       | 0.707 |
```

4.2.3 Exercises on eigenvalues and eigenvectors

1. Find the eigenvalues and eigenvectors of the following 2×2 *matrices.*

1. $\begin{pmatrix} 4 & 1 \\ 0 & 5 \end{pmatrix}$
2. $\begin{pmatrix} 0 & -2 \\ 0 & 2 \end{pmatrix}$
3. $\begin{pmatrix} -4 & -4 \\ 0 & 0 \end{pmatrix}$
4. $\begin{pmatrix} -9 & -6 \\ 12 & 9 \end{pmatrix}$
5. $\begin{pmatrix} 10 & 5 \\ -10 & -5 \end{pmatrix}$
6. $\begin{pmatrix} 0 & 8 \\ 0 & 4 \end{pmatrix}$
7. $\begin{pmatrix} 5 & 4 \\ -2 & -1 \end{pmatrix}$
8. $\begin{pmatrix} 4 & 0 \\ 6 & -2 \end{pmatrix}$
9. $\begin{pmatrix} 2 & 0 \\ 4 & 4 \end{pmatrix}$
10. $\begin{pmatrix} -2 & 2 \\ 0 & -3 \end{pmatrix}$
11. $\begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$
12. $\begin{pmatrix} 5 & 2 \\ -1 & 2 \end{pmatrix}$

13. $\begin{pmatrix} 1 & 0 \\ -3 & -2 \end{pmatrix}$

14. $\begin{pmatrix} 0 & -4 \\ 0 & -2 \end{pmatrix}$

15. $\begin{pmatrix} 5 & 0 \\ 3 & 2 \end{pmatrix}$

16. $\begin{pmatrix} -4 & -4 \\ 0 & -2 \end{pmatrix}$

17. $\begin{pmatrix} -1 & -6 \\ 0 & 5 \end{pmatrix}$

18. $\begin{pmatrix} 3 & 6 \\ 0 & 0 \end{pmatrix}$

19. $\begin{pmatrix} 5 & 0 \\ -7 & -2 \end{pmatrix}$

20. $\begin{pmatrix} -1 & 0 \\ 12 & 5 \end{pmatrix}$

21. $\begin{pmatrix} 1 & 0 \\ -4 & 3 \end{pmatrix}$

22. $\begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix}$

23. $\begin{pmatrix} 3 & 3 \\ -6 & -6 \end{pmatrix}$

24. $\begin{pmatrix} 5 & 8 \\ 0 & 1 \end{pmatrix}$

25. $\begin{pmatrix} -3 & 8 \\ 0 & 1 \end{pmatrix}$

26. $\begin{pmatrix} -2 & 1 \\ 0 & -1 \end{pmatrix}$

27. $\begin{pmatrix} 3 & -4 \\ 2 & -3 \end{pmatrix}$

28. $\begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix}$

29. $\begin{pmatrix} 5 & 0 \\ 10 & 0 \end{pmatrix}$

30. $\begin{pmatrix} -4 & -4 \\ 0 & 0 \end{pmatrix}$

2. Find the eigenvalues and eigenvectors of the following 3×3 *matrices.*

1. $\begin{pmatrix} -8 & 0 & -6 \\ -8 & -1 & -9 \\ 12 & 0 & 10 \end{pmatrix}$

2. $\begin{pmatrix} -2 & -5 & -10 \\ 0 & 1 & -2 \\ 0 & 1 & 4 \end{pmatrix}$

3. $\begin{pmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 1 & 6 & 4 \end{pmatrix}$

4. $\begin{pmatrix} 0 & 0 & 0 \\ 2 & -2 & 6 \\ 4 & 0 & 4 \end{pmatrix}$

5. $\begin{pmatrix} 1 & 0 & 0 \\ 6 & -2 & 0 \\ -6 & 2 & -1 \end{pmatrix}$

6. $\begin{pmatrix} -3 & -2 & 2 \\ 0 & -1 & -2 \\ -4 & -4 & 1 \end{pmatrix}$

7. $\begin{pmatrix} -1 & -2 & -2 \\ 2 & -2 & -2 \\ -2 & 0 & 0 \end{pmatrix}$

8. $\begin{pmatrix} 8 & 0 & -5 \\ 0 & -1 & 0 \\ 10 & 0 & -7 \end{pmatrix}$

9. $\begin{pmatrix} -2 & -12 & -12 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix}$

10. $\begin{pmatrix} -2 & 10 & -12 \\ 0 & 3 & -6 \\ 0 & 0 & -3 \end{pmatrix}$

11. $\begin{pmatrix} 0 & -9 & 1 \\ 0 & 4 & 0 \\ 0 & 3 & 1 \end{pmatrix}$

12. $\begin{pmatrix} 5 & 2 & 2 \\ -6 & -2 & -4 \\ 2 & 1 & 3 \end{pmatrix}$

13. $\begin{pmatrix} -3 & 5 & -8 \\ 0 & 2 & -8 \\ 0 & 0 & -2 \end{pmatrix}$

14. $\begin{pmatrix} -9 & 12 & 0 \\ -6 & 9 & 0 \\ 8 & -6 & 2 \end{pmatrix}$

15. $\begin{pmatrix} 2 & 1 & 6 \\ 4 & -1 & 6 \\ 1 & -1 & -3 \end{pmatrix}$

16. $\begin{pmatrix} 8 & 0 & 4 \\ 5 & 1 & 2 \\ -8 & 0 & -4 \end{pmatrix}$

17. $\begin{pmatrix} -4 & 8 & -6 \\ -2 & 4 & -4 \\ -1 & 2 & -3 \end{pmatrix}$

18. $\begin{pmatrix} -2 & 0 & 0 \\ 4 & 2 & 0 \\ -2 & 4 & 4 \end{pmatrix}$

19. $\begin{pmatrix} -3 & 5 & 5 \\ 2 & 0 & 2 \\ -2 & 2 & 0 \end{pmatrix}$

20. $\begin{pmatrix} 8 & -8 & -8 \\ 7 & -10 & -7 \\ -3 & 6 & 3 \end{pmatrix}$

21. $\begin{pmatrix} -2 & -8 & 4 \\ 0 & 0 & 2 \\ 0 & -4 & 6 \end{pmatrix}$

22. $\begin{pmatrix} 3 & 0 & -1 \\ -2 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$

23. $\begin{pmatrix} -6 & -6 & -5 \\ 0 & -2 & 0 \\ 10 & 12 & 9 \end{pmatrix}$

24. $\begin{pmatrix} 1 & -6 & 0 \\ 0 & 4 & 0 \\ 2 & 5 & 3 \end{pmatrix}$

25. $\begin{pmatrix} 2 & -5 & 9 \\ 0 & 2 & 2 \\ 0 & -1 & 5 \end{pmatrix}$

26. $\begin{pmatrix} 0 & -2 & 2 \\ 3 & 4 & -3 \\ -1 & -2 & 3 \end{pmatrix}$

27. $\begin{pmatrix} -2 & 2 & 3 \\ 0 & -8 & -5 \\ 0 & 10 & 7 \end{pmatrix}$

28. $\begin{pmatrix} 7 & 3 & 10 \\ -4 & 0 & -4 \\ -3 & -3 & -6 \end{pmatrix}$

29. $\begin{pmatrix} 2 & 6 & -7 \\ 2 & 3 & -4 \\ 2 & 4 & -5 \end{pmatrix}$

30. $\begin{pmatrix} -3 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & -6 & -2 \end{pmatrix}$

3. Further problems.

1. **(Characteristic polynomial.)** Find the characteristic polynomial of $A = \begin{pmatrix} 1 & -2 & -2 \\ -1 & 1 & 1 \\ -2 & 0 & -1 \end{pmatrix}$ and check how its coefficients are related to $\det A$ and $\operatorname{tr} A$.

2. **(Characteristic polynomial.)** Find the characteristic polynomial of $A = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 3 \\ 1 & 3 & -2 \end{pmatrix}$ and check how its coefficients are related to $\det A$ and $\operatorname{tr} A$.

3. **(Characteristic polynomial.)** Find the characteristic polynomial of $A = \begin{pmatrix} -1 & 2 & 3 \\ 3 & 0 & -2 \\ 2 & -1 & 3 \end{pmatrix}$ and check how its coefficients are related to $\det A$ and $\operatorname{tr} A$.

4. **(Characteristic polynomial.)** Find the characteristic polynomial of $A = \begin{pmatrix} 3 & 0 & -1 \\ 3 & -1 & 0 \\ 3 & 0 & 2 \end{pmatrix}$ and check how its coefficients are related to $\det A$ and $\operatorname{tr} A$.

5. **(Characteristic polynomial.)** Find the characteristic polynomial of $A = \begin{pmatrix} -1 & -1 & 3 \\ -1 & -1 & -1 \\ 3 & 3 & 1 \end{pmatrix}$ and check how its coefficients are related to $\det A$ and $\operatorname{tr} A$.

6. **(Characteristic polynomial.)** Find the characteristic polynomial of $A = \begin{pmatrix} -1 & 0 & -2 \\ 2 & 1 & -2 \\ -1 & 0 & -1 \end{pmatrix}$ and check how its coefficients are related to $\det A$ and $\operatorname{tr} A$.

7. **(Eigenvalues of related matrices.)** The eigenvalues of A are $1, 2, -3$. Find the eigenvalues of A^2, A^{-1} and $A + 2E$.

8. **(Triangular matrix.)** Write down the eigenvalues of $\left(\begin{smallmatrix} 2 & 5 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ without any computation.

9. **(Markov chain.)** Find the eigenvector of the transition matrix $P = \left(\begin{smallmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{smallmatrix}\right)$ corresponding to $\lambda = 1$ (the stationary distribution).

10. **(Symmetric matrix.)** Show that the eigenvectors of $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ are orthogonal.

4.3 Linear operators with a basis of eigenvectors

Theorem 4.3.1. *The matrix of an operator $\mathcal{A}$ on an n-dimensional space is diagonal in some basis if and only if $\mathcal{A}$ has n linearly independent eigenvectors, that is, if and only if its eigenvectors form a basis. In that case*

$$T^{-1}AT = D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where the columns of T are the eigenvectors and the diagonal of D carries the corresponding eigenvalues.

Sufficient conditions. If the n eigenvalues of A are pairwise distinct, then A is diagonalizable. Every real symmetric matrix is diagonalizable, and even by an orthogonal matrix Q: $Q^{T}AQ = D$. When A has a multiple eigenvalue λ, the matrix is diagonalizable if and only if, for every such λ, the *geometric* multiplicity $\dim\ker(A-\lambda E) = n - \operatorname{rank}(A-\lambda E)$ equals the *algebraic* multiplicity of λ as a root of the characteristic polynomial.

Example 4.3.1. Diagonalize $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ and find A^{n}.

Solution. In the previous section we found $\lambda_1 = 2$ with eigenvector $(1;-2)$ and $\lambda_2 = 5$ with eigenvector $(1;1)$. Therefore

$$T = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}, \quad T^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}, \quad T^{-1}AT = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}.$$

Since $A = TDT^{-1}$, we have $A^{n} = TD^{n}T^{-1}$, and multiplying out gives

$$A^{n} = \frac{1}{3}\begin{pmatrix} 2^{n} + 2\cdot 5^{n} & 5^{n} - 2^{n} \\ 2\cdot 5^{n} - 2^{n+1} & 2^{n+1} + 5^{n} \end{pmatrix}.$$

For instance, for $n = 2$ we obtain $A^{2} = \frac{1}{3}\begin{pmatrix} 54 & 21 \\ 42 & 33 \end{pmatrix} = \begin{pmatrix} 18 & 7 \\ 14 & 11 \end{pmatrix}$, which agrees with direct multiplication.

Example 4.3.2. Is $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ diagonalizable?

Solution. The only eigenvalue is $\lambda = 2$, of algebraic multiplicity two. The matrix $A - 2E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has rank 1, so the eigenspace is one-dimensional and is spanned by $(1;0)$. There is no pair of linearly independent eigenvectors, and A is **not diagonalizable**; it is a Jordan block.

Example 4.3.3. Diagonalize $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ by an orthogonal matrix.

Solution. The characteristic equation $\lambda^2 - 6\lambda + 8 = 0$ gives $\lambda_1 = 2$ with eigenvector $(1; -1)$ and $\lambda_2 = 4$ with eigenvector $(1; 1)$. These vectors are orthogonal; normalizing them, we obtain

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad Q^T A Q = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

4.3.1 Applications of diagonalization in information security

Fast computation of matrix powers. Many problems in cryptography and modelling require A^k for large k. For example, the state of a linear recurrent generator (such as an LFSR) is $s_k = A^k s_0$. If A is diagonalizable, then $A^k = T D^k T^{-1}$, and the work reduces to raising n numbers to the kth power.

Example 4.3.4. A Fibonacci-type generator is described by $\binom{F_{k+1}}{F_k} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^k \binom{1}{0}$. The eigenvalues of this matrix are $\lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}$, and diagonalization yields Binet's formula $F_k = \dfrac{\lambda_1^k - \lambda_2^k}{\sqrt{5}}$, which gives the state of the generator at any step without computing the previous ones.

Markov models. In a Markov model of user behaviour the state distribution after k steps is $x_k = P^k x_0$. Diagonalization gives the limiting distribution as $k \to \infty$, and a substantial deviation from it is treated as an anomaly.

4.3.2 Computing with Python, Maple and Mathcad

We diagonalize the matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ of the first example and use the result to compute A^{10}. SymPy's `diagonalize` returns the pair (T, D) directly.

Python

```
import sympy as sp
A = sp.Matrix([[4, 1], [2, 3]])
```

```
T, D = A.diagonalize()
print(T, D)
print(T.inv() * A * T == D)
print(T * D**10 * T.inv())
```

Output

```
Matrix([[-1, 1], [2, 1]]) Matrix([[2, 0], [0, 5]])
True
Matrix([[6510758, 3254867], [6509734, 3255891]])
```

SymPy chose the eigenvector $(-1;2)$ instead of $(1;-2)$; any nonzero multiple will do. The entry $6510758 = (2^{10}+2\cdot 5^{10})/3$ agrees with the general formula for A^n.

In Maple, `Eigenvectors` returns the eigenvalues together with the matrix T of eigenvectors, and the zero matrix in the output confirms that $T^{-1}AT$ is the diagonal matrix built from these eigenvalues.

Maple

```
with(LinearAlgebra):
A := Matrix([[4, 1], [2, 3]]):
v, T := Eigenvectors(A):
MatrixInverse(T) . A . T - DiagonalMatrix(v);
                        [0  0]
                        [0  0]
A^10;
                [6510758  3254867]
                [6509734  3255891]
```

In Mathcad the matrix of (normalized) eigenvectors is produced by `eigenvecs(A)`; the power A^{10} is typed with the usual power operator. (Mathcad and Maple may list the eigenvalues in a different order; the columns of T are then permuted accordingly.)

Mathcad

```
A := | 4  1 |        V := eigenvecs(A)
     | 2  3 |
V^-1 * A * V = | 2  0 |
               | 0  5 |
A^10 = | 6510758  3254867 |
       | 6509734  3255891 |
```

4.3.3 Exercises on diagonalization

1. Diagonalize the matrix, that is, find T and D such that $T^{-1}AT = D$.

1. $\begin{pmatrix} -2 & 0 \\ 4 & 0 \end{pmatrix}$
2. $\begin{pmatrix} -4 & -4 \\ 2 & 2 \end{pmatrix}$
3. $\begin{pmatrix} 7 & -10 \\ 5 & -8 \end{pmatrix}$
4. $\begin{pmatrix} -2 & 0 \\ 1 & -3 \end{pmatrix}$
5. $\begin{pmatrix} -6 & -6 \\ 3 & 3 \end{pmatrix}$
6. $\begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix}$
7. $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$
8. $\begin{pmatrix} -1 & 0 \\ -8 & 3 \end{pmatrix}$
9. $\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}$
10. $\begin{pmatrix} 1 & 0 \\ 8 & -3 \end{pmatrix}$
11. $\begin{pmatrix} 0 & -3 \\ 0 & -3 \end{pmatrix}$
12. $\begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$
13. $\begin{pmatrix} -1 & 2 \\ 0 & -3 \end{pmatrix}$
14. $\begin{pmatrix} 4 & 12 \\ 0 & -2 \end{pmatrix}$
15. $\begin{pmatrix} 0 & 0 \\ -4 & 4 \end{pmatrix}$
16. $\begin{pmatrix} -2 & -2 \\ 1 & 1 \end{pmatrix}$
17. $\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}$
18. $\begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$
19. $\begin{pmatrix} -5 & 2 \\ -1 & -2 \end{pmatrix}$
20. $\begin{pmatrix} 4 & 0 \\ 7 & -3 \end{pmatrix}$
21. $\begin{pmatrix} 4 & 8 \\ -4 & -8 \end{pmatrix}$
22. $\begin{pmatrix} 0 & 0 \\ 8 & 4 \end{pmatrix}$
23. $\begin{pmatrix} -1 & 0 \\ 4 & 3 \end{pmatrix}$
24. $\begin{pmatrix} 2 & 4 \\ 0 & 4 \end{pmatrix}$
25. $\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$
26. $\begin{pmatrix} -4 & 2 \\ -1 & -1 \end{pmatrix}$
27. $\begin{pmatrix} 0 & -8 \\ 0 & 4 \end{pmatrix}$
28. $\begin{pmatrix} -8 & 5 \\ -10 & 7 \end{pmatrix}$
29. $\begin{pmatrix} -1 & 0 \\ 0 & -3 \end{pmatrix}$
30. $\begin{pmatrix} -4 & 0 \\ 0 & -1 \end{pmatrix}$

2. Is the matrix diagonalizable? Find its eigenvalues and state your conclusion.

1. $\begin{pmatrix} -2 & -6 & 3 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{pmatrix}$

2. $\begin{pmatrix} -3 & 0 & 0 \\ 0 & 4 & 0 \\ 4 & -6 & 1 \end{pmatrix}$

3. $\begin{pmatrix} -2 & -4 & 0 \\ 3 & 5 & 1 \\ -1 & -1 & -1 \end{pmatrix}$

4. $\begin{pmatrix} 1 & -1 & 2 \\ 4 & 2 & 2 \\ -4 & 1 & -5 \end{pmatrix}$

5. $\begin{pmatrix} 4 & 2 & 0 \\ -1 & 1 & 0 \\ 2 & 2 & 2 \end{pmatrix}$

6. $\begin{pmatrix} 9 & 6 & -6 \\ -4 & -2 & 5 \\ 8 & 7 & -4 \end{pmatrix}$

7. $\begin{pmatrix} -1 & 4 & -1 \\ 0 & -3 & -7 \\ 0 & 0 & 4 \end{pmatrix}$

8. $\begin{pmatrix} 4 & -5 & 5 \\ 0 & -6 & 10 \\ 0 & -5 & 9 \end{pmatrix}$

9. $\begin{pmatrix} 1 & 1 & 2 \\ -10 & -6 & -8 \\ 2 & 1 & 0 \end{pmatrix}$

10. $\begin{pmatrix} 3 & 0 & 0 \\ -3 & 6 & -3 \\ -6 & 6 & -3 \end{pmatrix}$

11. $\begin{pmatrix} 8 & -4 & 4 \\ 2 & -2 & 0 \\ -10 & 4 & -6 \end{pmatrix}$

12. $\begin{pmatrix} -3 & 0 & 0 \\ 2 & -1 & -1 \\ 3 & 2 & -4 \end{pmatrix}$

13. $\begin{pmatrix} -3 & 0 & 0 \\ 1 & -2 & 0 \\ -9 & -4 & 2 \end{pmatrix}$

14. $\begin{pmatrix} -2 & -10 & 8 \\ 0 & 3 & -4 \\ 0 & 0 & -1 \end{pmatrix}$

15. $\begin{pmatrix} -1 & 1 & -2 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$

16. $\begin{pmatrix} 4 & 0 & 0 \\ 10 & -1 & 0 \\ -10 & 0 & -1 \end{pmatrix}$

17. $\begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$

18. $\begin{pmatrix} -2 & 0 & -1 \\ 0 & -2 & 0 \\ 0 & 1 & -3 \end{pmatrix}$

19. $\begin{pmatrix} 1 & 2 & 2 \\ 1 & 0 & 2 \\ -2 & -2 & -4 \end{pmatrix}$

20. $\begin{pmatrix} -3 & -4 & -8 \\ 0 & -1 & 4 \\ 0 & 0 & -3 \end{pmatrix}$

21. $\begin{pmatrix} 2 & 0 & 4 \\ -1 & 2 & -9 \\ 0 & 0 & -2 \end{pmatrix}$

22. $\begin{pmatrix} 3 & 0 & 0 \\ -3 & -2 & -1 \\ 12 & 0 & -3 \end{pmatrix}$

23. $\begin{pmatrix} 3 & 0 & 0 \\ -6 & 0 & 0 \\ -12 & 0 & -3 \end{pmatrix}$

24. $\begin{pmatrix} -1 & 0 & -2 \\ 0 & -3 & -1 \\ 1 & 0 & -4 \end{pmatrix}$

25. $\begin{pmatrix} -1 & 0 & 0 \\ -1 & -2 & 0 \\ -6 & -6 & 4 \end{pmatrix}$

26. $\begin{pmatrix} -3 & 3 & 0 \\ 0 & 0 & 0 \\ 0 & 6 & 3 \end{pmatrix}$

27. $\begin{pmatrix} -2 & 0 & 1 \\ -1 & -1 & 2 \\ 0 & 0 & -2 \end{pmatrix}$

28. $\begin{pmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 2 & 0 & -1 \end{pmatrix}$

29. $\begin{pmatrix} 4 & -3 & 5 \\ -4 & 2 & -4 \\ -4 & 3 & -5 \end{pmatrix}$

30. $\begin{pmatrix} 5 & -1 & -2 \\ 0 & 3 & 0 \\ 2 & -1 & 1 \end{pmatrix}$

3. Further problems.

1. **(Powers.)** For $A = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix}$ find A^4 by diagonalization.
2. **(Powers.)** For $A = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$ find A^3 by diagonalization.
3. **(Powers.)** For $A = \begin{pmatrix} 2 & 0 \\ -1 & 1 \end{pmatrix}$ find A^4 by diagonalization.
4. **(Powers.)** For $A = \begin{pmatrix} 3 & -8 \\ 0 & -1 \end{pmatrix}$ find A^4 by diagonalization.
5. **(Powers.)** For $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ find A^4 by diagonalization.
6. **(Powers.)** For $A = \begin{pmatrix} 7 & -4 \\ 8 & -5 \end{pmatrix}$ find A^5 by diagonalization.
7. **(Fibonacci numbers.)** Find the eigenvalues of $F = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and explain why the entries of F^n are Fibonacci numbers.
8. **(A non-diagonalizable matrix.)** Why is $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ not diagonalizable?
9. **(Symmetric matrix.)** Reduce $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ to diagonal form by an orthogonal matrix.
10. **(Trace and determinant.)** Verify the relations $\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n$ and $\det A = \lambda_1 \cdots \lambda_n$ for the diagonalizable matrix $\left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$.

Chapter 5

Quadratic Forms, Second-Order Curves and Iterative Methods

5.1 Quadratic forms and their reduction to canonical form

Definition 5.1.1. A **quadratic form** in n variables is a homogeneous polynomial of degree two,

$$\varphi(x_1, \dots, x_n) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j, \qquad a_{ij} = a_{ji}.$$

The symmetric matrix $A = (a_{ij})$ is called the **matrix of the form**; in matrix notation $\varphi = X^T A X$.

Remark 5.1.1. When writing down the matrix of a form, put the coefficient of x_i^2 in the diagonal position (i, i), and **half** the coefficient of $x_i x_j$ in each of the two symmetric positions $a_{ij} = a_{ji}$.

Example 5.1.1. The matrix of $\varphi = x_1^2 + 2x_2^2 - 3x_3^2 + 4x_1x_2 - 6x_2x_3$ is $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 2 & -3 \\ 0 & -3 & -3 \end{pmatrix}$.

5.1.1 Canonical form

Definition 5.1.2. A quadratic form that contains only the squares of the variables, $\varphi = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$, is said to be in **canonical form**.

Theorem 5.1.1 (Reduction by an orthogonal transformation)**.** *Every quadratic form can be reduced to canonical form by an orthogonal change of variables $X = QY$. The coefficients λ_i are then the eigenvalues of A, and the columns of Q are the corresponding normalized eigenvectors.*

Example 5.1.2. Reduce $\varphi = 2x^2 + 4xy + 5y^2$ to canonical form.

Solution. The matrix of the form is $A = \begin{pmatrix} 2 & 2 \\ 2 & 5 \end{pmatrix}$, and its characteristic equation $\lambda^2 - 7\lambda + 6 = 0$ has the roots $\lambda_1 = 1$ and $\lambda_2 = 6$. The corresponding eigenvectors are $(2; -1)$ and $(1; 2)$. After normalization they give the change of variables

$$x = \frac{2y_1 + y_2}{\sqrt{5}}, \qquad y = \frac{-y_1 + 2y_2}{\sqrt{5}},$$

which transforms the form into $\varphi = y_1^2 + 6y_2^2$.

Lagrange's method (completing the square). Instead of computing eigenvalues, one may group the terms containing x_1 into a perfect square, then do the same with x_2 in the remainder, and so on.

Example 5.1.3. For $\varphi = x_1^2 + 4x_1x_2 + 5x_2^2 - 2x_2x_3 + 2x_3^2$ we obtain

$$\varphi = (x_1 + 2x_2)^2 + x_2^2 - 2x_2x_3 + 2x_3^2 = (x_1 + 2x_2)^2 + (x_2 - x_3)^2 + x_3^2.$$

Setting $y_1 = x_1 + 2x_2$, $y_2 = x_2 - x_3$, $y_3 = x_3$, we arrive at $\varphi = y_1^2 + y_2^2 + y_3^2$. Note that this change of variables is not orthogonal, so the coefficients need not be the eigenvalues of A.

5.1.2 Definite forms. Sylvester's criterion

Definition 5.1.3. A quadratic form is called **positive definite** if $\varphi(X) > 0$ for all $X \neq 0$, and **negative definite** if $\varphi(X) < 0$ for all $X \neq 0$. A form that takes both positive and negative values is called **indefinite**.

Theorem 5.1.2 (Sylvester's criterion)**.** *A quadratic form is positive definite if and only if all leading principal minors of its matrix are positive:*

$$\Delta_1 = a_{11} > 0, \quad \Delta_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0, \quad \ldots, \quad \Delta_n = \det A > 0.$$

It is negative definite if and only if the signs of the minors alternate, starting with a negative one: $\Delta_1 < 0$, $\Delta_2 > 0$, $\Delta_3 < 0, \ldots$

Example 5.1.4. For $\varphi = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2$ the matrix is $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, and $\Delta_1 = 1 > 0, \Delta_2 = 1 > 0, \Delta_3 = 3 > 0$. Hence the form is positive definite.

5.1.3 Applications of quadratic forms in information security

The Mahalanobis distance. In anomaly detection, the "statistical" distance of an observation x from the normal profile μ is $d^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$, where Σ is the covariance matrix; this is a positive definite quadratic form. Reducing it to canonical form (which is what PCA does) passes to new coordinates in which the features are uncorrelated.

Lattice-based cryptography. For a lattice basis B the Gram matrix $G = B^T B$ defines a positive definite quadratic form, $\|Bk\|^2 = k^T G k$. The shortest vector problem asks for the minimum of this form over nonzero integer vectors k.

Optimization and risk assessment. In assessing the risk of a portfolio or a system one minimizes the quadratic form $w^T \Sigma w$; its positive definiteness guarantees that the minimum is unique.

5.1.4 Computing with Python, Maple and Mathcad

We check the form $\varphi = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2$ of the last example in two ways: by Sylvester's criterion and by the eigenvalues of its matrix, which are the coefficients of the canonical form. NumPy's `eigh` is designed for symmetric matrices.

Python

```
import numpy as np
A = np.array([[1, 1, 0], [1, 2, 0], [0, 0, 3]])
minors = [np.linalg.det(A[:k, :k]) for k in (1, 2, 3)]
print(np.round(minors, 6))
w, Q = np.linalg.eigh(A)
print(np.round(w, 4))
```

Output

```
[1. 1. 3.]
[0.382 2.618 3.   ]
```

All minors and all eigenvalues are positive, so the canonical form is $\varphi = \frac{3-\sqrt{5}}{2} y_1^2 + \frac{3+\sqrt{5}}{2} y_2^2 + 3y_3^2$ with positive coefficients.

Maple's `LinearAlgebra` package can test definiteness directly with `IsDefinite`; the minors are obtained with `SubMatrix`.

Maple

```
with(LinearAlgebra):
A := Matrix([[1, 1, 0], [1, 2, 0], [0, 0, 3]]):
seq(Determinant(SubMatrix(A, 1..k, 1..k)), k = 1..3);
                          1, 1, 3
IsDefinite(A, query = 'positive_definite');
                            true
```

In Mathcad Prime (with the default `ORIGIN = 0`) the leading minors are determinants of `submatrix(A, 0, k, 0, k)`, and the coefficients of the canonical form are given by `eigenvals(A)`.

Mathcad

```
A := | 1  1  0 |
     | 1  2  0 |
     | 0  0  3 |
|submatrix(A, 0, 1, 0, 1)| = 1        |A| = 3
eigenvals(A) = | 0.382 |
               | 2.618 |
               | 3     |
```

5.1.5 Exercises on quadratic forms

1. Write down the matrix of the quadratic form.

1. $\varphi = -2x_1^2 + 4x_1x_3 + 2x_2^2 + 4x_2x_3$
2. $\varphi = 3x_1^2 + 6x_1x_2 - 4x_1x_3 + 3x_2^2 + 6x_2x_3 + 4x_3^2$
3. $\varphi = -2x_1^2 + 2x_1x_2 + 2x_1x_3 - x_2^2 + 6x_2x_3 - 4x_3^2$
4. $\varphi = -4x_1^2 - 2x_1x_2 + 2x_1x_3 + 2x_2^2 - 6x_2x_3 - 2x_3^2$
5. $\varphi = -x_1^2 - 4x_1x_2 + 4x_1x_3 - 4x_2^2 + 2x_3^2$
6. $\varphi = 3x_1^2 - 6x_1x_2 - 2x_1x_3 + 4x_2^2 - 3x_3^2$
7. $\varphi = x_1^2 - 6x_1x_2 - 4x_1x_3 - 3x_2^2 - 6x_2x_3 - 4x_3^2$
8. $\varphi = 3x_1^2 - 4x_1x_2 + 2x_1x_3 + x_2^2 + 3x_3^2$
9. $\varphi = -2x_1^2 - 4x_1x_2 + 4x_1x_3 - 4x_2^2 - x_3^2$
10. $\varphi = -3x_1^2 + 6x_1x_2 - 4x_1x_3 - x_2^2 + 4x_2x_3 - 3x_3^2$
11. $\varphi = -3x_1^2 + 2x_1x_2 + 4x_2^2 + 6x_2x_3 + 2x_3^2$
12. $\varphi = -x_1^2 - 2x_1x_2 - 4x_1x_3 + 4x_2^2$
13. $\varphi = x_1^2 + 2x_1x_3 + 2x_2^2 - 2x_2x_3 + 3x_3^2$
14. $\varphi = 2x_1^2 - 2x_1x_2 - 6x_1x_3 - 3x_2^2 - 4x_2x_3 + 3x_3^2$
15. $\varphi = -4x_1^2 - 6x_1x_3 + 4x_2^2 + 2x_2x_3 + 4x_3^2$
16. $\varphi = -x_1^2 + 2x_1x_2 + 2x_1x_3 + x_2^2 + 2x_2x_3 - 3x_3^2$
17. $\varphi = -2x_1^2 + 4x_1x_2 - 4x_1x_3 + x_2^2 + 6x_2x_3 + x_3^2$
18. $\varphi = x_1^2 + 6x_1x_2 + 2x_1x_3 - 4x_2x_3 - 3x_3^2$
19. $\varphi = -2x_1^2 - 6x_1x_2 + 6x_1x_3 - 2x_2^2 + 6x_2x_3 + 2x_3^2$
20. $\varphi = x_1^2 - 4x_1x_2 + 4x_1x_3 - 4x_2^2 + 4x_2x_3 + 4x_3^2$
21. $\varphi = -x_1^2 - 4x_1x_3 - 4x_2^2 - 2x_2x_3 + 3x_3^2$
22. $\varphi = 4x_1^2 + 4x_1x_3 - x_2^2 - 4x_2x_3 - 2x_3^2$
23. $\varphi = 4x_1^2 - 6x_1x_2 + 2x_2^2 + 6x_2x_3 + 3x_3^2$
24. $\varphi = -4x_1^2 + 6x_1x_2 - 2x_1x_3 - 4x_2^2 + 6x_2x_3$
25. $\varphi = -6x_1x_2 - x_2^2 - 4x_2x_3 + 2x_3^2$
26. $\varphi = -3x_1^2 + 6x_1x_2 - 2x_1x_3 - 2x_2^2 - 4x_2x_3 + 3x_3^2$
27. $\varphi = -4x_1^2 - 2x_1x_2 - 4x_1x_3 + 4x_2^2 + 4x_2x_3 + 3x_3^2$
28. $\varphi = -4x_1^2 - 4x_1x_2 + 6x_1x_3 + x_2^2 + 6x_2x_3 - 2x_3^2$
29. $\varphi = x_1^2 + 2x_1x_2 + 6x_1x_3 + 2x_2^2 + 3x_3^2$
30. $\varphi = -4x_1x_2 + 2x_1x_3 - 4x_2x_3 - 4x_3^2$

2. Reduce the quadratic form to canonical form by an orthogonal transformation and determine its type.

1. $\varphi = -4xy - 3y^2$
2. $\varphi = -4x^2 + 6xy + 4y^2$
3. $\varphi = 5x^2 + 2xy + 5y^2$
4. $\varphi = -3x^2 + 4xy$
5. $\varphi = -8xy$
6. $\varphi = 3x^2 - 4xy + 3y^2$
7. $\varphi = -4xy - 3y^2$
8. $\varphi = -4xy$
9. $\varphi = 2x^2 + 4xy + 5y^2$
10. $\varphi = 5x^2 - 4xy + 2y^2$
11. $\varphi = 2x^2 - 4xy - y^2$
12. $\varphi = 10xy$
13. $\varphi = -3x^2 + 4xy$
14. $\varphi = 6xy$
15. $\varphi = -4x^2 - 2xy - 4y^2$
16. $\varphi = 2x^2 + 4xy - y^2$
17. $\varphi = 3x^2 - 8xy - 3y^2$
18. $\varphi = 4x^2 - 8xy - 2y^2$
19. $\varphi = x^2 - 4xy + y^2$
20. $\varphi = -10xy$
21. $\varphi = x^2 + 4xy - 2y^2$
22. $\varphi = 5x^2 + 2xy + 5y^2$
23. $\varphi = 4x^2 + 8xy - 2y^2$
24. $\varphi = -2x^2 + 2xy - 2y^2$
25. $\varphi = 8xy$
26. $\varphi = 3x^2 + 4xy + 3y^2$
27. $\varphi = -4xy$
28. $\varphi = x^2 - 4xy - 2y^2$
29. $\varphi = 6xy$
30. $\varphi = 3x^2 - 4xy + 3y^2$

3. Further problems.

1. **(Sylvester's criterion.)** Determine whether the form $\varphi = x_1^2 - 2x_1x_2 - 2x_1x_3 + 5x_2^2 - 4x_2x_3 + x_3^2$ is positive definite.
2. **(Sylvester's criterion.)** Determine whether the form $\varphi = 2x_1^2 + 4x_1x_2 + 4x_1x_3 + x_2^2 - 2x_2x_3 + 2x_3^2$ is positive definite.
3. **(Sylvester's criterion.)** Determine whether the form $\varphi = 5x_1^2 - 4x_1x_2 + 2x_1x_3 + 3x_2^2 + 3x_3^2$ is positive definite.

4. **(Sylvester's criterion.)** Determine whether the form $\varphi = 4x_1^2 - 4x_1x_2 + 4x_1x_3 + 4x_2^2 - 4x_2x_3 + 4x_3^2$ is positive definite.

5. **(Sylvester's criterion.)** Determine whether the form $\varphi = 5x_1^2 + 2x_1x_2 - 2x_1x_3 + x_2^2 - 4x_2x_3 + 3x_3^2$ is positive definite.

6. **(Sylvester's criterion.)** Determine whether the form $\varphi = 3x_1^2 + 4x_1x_2 - 2x_1x_3 + 3x_2^2 + 3x_3^2$ is positive definite.

7. **(Sylvester's criterion.)** Determine whether the form $\varphi = 3x_1^2 + 2x_1x_2 + x_2^2 - 4x_2x_3 + 3x_3^2$ is positive definite.

8. **(Sylvester's criterion.)** Determine whether the form $\varphi = 4x_1^2 - 2x_1x_3 + 2x_2^2 + 3x_3^2$ is positive definite.

9. **(Sylvester's criterion.)** Determine whether the form $\varphi = 3x_1^2 + 2x_1x_2 - 2x_1x_3 + x_2^2 + 4x_3^2$ is positive definite.

10. **(Sylvester's criterion.)** Determine whether the form $\varphi = x_1^2 + 2x_1x_2 - 2x_1x_3 + 5x_2^2 - 2x_3^2$ is positive definite.

5.2 Reducing a second-order equation to canonical form

Definition 5.2.1. The general equation of a second-order curve in the plane is

$$a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + 2a_{13}x + 2a_{23}y + a_{33} = 0.$$

Its **invariants**, that is, the quantities that do not change under rotations and translations of the coordinate system, are

$$I = a_{11} + a_{22}, \qquad \delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{vmatrix}, \qquad \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{vmatrix}.$$

Classification by type. If $\Delta \neq 0$, the curve is an ellipse when $\delta > 0$ (an imaginary ellipse if, in addition, $I \cdot \Delta > 0$), a hyperbola when $\delta < 0$, and a parabola when $\delta = 0$. If $\Delta = 0$, the curve is degenerate: a pair of lines (intersecting, parallel, coincident or imaginary) or a single point.

5.2.1 Translating the origin (completing the square)

When the equation contains no xy term, we complete the squares in x and in y and move the origin to the new centre (or vertex) by putting $X = x - x_0$, $Y = y - y_0$.

Example 5.2.1. Reduce $x^2 + 4y^2 - 4x + 8y - 8 = 0$ to canonical form.

Solution. Grouping the terms, we get $(x^2 - 4x + 4) + 4(y^2 + 2y + 1) = 8 + 4 + 4$, that is, $(x-2)^2 + 4(y+1)^2 = 16$. Dividing by 16 gives

$$\frac{X^2}{16} + \frac{Y^2}{4} = 1, \qquad X = x - 2,\ Y = y + 1.$$

This is an ellipse with centre $O'(2;-1)$ and semi-axes $a = 4$, $b = 2$.

Example 5.2.2. For $y^2 - 6y - 4x + 17 = 0$ we have $(y-3)^2 = 4x - 8 = 4(x-2)$, that is, $Y^2 = 4X$. The curve is a parabola with vertex $O'(2;3)$ and parameter $p = 2$.

5.2.2 Rotating the coordinate axes

If the equation contains an xy term, we rotate the axes,

$$x = x'\cos\alpha - y'\sin\alpha, \qquad y = x'\sin\alpha + y'\cos\alpha, \qquad \operatorname{ctg} 2\alpha = \frac{a_{11} - a_{22}}{2a_{12}}.$$

After the rotation the quadratic part becomes $\lambda_1 x'^2 + \lambda_2 y'^2$, where $\lambda_{1,2}$ are the eigenvalues of the matrix $\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$ (see the topic on quadratic forms).

Example 5.2.3. Determine the type of the curve $5x^2 + 4xy + 2y^2 = 6$ and reduce its equation to canonical form.

Solution. The matrix of the quadratic part is $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$, and its characteristic equation $\lambda^2 - 7\lambda + 6 = 0$ has the roots $\lambda_1 = 6$, $\lambda_2 = 1$.

The rotation angle is found from $\operatorname{ctg} 2\alpha = \frac{5-2}{4} = \frac{3}{4}$; hence $\cos 2\alpha = \frac{3}{5}$, $\cos\alpha = \frac{2}{\sqrt{5}}$, $\sin\alpha = \frac{1}{\sqrt{5}}$. In the new axes the equation reads $6x'^2 + y'^2 = 6$, or

$$x'^2 + \frac{y'^2}{6} = 1,$$

which is an ellipse, as expected from $\delta = 10 - 4 = 6 > 0$.

Example 5.2.4. For $x^2 - 4xy + y^2 = 3$ the characteristic equation is $\lambda^2 - 2\lambda - 3 = 0$, so $\lambda_1 = 3$, $\lambda_2 = -1$. The equation becomes $3x'^2 - y'^2 = 3$, that is, $x'^2 - \frac{y'^2}{3} = 1$. This is a hyperbola ($\delta = 1 - 4 = -3 < 0$).

5.2.3 Second-order curves in information security

Elliptic curve cryptography (ECC). Modern public-key systems (TLS, cryptocurrencies, digital signatures) use the group of points of a *third*-order curve $y^2 = x^3 + ax + b \pmod p$. A similar group can be built on second-order curves (conics), but there the discrete logarithm problem is much easier to solve, and for this reason elliptic curves are preferred in practice.

Example 5.2.5. On the curve $y^2 = x^3 + 2x + 3 \pmod 7$ take $x = 2$. Then $x^3 + 2x + 3 = 15 \equiv 1 \pmod 7$, and $y^2 \equiv 1$ gives $y = 1$ or $y = 6$. Thus the points $(2;1)$ and $(2;6)$ lie on the curve.

Ellipsoidal confidence regions. In anomaly detection the region of "normal" observations is described by an ellipse (ellipsoid) $(x-\mu)^T\Sigma^{-1}(x-\mu) \le c$. Reducing this equation to canonical form gives the principal axes and the semi-axes of the ellipse in terms of the eigenvectors and eigenvalues of Σ.

5.2.4 Computing with Python, Maple and Mathcad

For the curve $5x^2 + 4xy + 2y^2 - 6 = 0$ of the example above we compute the three invariants and the eigenvalues of the quadratic part. NumPy's `eigh`

returns the eigenvalues of a symmetric matrix in increasing order together with an orthonormal set of eigenvectors (the columns of `Q`, determined up to sign), which give the directions of the new axes:

Python

```
import numpy as np
# 5x^2 + 4xy + 2y^2 - 6 = 0
M = np.array([[5, 2, 0],
              [2, 2, 0],
              [0, 0, -6]], dtype=float)
A = M[:2, :2]
print("I =", np.trace(A))
print("delta =", round(np.linalg.det(A), 6))
print("Delta =", round(np.linalg.det(M), 6))
lam, Q = np.linalg.eigh(A)    # ascending order
print("eigenvalues:", lam)
print("Q =\n", Q.round(4))
```

Output

```
I = 7.0
delta = 6.0
Delta = -36.0
eigenvalues: [1. 6.]
Q =
 [[ 0.4472 -0.8944]
 [-0.8944 -0.4472]]
```

Since $\delta > 0$ and $I \cdot \Delta = -252 < 0$, the curve is a real ellipse, and the eigenvalues 1 and 6 are the coefficients of the canonical equation $6x'^2 + y'^2 = 6$. In Maple the same quantities are obtained exactly:

Maple

```
with(LinearAlgebra):
M := Matrix([[5, 2, 0], [2, 2, 0], [0, 0, -6]]):
A := SubMatrix(M, 1..2, 1..2):
Trace(A), Determinant(A), Determinant(M);
                          # 7, 6, -36
Eigenvectors(A);          # eigenvalues 6, 1 and
                          # vectors along (2,1), (-1,2)
```

In Mathcad the built-in functions `tr`, `eigenvals` and `eigenvecs` do the same job (`eigenvecs` returns normalized eigenvectors as columns):

Mathcad

```
     | 5  2  0 |
M := | 2  2  0 |        A := submatrix(M, 0, 1, 0, 1)
     | 0  0 -6 |
```

```
tr(A) = 7       |A| = 6       |M| = -36

eigenvals(A) = | 1 |
               | 6 |
```

5.2.5 Exercises on second-order curves

1. Reduce the equation of the curve to canonical form, state its type and find the new centre (vertex) O' ($X = x - x_0$, $Y = y - y_0$).

1. $16x^2 + 64x + 4y^2 + 8y + 4 = 0$
2. $x^2 + 4x - 16y^2 - 96y - 156 = 0$
3. $4x + y^2 + 4 = 0$
4. $x^2 + 4x + 25y^2 - 200y + 379 = 0$
5. $16x^2 + 128x - 16y^2 = 0$
6. $-6x + y^2 + 4y - 2 = 0$
7. $16x^2 - 32x + 9y^2 - 128 = 0$
8. $9x^2 - 9y^2 + 72y - 225 = 0$
9. $-6x + y^2 + 4y + 4 = 0$
10. $9x^2 - 72x + 25y^2 + 200y + 319 = 0$
11. $16x^2 - 128x - 25y^2 - 144 = 0$
12. $4x + y^2 + 8 = 0$
13. $16x^2 + 128x + 25y^2 - 144 = 0$
14. $9x^2 + 54x - 25y^2 - 150y - 369 = 0$
15. $-4x + y^2 - 4y - 8 = 0$
16. $x^2 - 8x + 25y^2 + 50y + 16 = 0$
17. $4x^2 + 24x - 9y^2 + 54y - 81 = 0$
18. $-4x + y^2 - 8y = 0$

19. $16x^2 - 32x + 9y^2 - 18y - 119 = 0$

20. $4x^2 + 32x - 9y^2 - 54y - 53 = 0$

21. $-2x + y^2 + 8y + 10 = 0$

22. $x^2 + 4x + 4y^2 - 24y + 36 = 0$

23. $9x^2 - 54x - 9y^2 + 72y - 144 = 0$

24. $-2x + y^2 - 6y + 15 = 0$

25. $9x^2 - 72x + 4y^2 - 8y + 112 = 0$

26. $9x^2 + 54x - 25y^2 - 100y - 244 = 0$

27. $-6x + y^2 + 8y + 4 = 0$

28. $9x^2 + 54x + 4y^2 + 16y + 61 = 0$

29. $16x^2 + 96x - 9y^2 + 54y - 81 = 0$

30. $-4x + y^2 + 6y + 13 = 0$

2. Eliminate the xy term by a rotation of the axes; find the canonical equation and the type of the curve.

1. $3x^2 - 4xy + 3y^2 = 20$

2. $5x^2 - 4xy + 2y^2 = 8$

3. $-10xy = 10$

4. $-3x^2 + 4xy - 3y^2 = 6$

5. $2x^2 - 4xy + 5y^2 = 20$

6. $-2x^2 - 4xy + y^2 = 6$

7. $4xy - 3y^2 = 20$

8. $-x^2 - 4xy + 2y^2 = 6$

9. $-6xy = 12$

10. $-3x^2 - 8xy + 3y^2 = 4$

11. $8xy = 6$

12. $2x^2 + 4xy - y^2 = 8$

13. $2x^2 - 6xy + 2y^2 = 6$

14. $-3x^2 - 2xy - 3y^2 = 6$

15. $x^2 + 4xy + y^2 = 10$

16. $-3x^2 + 4xy - 3y^2 = 6$

17. $-4x^2 - 2xy - 4y^2 = 12$

18. $x^2 + 4xy + y^2 = 12$

19. $5x^2 - 4xy + 2y^2 = 12$
20. $-2xy = 6$
21. $-x^2 - 4xy - y^2 = 8$
22. $-3x^2 - 4xy - 3y^2 = 6$
23. $x^2 - 4xy - 2y^2 = 6$
24. $4x^2 - 8xy - 2y^2 = 8$
25. $-x^2 + 6xy - y^2 = 6$
26. $-2x^2 - 8xy + 4y^2 = 12$
27. $-3x^2 - 8xy + 3y^2 = 4$
28. $2x^2 + 2xy + 2y^2 = 4$
29. $-4xy - 3y^2 = 12$
30. $2x^2 + 6xy + 2y^2 = 4$

3. Applied problems (10 problems).

1. **(Invariants.)** Use the invariants δ and Δ to determine the type of the curve $5x^2 + 4xy + 8y^2 - 32x - 56y + 80 = 0$.
2. **(Centre.)** Find the centre of the curve $x^2 - 4xy + y^2 - 2x + 4y + 1 = 0$.
3. **(Rotation angle.)** Find the rotation angle that eliminates the xy term in $3x^2 + 2\sqrt{3}\,xy + y^2 = 8$.
4. **(Circle.)** Find the centre and the radius of the circle $x^2+y^2-6x+4y-12 = 0$.
5. **(Parameters of an ellipse.)** Find the foci and the eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$.
6. **(Asymptotes of a hyperbola.)** Find the asymptotes and the eccentricity of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.
7. **(Parabola.)** Find the focus and the directrix of the parabola $y^2 = 8x$.
8. **(Intersecting lines.)** What curve is defined by the equation $x^2 - y^2 = 0$?
9. **(Elliptic curve.)** On the curve $y^2 = x^3 + 2x + 3 \pmod 7$, used in cryptography, find the ordinates of the point $P(2;\ ?)$.
10. **(Classification by type.)** Determine the type of the curve $x^2 + 2xy + y^2 - 4x + 4y = 0$.

5.3 The iterative method for the approximate solution of systems of linear algebraic equations

For large systems the direct methods (Cramer's rule, Gaussian elimination) require a very large number of operations, and rounding errors accumulate. In such cases one uses **iterative methods**, which produce the solution as the limit of a sequence of successive approximations.

5.3.1 The simple iteration (Jacobi) method

Definition 5.3.1. Solving the i-th equation of the system $Ax = b$ for x_i (we assume $a_{ii} \neq 0$), we bring the system to the form $x = Bx + c$:

$$x_i = \frac{1}{a_{ii}}\Big(b_i - \sum_{j\neq i} a_{ij}x_j\Big), \qquad b_{ij} = -\frac{a_{ij}}{a_{ii}}\ (j \neq i), \quad b_{ii} = 0, \quad c_i = \frac{b_i}{a_{ii}}.$$

We choose an initial approximation $x^{(0)}$ (usually $x^{(0)} = c$ or $x^{(0)} = 0$) and compute the successive approximations by the formula

$$x^{(k+1)} = Bx^{(k)} + c, \qquad k = 0, 1, 2, \ldots$$

Theorem 5.3.1 (A sufficient condition for convergence)**.** *If $\|B\| < 1$ (for instance, $\|B\|_\infty = \max\limits_i \sum\limits_j |b_{ij}| < 1$), then the iteration converges to the unique solution x^* for every choice of $x^{(0)}$, and*

$$\|x^{(k)} - x^*\| \leq \frac{\|B\|^k}{1 - \|B\|}\,\|x^{(1)} - x^{(0)}\|.$$

*In particular, the method converges whenever A is **diagonally dominant**, that is, $|a_{ii}| > \sum\limits_{j\neq i} |a_{ij}|$ in every row.*

Example 5.3.1. Solve the system $\begin{cases} 10x_1 + x_2 + x_3 = 12, \\ 2x_1 + 10x_2 + x_3 = 13, \\ 2x_1 + 2x_2 + 10x_3 = 14 \end{cases}$ by the simple iteration method.

Solution. The matrix is diagonally dominant ($10 > 2$, $10 > 3$, $10 > 4$). Solving each equation for its diagonal unknown, we obtain

$$\begin{cases} x_1 = 1.2 - 0.1x_2 - 0.1x_3, \\ x_2 = 1.3 - 0.2x_1 - 0.1x_3, \\ x_3 = 1.4 - 0.2x_1 - 0.2x_2, \end{cases} \qquad \|B\|_\infty = \max(0.2;\ 0.3;\ 0.4) = 0.4 < 1.$$

Starting from $x^{(0)} = (0; 0; 0)$, we get the following approximations:

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0	0	0
1	1.2	1.3	1.4
2	0.93	0.92	0.90
3	1.018	1.024	1.030

The approximations oscillate about the exact solution $x^* = (1; 1; 1)$; after three iterations the error does not exceed 0.03.

Remark 5.3.1. If the matrix is not diagonally dominant, it is advisable to bring the system to a diagonally dominant form by interchanging the equations or by replacing them with suitable linear combinations.

5.3.2 Iterative methods in information security

Large sparse systems. In cryptanalysis (for example, in the quadratic sieve and in index calculus methods) one meets sparse systems with hundreds of thousands of unknowns. Gaussian elimination is out of the question there, both in memory and in time, so iterative methods (Lanczos, Wiedemann and others) are used instead.

Reputation and PageRank. The trust scores of the nodes of a network form the solution of the equation $r = dPr + (1 - d)v$. It is found by the simple iteration $r^{(k+1)} = dPr^{(k)} + (1 - d)v$, which converges because $d < 1$.

Example 5.3.2. Let the trust scores in a network with three nodes satisfy $r_1 = 0.5r_2 + 0.2$, $r_2 = 0.5r_3 + 0.2$, $r_3 = 0.5r_1 + 0.2$. Starting from $r^{(0)} = (0;0;0)$, we get $r^{(1)} = (0.2;0.2;0.2)$, $r^{(2)} = (0.3;0.3;0.3)$, $r^{(3)} = (0.35;0.35;0.35)$, …, while the exact solution is $r_i = 0.4$.

5.3.3 Computing with Python, Maple and Mathcad

The iteration of the example above takes only a few lines of NumPy. We form B and c from A and b, check the sufficient condition and perform three iterations:

Python

```
import numpy as np
A = np.array([[10, 1, 1], [2, 10, 1], [2, 2, 10]],
             dtype=float)
b = np.array([12, 13, 14], dtype=float)
d = np.diag(A)
B = -A / d[:, None]
np.fill_diagonal(B, 0)
c = b / d
print("||B||_inf =", np.abs(B).sum(axis=1).max())
x = np.zeros(3)
for k in range(1, 4):
    x = B @ x + c
    print(k, x.round(4))
```

Output

```
||B||_inf = 0.4
1 [1.2 1.3 1.4]
2 [0.93 0.92 0.9 ]
3 [1.018 1.024 1.03 ]
```

In Maple we write $B = E - D^{-1}A$ and $c = D^{-1}b$, where D is the diagonal part of A (the name `D` is reserved in Maple, so we use `D1`):

Maple

```
with(LinearAlgebra):
A := Matrix([[10, 1, 1], [2, 10, 1], [2, 2, 10]]):
b := Vector([12, 13, 14]):
D1 := DiagonalMatrix(Diagonal(A)):
B := IdentityMatrix(3) - D1^(-1) . A:
c := D1^(-1) . b:
Norm(B, infinity);              # 2/5
x := Vector(3):
for k to 3 do x := B . x + c end do:
evalf(x);                       # 1.018, 1.024, 1.030
```

In Mathcad (with `ORIGIN = 0`) the iteration can be written with a range variable: each column of the matrix X is one approximation. The function `lsolve` gives the exact solution for comparison, and `normi` is the ∞-norm:

Mathcad

```
     | 10  1  1 |        | 12 |
A := |  2 10  1 |   b := | 13 |
     |  2  2 10 |        | 14 |
i := 0..2     d_i := A_(i,i)     D := diag(d)
B := identity(3) - D^-1 * A      c := D^-1 * b
normi(B) = 0.4
X<0> := 0 * c     k := 0..2
X<k+1> := B * X<k> + c
          | 0   1.2   0.93   1.018 |
     X =  | 0   1.3   0.92   1.024 |
          | 0   1.4   0.90   1.030 |
lsolve(A, b) = (1 1 1)^T
```

5.3.4 Exercises on the simple iteration method

1. Perform two iterations of the simple iteration (Jacobi) method for the system, starting from $x^{(0)} = (0; 0; 0)$*, and compare the result with the exact solution.*

1. $\begin{cases} -8x_1 - x_2 + 3x_3 = 15 \\ -3x_1 + 9x_2 + x_3 = 15 \\ -x_1 - 4x_3 = 2 \end{cases}$

2. $\begin{cases} -11x_1 - 3x_2 - 3x_3 = -19 \\ 2x_1 + 6x_2 = -8 \\ -3x_1 + x_2 + 9x_3 = 1 \end{cases}$

3. $\begin{cases} -6x_1 + 2x_2 - x_3 = -6 \\ x_1 - 8x_2 + 3x_3 = -20 \\ -x_1 + x_2 - 8x_3 = 16 \end{cases}$

4. $\begin{cases} 8x_1 + 3x_2 - 2x_3 = -1 \\ x_1 + 9x_2 + 2x_3 = -2 \\ 3x_1 + x_2 + 9x_3 = 29 \end{cases}$

5. $\begin{cases} 9x_1 - x_2 - 3x_3 = -20 \\ -x_1 + 5x_2 + x_3 = -2 \\ -x_1 + 3x_2 + 10x_3 = 9 \end{cases}$

6. $\begin{cases} -9x_1 + 2x_2 + x_3 = 26 \\ -3x_1 + 9x_2 - 2x_3 = 29 \\ -3x_2 + 6x_3 = 3 \end{cases}$

7. $\begin{cases} -5x_1 + x_2 = -7 \\ 2x_1 + 6x_2 - x_3 = -13 \\ -3x_1 + 3x_2 - 12x_3 = -45 \end{cases}$

8. $\begin{cases} 8x_1 - x_2 - 2x_3 = -1 \\ -3x_1 + 10x_2 + x_3 = 29 \\ x_1 + 2x_2 + 6x_3 = 0 \end{cases}$

9. $\begin{cases} 8x_1 + 3x_3 = 15 \\ -2x_1 - 10x_2 + 2x_3 = -2 \\ 3x_1 - 6x_3 = 27 \end{cases}$

10. $\begin{cases} 7x_1 + 2x_2 - 2x_3 = 17 \\ -x_1 - 6x_2 + 2x_3 = 1 \\ 3x_1 + 9x_3 = 27 \end{cases}$

11. $\begin{cases} -8x_1 - 3x_2 - x_3 = 22 \\ 7x_2 + 3x_3 = 6 \\ -2x_1 + 2x_2 - 10x_3 = -14 \end{cases}$

12. $\begin{cases} -6x_1 + x_3 = -3 \\ 2x_1 + 8x_2 + 2x_3 = 18 \\ x_1 - 2x_2 + 9x_3 = -33 \end{cases}$

13. $\begin{cases} -6x_1 + x_3 = 2 \\ -6x_2 + 2x_3 = -14 \\ -x_1 - 4x_3 = -8 \end{cases}$

14. $\begin{cases} 9x_1 + 3x_2 + x_3 = -29 \\ -x_1 + 7x_2 - x_3 = -5 \\ 2x_1 - 3x_2 + 11x_3 = 8 \end{cases}$

15. $\begin{cases} 9x_1 - 3x_3 = -21 \\ 3x_1 + 10x_2 - x_3 = -17 \\ 3x_1 + 3x_2 + 11x_3 = 2 \end{cases}$

16. $\begin{cases} -9x_1 + x_2 + 3x_3 = -29 \\ 3x_1 + 9x_2 - 2x_3 = -6 \\ 2x_1 + x_2 + 6x_3 = -16 \end{cases}$

17. $\begin{cases} 6x_1 + x_2 + 2x_3 = -13 \\ -2x_1 + 8x_2 - x_3 = -20 \\ 2x_2 + 6x_3 = -18 \end{cases}$

18. $\begin{cases} -9x_1 - 2x_2 - 2x_3 = 3 \\ 8x_2 + 3x_3 = 9 \\ x_1 - 2x_2 - 7x_3 = -22 \end{cases}$

19. $\begin{cases} 10x_1 + 3x_2 - 3x_3 = -39 \\ -7x_2 - x_3 = 5 \\ 2x_1 - 2x_2 - 7x_3 = -18 \end{cases}$

20. $\begin{cases} 8x_1 + 3x_3 = -1 \\ -2x_1 - 6x_2 = -2 \\ -x_1 - 2x_2 - 8x_3 = 23 \end{cases}$

21. $\begin{cases} -6x_1 + x_2 + 2x_3 = 7 \\ -4x_2 + x_3 = 14 \\ -2x_1 - 8x_3 = -14 \end{cases}$

22. $\begin{cases} -4x_1 = 12 \\ -x_1 + 6x_2 + 2x_3 = 3 \\ 2x_1 + 2x_2 - 7x_3 = -6 \end{cases}$

23. $\begin{cases} 8x_1 - 2x_2 + 2x_3 = 2 \\ 3x_1 - 11x_2 - 3x_3 = -3 \\ x_1 - 3x_2 - 9x_3 = -9 \end{cases}$

24. $\begin{cases} 5x_1 + x_2 - x_3 = -13 \\ -x_1 + 7x_2 - 3x_3 = 21 \\ -2x_1 + 3x_2 - 10x_3 = 5 \end{cases}$

25. $\begin{cases} -11x_1 + 3x_2 - 3x_3 = -7 \\ 3x_1 - 8x_2 + 2x_3 = -22 \\ 2x_2 + 5x_3 = -4 \end{cases}$

26. $\begin{cases} 11x_1 + 3x_2 - 2x_3 = -34 \\ 8x_2 + 2x_3 = -10 \\ 2x_1 - 2x_2 + 7x_3 = 21 \end{cases}$

27. $\begin{cases} -6x_1 - x_2 + 2x_3 = -20 \\ x_1 - 7x_2 + x_3 = 15 \\ -3x_2 - 6x_3 = 18 \end{cases}$

28. $\begin{cases} 12x_1 + 3x_2 + 3x_3 = -3 \\ 3x_1 + 9x_2 - 2x_3 = 13 \\ -2x_1 - 2x_2 + 7x_3 = 5 \end{cases}$

29. $\begin{cases} -9x_1 + 3x_2 + x_3 = -21 \\ 2x_1 + 9x_2 + 3x_3 = 24 \\ 3x_1 - 2x_2 - 10x_3 = 33 \end{cases}$

30. $\begin{cases} 9x_1 - 2x_2 - 2x_3 = -27 \\ -x_1 + 4x_2 = 7 \\ -2x_2 + 6x_3 = -8 \end{cases}$

2. Check whether the sufficient condition for the convergence of the simple iteration method ($\|B\|_\infty < 1$) holds.

1. $A = \begin{pmatrix} -10 & -3 & 3 \\ -3 & -11 & 3 \\ -3 & -2 & -8 \end{pmatrix}$

2. $A = \begin{pmatrix} 10 & 3 & -3 \\ 0 & 5 & 2 \\ 0 & 0 & 5 \end{pmatrix}$

3. $A = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 2 & -2 \\ 4 & 0 & -2 \end{pmatrix}$

4. $A = \begin{pmatrix} -6 & 2 & -1 \\ 2 & -11 & 3 \\ -3 & -1 & -7 \end{pmatrix}$

5. $A = \begin{pmatrix} 11 & 3 & -3 \\ -1 & -8 & -3 \\ -3 & 1 & 9 \end{pmatrix}$

6. $A = \begin{pmatrix} 2 & 4 & -2 \\ 3 & 3 & 2 \\ 1 & 1 & 1 \end{pmatrix}$

7. $A = \begin{pmatrix} 4 & 0 & -1 \\ 1 & 6 & 2 \\ -3 & -3 & -11 \end{pmatrix}$

8. $A = \begin{pmatrix} -9 & -3 & 2 \\ -1 & -7 & -1 \\ 1 & 3 & 8 \end{pmatrix}$

9. $A = \begin{pmatrix} 3 & 3 & 4 \\ 2 & 2 & -2 \\ 3 & -4 & 2 \end{pmatrix}$

10. $A = \begin{pmatrix} 9 & 2 & -2 \\ 2 & -6 & -1 \\ 3 & 1 & -9 \end{pmatrix}$

11. $A = \begin{pmatrix} 10 & 3 & 1 \\ 0 & 9 & 3 \\ -2 & -3 & 10 \end{pmatrix}$

12. $A = \begin{pmatrix} 2 & -1 & -1 \\ -4 & 3 & 4 \\ -1 & 2 & 2 \end{pmatrix}$

13. $A = \begin{pmatrix} 11 & -3 & 3 \\ 1 & 7 & -3 \\ 2 & -3 & -10 \end{pmatrix}$

14. $A = \begin{pmatrix} 9 & -3 & 3 \\ -2 & 8 & -3 \\ 3 & -2 & 9 \end{pmatrix}$

15. $A = \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & 4 \\ -4 & -4 & -2 \end{pmatrix}$

16. $A = \begin{pmatrix} 9 & 3 & 1 \\ 0 & 8 & 3 \\ -3 & 1 & 10 \end{pmatrix}$

17. $A = \begin{pmatrix} -9 & -1 & 3 \\ 2 & 9 & -1 \\ 2 & 2 & 7 \end{pmatrix}$

18. $A = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 3 & -3 \\ 4 & -2 & 1 \end{pmatrix}$

19. $A = \begin{pmatrix} -6 & -3 & 0 \\ 1 & 7 & -3 \\ 0 & -1 & -5 \end{pmatrix}$

20. $A = \begin{pmatrix} 8 & 2 & 2 \\ 1 & -9 & -2 \\ -2 & -1 & 9 \end{pmatrix}$

21. $A = \begin{pmatrix} 3 & 2 & -1 \\ -1 & 2 & -2 \\ 3 & -4 & 2 \end{pmatrix}$

22. $A = \begin{pmatrix} 9 & 3 & 3 \\ 1 & -6 & 0 \\ 3 & 1 & -9 \end{pmatrix}$

23. $A = \begin{pmatrix} 8 & -1 & -2 \\ 2 & -8 & -1 \\ 3 & 3 & 10 \end{pmatrix}$

24. $A = \begin{pmatrix} 3 & -4 & 1 \\ -3 & 1 & -3 \\ 4 & -3 & -2 \end{pmatrix}$

25. $A = \begin{pmatrix} 8 & -1 & 1 \\ 2 & 9 & -2 \\ -1 & -1 & -5 \end{pmatrix}$

26. $A = \begin{pmatrix} -8 & -2 & -2 \\ 1 & 9 & 2 \\ 0 & 1 & -7 \end{pmatrix}$

27. $A = \begin{pmatrix} 1 & -4 & 3 \\ -3 & 3 & -3 \\ -4 & 0 & 2 \end{pmatrix}$

28. $A = \begin{pmatrix} -9 & 2 & -1 \\ 0 & -7 & 2 \\ 1 & 3 & 10 \end{pmatrix}$

29. $A = \begin{pmatrix} -5 & 0 & -2 \\ -2 & -10 & 2 \\ 2 & 0 & 6 \end{pmatrix}$

30. $A = \begin{pmatrix} 3 & -3 & -1 \\ -4 & -2 & 0 \\ 1 & -3 & 1 \end{pmatrix}$

3. Applied problems (10 problems).

1. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 8x_1 + 3x_2 = 25 \\ 3x_1 + 6x_2 = 24 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

2. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 8x_1 + 3x_2 = 27 \\ -2x_1 + 5x_2 = -1 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

3. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 8x_1 - 3x_2 = 22 \\ -2x_1 + 5x_2 = -14 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

4. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 5x_1 + x_2 = -23 \\ 3x_1 + 7x_2 = -33 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

5. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 7x_1 + 3x_2 = 19 \\ 3x_1 + 5x_2 = -3 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

6. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 9x_1 + x_2 = -6 \\ -x_1 + 6x_2 = 19 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

7. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 9x_1 - x_2 = -17 \\ -x_1 + 9x_2 = -7 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

8. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 7x_1 - 3x_2 = -6 \\ -2x_1 + 9x_2 = 18 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

9. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 5x_1 + 2x_2 = 18 \\ 3x_1 + 8x_2 = 4 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

10. **(Three iterations.)** Perform three iterations of the simple iteration method for the system $\begin{cases} 6x_1 + 2x_2 = 32 \\ -3x_1 + 5x_2 = 8 \end{cases}$, starting from $x^{(0)} = (0; 0)$, and compare the result with the exact solution.

5.4 The Seidel method for the approximate solution of systems of linear algebraic equations

Definition 5.4.1. The **Seidel (Gauss–Seidel) method** is a modification of the simple iteration method: when $x_i^{(k+1)}$ is computed, the values $x_1^{(k+1)}, \dots, x_{i-1}^{(k+1)}$ already found in the current iteration are used at once:

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\Big), \qquad i = 1, \dots, n.$$

Theorem 5.4.1. *If A is diagonally dominant, or if A is symmetric and positive definite, the Seidel method converges to the unique solution for every initial approximation.*

As a rule, the Seidel method converges faster than simple iteration, and only one vector has to be stored during the computation.

Example 5.4.1. Solve the system $\begin{cases} 10x_1 + x_2 + x_3 = 12, \\ 2x_1 + 10x_2 + x_3 = 13, \\ 2x_1 + 2x_2 + 10x_3 = 14 \end{cases}$ of the previous topic by the Seidel method.

Solution. We start from $x^{(0)} = (0; 0; 0)$. In the first iteration each new value is substituted immediately into the next equation:

$$x_1^{(1)} = \frac{12 - 0 - 0}{10} = 1.2, \quad x_2^{(1)} = \frac{13 - 2 \cdot 1.2 - 0}{10} = 1.06,$$
$$x_3^{(1)} = \frac{14 - 2 \cdot 1.2 - 2 \cdot 1.06}{10} = 0.948.$$

The second iteration uses $x^{(1)}$ in the same way:

$$x_1^{(2)} = \frac{12 - 1.06 - 0.948}{10} = 0.9992,$$
$$x_2^{(2)} = \frac{13 - 1.9984 - 0.948}{10} = 1.00536,$$
$$x_3^{(2)} = \frac{14 - 1.9984 - 2.01072}{10} = 0.99909.$$

Continuing, we obtain the table below; the last column gives the error with respect to the exact solution $x^* = (1; 1; 1)$.

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$	$\|x^{(k)} - x^*\|_\infty$
1	1.2	1.06	0.948	0.2
2	0.9992	1.00536	0.99909	0.0054
3	0.99956	1.00018	1.00005	0.00044

After three iterations the error of the Seidel method is about $4.4 \cdot 10^{-4}$, whereas simple iteration gave 0.03: the Seidel method converges considerably faster here.

Remark 5.4.1. The iteration is usually stopped as soon as $\|x^{(k+1)} - x^{(k)}\| < \varepsilon$, where ε is the prescribed accuracy.

5.4.1 The Seidel method in information security

Devices with limited resources. IoT devices and smart cards have very little memory. The Seidel method overwrites the old values with the new ones as it goes ("in-place" computation), which makes it convenient for approximate computations in embedded cryptographic modules.

Network models and traffic distribution. The flow balance equations between the nodes of a network are diagonally dominant: the load of each node exceeds the influence of its neighbours. Redistributing the load during a DDoS attack requires such systems to be solved again quickly, and the Seidel method suits this well, since the previous solution is a good initial approximation.

Iterative methods and side-channel attacks. If the number of iterations depends on the data, secret information may "leak" through the running time. For this reason cryptographic software should use a number of iterations that is fixed in advance.

5.4.2 Computing with Python, Maple and Mathcad

The Seidel iteration is easy to program, because the vector `x` is updated in place. The following code repeats the example above and prints the error of each approximation:

Python

```
import numpy as np
A = np.array([[10, 1, 1], [2, 10, 1], [2, 2, 10]],
```

```
            dtype=float)
b = np.array([12, 13, 14], dtype=float)
x = np.zeros(3)
for k in range(1, 4):
    for i in range(3):
        s = A[i] @ x - A[i, i] * x[i]
        x[i] = (b[i] - s) / A[i, i]
    err = np.abs(x - 1).max()
    print(k, x.round(5), f"{err:.1e}")
```

Output

```
1 [1.2   1.06  0.948] 2.0e-01
2 [0.9992  1.00536 0.99909] 5.4e-03
3 [0.99956 1.00018 1.00005] 4.4e-04
```

The same loop in Maple works with exact rational numbers, and `evalf` converts the result to decimals at the end:

Maple

```
with(LinearAlgebra):
A := Matrix([[10, 1, 1], [2, 10, 1], [2, 2, 10]]):
b := Vector([12, 13, 14]):
x := Vector(3):
for k to 3 do
  for i to 3 do
    x[i] := (b[i] - add(A[i, j]*x[j], j = 1..3)
             + A[i, i]*x[i]) / A[i, i];
  end do;
end do:
evalf(x);        # 0.9995552, 1.000180160, 1.000052928
```

In Mathcad (with `ORIGIN` = 0) the method is written as a small program; the operator ← is the local assignment of the programming toolbar:

Mathcad

```
Seidel(A, b, x, m) :=
    | for k in 1..m
    |   for i in 0..rows(A)-1
    |     s <- sum(A_(i,j)*x_j, j) - A_(i,i)*x_i
    |     x_i <- (b_i - s)/A_(i,i)
    | return x
x0 := (0 0 0)^T
Seidel(A, b, x0, 3) = (0.99956 1.00018 1.00005)^T
```

Here `A` and `b` are entered as in the previous topic, and the sum is entered with the summation operator over $j = 0, 1, 2$.

5.4.3 Exercises on the Seidel method

1. Perform two iterations of the Seidel method for the system, starting from $x^{(0)} = (0; 0; 0)$.

1. $\begin{cases} 7x_1 + 3x_3 = 27 \\ x_1 - 6x_2 - 2x_3 = -13 \\ -2x_1 - 2x_2 - 10x_3 = -30 \end{cases}$

2. $\begin{cases} -10x_1 - 3x_2 + x_3 = 20 \\ -2x_1 + 9x_2 + 2x_3 = -23 \\ 3x_1 + 8x_3 = 5 \end{cases}$

3. $\begin{cases} -6x_1 - x_2 + 2x_3 = 9 \\ -3x_1 + 8x_2 - x_3 = 9 \\ -x_1 + 2x_2 - 8x_3 = -13 \end{cases}$

4. $\begin{cases} -11x_1 - 2x_2 - 3x_3 = -11 \\ x_1 + 8x_2 + 3x_3 = 19 \\ -2x_1 + 2x_2 - 9x_3 = 22 \end{cases}$

5. $\begin{cases} 11x_1 + 3x_2 - 2x_3 = -9 \\ -3x_1 + 10x_2 - 2x_3 = -16 \\ -x_1 - x_2 - 5x_3 = -14 \end{cases}$

6. $\begin{cases} 9x_1 - x_2 + 3x_3 = 9 \\ -2x_1 + 6x_2 + x_3 = 3 \\ 3x_1 - 8x_3 = -24 \end{cases}$

7. $\begin{cases} -12x_1 - 3x_2 - 3x_3 = 21 \\ -3x_1 - 9x_2 + 2x_3 = 8 \\ 3x_2 - 8x_3 = 13 \end{cases}$

8. $\begin{cases} 11x_1 + 2x_2 - 3x_3 = -6 \\ 3x_1 - 11x_2 + 2x_3 = 4 \\ 3x_1 - 3x_2 - 11x_3 = -22 \end{cases}$

9. $\begin{cases} -5x_1 - x_2 - x_3 = 4 \\ 3x_1 + 10x_2 - 3x_3 = -14 \\ -3x_2 + 7x_3 = -8 \end{cases}$

10. $\begin{cases} -9x_1 - 3x_2 - 3x_3 = -12 \\ 3x_1 + 11x_2 - 3x_3 = -4 \\ -2x_1 - 3x_2 - 8x_3 = 24 \end{cases}$

11. $\begin{cases} -9x_1 + 3x_2 + x_3 = -7 \\ 3x_1 - 10x_2 + x_3 = -22 \\ 3x_1 + x_2 - 10x_3 = -11 \end{cases}$

12. $\begin{cases} -8x_1 - 2x_2 - x_3 = 24 \\ -3x_1 - 8x_2 + x_3 = 19 \\ -2x_1 - 2x_2 + 8x_3 = 24 \end{cases}$

13. $\begin{cases} -8x_1 + x_2 + x_3 = 17 \\ -2x_1 - 8x_2 = 20 \\ -3x_1 - 2x_2 + 8x_3 = 34 \end{cases}$

14. $\begin{cases} 10x_1 + 2x_2 + 3x_3 = -1 \\ -2x_1 - 8x_2 - 2x_3 = 12 \\ -x_1 - x_2 - 8x_3 = 24 \end{cases}$

15. $\begin{cases} 9x_1 - x_2 + 2x_3 = -3 \\ x_1 - 5x_2 - x_3 = -15 \\ -2x_1 - x_2 + 8x_3 = -3 \end{cases}$

16. $\begin{cases} 5x_1 = -5 \\ x_1 + 8x_2 - 2x_3 = -11 \\ -3x_1 - 3x_2 + 10x_3 = -21 \end{cases}$

17. $\begin{cases} -8x_1 + 3x_2 + x_3 = 18 \\ -2x_1 - 9x_2 - x_3 = -26 \\ x_1 + 2x_2 - 7x_3 = -2 \end{cases}$

18. $\begin{cases} 7x_1 + x_3 = -19 \\ -3x_1 - 10x_2 - 3x_3 = -7 \\ -2x_1 + 7x_3 = 20 \end{cases}$

19. $\begin{cases} 8x_1 - 3x_2 - x_3 = 27 \\ 2x_1 - 8x_2 - 2x_3 = 14 \\ 2x_1 + x_2 - 7x_3 = 5 \end{cases}$

20. $\begin{cases} 8x_1 + x_2 - 3x_3 = 6 \\ 6x_2 - x_3 = 5 \\ -x_1 - 2x_2 + 6x_3 = 3 \end{cases}$

21. $\begin{cases} 7x_1 + x_2 - 2x_3 = -8 \\ 2x_1 + 7x_2 = 10 \\ 3x_1 + 2x_2 - 8x_3 = 14 \end{cases}$

22. $\begin{cases} -7x_1 + 2x_2 - x_3 = -5 \\ -6x_2 + 2x_3 = 14 \\ -3x_1 - 2x_2 + 10x_3 = 14 \end{cases}$

23. $\begin{cases} -8x_1 + 2x_2 + 3x_3 = 9 \\ x_1 + 7x_2 + 2x_3 = -1 \\ 3x_1 + x_2 - 10x_3 = 25 \end{cases}$

24. $\begin{cases} -7x_1 + 2x_2 - 2x_3 = 18 \\ 4x_2 - x_3 = 5 \\ 2x_1 - 2x_2 - 9x_3 = 3 \end{cases}$

25. $\begin{cases} -9x_1 - 3x_2 + 3x_3 = -3 \\ -2x_1 + 7x_2 - 2x_3 = 7 \\ 3x_1 - 2x_2 - 11x_3 = -2 \end{cases}$

26. $\begin{cases} 6x_1 - x_2 - x_3 = 12 \\ 2x_1 - 6x_2 + x_3 = -9 \\ -3x_3 = -9 \end{cases}$

27. $\begin{cases} 9x_1 - 3x_2 + x_3 = -18 \\ 3x_1 + 9x_2 + 2x_3 = -21 \\ 3x_1 - 9x_3 = 21 \end{cases}$

28. $\begin{cases} -9x_1 + 3x_2 + x_3 = 25 \\ -x_1 - 8x_2 + 2x_3 = -26 \\ -x_1 + 2x_2 + 7x_3 = -6 \end{cases}$

29. $\begin{cases} -8x_1 - 2x_2 = 6 \\ 8x_2 + 3x_3 = -27 \\ x_1 - 3x_2 - 8x_3 = 17 \end{cases}$

30. $\begin{cases} 6x_1 - 2x_2 = -20 \\ 3x_1 + 10x_2 + 2x_3 = 3 \\ -x_1 - 2x_2 + 7x_3 = 8 \end{cases}$

2. Perform three iterations of the Seidel method for the 2×2 *system, starting from* $x^{(0)} = (0; 0)$.

1. $\begin{cases} 8x_1 + 2x_2 = -6 \\ -2x_1 + 5x_2 = 7 \end{cases}$

2. $\begin{cases} 8x_1 = 24 \\ -2x_1 + 8x_2 = -14 \end{cases}$

3. $\begin{cases} 7x_1 + 2x_2 = 1 \\ x_1 + 6x_2 = 23 \end{cases}$

4. $\begin{cases} 7x_1 - 3x_2 = 13 \\ 2x_1 + 4x_2 = -6 \end{cases}$

5. $\begin{cases} 5x_1 + 3x_2 = -13 \\ 5x_2 = -5 \end{cases}$

6. $\begin{cases} 6x_1 - x_2 = 18 \\ -x_1 + 4x_2 = -3 \end{cases}$

7. $\begin{cases} 6x_1 + 3x_2 = -6 \\ -2x_1 + 4x_2 = -18 \end{cases}$

8. $\begin{cases} 7x_1 + x_2 = 22 \\ 7x_2 = 7 \end{cases}$

9. $\begin{cases} 5x_1 - 2x_2 = 8 \\ 3x_1 + 8x_2 = -32 \end{cases}$

10. $\begin{cases} 7x_1 = 14 \\ 3x_1 + 4x_2 = 14 \end{cases}$

11. $\begin{cases} 4x_1 - 2x_2 = -16 \\ x_1 + 5x_2 = -4 \end{cases}$

12. $\begin{cases} 6x_1 - 3x_2 = -3 \\ 2x_1 + 5x_2 = -19 \end{cases}$

13. $\begin{cases} 5x_1 - 3x_2 = 4 \\ 3x_1 + 6x_2 = -21 \end{cases}$

14. $\begin{cases} 7x_1 - 3x_2 = -13 \\ 2x_1 + 8x_2 = 14 \end{cases}$

15. $\begin{cases} 5x_1 - x_2 = -1 \\ 9x_2 = 9 \end{cases}$

16. $\begin{cases} 9x_1 - 3x_2 = -21 \\ -3x_1 + 4x_2 = 10 \end{cases}$

17. $\begin{cases} 6x_1 = -12 \\ 5x_2 = -15 \end{cases}$

18. $\begin{cases} 6x_1 - 3x_2 = -12 \\ -x_1 + 6x_2 = 2 \end{cases}$

19. $\begin{cases} 6x_1 - 2x_2 = 14 \\ x_1 + 7x_2 = 17 \end{cases}$

20. $\begin{cases} 4x_1 + 3x_2 = -16 \\ -3x_1 + 6x_2 = 12 \end{cases}$

21. $\begin{cases} 4x_1 = -8 \\ 2x_1 + 9x_2 = -13 \end{cases}$

22. $\begin{cases} 9x_1 - 3x_2 = -6 \\ -2x_1 + 9x_2 = -32 \end{cases}$

23. $\begin{cases} 5x_1 + 3x_2 = 6 \\ -3x_1 + 8x_2 = -33 \end{cases}$

24. $\begin{cases} 8x_1 + x_2 = 4 \\ x_1 + 9x_2 = -35 \end{cases}$

25. $\begin{cases} 5x_1 - 3x_2 = -3 \\ -3x_1 + 5x_2 = -11 \end{cases}$

26. $\begin{cases} 4x_1 - x_2 = 7 \\ 3x_1 + 7x_2 = -18 \end{cases}$

27. $\begin{cases} 6x_1 + 3x_2 = -33 \\ 5x_2 = -15 \end{cases}$

28. $\begin{cases} 9x_1 = -36 \\ -3x_1 + 5x_2 = 12 \end{cases}$

29. $\begin{cases} 6x_1 + 3x_2 = 18 \\ 3x_1 + 8x_2 = 35 \end{cases}$

30. $\begin{cases} 5x_1 - 2x_2 = -3 \\ 4x_2 = -4 \end{cases}$

3. Applied problems (10 problems).

1. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} 7x_1 - x_2 = 2 \\ 3x_1 - 11x_2 - 3x_3 = 25 \\ 3x_1 - 9x_3 = 9 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

2. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} -6x_1 + 3x_3 = 0 \\ 2x_1 - 9x_2 - 2x_3 = -16 \\ -x_1 - 2x_2 + 9x_3 = -21 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

3. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} 11x_1 - 3x_2 + 2x_3 = -35 \\ 2x_1 - 8x_2 = -6 \\ -3x_1 + 2x_2 + 10x_3 = -1 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

4. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} 5x_1 - x_2 - x_3 = -8 \\ -3x_1 - 11x_2 + 2x_3 = 15 \\ x_1 + 3x_2 + 7x_3 = -12 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

5. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} 7x_1 + x_2 + x_3 = 9 \\ 2x_1 - 10x_2 - 2x_3 = 6 \\ -3x_1 + 3x_2 + 9x_3 = 21 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

6. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} 10x_1 + 2x_2 - 2x_3 = 30 \\ x_1 - 7x_2 + x_3 = 15 \\ 2x_1 - x_2 + 9x_3 = -10 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

7. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} -6x_1 - x_2 + x_3 = 21 \\ x_1 - 8x_2 - x_3 = 12 \\ x_1 + 3x_2 + 7x_3 = -2 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

8. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} 6x_1 + x_3 = 1 \\ -x_1 + 6x_2 - 2x_3 = -8 \\ -2x_1 + 3x_2 + 11x_3 = 8 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

9. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} -7x_1 + 2x_2 = -20 \\ -3x_1 + 8x_2 = -30 \\ 2x_1 + 3x_2 + 11x_3 = -16 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

10. **(Comparing the methods.)** Perform three iterations of the Jacobi and of the Seidel method for the system $\begin{cases} -6x_1 - x_2 = 13 \\ 2x_1 + 8x_2 + x_3 = -14 \\ x_1 + 2x_2 - 6x_3 = 8 \end{cases}$, starting from $x^{(0)} = (0; 0; 0)$, and compare the errors $\|x^{(3)} - x^*\|_\infty$.

Conclusion

This study guide has presented the theory, worked examples and exercises needed for the practical part of a course in linear algebra. The material starts with matrices and determinants and proceeds through exact methods for systems of linear equations (Cramer's rule, the matrix method, the Gauss and Gauss–Jordan methods) and approximate ones (simple iteration and the Gauss–Seidel method), vector algebra, elements of analytic geometry, linear and Euclidean spaces, linear operators, eigenvalues and eigenvectors, and quadratic forms.

Every topic has been connected with its use in information security: the Hill cipher, error-correcting codes, secret sharing schemes, linear cryptanalysis, lattice-based cryptography, network analysis and anomaly detection. These applications are meant to show students that linear algebra is not only an abstract theory but a tool they will use directly in their profession. The short Python, Maple and Mathcad examples show how the same computations are carried out in practice, where matrices are large and hand calculation is no longer an option.

The more than two thousand exercises, all supplied with answers, give students ample material for independent work and allow teachers to prepare individual assignments. The knowledge and skills acquired here form the basis for subsequent courses such as discrete mathematics, probability theory, numerical methods, cryptography and coding theory.

Answers to the Exercises

The answers are arranged by section, exercise set and exercise number. Inverse matrices are written with the common denominator taken outside the matrix. Approximate values are rounded to four decimal places.

Chapter 1. Matrices and Determinants

1.1. The concept of a matrix

Exercise set 1.

1. $\begin{pmatrix} -13 & -51 \\ 68 & 4 \end{pmatrix}$
2. $\begin{pmatrix} -21 & -15 \\ 10 & 9 \end{pmatrix}$
3. $\begin{pmatrix} -15 & -2 \\ 4 & -8 \end{pmatrix}$
4. $\begin{pmatrix} -21 & -2 \\ -8 & -5 \end{pmatrix}$
5. $\begin{pmatrix} 28 & 32 \\ -24 & 44 \end{pmatrix}$
6. $\begin{pmatrix} -3 & -1 \\ 3 & -8 \end{pmatrix}$
7. $\begin{pmatrix} -13 & -12 \\ -6 & -7 \end{pmatrix}$
8. $\begin{pmatrix} -2 & -8 \\ -16 & -10 \end{pmatrix}$
9. $\begin{pmatrix} 8 & 21 \\ 0 & -20 \end{pmatrix}$
10. $\begin{pmatrix} 21 & 22 \\ -33 & 10 \end{pmatrix}$
11. $\begin{pmatrix} 2 & -3 \\ -9 & -4 \end{pmatrix}$
12. $\begin{pmatrix} -15 & 6 \\ 8 & -9 \end{pmatrix}$
13. $\begin{pmatrix} 14 & -15 \\ 60 & -1 \end{pmatrix}$
14. $\begin{pmatrix} -19 & 12 \\ 3 & -10 \end{pmatrix}$
15. $\begin{pmatrix} 21 & 6 \\ -6 & 13 \end{pmatrix}$
16. $\begin{pmatrix} -53 & 24 \\ 48 & -29 \end{pmatrix}$
17. $\begin{pmatrix} -18 & -26 \\ -39 & -31 \end{pmatrix}$
18. $\begin{pmatrix} -26 & 30 \\ -20 & 14 \end{pmatrix}$
19. $\begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix}$
20. $\begin{pmatrix} -3 & -8 \\ -8 & 5 \end{pmatrix}$
21. $\begin{pmatrix} 10 & -20 \\ -5 & -15 \end{pmatrix}$
22. $\begin{pmatrix} -26 & 26 \\ -104 & -26 \end{pmatrix}$
23. $\begin{pmatrix} -27 & -4 \\ -3 & -29 \end{pmatrix}$
24. $\begin{pmatrix} -5 & -7 \\ -7 & -26 \end{pmatrix}$
25. $\begin{pmatrix} -5 & -6 \\ -6 & 1 \end{pmatrix}$
26. $\begin{pmatrix} -3 & 0 \\ 9 & -12 \end{pmatrix}$
27. $\begin{pmatrix} 7 & -3 \\ 6 & 7 \end{pmatrix}$
28. $\begin{pmatrix} -15 & 28 \\ 7 & -29 \end{pmatrix}$
29. $\begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix}$
30. $\begin{pmatrix} -22 & 0 \\ 0 & -22 \end{pmatrix}$

Exercise set 2.

1. $A + A^T = \begin{pmatrix} -4 & 0 & 11 \\ 0 & -2 & 6 \\ 11 & 6 & -2 \end{pmatrix}$
2. $A + A^T = \begin{pmatrix} -10 & -8 \\ -8 & -6 \end{pmatrix}$

3. $A + A^T = \begin{pmatrix} -6 & 0 & -9 \\ 0 & 6 & 4 \\ -9 & 4 & 8 \end{pmatrix}$
4. $A + A^T = \begin{pmatrix} 12 & -1 \\ -1 & 8 \end{pmatrix}$
5. $A + A^T = \begin{pmatrix} 0 & 11 \\ 11 & 12 \end{pmatrix}$
6. $A + A^T = \begin{pmatrix} 2 & -4 \\ -4 & 12 \end{pmatrix}$
7. $A + A^T = \begin{pmatrix} -10 & 11 \\ 11 & 0 \end{pmatrix}$
8. $A + A^T = \begin{pmatrix} 0 & 10 & -9 \\ 10 & -8 & -10 \\ -9 & -10 & -8 \end{pmatrix}$
9. $A + A^T = \begin{pmatrix} 2 & 2 & 5 \\ 2 & 6 & 7 \\ 5 & 7 & -2 \end{pmatrix}$
10. $A + A^T = \begin{pmatrix} 4 & -2 \\ -2 & -12 \end{pmatrix}$
11. $A + A^T = \begin{pmatrix} 12 & 9 \\ 9 & -10 \end{pmatrix}$
12. $A + A^T = \begin{pmatrix} 10 & -7 & -2 \\ -7 & -6 & -9 \\ -2 & -9 & -4 \end{pmatrix}$
13. $A + A^T = \begin{pmatrix} -6 & 5 & 3 \\ 5 & -4 & -2 \\ 3 & -2 & -12 \end{pmatrix}$
14. $A + A^T = \begin{pmatrix} 2 & 7 \\ 7 & 4 \end{pmatrix}$
15. $A + A^T = \begin{pmatrix} 4 & -8 & -4 \\ -8 & 4 & 3 \\ -4 & 3 & 12 \end{pmatrix}$
16. $A + A^T = \begin{pmatrix} 6 & 0 \\ 0 & 12 \end{pmatrix}$
17. $A + A^T = \begin{pmatrix} -8 & -3 \\ -3 & 6 \end{pmatrix}$
18. $A + A^T = \begin{pmatrix} 4 & -7 \\ -7 & 8 \end{pmatrix}$
19. $A + A^T = \begin{pmatrix} 4 & 8 & 3 \\ 8 & 12 & -11 \\ 3 & -11 & -6 \end{pmatrix}$
20. $A + A^T = \begin{pmatrix} -4 & 0 \\ 0 & -10 \end{pmatrix}$
21. $A + A^T = \begin{pmatrix} 2 & 8 & -7 \\ 8 & -10 & 4 \\ -7 & 4 & 4 \end{pmatrix}$
22. $A + A^T = \begin{pmatrix} 4 & -5 & 7 \\ -5 & 2 & 8 \\ 7 & 8 & 2 \end{pmatrix}$
23. $A + A^T = \begin{pmatrix} -6 & 3 & 3 \\ 3 & 4 & -7 \\ 3 & -7 & 0 \end{pmatrix}$
24. $A + A^T = \begin{pmatrix} 0 & 0 \\ 0 & -10 \end{pmatrix}$
25. $A + A^T = \begin{pmatrix} 0 & -8 \\ -8 & 8 \end{pmatrix}$
26. $A + A^T = \begin{pmatrix} 12 & 1 \\ 1 & -8 \end{pmatrix}$
27. $A + A^T = \begin{pmatrix} -8 & -6 \\ -6 & 2 \end{pmatrix}$
28. $A + A^T = \begin{pmatrix} 10 & -5 \\ -5 & 2 \end{pmatrix}$
29. $A + A^T = \begin{pmatrix} 8 & -7 \\ -7 & 10 \end{pmatrix}$
30. $A + A^T = \begin{pmatrix} 4 & 0 & -2 \\ 0 & -6 & -6 \\ -2 & -6 & 10 \end{pmatrix}$

Exercise set 3.

1. $Z = -4X + 2Y$
2. $Z = -3X - Y$
3. $Z = -2X + 2Y$
4. $Z = 3X + Y$
5. $Z = -4X - 3Y$
6. $Z = -3X + 3Y$
7. $Z = -2X - 4Y$
8. $Z = -2X - Y$
9. $Z = 3X + 4Y$
10. $Z = -4X - 4Y$
11. $Z = -X + 3Y$
12. $Z = 4X + 2Y$
13. $Z = -2X + 2Y$
14. $Z = 2X - 2Y$
15. $Z = -X - 4Y$
16. $Z = X + Y$
17. $Z = X - 2Y$
18. $Z = 4X - Y$
19. $Z = -2X + 2Y$
20. $Z = -X - 3Y$
21. $Z = X + 3Y$
22. $Z = 4X + 2Y$
23. $Z = -X - 3Y$
24. $Z = 2X + 3Y$
25. $Z = -X + 3Y$
26. $Z = -3X + 4Y$
27. $Z = -X - Y$
28. $Z = -4X - Y$
29. $Z = -2X - 4Y$
30. $Z = -X + 3Y$

Exercise set 4.

1. $a = 9,\ b = 9$
2. $a = 3,\ b = 9$
3. $a = 9,\ b = 6$
4. $a = 3,\ b = 4$
5. $a = 6,\ b = 3$
6. $a = 8,\ b = 2$
7. $a = 9,\ b = 8$
8. $a = 8,\ b = 7$
9. $a = 7,\ b = 8$
10. $a = 6,\ b = 7$
11. $a = 3,\ b = 7$
12. $a = 3,\ b = 2$
13. $a = 5,\ b = 6$
14. $a = 7,\ b = 8$
15. $a = 5,\ b = 3$
16. $a = 4,\ b = 6$
17. $a = 4,\ b = 9$
18. $a = 6,\ b = 6$
19. $a = 9,\ b = 8$
20. $a = 3,\ b = 5$
21. $a = 9,\ b = 7$
22. $a = 5,\ b = 5$
23. $a = 3,\ b = 7$
24. $a = 5,\ b = 2$
25. $a = 5,\ b = 8$
26. $a = 6,\ b = 7$
27. $a = 5,\ b = 3$
28. $a = 9,\ b = 8$
29. $a = 8,\ b = 9$
30. $a = 8,\ b = 9$

Exercise set 5.

1. 12 sizes: $1 \times 72,\ 2 \times 36,\ 3 \times 24,\ 4 \times 18,\ 6 \times 12,\ 8 \times 9,\ 9 \times 8,\ 12 \times 6,\ 18 \times 4,\ 24 \times 3,\ 36 \times 2,\ 72 \times 1$
2. 9 sizes: $1 \times 36,\ 2 \times 18,\ 3 \times 12,\ 4 \times 9,\ 6 \times 6,\ 9 \times 4,\ 12 \times 3,\ 18 \times 2,\ 36 \times 1$
3. 8 sizes: $1 \times 30,\ 2 \times 15,\ 3 \times 10,\ 5 \times 6,\ 6 \times 5,\ 10 \times 3,\ 15 \times 2,\ 30 \times 1$
4. 8 sizes: $1 \times 42,\ 2 \times 21,\ 3 \times 14,\ 6 \times 7,\ 7 \times 6,\ 14 \times 3,\ 21 \times 2,\ 42 \times 1$
5. 7 sizes: $1 \times 64,\ 2 \times 32,\ 4 \times 16,\ 8 \times 8,\ 16 \times 4,\ 32 \times 2,\ 64 \times 1$
6. 6 sizes: $1 \times 28,\ 2 \times 14,\ 4 \times 7,\ 7 \times 4,\ 14 \times 2,\ 28 \times 1$

7. 8 sizes: $1\times88,\ 2\times44,\ 4\times22,\ 8\times11,\ 11\times8,\ 22\times4,\ 44\times2,\ 88\times1$
8. 6 sizes: $1\times44,\ 2\times22,\ 4\times11,\ 11\times4,\ 22\times2,\ 44\times1$
9. 10 sizes: $1\times48,\ 2\times24,\ 3\times16,\ 4\times12,\ 6\times8,\ 8\times6,\ 12\times4,\ 16\times3,\ 24\times2,\ 48\times1$
10. 6 sizes: $1\times75,\ 3\times25,\ 5\times15,\ 15\times5,\ 25\times3,\ 75\times1$
11. 8 sizes: $1\times54,\ 2\times27,\ 3\times18,\ 6\times9,\ 9\times6,\ 18\times3,\ 27\times2,\ 54\times1$
12. 8 sizes: $1\times40,\ 2\times20,\ 4\times10,\ 5\times8,\ 8\times5,\ 10\times4,\ 20\times2,\ 40\times1$
13. 8 sizes: $1\times66,\ 2\times33,\ 3\times22,\ 6\times11,\ 11\times6,\ 22\times3,\ 33\times2,\ 66\times1$
14. 6 sizes: $1\times63,\ 3\times21,\ 7\times9,\ 9\times7,\ 21\times3,\ 63\times1$
15. 6 sizes: $1\times45,\ 3\times15,\ 5\times9,\ 9\times5,\ 15\times3,\ 45\times1$
16. 6 sizes: $1\times12,\ 2\times6,\ 3\times4,\ 4\times3,\ 6\times2,\ 12\times1$
17. 5 sizes: $1\times16,\ 2\times8,\ 4\times4,\ 8\times2,\ 16\times1$
18. 12 sizes: $1\times60,\ 2\times30,\ 3\times20,\ 4\times15,\ 5\times12,\ 6\times10,\ 10\times6,\ 12\times5,\ 15\times4,\ 20\times3,\ 30\times2,\ 60\times1$
19. 6 sizes: $1\times18,\ 2\times9,\ 3\times6,\ 6\times3,\ 9\times2,\ 18\times1$
20. 12 sizes: $1\times96,\ 2\times48,\ 3\times32,\ 4\times24,\ 6\times16,\ 8\times12,\ 12\times8,\ 16\times6,\ 24\times4,\ 32\times3,\ 48\times2,\ 96\times1$
21. 12 sizes: $1\times84,\ 2\times42,\ 3\times28,\ 4\times21,\ 6\times14,\ 7\times12,\ 12\times7,\ 14\times6,\ 21\times4,\ 28\times3,\ 42\times2,\ 84\times1$
22. 8 sizes: $1\times70,\ 2\times35,\ 5\times14,\ 7\times10,\ 10\times7,\ 14\times5,\ 35\times2,\ 70\times1$
23. 8 sizes: $1\times56,\ 2\times28,\ 4\times14,\ 7\times8,\ 8\times7,\ 14\times4,\ 28\times2,\ 56\times1$
24. 10 sizes: $1\times80,\ 2\times40,\ 4\times20,\ 5\times16,\ 8\times10,\ 10\times8,\ 16\times5,\ 20\times4,\ 40\times2,\ 80\times1$
25. 8 sizes: $1\times24,\ 2\times12,\ 3\times8,\ 4\times6,\ 6\times4,\ 8\times3,\ 12\times2,\ 24\times1$
26. 12 sizes: $1\times90,\ 2\times45,\ 3\times30,\ 5\times18,\ 6\times15,\ 9\times10,\ 10\times9,\ 15\times6,\ 18\times5,\ 30\times3,\ 45\times2,\ 90\times1$
27. 6 sizes: $1\times20,\ 2\times10,\ 4\times5,\ 5\times4,\ 10\times2,\ 20\times1$
28. 5 sizes: $1\times81,\ 3\times27,\ 9\times9,\ 27\times3,\ 81\times1$
29. 6 sizes: $1\times32,\ 2\times16,\ 4\times8,\ 8\times4,\ 16\times2,\ 32\times1$
30. 6 sizes: $1\times50,\ 2\times25,\ 5\times10,\ 10\times5,\ 25\times2,\ 50\times1$

Exercise set 6.

1. $\begin{pmatrix} -1 & 9 \\ 3 & -2 \\ 0 & -4 \end{pmatrix}$
2. $\begin{pmatrix} 1 & -6 & 8 \\ 1 & -5 & 3 \\ 6 & -3 & -12 \end{pmatrix}$
3. $\begin{pmatrix} 1 & -1 & -15 \\ -4 & 9 & -17 \end{pmatrix}$
4. $\begin{pmatrix} -2 & -10 & 14 \\ 4 & 14 & -8 \end{pmatrix}$

5. $\begin{pmatrix} -9 & 13 & -2 \\ 19 & 3 & 13 \\ 2 & 15 & -4 \end{pmatrix}$
6. $\begin{pmatrix} 7 & -9 \\ -13 & -6 \\ -4 & 4 \end{pmatrix}$
7. $\begin{pmatrix} -7 & -6 \\ -5 & -2 \end{pmatrix}$
8. $\begin{pmatrix} 2 & 20 \\ 16 & -8 \\ 8 & 4 \end{pmatrix}$
9. $\begin{pmatrix} 1 & -4 & -6 \\ -1 & -9 & 6 \end{pmatrix}$
10. $\begin{pmatrix} 3 & 11 \\ -7 & -14 \\ 12 & 16 \end{pmatrix}$
11. $\begin{pmatrix} -10 & -15 & 19 \\ -11 & -6 & 9 \\ 2 & 9 & 31 \end{pmatrix}$
12. $\begin{pmatrix} 17 & -12 & -13 \\ -7 & -8 & 19 \end{pmatrix}$
13. $\begin{pmatrix} 6 & -9 \\ 6 & 2 \end{pmatrix}$
14. $\begin{pmatrix} -4 & 6 & 5 \\ 6 & -1 & -1 \end{pmatrix}$
15. $\begin{pmatrix} 10 & -13 & -2 \\ 2 & -21 & -15 \\ 20 & -26 & 16 \end{pmatrix}$
16. $\begin{pmatrix} 15 & 14 \\ 10 & 2 \end{pmatrix}$
17. $\begin{pmatrix} -15 & -10 & 10 \\ 11 & 5 & -14 \end{pmatrix}$
18. $\begin{pmatrix} -7 & -10 & 8 \\ 7 & -9 & -12 \end{pmatrix}$
19. $\begin{pmatrix} 2 & 20 \\ 6 & -24 \\ 0 & -26 \end{pmatrix}$
20. $\begin{pmatrix} -21 & -11 & 15 \\ -15 & 11 & -3 \end{pmatrix}$
21. $\begin{pmatrix} -4 & 16 & -2 \\ -2 & 12 & -4 \\ -10 & 4 & 6 \end{pmatrix}$
22. $\begin{pmatrix} -1 & -13 \\ 13 & 21 \end{pmatrix}$
23. $\begin{pmatrix} 22 & 16 & -2 \\ -2 & 2 & 10 \end{pmatrix}$
24. $\begin{pmatrix} -4 & 28 & -14 \\ -12 & -8 & 20 \\ -22 & 20 & -24 \end{pmatrix}$
25. $\begin{pmatrix} -13 & 6 & 22 \\ -1 & 16 & 6 \end{pmatrix}$
26. $\begin{pmatrix} 12 & 30 & -3 \\ 21 & 12 & 3 \\ 3 & -21 & -12 \end{pmatrix}$
27. $\begin{pmatrix} -18 & -24 & -15 \\ 9 & -3 & -3 \end{pmatrix}$
28. $\begin{pmatrix} -20 & 26 \\ -20 & -26 \end{pmatrix}$
29. $\begin{pmatrix} -9 & -2 & 10 \\ -14 & 2 & 5 \end{pmatrix}$
30. $\begin{pmatrix} -2 & -1 \\ 2 & 2 \\ 5 & 0 \end{pmatrix}$

Exercise set 7.

1. $c = \begin{pmatrix} 0 \\ 25 \end{pmatrix}$ $(Kp = (78; 103)^T)$
2. $c = \begin{pmatrix} 16 \\ 3 \end{pmatrix}$
3. $c = \begin{pmatrix} 0 \\ 19 \end{pmatrix}$

4. $c = \begin{pmatrix} 3 \\ 15 \end{pmatrix}$
5. $\det K = -2$, $\gcd(-2, 26) = 2$; since $\gcd(\det K, 26) \neq 1$, K is not invertible modulo 26 and cannot serve as a key
6. $\det K = 0$; K is singular (has no inverse), so it cannot serve as a key
7. $c = (1; 0; 1; 1; 0; 1; 0)$
8. $s = (0; 0; 1)^T \neq 0$; the word contains an error. The syndrome equals the 4th column of H, so the 4th bit is wrong; the corrected word is $(1; 0; 1; 0; 1; 0; 1)$
9. $A^2 = \begin{pmatrix} 2 & 0 & 0 & 2 & 1 \\ 0 & 2 & 2 & 0 & 1 \\ 0 & 2 & 3 & 1 & 1 \\ 2 & 0 & 1 & 3 & 1 \\ 1 & 1 & 1 & 1 & 2 \end{pmatrix}$; the most connected nodes are S_3 and S_4 (degree 3, the diagonal entries of A^2)
10. $K^2 - 3K + E = O$, hence $K(3E - K) = E$ and $K^{-1} = 3E - K = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$

1.2. The concept of a determinant. Determinants of the second and third order

Exercise set 1.

1. 22	7. 35	13. 27	19. 28	25. 42
2. −11	8. −14	14. 18	20. −3	26. 7
3. 20	9. 26	15. 13	21. 19	27. −37
4. 42	10. 18	16. 0	22. −43	28. 35
5. 32	11. 58	17. −41	23. −26	29. −46
6. 16	12. 3	18. 84	24. 31	30. 1

Exercise set 2.

1. 6	7. −89	13. 22	19. 117	25. 20
2. 83	8. 77	14. 17	20. −129	26. 49
3. 17	9. 89	15. −52	21. 39	27. 87
4. 38	10. 75	16. −17	22. 51	28. 24
5. 65	11. −32	17. 16	23. 71	29. 66
6. 90	12. −47	18. 22	24. 7	30. −55

Exercise set 3.

1. −77	7. 102	13. 83	19. 73	25. 63
2. 23	8. 50	14. 93	20. −86	26. 27
3. 100	9. −79	15. 5	21. 9	27. −20
4. 51	10. 13	16. −18	22. 1	28. 42
5. 90	11. −75	17. −31	23. −36	29. 20
6. 106	12. −70	18. 63	24. 24	30. 66

Exercise set 4.

1. 0	7. 0	13. 0	19. −21	25. −5
2. 0	8. 210	14. 0	20. 0	26. 0
3. −12	9. 0	15. 0	21. 140	27. 0
4. 0	10. 0	16. 0	22. −42	28. 0
5. 0	11. −15	17. 0	23. 0	29. 0
6. 0	12. 0	18. 0	24. 0	30. 396

Exercise set 5.

1. $\det K = 11,\ \gcd(11, 26) = 1$ – admissible
2. $\det K = 0,\ \gcd(0, 26) = 26$ – not admissible
3. $\det K = 1,\ \gcd(1, 26) = 1$ – admissible
4. $\det K = 19,\ \gcd(19, 26) = 1$ – admissible
5. $\det K = 43,\ \gcd(43, 26) = 1$ – admissible
6. $\det K = 0,\ \gcd(0, 26) = 26$ – not admissible
7. $\det K = -1,\ \gcd(-1, 26) = 1$ – admissible
8. $\det K = 19,\ \gcd(19, 26) = 1$ – admissible
9. $\det K = 51,\ \gcd(51, 26) = 1$ – admissible
10. $\det K = 1,\ \gcd(1, 26) = 1$ – admissible
11. $\det K = 89,\ \gcd(89, 26) = 1$ – admissible
12. $\det K = -2,\ \gcd(-2, 26) = 2$ – not admissible
13. $\det K = 9,\ \gcd(9, 26) = 1$ – admissible
14. $\det K = -11,\ \gcd(-11, 26) = 1$ – admissible
15. $\det K = 1,\ \gcd(1, 26) = 1$ – admissible
16. $\det K = 2,\ \gcd(2, 26) = 2$ – not admissible
17. $\det K = 24,\ \gcd(24, 26) = 2$ – not admissible
18. $\det K = 31,\ \gcd(31, 26) = 1$ – admissible

19. $\det K = 0,\ \gcd(0, 26) = 26$ – not admissible
20. $\det K = 172,\ \gcd(172, 26) = 2$ – not admissible
21. $\det K = 1,\ \gcd(1, 26) = 1$ – admissible
22. $\det K = 26,\ \gcd(26, 26) = 26$ – not admissible
23. $\det K = 17,\ \gcd(17, 26) = 1$ – admissible
24. $\det K = 22,\ \gcd(22, 26) = 2$ – not admissible
25. $\det K = 1,\ \gcd(1, 26) = 1$ – admissible
26. $\det K = -3,\ \gcd(-3, 26) = 1$ – admissible
27. $\det K = 1,\ \gcd(1, 26) = 1$ – admissible
28. $\det K = -1,\ \gcd(-1, 26) = 1$ – admissible
29. $\det K = -29,\ \gcd(-29, 26) = 1$ – admissible
30. $\det K = 53,\ \gcd(53, 26) = 1$ – admissible

1.3. Minors and cofactors. Evaluation of determinants of order n

Exercise set 1.

1. $M_{11} = 22,\ A_{11} = 22;\ M_{23} = 7,\ A_{23} = -7$
2. $M_{12} = -11,\ A_{12} = 11;\ M_{31} = -1,\ A_{31} = -1$
3. $M_{22} = 11,\ A_{22} = 11;\ M_{13} = -2,\ A_{13} = -2$
4. $M_{32} = -20,\ A_{32} = 20;\ M_{11} = 33,\ A_{11} = 33$
5. $M_{21} = -4,\ A_{21} = 4;\ M_{33} = -6,\ A_{33} = -6$
6. $M_{13} = -6,\ A_{13} = -6;\ M_{22} = -8,\ A_{22} = -8$
7. $M_{31} = 15,\ A_{31} = 15;\ M_{12} = -7,\ A_{12} = 7$
8. $M_{23} = -7,\ A_{23} = 7;\ M_{11} = 5,\ A_{11} = 5$
9. $M_{32} = -12,\ A_{32} = 12;\ M_{21} = -4,\ A_{21} = 4$
10. $M_{11} = 23,\ A_{11} = 23;\ M_{33} = 24,\ A_{33} = 24$
11. $M_{12} = 18,\ A_{12} = -18;\ M_{31} = -12,\ A_{31} = -12$
12. $M_{22} = -9,\ A_{22} = -9;\ M_{13} = -10,\ A_{13} = -10$
13. $M_{21} = 4,\ A_{21} = -4;\ M_{32} = 2,\ A_{32} = -2$
14. $M_{13} = 6,\ A_{13} = 6;\ M_{23} = -4,\ A_{23} = 4$
15. $M_{11} = -18,\ A_{11} = -18;\ M_{32} = -26,\ A_{32} = 26$
16. $M_{31} = 10,\ A_{31} = 10;\ M_{12} = -8,\ A_{12} = 8$
17. $M_{22} = 19,\ A_{22} = 19;\ M_{33} = 3,\ A_{33} = 3$
18. $M_{13} = 2,\ A_{13} = 2;\ M_{21} = -4,\ A_{21} = 4$
19. $M_{23} = -11,\ A_{23} = 11;\ M_{11} = 12,\ A_{11} = 12$
20. $M_{32} = -12,\ A_{32} = 12;\ M_{12} = 3,\ A_{12} = -3$

21. $M_{11} = 6,\ A_{11} = 6;\ M_{23} = -10,\ A_{23} = 10$
22. $M_{21} = -4,\ A_{21} = 4;\ M_{33} = 12,\ A_{33} = 12$
23. $M_{31} = 22,\ A_{31} = 22;\ M_{22} = -20,\ A_{22} = -20$
24. $M_{12} = 17,\ A_{12} = -17;\ M_{33} = 20,\ A_{33} = 20$
25. $M_{13} = 4,\ A_{13} = 4;\ M_{21} = 4,\ A_{21} = -4$
26. $M_{22} = 5,\ A_{22} = 5;\ M_{31} = 8,\ A_{31} = 8$
27. $M_{32} = -6,\ A_{32} = 6;\ M_{11} = -8,\ A_{11} = -8$
28. $M_{23} = -9,\ A_{23} = 9;\ M_{12} = -21,\ A_{12} = 21$
29. $M_{11} = -6,\ A_{11} = -6;\ M_{13} = 4,\ A_{13} = 4$
30. $M_{21} = 4,\ A_{21} = -4;\ M_{32} = 8,\ A_{32} = -8$

Exercise set 2.

1. 12	7. 66	13. −46	19. −30	25. 53
2. −79	8. −62	14. −54	20. −64	26. 56
3. 24	9. 46	15. −58	21. 32	27. −68
4. −62	10. −49	16. 80	22. −62	28. −39
5. 38	11. 70	17. −64	23. 85	29. 14
6. −36	12. 62	18. −58	24. 62	30. 32

Exercise set 3.

1. −10	7. 92	13. −198	19. −56	25. 154
2. −65	8. 114	14. 35	20. 6	26. 140
3. 21	9. 110	15. −160	21. 152	27. 110
4. −8	10. 83	16. −18	22. 49	28. 91
5. −9	11. 1	17. −52	23. 23	29. 17
6. −134	12. 212	18. 28	24. 63	30. 0

Exercise set 4.

1. 79 and 0	7. −38 and 0	13. 64 and 0	19. 3 and 0	25. 54 and 0
2. −70 and 0	8. −1 and 0	14. −41 and 0	20. 45 and 0	26. 21 and 0
3. −60 and 0	9. 50 and 0	15. 46 and 0	21. 84 and 0	27. −40 and 0
4. −90 and 0	10. 60 and 0	16. 75 and 0	22. 23 and 0	28. −5 and 0
5. −54 and 0	11. 7 and 0	17. 10 and 0	23. 102 and 0	29. −55 and 0
6. 60 and 0	12. 35 and 0	18. 48 and 0	24. 49 and 0	30. 10 and 0

1.4. Computing the rank of a matrix. Singular and nonsingular matrices

Exercise set 1.

1. rg = 2
2. rg = 2
3. rg = 1
4. rg = 2
5. rg = 2
6. rg = 3
7. rg = 1
8. rg = 2
9. rg = 1
10. rg = 2
11. rg = 1
12. rg = 2
13. rg = 2
14. rg = 2
15. rg = 3
16. rg = 1
17. rg = 2
18. rg = 3
19. rg = 1
20. rg = 2
21. rg = 1
22. rg = 2
23. rg = 2
24. rg = 1
25. rg = 2
26. rg = 2
27. rg = 1
28. rg = 3
29. rg = 1
30. rg = 2

Exercise set 2.

1. rg = 2
2. rg = 2
3. rg = 1
4. rg = 2
5. rg = 2
6. rg = 1
7. rg = 2
8. rg = 2
9. rg = 2
10. rg = 1
11. rg = 4
12. rg = 2
13. rg = 1
14. rg = 2
15. rg = 2
16. rg = 2
17. rg = 2
18. rg = 1
19. rg = 3
20. rg = 4
21. rg = 1
22. rg = 1
23. rg = 2
24. rg = 3
25. rg = 2
26. rg = 1
27. rg = 2
28. rg = 3
29. rg = 2
30. rg = 1

Exercise set 3.

1. det = −2, nonsingular
2. det = 0, singular
3. det = 13, nonsingular
4. det = 0, singular
5. det = −8, nonsingular
6. det = 0, singular
7. det = 0, singular
8. det = 0, singular
9. det = 0, singular
10. det = 60, nonsingular
11. det = 1, nonsingular
12. det = 0, singular
13. det = 0, singular
14. det = 1, nonsingular
15. det = 0, singular
16. det = 0, singular
17. det = −30, nonsingular
18. det = 0, singular
19. det = 1, nonsingular
20. det = 0, singular
21. det = 0, singular
22. det = −6, nonsingular
23. det = 0, singular
24. det = 0, singular
25. det = 0, singular
26. det = 4, nonsingular
27. det = 0, singular
28. det = 0, singular
29. det = 0, singular
30. det = 0, singular

Exercise set 4.

1. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
2. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
3. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
4. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
5. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
6. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
7. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
8. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
9. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (unique solution)
10. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
11. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
12. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
13. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
14. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
15. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
16. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
17. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 3$; consistent (unique solution)
18. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
19. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (unique solution)
20. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 3$; consistent (infinitely many solutions)
21. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
22. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
23. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (infinitely many solutions)
24. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 2$; consistent (unique solution)
25. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
26. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
27. $\operatorname{rg} A = 2,\ \operatorname{rg} \bar{A} = 3$; inconsistent
28. $\operatorname{rg} A = \operatorname{rg} \bar{A} = 1$; consistent (infinitely many solutions)
29. $\operatorname{rg} A = 1,\ \operatorname{rg} \bar{A} = 2$; inconsistent
30. $\operatorname{rg} A = 2,\ \operatorname{rg} \bar{A} = 3$; inconsistent

1.5. The inverse matrix

Exercise set 1.

1. $\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$
2. does not exist (det = 0)
3. $\begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$
4. $\begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}$
5. $\frac{1}{15}\begin{pmatrix} 3 & 0 \\ 0 & -5 \end{pmatrix}$
6. does not exist (det = 0)
7. $\begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}$
8. $\begin{pmatrix} 1 & -2 \\ -3 & 7 \end{pmatrix}$

9. does not exist (det = 0)
10. $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$
11. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
12. does not exist (det = 0)
13. $\frac{1}{14}\begin{pmatrix} -3 & 1 \\ 2 & 4 \end{pmatrix}$
14. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
15. does not exist (det = 0)
16. $\frac{1}{17}\begin{pmatrix} 3 & -2 \\ 1 & 5 \end{pmatrix}$
17. does not exist (det = 0)
18. $\begin{pmatrix} 2 & -5 \\ -3 & 8 \end{pmatrix}$
19. $\begin{pmatrix} -7 & 4 \\ 2 & -1 \end{pmatrix}$
20. $\frac{1}{10}\begin{pmatrix} -5 & 0 \\ 0 & -2 \end{pmatrix}$
21. does not exist (det = 0)
22. $\frac{1}{13}\begin{pmatrix} 1 & 1 \\ -4 & 9 \end{pmatrix}$
23. does not exist (det = 0)
24. $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$
25. $\begin{pmatrix} 2 & -5 \\ 1 & -3 \end{pmatrix}$
26. $\frac{1}{6}\begin{pmatrix} 0 & 2 \\ -3 & 0 \end{pmatrix}$
27. $\begin{pmatrix} 11 & 4 \\ -3 & -1 \end{pmatrix}$
28. $\frac{1}{20}\begin{pmatrix} 3 & -1 \\ 2 & 6 \end{pmatrix}$
29. does not exist (det = 0)
30. $\frac{1}{13}\begin{pmatrix} 1 & 3 \\ -2 & 7 \end{pmatrix}$

Exercise set 2.

1. $\begin{pmatrix} 1 & -1 & 1 \\ -38 & 41 & -34 \\ 27 & -29 & 24 \end{pmatrix}$
2. $\frac{1}{9}\begin{pmatrix} -1 & -2 & 5 \\ 2 & 4 & -1 \\ 6 & 3 & -3 \end{pmatrix}$
3. $\frac{1}{73}\begin{pmatrix} 26 & 5 & -2 \\ 3 & 9 & 11 \\ -5 & -15 & 6 \end{pmatrix}$
4. $\frac{1}{41}\begin{pmatrix} 2 & 11 & 1 \\ 21 & 13 & -10 \\ 8 & 3 & 4 \end{pmatrix}$
5. $\frac{1}{10}\begin{pmatrix} 4 & -8 & -6 \\ 3 & 4 & 3 \\ 1 & -2 & 1 \end{pmatrix}$
6. $\frac{1}{43}\begin{pmatrix} 5 & 1 & -10 \\ -3 & 8 & 6 \\ 10 & 2 & 23 \end{pmatrix}$
7. $\frac{1}{16}\begin{pmatrix} 8 & 4 & -4 \\ 4 & 2 & 6 \\ -6 & 1 & 3 \end{pmatrix}$
8. $\frac{1}{8}\begin{pmatrix} 2 & -9 & -3 \\ 4 & -10 & -6 \\ 2 & 3 & 1 \end{pmatrix}$
9. $\frac{1}{16}\begin{pmatrix} 4 & 2 & 14 \\ 10 & 1 & 15 \\ 8 & 4 & 12 \end{pmatrix}$
10. $\frac{1}{54}\begin{pmatrix} 10 & 4 & 2 \\ -2 & 10 & 5 \\ 4 & -20 & 17 \end{pmatrix}$
11. $\frac{1}{20}\begin{pmatrix} 15 & 5 & -10 \\ -2 & 6 & 4 \\ -3 & -1 & 6 \end{pmatrix}$
12. $\frac{1}{48}\begin{pmatrix} -4 & 8 & 10 \\ 2 & -4 & 7 \\ 22 & 4 & 5 \end{pmatrix}$
13. $\frac{1}{2}\begin{pmatrix} -20 & -11 & 12 \\ 4 & 2 & -2 \\ -10 & -5 & 6 \end{pmatrix}$
14. $\frac{1}{49}\begin{pmatrix} 10 & 6 & 3 \\ -5 & -3 & -26 \\ -3 & 8 & 4 \end{pmatrix}$
15. $\frac{1}{53}\begin{pmatrix} 6 & 9 & -2 \\ -15 & 4 & 5 \\ 8 & 12 & 15 \end{pmatrix}$

16. $\frac{1}{61}\begin{pmatrix} 17 & -5 & -1 \\ 10 & 15 & 3 \\ 6 & 9 & 14 \end{pmatrix}$
17. $\frac{1}{39}\begin{pmatrix} 5 & 11 & 2 \\ -8 & -2 & -11 \\ -20 & -5 & -8 \end{pmatrix}$
18. $\frac{1}{17}\begin{pmatrix} 4 & 1 & -3 \\ 12 & 3 & 8 \\ -9 & 2 & -6 \end{pmatrix}$
19. $\frac{1}{26}\begin{pmatrix} -2 & -4 & -20 \\ -7 & -1 & -5 \\ -4 & -8 & -14 \end{pmatrix}$
20. $\frac{1}{54}\begin{pmatrix} 9 & 6 & 15 \\ 12 & 8 & 2 \\ 3 & 20 & 5 \end{pmatrix}$
21. $\frac{1}{14}\begin{pmatrix} 3 & 20 & 12 \\ 1 & 2 & 4 \\ 5 & 10 & 6 \end{pmatrix}$
22. $\frac{1}{46}\begin{pmatrix} 2 & 4 & 8 \\ -20 & 6 & 12 \\ -5 & -10 & 3 \end{pmatrix}$
23. $\frac{1}{37}\begin{pmatrix} 12 & 5 & 20 \\ 9 & 13 & 15 \\ -5 & 1 & 4 \end{pmatrix}$
24. $\frac{1}{62}\begin{pmatrix} 5 & 1 & -15 \\ -2 & 12 & 6 \\ 20 & 4 & 2 \end{pmatrix}$
25. $\frac{1}{13}\begin{pmatrix} 0 & 0 & 13 \\ 5 & 1 & -20 \\ 2 & 3 & -8 \end{pmatrix}$
26. $\frac{1}{62}\begin{pmatrix} -20 & 2 & 8 \\ 5 & 15 & -2 \\ 1 & 3 & 12 \end{pmatrix}$
27. $\frac{1}{27}\begin{pmatrix} 12 & 20 & -3 \\ 0 & 9 & 0 \\ 3 & 5 & 6 \end{pmatrix}$
28. $\frac{1}{2}\begin{pmatrix} -1 & -4 & -2 \\ -5 & -20 & -12 \\ -3 & -10 & -6 \end{pmatrix}$
29. $\frac{1}{18}\begin{pmatrix} -2 & 1 & 7 \\ 8 & 5 & -1 \\ -6 & 3 & 3 \end{pmatrix}$
30. $\frac{1}{38}\begin{pmatrix} -5 & 9 & 15 \\ 6 & 12 & 20 \\ -2 & -4 & 6 \end{pmatrix}$

Exercise set 3.

1. $X = \begin{pmatrix} 3 \\ -8 \end{pmatrix}$
2. $X = \begin{pmatrix} -16 & -11 \\ 10 & 6 \end{pmatrix}$
3. $X = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$
4. $X = \frac{1}{17}\begin{pmatrix} 7 & 2 \\ 8 & -5 \end{pmatrix}$
5. $X = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$
6. $X = \begin{pmatrix} -5 & 10 \\ 17 & -33 \end{pmatrix}$
7. $X = \frac{1}{14}\begin{pmatrix} 5 \\ 20 \end{pmatrix}$
8. $X = \begin{pmatrix} 0 & -10 \\ 1 & 18 \end{pmatrix}$
9. $X = \begin{pmatrix} 9 \\ -14 \end{pmatrix}$
10. $X = \frac{1}{13}\begin{pmatrix} 3 & 3 \\ -12 & -38 \end{pmatrix}$
11. $X = \begin{pmatrix} -22 \\ -13 \end{pmatrix}$
12. $X = \frac{1}{10}\begin{pmatrix} 3 & -4 \\ 2 & 14 \end{pmatrix}$
13. $X = \begin{pmatrix} -29 \\ 8 \end{pmatrix}$
14. $X = \begin{pmatrix} 1 & 2 \\ 5 & -3 \end{pmatrix}$
15. $X = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$
16. $X = \begin{pmatrix} 5 & -9 \\ -6 & 12 \end{pmatrix}$
17. $X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$
18. $X = \begin{pmatrix} 2 & -1 \\ -2 & 3 \end{pmatrix}$
19. $X = \begin{pmatrix} 7 \\ -3 \end{pmatrix}$
20. $X = \begin{pmatrix} 3 & 1 \\ -2 & 3 \end{pmatrix}$
21. $X = \frac{1}{13}\begin{pmatrix} 1 \\ -2 \end{pmatrix}$
22. $\det A = 0$; infinitely many solutions
    $X = \begin{pmatrix} t \\ -2-3t \end{pmatrix}, t \in \mathbb{R}$

23. $X = \begin{pmatrix} 2 & -8 \\ -1 & 5 \end{pmatrix}$
24. $X = \begin{pmatrix} -13 \\ -8 \end{pmatrix}$
25. $X = \begin{pmatrix} -4 & 42 \\ 1 & -11 \end{pmatrix}$
26. $X = \frac{1}{2}\begin{pmatrix} 3 \\ 0 \end{pmatrix}$
27. $X = \begin{pmatrix} 19 & -5 \\ -26 & 8 \end{pmatrix}$
28. $\det A = 0$; no solution (the system is inconsistent)
29. $X = \begin{pmatrix} -5 & 17 \\ 4 & -13 \end{pmatrix}$
30. $X = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ -7 & 2 \end{pmatrix}$

Exercise set 4.

1. $X = \begin{pmatrix} -23 & 9 \\ -1 & 1 \end{pmatrix}$
2. $X = \begin{pmatrix} -9 & 5 \\ -15 & 9 \end{pmatrix}$
3. $X = \begin{pmatrix} 5 & 3 \\ -11 & -7 \end{pmatrix}$
4. $X = \begin{pmatrix} 5 & -2 \\ 2 & 0 \end{pmatrix}$
5. $X = \begin{pmatrix} 2 & -3 \\ -2 & 4 \end{pmatrix}$
6. $X = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}$
7. $X = \frac{1}{17}\begin{pmatrix} -1 & 12 \\ 4 & -31 \end{pmatrix}$
8. $X = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$
9. $X = \begin{pmatrix} 5 & -13 \\ -1 & 4 \end{pmatrix}$
10. $X = \begin{pmatrix} -3 & 2 \\ 1 & 5 \end{pmatrix}$
11. $X = \frac{1}{14}\begin{pmatrix} 18 & 6 \\ -13 & 19 \end{pmatrix}$
12. $X = \frac{1}{13}\begin{pmatrix} 10 & -16 \\ -15 & 37 \end{pmatrix}$
13. $X = \begin{pmatrix} 3 & -9 \\ 0 & -1 \end{pmatrix}$
14. $X = \frac{1}{20}\begin{pmatrix} -7 & 9 \\ 6 & -2 \end{pmatrix}$
15. $X = \begin{pmatrix} -23 & 13 \\ 4 & -2 \end{pmatrix}$
16. $X = \frac{1}{14}\begin{pmatrix} -5 & -3 \\ -11 & 13 \end{pmatrix}$
17. $X = \begin{pmatrix} 8 & 4 \\ 5 & 4 \end{pmatrix}$
18. $X = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$
19. $X = \begin{pmatrix} 11 & -8 \\ 9 & -6 \end{pmatrix}$
20. $X = \begin{pmatrix} -2 & 2 \\ -3 & -1 \end{pmatrix}$
21. $X = \frac{1}{13}\begin{pmatrix} 2 & -7 \\ -8 & 41 \end{pmatrix}$
22. $X = \begin{pmatrix} -10 & 4 \\ 12 & -5 \end{pmatrix}$
23. $X = \begin{pmatrix} -5 & 9 \\ 10 & -16 \end{pmatrix}$
24. $X = \begin{pmatrix} 55 & 20 \\ -20 & -7 \end{pmatrix}$
25. $X = \begin{pmatrix} -1 & 4 \\ -6 & 22 \end{pmatrix}$
26. $X = \begin{pmatrix} 18 & -7 \\ 4 & -1 \end{pmatrix}$
27. $X = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix}$
28. $X = \frac{1}{2}\begin{pmatrix} -4 & 0 \\ 1 & 9 \end{pmatrix}$
29. $X = \begin{pmatrix} -4 & -7 \\ 2 & 4 \end{pmatrix}$
30. $X = \begin{pmatrix} -2 & -5 \\ 5 & 10 \end{pmatrix}$

1.6. Computing the inverse matrix by elementary row operations

Exercise set 1.

1. $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$
2. $\begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$
3. $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$
4. $\frac{1}{2}\begin{pmatrix} 4 & -2 \\ -7 & 4 \end{pmatrix}$
5. $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}$
6. $\frac{1}{20}\begin{pmatrix} 3 & -1 \\ 2 & 6 \end{pmatrix}$
7. $\begin{pmatrix} -7 & -4 \\ -2 & -1 \end{pmatrix}$
8. $\begin{pmatrix} -1 & 3 \\ 2 & -5 \end{pmatrix}$
9. $\begin{pmatrix} -3 & -2 \\ -4 & -3 \end{pmatrix}$
10. $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$
11. $\begin{pmatrix} 1 & -2 \\ 3 & -7 \end{pmatrix}$
12. $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$
13. $\begin{pmatrix} 2 & -5 \\ -3 & 8 \end{pmatrix}$
14. $\begin{pmatrix} -3 & -1 \\ 5 & 2 \end{pmatrix}$

15. $\begin{pmatrix} 1 & -3 \\ 1 & -4 \end{pmatrix}$
16. $\begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}$
17. $\frac{1}{13}\begin{pmatrix} 1 & 1 \\ -4 & 9 \end{pmatrix}$
18. $\begin{pmatrix} 5 & 3 \\ -2 & -1 \end{pmatrix}$
19. $\begin{pmatrix} -1 & -2 \\ -3 & -5 \end{pmatrix}$
20. $\begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$
21. $\begin{pmatrix} 4 & -7 \\ -1 & 2 \end{pmatrix}$
22. $\begin{pmatrix} -1 & -1 \\ -3 & -4 \end{pmatrix}$
23. $\begin{pmatrix} -5 & 2 \\ -3 & 1 \end{pmatrix}$
24. $\begin{pmatrix} 1 & 5 \\ 1 & 6 \end{pmatrix}$
25. $\begin{pmatrix} 3 & 2 \\ -4 & -3 \end{pmatrix}$
26. $\frac{1}{6}\begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}$
27. $\begin{pmatrix} 1 & -3 \\ -2 & 7 \end{pmatrix}$
28. $\begin{pmatrix} 2 & 1 \\ -7 & -3 \end{pmatrix}$
29. $\frac{1}{5}\begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$
30. $\begin{pmatrix} -1 & 3 \\ -1 & 2 \end{pmatrix}$

Exercise set 2.

1. $\begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix}$
2. $\frac{1}{8}\begin{pmatrix} 2 & 1 & 1 \\ -4 & 2 & 2 \\ 2 & -7 & 1 \end{pmatrix}$
3. $\frac{1}{5}\begin{pmatrix} -2 & 4 & -7 \\ 3 & -1 & 3 \\ -1 & 2 & -1 \end{pmatrix}$
4. $\frac{1}{22}\begin{pmatrix} 6 & -4 & -2 \\ 3 & 9 & -1 \\ 4 & 12 & 6 \end{pmatrix}$
5. $\frac{1}{41}\begin{pmatrix} 2 & 11 & 1 \\ 21 & 13 & -10 \\ 8 & 3 & 4 \end{pmatrix}$
6. $\frac{1}{13}\begin{pmatrix} 1 & 2 & 6 \\ 2 & 4 & -1 \\ -6 & 1 & 3 \end{pmatrix}$
7. $\frac{1}{18}\begin{pmatrix} 8 & 7 & -4 \\ 6 & 3 & 6 \\ -4 & 1 & 2 \end{pmatrix}$
8. $\frac{1}{43}\begin{pmatrix} 5 & 1 & -10 \\ -3 & 8 & 6 \\ 10 & 2 & 23 \end{pmatrix}$
9. $\frac{1}{8}\begin{pmatrix} 2 & -9 & -3 \\ 4 & -10 & -6 \\ 2 & 3 & 1 \end{pmatrix}$
10. $\frac{1}{16}\begin{pmatrix} 4 & 2 & 14 \\ 10 & 1 & 15 \\ 8 & 4 & 12 \end{pmatrix}$
11. $\frac{1}{54}\begin{pmatrix} 10 & 4 & 2 \\ -2 & 10 & 5 \\ 4 & -20 & 17 \end{pmatrix}$
12. $\frac{1}{20}\begin{pmatrix} 15 & 5 & -10 \\ -2 & 6 & 4 \\ -3 & -1 & 6 \end{pmatrix}$
13. $\frac{1}{48}\begin{pmatrix} -4 & 8 & 10 \\ 2 & -4 & 7 \\ 22 & 4 & 5 \end{pmatrix}$
14. $\frac{1}{2}\begin{pmatrix} -20 & -11 & 12 \\ 4 & 2 & -2 \\ -10 & -5 & 6 \end{pmatrix}$
15. $\frac{1}{49}\begin{pmatrix} 10 & 6 & 3 \\ -5 & -3 & -26 \\ -3 & 8 & 4 \end{pmatrix}$
16. $\frac{1}{53}\begin{pmatrix} 6 & 9 & -2 \\ -15 & 4 & 5 \\ 8 & 12 & 15 \end{pmatrix}$
17. $\frac{1}{61}\begin{pmatrix} 17 & -5 & -1 \\ 10 & 15 & 3 \\ 6 & 9 & 14 \end{pmatrix}$

18. $\frac{1}{39}\begin{pmatrix} 5 & 11 & 2 \\ -8 & -2 & -11 \\ -20 & -5 & -8 \end{pmatrix}$
19. $\frac{1}{17}\begin{pmatrix} 4 & 1 & -3 \\ 12 & 3 & 8 \\ -9 & 2 & -6 \end{pmatrix}$
20. $\frac{1}{26}\begin{pmatrix} -2 & -4 & -20 \\ -7 & -1 & -5 \\ -4 & -8 & -14 \end{pmatrix}$
21. $\frac{1}{54}\begin{pmatrix} 9 & 6 & 15 \\ 12 & 8 & 2 \\ 3 & 20 & 5 \end{pmatrix}$
22. $\frac{1}{14}\begin{pmatrix} 3 & 20 & 12 \\ 1 & 2 & 4 \\ 5 & 10 & 6 \end{pmatrix}$
23. $\frac{1}{46}\begin{pmatrix} 2 & 4 & 8 \\ -20 & 6 & 12 \\ -5 & -10 & 3 \end{pmatrix}$
24. $\frac{1}{37}\begin{pmatrix} 12 & 5 & 20 \\ 9 & 13 & 15 \\ -5 & 1 & 4 \end{pmatrix}$
25. $\frac{1}{62}\begin{pmatrix} 5 & 1 & -15 \\ -2 & 12 & 6 \\ 20 & 4 & 2 \end{pmatrix}$
26. $\frac{1}{13}\begin{pmatrix} 0 & 0 & 13 \\ 5 & 1 & -20 \\ 2 & 3 & -8 \end{pmatrix}$
27. $\frac{1}{62}\begin{pmatrix} -20 & 2 & 8 \\ 5 & 15 & -2 \\ 1 & 3 & 12 \end{pmatrix}$
28. $\frac{1}{27}\begin{pmatrix} 12 & 20 & -3 \\ 0 & 9 & 0 \\ 3 & 5 & 6 \end{pmatrix}$
29. $\frac{1}{18}\begin{pmatrix} -2 & 1 & 7 \\ 8 & 5 & -1 \\ -6 & 3 & 3 \end{pmatrix}$
30. $\frac{1}{38}\begin{pmatrix} -5 & 9 & 15 \\ 6 & 12 & 20 \\ -2 & -4 & 6 \end{pmatrix}$

Exercise set 3.

1. $X = \frac{1}{5}\begin{pmatrix} 13 \\ -7 \end{pmatrix}$
2. $X = \begin{pmatrix} 1 & -7 \\ 1 & 10 \end{pmatrix}$
3. $X = \begin{pmatrix} 10 \\ 7 \end{pmatrix}$
4. $X = \frac{1}{5}\begin{pmatrix} -24 \\ 16 \\ -7 \end{pmatrix}$
5. $X = \frac{1}{17}\begin{pmatrix} 1 & 10 \\ -5 & 18 \end{pmatrix}$
6. $X = \frac{1}{13}\begin{pmatrix} -17 \\ 5 \\ -15 \end{pmatrix}$
7. $X = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
8. $X = \frac{1}{22}\begin{pmatrix} 16 \\ 19 \\ -4 \end{pmatrix}$
9. $X = \begin{pmatrix} -1 & 3 \\ -2 & 4 \end{pmatrix}$
10. $X = \frac{1}{8}\begin{pmatrix} 3 \\ 6 \\ -37 \end{pmatrix}$
11. $X = \begin{pmatrix} 23 \\ 7 \end{pmatrix}$
12. $X = \frac{1}{41}\begin{pmatrix} -29 \\ -38 \\ 7 \end{pmatrix}$
13. $X = \begin{pmatrix} 3 & -9 \\ 3 & -11 \end{pmatrix}$
14. $X = \frac{1}{43}\begin{pmatrix} -1 \\ 35 \\ -2 \end{pmatrix}$
15. $X = \begin{pmatrix} 25 \\ -7 \end{pmatrix}$
16. $X = \frac{1}{8}\begin{pmatrix} 25 \\ 34 \\ 5 \end{pmatrix}$
17. $X = \begin{pmatrix} -10 & 9 \\ 7 & -6 \end{pmatrix}$
18. $X = \frac{1}{4}\begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix}$
19. $X = \frac{1}{13}\begin{pmatrix} 7 \\ -2 \end{pmatrix}$
20. $X = \frac{1}{27}\begin{pmatrix} 17 \\ 2 \\ 50 \end{pmatrix}$
21. $X = \begin{pmatrix} 6 & 7 \\ -2 & -2 \end{pmatrix}$
22. $X = \frac{1}{9}\begin{pmatrix} 11 \\ 24 \\ -10 \end{pmatrix}$
23. $X = \begin{pmatrix} -13 \\ 30 \end{pmatrix}$
24. $X = \frac{1}{4}\begin{pmatrix} -1 \\ -6 \\ 1 \end{pmatrix}$

25. $X = \begin{pmatrix} -1 & 1 \\ 0 & -3 \end{pmatrix}$
26. $X = \frac{1}{48}\begin{pmatrix} 86 \\ 41 \\ -53 \end{pmatrix}$
27. $X = \begin{pmatrix} -7 & -3 \\ -18 & -7 \end{pmatrix}$
28. $X = \frac{1}{2}\begin{pmatrix} -47 \\ 8 \\ -23 \end{pmatrix}$
29. $X = \begin{pmatrix} -7 \\ -6 \end{pmatrix}$
30. $X = \frac{1}{46}\begin{pmatrix} 38 \\ 80 \\ -49 \end{pmatrix}$

Exercise set 4.

1. $K^{-1} \equiv \begin{pmatrix} 5 & 12 \\ 15 & 25 \end{pmatrix} \pmod{26}$; $p = \begin{pmatrix} 25 \\ 9 \end{pmatrix}$
2. $\det K = 441 \equiv 25 \pmod{26}$, $\gcd(25, 26) = 1$, so K^{-1} exists; $K^{-1} \equiv \begin{pmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{pmatrix} \pmod{26}$
3. $X = A^{-1}(C - B) = \frac{1}{5}\begin{pmatrix} 10 & -6 \\ -5 & 9 \end{pmatrix}$
4. $\det A = -5$, $A^{-1} = \frac{1}{5}\begin{pmatrix} -2 & 4 & -7 \\ 3 & -1 & 3 \\ -1 & 2 & -1 \end{pmatrix}$
5. $A^{-1} = \frac{1}{5}\begin{pmatrix} 1 & 1 \\ -3 & 2 \end{pmatrix}$ (the same by both methods)
6. $G' = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$
7. every row sum is 0, so the sum of the columns is the zero vector; hence $\det L = 0$ and L^{-1} does not exist
8. $\det A = 1 + a^3 \neq 0$, i.e. $a \neq -1$; for $a = 1$, $A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}$
9. $A^{-1} = \frac{1}{5}\begin{pmatrix} 3 & 1 & 1 \\ -6 & 3 & -2 \\ -2 & 1 & 1 \end{pmatrix}$, $B^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, $(AB)^{-1} = \frac{1}{5}\begin{pmatrix} 11 & -3 & 2 \\ -8 & 4 & -1 \\ -2 & 1 & 1 \end{pmatrix}$
10. $\det M = 0 \pmod 2$, so M is not invertible

Chapter 2. Systems of Linear Algebraic Equations, Vectors and Lines in the Plane

2.1. Cramer's rule for systems of linear algebraic equations

Exercise set 1.

1. $\Delta = -4$, $\Delta_1 = 24$, $\Delta_2 = 20$; $x_1 = -6$, $x_2 = -5$
2. $\Delta = 1$, $\Delta_1 = -6$, $\Delta_2 = 2$; $x_1 = -6$, $x_2 = 2$
3. $\Delta = 6$, $\Delta_1 = -30$, $\Delta_2 = -18$; $x_1 = -5$, $x_2 = -3$
4. $\Delta = -4$, $\Delta_1 = 20$, $\Delta_2 = -12$; $x_1 = -5$, $x_2 = 3$
5. $\Delta = -11$, $\Delta_1 = -55$, $\Delta_2 = 55$; $x_1 = 5$, $x_2 = -5$
6. $\Delta = 9$, $\Delta_1 = 9$, $\Delta_2 = 36$; $x_1 = 1$, $x_2 = 4$
7. $\Delta = -2$, $\Delta_1 = -2$, $\Delta_2 = -6$; $x_1 = 1$, $x_2 = 3$
8. $\Delta = -5$, $\Delta_1 = -30$, $\Delta_2 = 20$; $x_1 = 6$, $x_2 = -4$

9. $\Delta = -7,\ \Delta_1 = -21,\ \Delta_2 = 14;\ x_1 = 3,\ x_2 = -2$
10. $\Delta = 11,\ \Delta_1 = 11,\ \Delta_2 = -22;\ x_1 = 1,\ x_2 = -2$
11. $\Delta = 3,\ \Delta_1 = 15,\ \Delta_2 = -3;\ x_1 = 5,\ x_2 = -1$
12. $\Delta = -8,\ \Delta_1 = 16,\ \Delta_2 = -40;\ x_1 = -2,\ x_2 = 5$
13. $\Delta = 4,\ \Delta_1 = 4,\ \Delta_2 = -4;\ x_1 = 1,\ x_2 = -1$
14. $\Delta = 11,\ \Delta_1 = -66,\ \Delta_2 = -33;\ x_1 = -6,\ x_2 = -3$
15. $\Delta = -3,\ \Delta_1 = 15,\ \Delta_2 = 12;\ x_1 = -5,\ x_2 = -4$
16. $\Delta = -2,\ \Delta_1 = 8,\ \Delta_2 = 0;\ x_1 = -4,\ x_2 = 0$
17. $\Delta = 10,\ \Delta_1 = -10,\ \Delta_2 = 40;\ x_1 = -1,\ x_2 = 4$
18. $\Delta = -12,\ \Delta_1 = 48,\ \Delta_2 = 48;\ x_1 = -4,\ x_2 = -4$
19. $\Delta = 10,\ \Delta_1 = -20,\ \Delta_2 = -60;\ x_1 = -2,\ x_2 = -6$
20. $\Delta = 4,\ \Delta_1 = 12,\ \Delta_2 = 12;\ x_1 = 3,\ x_2 = 3$
21. $\Delta = -1,\ \Delta_1 = 6,\ \Delta_2 = -1;\ x_1 = -6,\ x_2 = 1$
22. $\Delta = -12,\ \Delta_1 = 0,\ \Delta_2 = 0;\ x_1 = 0,\ x_2 = 0$
23. $\Delta = -12,\ \Delta_1 = 60,\ \Delta_2 = 36;\ x_1 = -5,\ x_2 = -3$
24. $\Delta = -7,\ \Delta_1 = 7,\ \Delta_2 = -21;\ x_1 = -1,\ x_2 = 3$
25. $\Delta = 2,\ \Delta_1 = -10,\ \Delta_2 = -10;\ x_1 = -5,\ x_2 = -5$
26. $\Delta = -10,\ \Delta_1 = -50,\ \Delta_2 = 10;\ x_1 = 5,\ x_2 = -1$
27. $\Delta = 6,\ \Delta_1 = 12,\ \Delta_2 = -36;\ x_1 = 2,\ x_2 = -6$
28. $\Delta = -10,\ \Delta_1 = 30,\ \Delta_2 = -20;\ x_1 = -3,\ x_2 = 2$
29. $\Delta = 3,\ \Delta_1 = -3,\ \Delta_2 = 15;\ x_1 = -1,\ x_2 = 5$
30. $\Delta = -1,\ \Delta_1 = -3,\ \Delta_2 = 1;\ x_1 = 3,\ x_2 = -1$

Exercise set 2.

1. $\Delta = 20;\ x_1 = 1,\ x_2 = -1,\ x_3 = 3$
2. $\Delta = -36;\ x_1 = 2,\ x_2 = 1,\ x_3 = -3$
3. $\Delta = 2;\ x_1 = 3,\ x_2 = -2,\ x_3 = 3$
4. $\Delta = -12;\ x_1 = 4,\ x_2 = 2,\ x_3 = 4$
5. $\Delta = 21;\ x_1 = -4,\ x_2 = -3,\ x_3 = 4$
6. $\Delta = 18;\ x_1 = -3,\ x_2 = -1,\ x_3 = 2$
7. $\Delta = -4;\ x_1 = 3,\ x_2 = -1,\ x_3 = -3$
8. $\Delta = 3;\ x_1 = 2,\ x_2 = -1,\ x_3 = 1$
9. $\Delta = -19;\ x_1 = 2,\ x_2 = 1,\ x_3 = 4$
10. $\Delta = 23;\ x_1 = 0,\ x_2 = -4,\ x_3 = -3$
11. $\Delta = 6;\ x_1 = 4,\ x_2 = 2,\ x_3 = 0$
12. $\Delta = 28;\ x_1 = -1,\ x_2 = 0,\ x_3 = 0$
13. $\Delta = 16;\ x_1 = -1,\ x_2 = 3,\ x_3 = -3$
14. $\Delta = 10;\ x_1 = -1,\ x_2 = -2,\ x_3 = 2$
15. $\Delta = -22;\ x_1 = -2,\ x_2 = 0,\ x_3 = 3$
16. $\Delta = -16;\ x_1 = -1,\ x_2 = 1,\ x_3 = -2$
17. $\Delta = -9;\ x_1 = 4,\ x_2 = -4,\ x_3 = -3$
18. $\Delta = -19;\ x_1 = -2,\ x_2 = -4,\ x_3 = 4$
19. $\Delta = 40;\ x_1 = -4,\ x_2 = -2,\ x_3 = 1$
20. $\Delta = 28;\ x_1 = 0,\ x_2 = -4,\ x_3 = 3$
21. $\Delta = -6;\ x_1 = 0,\ x_2 = -4,\ x_3 = -1$
22. $\Delta = -34;\ x_1 = 3,\ x_2 = 0,\ x_3 = -3$

23. $\Delta = 21;\ x_1 = 4,\ x_2 = 3,\ x_3 = 0$
24. $\Delta = -13;\ x_1 = -2,\ x_2 = 4,\ x_3 = 0$
25. $\Delta = 22;\ x_1 = 3,\ x_2 = 3,\ x_3 = 2$
26. $\Delta = 12;\ x_1 = 1,\ x_2 = 1,\ x_3 = 2$
27. $\Delta = 9;\ x_1 = -3,\ x_2 = 1,\ x_3 = 2$
28. $\Delta = -14;\ x_1 = 4,\ x_2 = -1,\ x_3 = -3$
29. $\Delta = -8;\ x_1 = -2,\ x_2 = 3,\ x_3 = 2$
30. $\Delta = -11;\ x_1 = -1,\ x_2 = -3,\ x_3 = -2$

Exercise set 3.

1. $\Delta = 11,\ p = (7;\ 4)$
2. $\Delta = 5,\ p = (11;\ 5)$
3. $\Delta = 7,\ p = (8;\ 19)$
4. $\Delta = 4 - k^2 \neq 0$, i.e. $k \neq \pm 2$; for $k = 1,\ x_1 = 2,\ x_2 = 1$
5. $\Delta = a^2 + a - 6 = 0$, i.e. for $a = 2$ or $a = -3$
6. $\Delta = 30,\ \Delta_1 = 30,\ \Delta_2 = -30,\ \Delta_3 = 60;\ (1; -1; 2)$
7. $\Delta = 7;\ x_1 = \frac{25}{7},\ x_2 = \frac{23}{7},\ x_3 = \frac{36}{7}$
8. $\Delta = -23;\ x_1 = \frac{41}{23},\ x_2 = \frac{19}{23}$
9. $\Delta = 0$; the first system has infinitely many solutions ($x_1 = 3 + 2x_2$), the second is inconsistent
10. $a = 3,\ b = 2,\ c = 5,\ d = 7$, i.e. $K = \left(\begin{smallmatrix} 3 & 2 \\ 5 & 7 \end{smallmatrix}\right)$

2.2. The matrix method for systems of linear equations

Exercise set 1.

1. $x_1 = 3,\ x_2 = -4$
2. $x_1 = 13,\ x_2 = 9$
3. $x_1 = 2,\ x_2 = 3$
4. $x_1 = \frac{13}{7},\ x_2 = \frac{10}{7}$
5. $x_1 = \frac{22}{7},\ x_2 = \frac{17}{7}$
6. $x_1 = 1,\ x_2 = 2$
7. $x_1 = 5,\ x_2 = 3$
8. $x_1 = \frac{34}{11},\ x_2 = \frac{13}{11}$
9. $x_1 = 1,\ x_2 = 5$
10. $x_1 = \frac{7}{2},\ x_2 = 4$
11. $x_1 = 3,\ x_2 = 1$
12. $x_1 = 1,\ x_2 = 1$
13. $x_1 = \frac{11}{3},\ x_2 = \frac{10}{3}$
14. $x_1 = 1,\ x_2 = 2$
15. $x_1 = \frac{13}{5},\ x_2 = \frac{11}{5}$
16. $x_1 = 1,\ x_2 = 1$
17. $x_1 = 1,\ x_2 = 2$
18. $x_1 = \frac{9}{7},\ x_2 = \frac{17}{7}$
19. $x_1 = 3,\ x_2 = 2$
20. $x_1 = 1,\ x_2 = 1$
21. $x_1 = 2,\ x_2 = 4$
22. $x_1 = \frac{11}{5},\ x_2 = \frac{23}{5}$
23. $x_1 = \frac{2}{17},\ x_2 = \frac{31}{17}$
24. $x_1 = \frac{13}{11},\ x_2 = \frac{25}{11}$
25. $x_1 = \frac{13}{7},\ x_2 = \frac{19}{7}$
26. $x_1 = -\frac{1}{7},\ x_2 = \frac{17}{7}$
27. $x_1 = \frac{26}{11},\ x_2 = \frac{23}{11}$
28. $x_1 = -\frac{5}{7},\ x_2 = \frac{17}{7}$
29. $x_1 = \frac{13}{7},\ x_2 = \frac{23}{7}$
30. $x_1 = 1,\ x_2 = 2$

Exercise set 2.

1. $x_1 = -2,\ x_2 = 3,\ x_3 = 5$
2. $x_1 = \frac{11}{14},\ x_2 = \frac{13}{14},\ x_3 = \frac{39}{14}$
3. $x_1 = -3,\ x_2 = 5,\ x_3 = 4$
4. $x_1 = 4,\ x_2 = 3,\ x_3 = 2$
5. $x_1 = \frac{1}{10},\ x_2 = \frac{3}{2},\ x_3 = \frac{31}{10}$
6. $x_1 = 1,\ x_2 = 2,\ x_3 = 3$
7. $x_1 = 2,\ x_2 = 1,\ x_3 = 3$
8. $x_1 = \frac{20}{11},\ x_2 = \frac{78}{55},\ x_3 = \frac{59}{55}$
9. $x_1 = \frac{7}{6},\ x_2 = \frac{1}{3},\ x_3 = \frac{13}{6}$
10. $x_1 = \frac{29}{22},\ x_2 = \frac{3}{2},\ x_3 = \frac{67}{22}$
11. $x_1 = \frac{6}{11},\ x_2 = \frac{8}{11},\ x_3 = \frac{29}{11}$
12. $x_1 = \frac{5}{3},\ x_2 = \frac{5}{3},\ x_3 = 2$
13. $x_1 = \frac{59}{78},\ x_2 = \frac{103}{78},\ x_3 = \frac{145}{78}$
14. $x_1 = \frac{55}{32},\ x_2 = \frac{79}{32},\ x_3 = \frac{53}{32}$
15. $x_1 = \frac{8}{7},\ x_2 = 1,\ x_3 = \frac{18}{7}$
16. $x_1 = \frac{1}{2},\ x_2 = \frac{13}{10},\ x_3 = \frac{29}{10}$
17. $x_1 = \frac{15}{8},\ x_2 = \frac{13}{8},\ x_3 = \frac{11}{8}$
18. $x_1 = \frac{36}{25},\ x_2 = \frac{13}{5},\ x_3 = \frac{34}{25}$
19. $x_1 = 7,\ x_2 = -2,\ x_3 = 4$
20. $x_1 = \frac{16}{11},\ x_2 = \frac{23}{11},\ x_3 = \frac{21}{11}$
21. $x_1 = \frac{27}{11},\ x_2 = \frac{20}{11},\ x_3 = \frac{52}{11}$
22. $x_1 = \frac{9}{5},\ x_2 = \frac{46}{25},\ x_3 = \frac{53}{25}$
23. $x_1 = \frac{16}{13},\ x_2 = \frac{11}{13},\ x_3 = \frac{31}{13}$
24. $x_1 = \frac{29}{16},\ x_2 = \frac{25}{16},\ x_3 = \frac{21}{16}$
25. $x_1 = \frac{13}{5},\ x_2 = \frac{6}{5},\ x_3 = 1$
26. $x_1 = \frac{1}{2},\ x_2 = \frac{3}{2},\ x_3 = \frac{5}{2}$
27. $x_1 = 1,\ x_2 = 2,\ x_3 = 3$
28. $x_1 = \frac{4}{3},\ x_2 = \frac{5}{3},\ x_3 = \frac{8}{3}$
29. $x_1 = \frac{41}{25},\ x_2 = \frac{53}{25},\ x_3 = \frac{54}{25}$
30. $x_1 = \frac{41}{15},\ x_2 = \frac{5}{3},\ x_3 = \frac{2}{5}$

Exercise set 3.

1. (a) $x = (1;\ 1)$; (b) $x = (1;\ 3)$; (c) $x = (-4;\ 7)$
2. (a) $x = (\frac{9}{5};\ -\frac{11}{5})$; (b) $x = (1;\ 1)$; (c) $x = (\frac{1}{5};\ \frac{16}{5})$
3. (a) $x = (\frac{13}{17};\ \frac{27}{17})$; (b) $x = (\frac{37}{17};\ \frac{1}{17})$; (c) $x = (\frac{2}{17};\ -\frac{5}{17})$
4. (a) $x = (\frac{11}{7};\ \frac{19}{7})$; (b) $x = (\frac{2}{7};\ \frac{13}{7})$; (c) $x = (\frac{8}{7};\ -\frac{32}{7})$
5. (a) $x = (1;\ 2)$; (b) $x = (-\frac{16}{7};\ \frac{3}{7})$; (c) $x = (2;\ 0)$
6. (a) $x = (-2;\ 3;\ 5)$; (b) $x = (-4;\ 3;\ 9)$
7. (a) $x = (-\frac{1}{4};\ \frac{5}{8};\ -\frac{9}{8})$; (b) $x = (-3;\ 3;\ 1)$
8. (a) $x = (\frac{7}{11};\ -\frac{2}{11};\ \frac{23}{11})$; (b) $x = (-\frac{12}{11};\ \frac{21}{22};\ \frac{14}{11})$
9. (a) $x = (\frac{13}{7};\ \frac{10}{7})$; (b) $x = (\frac{1}{7};\ \frac{25}{7})$; (c) $x = (-\frac{1}{7};\ -\frac{11}{7})$
10. (a) $x = (\frac{7}{13};\ \frac{1}{13};\ -\frac{29}{13})$; (b) $x = (-\frac{10}{13};\ \frac{19}{13};\ -\frac{5}{13})$
11. (a) $x = (1;\ 2)$; (b) $x = (-5;\ -15)$; (c) $x = (1;\ -1)$
12. (a) $x = (-3;\ -1)$; (b) $x = (35;\ 21)$; (c) $x = (2;\ 0)$
13. (a) $x = (\frac{5}{8};\ \frac{1}{4};\ \frac{13}{8})$; (b) $x = (\frac{1}{2};\ -2;\ \frac{1}{2})$
14. (a) $x = (2;\ -3)$; (b) $x = (1;\ 1)$; (c) $x = (1;\ -3)$
15. (a) $x = (\frac{32}{41};\ \frac{49}{41};\ \frac{5}{41})$; (b) $x = (-\frac{8}{41};\ \frac{39}{41};\ \frac{50}{41})$
16. (a) $x = (1;\ 2)$; (b) $x = (1;\ -1)$; (c) $x = (5;\ 5)$

17. (a) $x = (\frac{4}{43}; -\frac{11}{43}; \frac{137}{43})$; (b) $x = (\frac{48}{43}; \frac{40}{43}; -\frac{76}{43})$
18. (a) $x = (3; -2)$; (b) $x = (1; 1)$; (c) $x = (-18; 13)$
19. (a) $x = (\frac{59}{53}; -\frac{15}{53}; \frac{61}{53})$; (b) $x = (-\frac{32}{53}; \frac{80}{53}; \frac{81}{53})$
20. (a) $x = (1; 1)$; (b) $x = (\frac{11}{5}; -\frac{1}{5})$; (c) $x = (-\frac{3}{10}; \frac{9}{5})$
21. (a) $x = (11; -1; 4)$; (b) $x = (41; -7; 21)$
22. (a) $x = (-1; -3)$; (b) $x = (-2; -3)$; (c) $x = (3; 2)$
23. (a) $x = (-\frac{11}{61}; \frac{94}{61}; \frac{32}{61})$; (b) $x = (-\frac{88}{61}; -\frac{41}{61}; \frac{12}{61})$
24. (a) $x = (2; 2)$; (b) $x = (-3; -1)$; (c) $x = (18; 5)$
25. (a) $x = (\frac{89}{39}; \frac{37}{39}; -\frac{5}{39})$; (b) $x = (\frac{8}{39}; -\frac{44}{39}; -\frac{71}{39})$
26. (a) $x = (1; 1)$; (b) $x = (-\frac{3}{13}; -\frac{27}{13})$; (c) $x = (2; 1)$
27. (a) $x = (-\frac{10}{17}; \frac{38}{17}; -\frac{37}{17})$; (b) $x = (-\frac{7}{17}; -\frac{21}{17}; \frac{37}{17})$
28. (a) $x = (-1; -1)$; (b) $x = (-5; -4)$; (c) $x = (7; 8)$
29. (a) $x = (-\frac{6}{13}; -\frac{8}{13}; \frac{1}{13})$; (b) $x = (-\frac{11}{13}; \frac{1}{26}; -\frac{22}{13})$
30. (a) $x = (3; 3)$; (b) $x = (3; 8)$; (c) $x = (3; -1)$

2.3. The Gauss and Gauss–Jordan methods for systems of linear equations

Exercise set 1.

1. $x_1 = 1,\ x_2 = 2$
2. $x_1 = 2,\ x_2 = 1$
3. $x_1 = \frac{23}{9},\ x_2 = \frac{17}{9}$
4. $x_1 = \frac{31}{13},\ x_2 = \frac{38}{13}$
5. $x_1 = 2,\ x_2 = 3$
6. $x_1 = \frac{25}{11},\ x_2 = \frac{18}{11}$
7. $x_1 = \frac{8}{11},\ x_2 = \frac{27}{11}$
8. $x_1 = \frac{7}{5},\ x_2 = \frac{8}{5}$
9. $x_1 = \frac{13}{9},\ x_2 = \frac{19}{9}$
10. $x_1 = 3,\ x_2 = 1$
11. $x_1 = \frac{35}{19},\ x_2 = \frac{37}{19}$
12. $x_1 = -1,\ x_2 = -1$
13. $x_1 = 3,\ x_2 = 2$
14. $x_1 = \frac{9}{7},\ x_2 = \frac{12}{7}$
15. $x_1 = \frac{34}{19},\ x_2 = \frac{51}{19}$
16. $x_1 = \frac{19}{9},\ x_2 = \frac{23}{9}$
17. $x_1 = \frac{23}{7},\ x_2 = \frac{16}{7}$
18. $x_1 = \frac{20}{13},\ x_2 = \frac{27}{13}$
19. $x_1 = \frac{23}{11},\ x_2 = \frac{15}{11}$
20. $x_1 = 3,\ x_2 = 2$
21. $x_1 = \frac{15}{7},\ x_2 = \frac{22}{7}$
22. $x_1 = \frac{37}{13},\ x_2 = \frac{19}{13}$
23. $x_1 = 2,\ x_2 = 1$
24. $x_1 = 1,\ x_2 = -1$
25. $x_1 = \frac{21}{19},\ x_2 = \frac{37}{19}$
26. $x_1 = \frac{13}{8},\ x_2 = \frac{17}{8}$
27. $x_1 = \frac{13}{5},\ x_2 = \frac{11}{5}$
28. $x_1 = \frac{25}{11},\ x_2 = \frac{20}{11}$
29. $x_1 = \frac{25}{13},\ x_2 = \frac{30}{13}$
30. $x_1 = \frac{17}{11},\ x_2 = \frac{16}{11}$

Exercise set 2.

1. $x_1 = 1,\ x_2 = 2,\ x_3 = 3$
2. no solution (rank $A = 2 <$ rank $\bar{A} = 3$)
3. $x_1 = -3,\ x_2 = 5,\ x_3 = 4$
4. $x_1 = 4,\ x_2 = 3,\ x_3 = 2$
5. $x_1 = \frac{1}{10},\ x_2 = \frac{3}{2},\ x_3 = \frac{31}{10}$
6. $x_1 = \frac{6}{11},\ x_2 = \frac{8}{11},\ x_3 = \frac{29}{11}$
7. $x_1 = \frac{20}{11},\ x_2 = \frac{78}{55},\ x_3 = \frac{59}{55}$
8. $x_1 = \frac{7}{6},\ x_2 = \frac{1}{3},\ x_3 = \frac{13}{6}$
9. $x_1 = 2,\ x_2 = 1,\ x_3 = 3$
10. $x_1 = \frac{5}{3},\ x_2 = \frac{5}{3},\ x_3 = 2$
11. $x_1 = \frac{59}{78},\ x_2 = \frac{103}{78},\ x_3 = \frac{145}{78}$
12. $x_1 = \frac{8}{7},\ x_2 = 1,\ x_3 = \frac{18}{7}$
13. $x_1 = \frac{55}{32},\ x_2 = \frac{79}{32},\ x_3 = \frac{53}{32}$
14. $x_1 = \frac{1}{2},\ x_2 = \frac{13}{10},\ x_3 = \frac{29}{10}$
15. $x_1 = \frac{15}{8},\ x_2 = \frac{13}{8},\ x_3 = \frac{11}{8}$
16. $x_1 = \frac{36}{25},\ x_2 = \frac{13}{5},\ x_3 = \frac{34}{25}$
17. no solution (rank $A = 2 <$ rank $\bar{A} = 3$)
18. $x_1 = \frac{16}{11},\ x_2 = \frac{23}{11},\ x_3 = \frac{21}{11}$
19. $x_1 = \frac{3}{2},\ x_2 = \frac{1}{2},\ x_3 = \frac{5}{2}$
20. $x_1 = \frac{27}{11},\ x_2 = \frac{20}{11},\ x_3 = \frac{52}{11}$
21. $x_1 = \frac{9}{5},\ x_2 = \frac{46}{25},\ x_3 = \frac{53}{25}$
22. $x_1 = \frac{16}{13},\ x_2 = \frac{11}{13},\ x_3 = \frac{31}{13}$
23. $x_1 = \frac{13}{5},\ x_2 = \frac{6}{5},\ x_3 = 1$
24. $x_1 = \frac{29}{16},\ x_2 = \frac{25}{16},\ x_3 = \frac{21}{16}$
25. $x_1 = \frac{1}{2},\ x_2 = \frac{3}{2},\ x_3 = \frac{5}{2}$
26. no solution (rank $A = 2 <$ rank $\bar{A} = 3$)
27. $x_1 = \frac{4}{3},\ x_2 = \frac{5}{3},\ x_3 = \frac{8}{3}$
28. $x_1 = \frac{41}{25},\ x_2 = \frac{53}{25},\ x_3 = \frac{54}{25}$
29. $x_1 = \frac{41}{15},\ x_2 = \frac{5}{3},\ x_3 = \frac{2}{5}$
30. $x_1 = \frac{11}{7},\ x_2 = \frac{15}{7},\ x_3 = \frac{16}{7}$

Exercise set 3.

1. $x_1 = -\frac{11}{5},\ x_2 = \frac{24}{5},\ x_3 = \frac{17}{5}$
2. $x_1 = 2,\ x_2 = 1$
3. $x_1 = \frac{43}{19},\ x_2 = \frac{32}{19},\ x_3 = \frac{23}{19}$
4. $x_1 = \frac{17}{7},\ x_2 = \frac{15}{7}$
5. $x_1 = \frac{11}{5},\ x_2 = \frac{6}{5},\ x_3 = -\frac{1}{5}$
6. $x_1 = 2,\ x_2 = 1,\ x_3 = 2$
7. $x_1 = 1,\ x_2 = 2$
8. $x_1 = 1,\ x_2 = 2,\ x_3 = 3$
9. $x_1 = \frac{41}{25},\ x_2 = \frac{53}{25},\ x_3 = \frac{54}{25}$
10. $x_1 = \frac{18}{7},\ x_2 = \frac{17}{7}$
11. no solution (rank $A = 2 <$ rank $\bar{A} = 3$)
12. $x_1 = 1,\ x_2 = 3$
13. $x_1 = \frac{16}{13},\ x_2 = \frac{11}{13},\ x_3 = \frac{31}{13}$
14. $x_1 = 1,\ x_2 = 5$
15. $x_1 = \frac{8}{7},\ x_2 = 1,\ x_3 = \frac{18}{7}$
16. $x_1 = \frac{5}{3},\ x_2 = \frac{5}{3},\ x_3 = 2$
17. $x_1 = \frac{15}{7},\ x_2 = \frac{11}{7}$
18. $x_1 = \frac{16}{11},\ x_2 = \frac{23}{11},\ x_3 = \frac{21}{11}$
19. $x_1 = \frac{27}{11},\ x_2 = \frac{20}{11},\ x_3 = \frac{52}{11}$
20. $x_1 = \frac{7}{3},\ x_2 = \frac{5}{3}$
21. $x_1 = \frac{1}{2},\ x_2 = \frac{3}{2},\ x_3 = \frac{5}{2}$
22. $x_1 = \frac{3}{2},\ x_2 = \frac{1}{2},\ x_3 = \frac{5}{2}$
23. $x_1 = \frac{9}{4},\ x_2 = \frac{7}{4}$
24. $x_1 = \frac{36}{25},\ x_2 = \frac{13}{5},\ x_3 = \frac{34}{25}$
25. $x_1 = \frac{13}{5},\ x_2 = \frac{6}{5},\ x_3 = 1$
26. $x_1 = \frac{35}{19},\ x_2 = \frac{37}{19}$
27. no solution (rank $A = 2 <$ rank $\bar{A} = 3$)
28. $x_1 = \frac{29}{16},\ x_2 = \frac{25}{16},\ x_3 = \frac{21}{16}$
29. $x_1 = \frac{13}{11},\ x_2 = \frac{25}{11}$
30. $x_1 = \frac{4}{3},\ x_2 = \frac{5}{3},\ x_3 = \frac{8}{3}$

Exercise set 4.

1. infinitely many solutions: $x_1 = -3x_3 - 1,\ x_2 = 2x_3 + 2$ (x_3 arbitrary)
2. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
3. infinitely many solutions: $x_1 = -x_2 - x_3 + 1$ (x_2, x_3 arbitrary)
4. infinitely many solutions: $x_1 = 5 - 2x_2$ (x_2 arbitrary)
5. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
6. infinitely many solutions: $x_1 = 1,\ x_2 = x_3 - x_4 - 1$ (x_3, x_4 arbitrary)
7. infinitely many solutions: $x_1 = \frac{x_2}{3} + \frac{5}{3}$ (x_2 arbitrary)
8. infinitely many solutions: $x_1 = x_3 - \frac{1}{3},\ x_2 = \frac{2}{3} - 2x_3$ (x_3 arbitrary)
9. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
10. infinitely many solutions: $x_1 = 2 - x_4,\ x_2 = x_3 - x_4 + 1$ (x_3, x_4 arbitrary)
11. infinitely many solutions: $x_1 = -3x_2 + 2x_3$ (x_2, x_3 arbitrary)
12. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
13. infinitely many solutions: $x_1 = 2 - \frac{x_3}{2},\ x_2 = \frac{3x_3}{2} - 1$ (x_3 arbitrary)
14. infinitely many solutions: $x_1 = -\frac{x_2}{2} + \frac{x_3}{2} + 2$ (x_2, x_3 arbitrary)
15. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
16. infinitely many solutions: $x_1 = -2x_2 - x_3 - x_4 + 5$ (x_2, x_3, x_4 arbitrary)
17. $x_1 = 3,\ x_2 = 1$
18. infinitely many solutions: $x_1 = -2x_2 + x_3 + 1$ (x_2, x_3 arbitrary)
19. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
20. infinitely many solutions: $x_1 = -\frac{11x_4}{5} - 1,\ x_2 = -\frac{x_4}{5} - 1,\ x_3 = \frac{7x_4}{5} + 2$ (x_4 arbitrary)
21. infinitely many solutions: $x_1 = -2x_2 - 3x_3$ (x_2, x_3 arbitrary)
22. infinitely many solutions: $x_1 = x_2 + 2$ (x_2 arbitrary)
23. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
24. infinitely many solutions: $x_1 = -x_2 + x_3 + 3$ (x_2, x_3 arbitrary)
25. infinitely many solutions: $x_1 = -2x_2 + x_3 + 1,\ x_4 = 1$ (x_2, x_3 arbitrary)
26. infinitely many solutions: $x_1 = 3x_2 + 5$ (x_2 arbitrary)
27. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
28. infinitely many solutions: $x_1 = 3 - 2x_2$ (x_2 arbitrary)
29. no solution (rank $A = 1 <$ rank $\bar{A} = 2$)
30. infinitely many solutions: $x_1 = -4x_3 - 1,\ x_2 = 3x_3 + 2$ (x_3 arbitrary)

2.4. Arithmetic operations on vectors: multiplication of a vector by a number, addition and subtraction of vectors

Exercise set 1.

1. $(-2; 22; -17)$
2. $(7; 16; -14)$
3. $(1; 3; 1)$
4. $(13; -7; 5)$
5. $(11; -2; 7)$
6. $(-1; -1; 12)$
7. $(12; 4; -13)$
8. $(1; 6; -9)$
9. $(7; 17; -13)$
10. $(-4; 6; -14)$
11. $(4; -4; 8)$
12. $(-2; 6; 0)$
13. $(-6; 18; 13)$
14. $(5; -4; 7)$
15. $(10; 11; 14)$
16. $(-7; 8; -11)$
17. $(-1; 3; 2)$
18. $(6; -18; 6)$
19. $(-4; 17; 13)$
20. $(-3; -6; 3)$
21. $(-5; 2; 2)$
22. $(-13; -9; 17)$
23. $(18; 0; -6)$
24. $(10; 2; -7)$
25. $(-3; 3; 0)$
26. $(4; 7; 6)$
27. $(13; -8; -8)$
28. $(2; 2; -4)$
29. $(15; -19; -1)$
30. $(15; 12; 6)$

Exercise set 2.

1. $\overrightarrow{AB} = (2; 2; 7)$, $|\overrightarrow{AB}| = \sqrt{57}$, $M(0; -1; \frac{5}{2})$
2. $\overrightarrow{AB} = (1; -3; 0)$, $|\overrightarrow{AB}| = \sqrt{10}$, $M(-\frac{5}{2}; \frac{5}{2}; 5)$
3. $\overrightarrow{AB} = (4; -4; -1)$, $|\overrightarrow{AB}| = \sqrt{33}$, $M(0; 0; \frac{1}{2})$
4. $\overrightarrow{AB} = (-2; 1; -1)$, $|\overrightarrow{AB}| = \sqrt{6}$, $M(0; \frac{9}{2}; -\frac{3}{2})$
5. $\overrightarrow{AB} = (2; 0; -6)$, $|\overrightarrow{AB}| = 2\sqrt{10}$, $M(2; 1; -2)$
6. $\overrightarrow{AB} = (-6; 4; 0)$, $|\overrightarrow{AB}| = 2\sqrt{13}$, $M(2; 4; 5)$
7. $\overrightarrow{AB} = (3; -6; 9)$, $|\overrightarrow{AB}| = 3\sqrt{14}$, $M(\frac{3}{2}; 3; \frac{1}{2})$
8. $\overrightarrow{AB} = (2; 3; -9)$, $|\overrightarrow{AB}| = \sqrt{94}$, $M(4; -\frac{7}{2}; -\frac{3}{2})$
9. $\overrightarrow{AB} = (-4; -4; 1)$, $|\overrightarrow{AB}| = \sqrt{33}$, $M(1; -3; \frac{9}{2})$
10. $\overrightarrow{AB} = (-2; 0; -3)$, $|\overrightarrow{AB}| = \sqrt{13}$, $M(-1; 3; -\frac{5}{2})$
11. $\overrightarrow{AB} = (-2; 6; 8)$, $|\overrightarrow{AB}| = 2\sqrt{26}$, $M(-4; -1; 2)$
12. $\overrightarrow{AB} = (11; -2; -3)$, $|\overrightarrow{AB}| = \sqrt{134}$, $M(-\frac{1}{2}; -4; \frac{3}{2})$
13. $\overrightarrow{AB} = (4; 3; 0)$, $|\overrightarrow{AB}| = 5$, $M(-1; \frac{9}{2}; 3)$
14. $\overrightarrow{AB} = (5; 3; 0)$, $|\overrightarrow{AB}| = \sqrt{34}$, $M(-\frac{7}{2}; \frac{3}{2}; -3)$
15. $\overrightarrow{AB} = (4; 0; -4)$, $|\overrightarrow{AB}| = 4\sqrt{2}$, $M(4; 6; 1)$
16. $\overrightarrow{AB} = (9; -10; -11)$, $|\overrightarrow{AB}| = \sqrt{302}$, $M(-\frac{3}{2}; -1; -\frac{1}{2})$
17. $\overrightarrow{AB} = (-4; 4; 4)$, $|\overrightarrow{AB}| = 4\sqrt{3}$, $M(1; 1; 2)$
18. $\overrightarrow{AB} = (-1; -3; 10)$, $|\overrightarrow{AB}| = \sqrt{110}$, $M(-\frac{9}{2}; \frac{9}{2}; 1)$
19. $\overrightarrow{AB} = (6; 6; 8)$, $|\overrightarrow{AB}| = 2\sqrt{34}$, $M(2; -1; -2)$
20. $\overrightarrow{AB} = (-6; 2; 8)$, $|\overrightarrow{AB}| = 2\sqrt{26}$, $M(3; 0; -1)$
21. $\overrightarrow{AB} = (2; 11; 5)$, $|\overrightarrow{AB}| = 5\sqrt{6}$, $M(0; -\frac{1}{2}; -\frac{3}{2})$
22. $\overrightarrow{AB} = (-2; 2; -9)$, $|\overrightarrow{AB}| = \sqrt{89}$, $M(2; 2; \frac{1}{2})$

23. $\overrightarrow{AB} = (3; 2; 2)$, $|\overrightarrow{AB}| = \sqrt{17}$, $M(\frac{1}{2}; 1; 0)$
24. $\overrightarrow{AB} = (0; 7; 1)$, $|\overrightarrow{AB}| = 5\sqrt{2}$, $M(0; \frac{1}{2}; -\frac{9}{2})$
25. $\overrightarrow{AB} = (10; 1; -1)$, $|\overrightarrow{AB}| = \sqrt{102}$, $M(-1; -\frac{7}{2}; \frac{7}{2})$
26. $\overrightarrow{AB} = (2; -6; -5)$, $|\overrightarrow{AB}| = \sqrt{65}$, $M(1; 3; -\frac{7}{2})$
27. $\overrightarrow{AB} = (-5; 3; -2)$, $|\overrightarrow{AB}| = \sqrt{38}$, $M(\frac{1}{2}; -\frac{9}{2}; -5)$
28. $\overrightarrow{AB} = (1; 7; 4)$, $|\overrightarrow{AB}| = \sqrt{66}$, $M(\frac{7}{2}; -\frac{1}{2}; -3)$
29. $\overrightarrow{AB} = (4; 5; 12)$, $|\overrightarrow{AB}| = \sqrt{185}$, $M(-2; \frac{1}{2}; 0)$
30. $\overrightarrow{AB} = (6; 2; 2)$, $|\overrightarrow{AB}| = 2\sqrt{11}$, $M(3; -4; 0)$

Exercise set 3.

1. $\vec{a}^0 = (\frac{2}{3}; -\frac{1}{3}; \frac{2}{3})$
2. yes, $\vec{b} = -\frac{3}{2}\vec{a}$; opposite directions
3. $m = 2,\ n = 6$
4. $D(2; -3; 3)$
5. $G(2; 1; 1)$
6. $\cos\alpha = \frac{1}{3},\ \cos\beta = \frac{2}{3},\ \cos\gamma = -\frac{2}{3}$
7. $\vec{u} - \vec{v} = (4; -2; -2; 0)$, $\frac{1}{2}(\vec{u} + \vec{v}) = (3; 3; 1; 3)$
8. $\vec{c} = 2\vec{a} + 3\vec{b}$
9. $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| = 5$
10. $M(3; 0; 2)$

2.5. Scalar, vector and mixed products of vectors

Exercise set 1.

1. $\vec{a} \cdot \vec{b} = 21$, $\cos\varphi = \frac{21\sqrt{493}}{493}$
2. $\vec{a} \cdot \vec{b} = -7$, $\cos\varphi = -\frac{7\sqrt{110}}{330}$
3. $\vec{a} \cdot \vec{b} = 7$, $\cos\varphi = \frac{7\sqrt{754}}{754}$
4. $\vec{a} \cdot \vec{b} = 7$, $\cos\varphi = \frac{\sqrt{1239}}{177}$
5. $\vec{a} \cdot \vec{b} = -24$, $\cos\varphi = -\frac{4\sqrt{33}}{55}$
6. $\vec{a} \cdot \vec{b} = 3$, $\cos\varphi = \frac{\sqrt{66}}{110}$
7. $\vec{a} \cdot \vec{b} = -2$, $\cos\varphi = -\frac{1}{15}$
8. $\vec{a} \cdot \vec{b} = -17$, $\cos\varphi = -\frac{\sqrt{510}}{30}$
9. $\vec{a} \cdot \vec{b} = 19$, $\cos\varphi = \frac{19\sqrt{55}}{330}$
10. $\vec{a} \cdot \vec{b} = 7$, $\cos\varphi = \frac{7\sqrt{130}}{130}$
11. $\vec{a} \cdot \vec{b} = 4$, $\cos\varphi = \frac{\sqrt{105}}{105}$
12. $\vec{a} \cdot \vec{b} = 22$, $\cos\varphi = \frac{11\sqrt{21}}{63}$
13. $\vec{a} \cdot \vec{b} = 11$, $\cos\varphi = \frac{\sqrt{22}}{18}$
14. $\vec{a} \cdot \vec{b} = 5$, $\cos\varphi = \frac{5\sqrt{19}}{57}$
15. $\vec{a} \cdot \vec{b} = -19$, $\cos\varphi = -\frac{19\sqrt{510}}{510}$
16. $\vec{a} \cdot \vec{b} = 13$, $\cos\varphi = \frac{\sqrt{754}}{58}$
17. $\vec{a} \cdot \vec{b} = -3$, $\cos\varphi = -\frac{\sqrt{1122}}{374}$
18. $\vec{a} \cdot \vec{b} = 23$, $\cos\varphi = \frac{23\sqrt{646}}{646}$
19. $\vec{a} \cdot \vec{b} = -2$, $\cos\varphi = -\frac{\sqrt{70}}{105}$
20. $\vec{a} \cdot \vec{b} = -4$, $\cos\varphi = -\frac{2\sqrt{15}}{45}$
21. $\vec{a} \cdot \vec{b} = -28$, $\cos\varphi = -\frac{2\sqrt{399}}{57}$
22. $\vec{a} \cdot \vec{b} = -14$, $\cos\varphi = -\frac{14\sqrt{969}}{969}$
23. $\vec{a} \cdot \vec{b} = 9$, $\cos\varphi = \frac{3\sqrt{21}}{14}$
24. $\vec{a} \cdot \vec{b} = -28$, $\cos\varphi = -\frac{7\sqrt{15}}{45}$
25. $\vec{a} \cdot \vec{b} = 16$, $\cos\varphi = \frac{4\sqrt{5}}{15}$
26. $\vec{a} \cdot \vec{b} = -20$, $\cos\varphi = -\frac{20}{29}$

27. $\vec{a}\cdot\vec{b}=-10$, $\cos\varphi=-\frac{\sqrt{10}}{7}$
28. $\vec{a}\cdot\vec{b}=-28$, $\cos\varphi=-\frac{7\sqrt{258}}{129}$
29. $\vec{a}\cdot\vec{b}=12$, $\cos\varphi=\frac{\sqrt{30}}{15}$
30. $\vec{a}\cdot\vec{b}=9$, $\cos\varphi=\frac{3\sqrt{21}}{70}$

Exercise set 2.

1. $\vec{a}\times\vec{b}=(-7;\,0;\,0)$, $S=7$
2. $\vec{a}\times\vec{b}=(-8;\,6;\,12)$, $S=2\sqrt{61}$
3. $\vec{a}\times\vec{b}=(-7;\,3;\,-3)$, $S=\sqrt{67}$
4. $\vec{a}\times\vec{b}=(3;\,-4;\,-6)$, $S=\sqrt{61}$
5. $\vec{a}\times\vec{b}=(-14;\,16;\,-4)$, $S=6\sqrt{13}$
6. $\vec{a}\times\vec{b}=(-6;\,0;\,-12)$, $S=6\sqrt{5}$
7. $\vec{a}\times\vec{b}=(10;\,12;\,4)$, $S=2\sqrt{65}$
8. $\vec{a}\times\vec{b}=(24;\,12;\,-12)$, $S=12\sqrt{6}$
9. $\vec{a}\times\vec{b}=(-3;\,6;\,3)$, $S=3\sqrt{6}$
10. $\vec{a}\times\vec{b}=(5;\,4;\,6)$, $S=\sqrt{77}$
11. $\vec{a}\times\vec{b}=(-3;\,3;\,6)$, $S=3\sqrt{6}$
12. $\vec{a}\times\vec{b}=(0;\,-20;\,20)$, $S=20\sqrt{2}$
13. $\vec{a}\times\vec{b}=(-6;\,6;\,3)$, $S=9$
14. $\vec{a}\times\vec{b}=(21;\,0;\,-21)$, $S=21\sqrt{2}$
15. $\vec{a}\times\vec{b}=(16;\,16;\,0)$, $S=16\sqrt{2}$
16. $\vec{a}\times\vec{b}=(12;\,5;\,9)$, $S=5\sqrt{10}$
17. $\vec{a}\times\vec{b}=(8;\,4;\,4)$, $S=4\sqrt{6}$
18. $\vec{a}\times\vec{b}=(3;\,-5;\,4)$, $S=5\sqrt{2}$
19. $\vec{a}\times\vec{b}=(0;\,-2;\,-1)$, $S=\sqrt{5}$
20. $\vec{a}\times\vec{b}=(1;\,5;\,4)$, $S=\sqrt{42}$
21. $\vec{a}\times\vec{b}=(6;\,3;\,-2)$, $S=7$
22. $\vec{a}\times\vec{b}=(6;\,-6;\,9)$, $S=3\sqrt{17}$
23. $\vec{a}\times\vec{b}=(-10;\,15;\,-5)$, $S=5\sqrt{14}$
24. $\vec{a}\times\vec{b}=(4;\,0;\,-4)$, $S=4\sqrt{2}$
25. $\vec{a}\times\vec{b}=(10;\,-10;\,10)$, $S=10\sqrt{3}$
26. $\vec{a}\times\vec{b}=(-9;\,8;\,-4)$, $S=\sqrt{161}$
27. $\vec{a}\times\vec{b}=(-3;\,-6;\,15)$, $S=3\sqrt{30}$
28. $\vec{a}\times\vec{b}=(6;\,-4;\,9)$, $S=\sqrt{133}$
29. $\vec{a}\times\vec{b}=(-7;\,16;\,-4)$, $S=\sqrt{321}$
30. $\vec{a}\times\vec{b}=(-14;\,8;\,-12)$, $S=2\sqrt{101}$

Exercise set 3.

1. $(\vec{a},\vec{b},\vec{c})=-12$; not coplanar, $V=12$
2. $(\vec{a},\vec{b},\vec{c})=4$; not coplanar, $V=4$
3. $(\vec{a},\vec{b},\vec{c})=65$; not coplanar, $V=65$
4. $(\vec{a},\vec{b},\vec{c})=-17$; not coplanar, $V=17$
5. $(\vec{a},\vec{b},\vec{c})=-26$; not coplanar, $V=26$
6. $(\vec{a},\vec{b},\vec{c})=0$; coplanar, $V=0$
7. $V=\frac{1}{6}|-4|=\frac{2}{3}$
8. $\lambda=5$
9. $7m-10=0$, $m=\frac{10}{7}$
10. $S=\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|=\frac{7}{2}$

2.6. Equations of a straight line in the plane in various forms. The angle between two lines. The distance from a point to a line

Exercise set 1.

1. $4x-3y-10=0$
2. $x+8y+4=0$
3. $x+7y+13=0$
4. $x-3y+14=0$
5. $3x-y-12=0$
6. $x-4y+8=0$
7. $x+1=0$
8. $8x+3y+39=0$
9. $y-4=0$
10. $12x-5y+30=0$
11. $7x+8y+10=0$
12. $5x+6y+25=0$
13. $x+4y-4=0$
14. $7x+3y-2=0$
15. $2x+3y+8=0$
16. $4x+y-3=0$
17. $x+y-4=0$
18. $x-y+2=0$
19. $9x+11y+10=0$
20. $8x+7y-18=0$
21. $x-6y-5=0$
22. $y+2=0$
23. $y+1=0$
24. $3x-4y+6=0$
25. $x+2y+6=0$
26. $y+6=0$
27. $10x+3y+18=0$
28. $x-y-3=0$
29. $9x+7y+3=0$
30. $x-3y-11=0$

Exercise set 2.

1. $d=\frac{\sqrt{10}}{10}$
2. $d=\frac{14\sqrt{41}}{41}$
3. $d=\frac{2\sqrt{10}}{5}$
4. $d=\frac{11\sqrt{26}}{26}$
5. $d=\sqrt{5}$
6. $d=\frac{3\sqrt{61}}{61}$
7. $d=\frac{\sqrt{10}}{4}$
8. $d=\frac{11\sqrt{2}}{12}$
9. $d=\frac{2\sqrt{34}}{17}$
10. $d=\frac{\sqrt{13}}{2}$
11. $d=\frac{5\sqrt{41}}{41}$
12. $d=3\sqrt{2}$
13. $d=\frac{7\sqrt{26}}{26}$
14. $d=\frac{19\sqrt{5}}{15}$
15. $d=\frac{8\sqrt{41}}{41}$
16. $d=\frac{11\sqrt{2}}{4}$
17. $d=\frac{16\sqrt{61}}{61}$
18. $d=\frac{13}{5}$
19. $d=\frac{17\sqrt{10}}{20}$
20. $d=\frac{7\sqrt{2}}{6}$
21. $d=\frac{5\sqrt{61}}{61}$
22. $d=\sqrt{2}$
23. $d=0$ (M lies on the line)
24. $d=\frac{16\sqrt{29}}{29}$
25. $d=\frac{7\sqrt{26}}{26}$
26. $d=\frac{40\sqrt{61}}{61}$
27. $d=\frac{7\sqrt{34}}{17}$
28. $d=\frac{15\sqrt{41}}{41}$
29. $d=\frac{8\sqrt{26}}{13}$
30. $d=\frac{2\sqrt{17}}{17}$

Exercise set 3.

1. perpendicular ($\varphi=90^\circ$)
2. parallel ($\varphi=0$)
3. $\cos\varphi=\frac{\sqrt{2}}{10}$, $\varphi\approx 81.9^\circ$
4. perpendicular ($\varphi=90^\circ$)
5. $\cos\varphi=\frac{5\sqrt{221}}{221}$, $\varphi\approx 70.3^\circ$
6. $3x-4y-18=0$
7. $x-2y+7=0$
8. $\frac{x}{-4}+\frac{y}{6}=1$
9. $d=5$
10. $AB:\ 4x-3y-1=0$, $h=\frac{24}{5}$

Chapter 3. Lines and Planes in Space; Linear and Euclidean Spaces

3.1. Lines in space. Equations of a line in space, the angle between lines, the distance between lines

Exercise set 1.

1. $\frac{x+1}{6} = \frac{y+1}{-2} = \frac{z-5}{0}$
2. $\frac{x+3}{1} = \frac{y+4}{3} = \frac{z-3}{-8}$
3. $\frac{x-1}{3} = \frac{y+3}{-2} = \frac{z-5}{-6}$
4. $\frac{x+3}{5} = \frac{y+4}{5} = \frac{z+1}{-2}$
5. $\frac{x+1}{3} = \frac{y}{3} = \frac{z+2}{6}$
6. $\frac{x-5}{-1} = \frac{y+4}{9} = \frac{z}{-1}$
7. $\frac{x-2}{-5} = \frac{y-3}{-1} = \frac{z-4}{1}$
8. $\frac{x-2}{-3} = \frac{y-3}{-5} = \frac{z+3}{0}$
9. $\frac{x-3}{-3} = \frac{y}{2} = \frac{z+1}{0}$
10. $\frac{x-4}{-7} = \frac{y+1}{5} = \frac{z-1}{1}$
11. $\frac{x-3}{-5} = \frac{y+2}{0} = \frac{z-4}{-4}$
12. $\frac{x+3}{8} = \frac{y+4}{9} = \frac{z-1}{1}$
13. $\frac{x-1}{-6} = \frac{y+5}{3} = \frac{z-1}{-4}$
14. $\frac{x-2}{-3} = \frac{y-2}{-5} = \frac{z-5}{-4}$
15. $\frac{x+1}{6} = \frac{y}{0} = \frac{z+2}{5}$
16. $\frac{x-5}{-7} = \frac{y+3}{5} = \frac{z-4}{-6}$
17. $\frac{x-4}{0} = \frac{y+5}{0} = \frac{z-4}{-4}$
18. $\frac{x-4}{-7} = \frac{y-4}{-5} = \frac{z}{-2}$
19. $\frac{x+3}{-1} = \frac{y-5}{-7} = \frac{z-3}{-6}$
20. $\frac{x+4}{8} = \frac{y-1}{2} = \frac{z-5}{-2}$
21. $\frac{x+5}{10} = \frac{y-2}{-6} = \frac{z}{3}$
22. $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z+1}{0}$
23. $\frac{x+1}{-3} = \frac{y-4}{-2} = \frac{z-4}{-1}$
24. $\frac{x}{0} = \frac{y+5}{5} = \frac{z+5}{2}$
25. $\frac{x-2}{1} = \frac{y-3}{-1} = \frac{z-3}{-5}$
26. $\frac{x-5}{-10} = \frac{y+2}{3} = \frac{z+1}{4}$
27. $\frac{x+3}{4} = \frac{y+3}{8} = \frac{z+1}{-1}$
28. $\frac{x+1}{-1} = \frac{y-1}{-1} = \frac{z-4}{-6}$
29. $\frac{x-4}{-6} = \frac{y+1}{6} = \frac{z-1}{-1}$
30. $\frac{x-1}{0} = \frac{y+5}{7} = \frac{z}{-4}$

Exercise set 2.

1. $\cos\varphi = \frac{4\sqrt{410}}{205}$
2. $\cos\varphi = \frac{5\sqrt{174}}{174}$
3. $\cos\varphi = \frac{\sqrt{70}}{10}$
4. $\cos\varphi = \frac{3\sqrt{165}}{55}$
5. $\cos\varphi = \frac{2\sqrt{266}}{133}$
6. $\cos\varphi = \frac{\sqrt{7}}{7}$
7. $\varphi = 90°$
8. $\cos\varphi = \frac{10\sqrt{41}}{123}$
9. $\cos\varphi = \frac{\sqrt{17}}{34}$
10. $\cos\varphi = \frac{5\sqrt{34}}{34}$
11. $\cos\varphi = \frac{3\sqrt{102}}{34}$
12. $\cos\varphi = \frac{19\sqrt{462}}{462}$
13. $\cos\varphi = \frac{3\sqrt{170}}{170}$
14. $\varphi = 90°$
15. $\cos\varphi = \frac{\sqrt{130}}{130}$
16. $\cos\varphi = \frac{2\sqrt{858}}{429}$
17. $\cos\varphi = \frac{2}{7}$
18. $\cos\varphi = \frac{12\sqrt{221}}{221}$
19. $\cos\varphi = \frac{5\sqrt{574}}{574}$
20. $\cos\varphi = \frac{11\sqrt{14}}{70}$
21. $\cos\varphi = \frac{6\sqrt{377}}{377}$
22. $\cos\varphi = \frac{8\sqrt{357}}{357}$
23. $\cos\varphi = \frac{7\sqrt{374}}{187}$
24. $\cos\varphi = \frac{7\sqrt{6}}{18}$
25. $\cos\varphi = \frac{5\sqrt{638}}{638}$
26. $\cos\varphi = \frac{\sqrt{58}}{58}$
27. $\cos\varphi = \frac{\sqrt{2}}{10}$
28. $\cos\varphi = \frac{9\sqrt{130}}{130}$
29. $\cos\varphi = \frac{\sqrt{114}}{57}$
30. $\cos\varphi = \frac{\sqrt{3}}{15}$

Exercise set 3.

1. $d = \sqrt{13}$
2. $d = \frac{13\sqrt{34}}{17}$
3. $d = \frac{\sqrt{1115}}{5}$
4. $d = \frac{\sqrt{70}}{5}$
5. $d = \frac{2\sqrt{66}}{11}$
6. $d = \frac{2\sqrt{874}}{19}$
7. $d = \frac{7\sqrt{30}}{5}$
8. $x = 1 + 2t,\ y = -3 - t,\ z = 4t$; $(5; -5; 8)$
9. yes, at the point $(2; 4; 4)$
10. $d = 2$

3.2. Planes in space. The general equation of a plane. Relative position of lines and planes in space

Exercise set 1.

1. $x + y + z = 0$
2. $x - 6y + 2z + 5 = 0$
3. $16x - 9y + 24z + 68 = 0$
4. $2x - 3y - z + 5 = 0$
5. $4x - 15y - 2z - 31 = 0$
6. $x - z = 0$
7. $20x + 15y + 22z + 66 = 0$
8. $5x + 20y + 2z - 66 = 0$
9. $x - 3y + z + 7 = 0$
10. $7x + 14y + 13z + 10 = 0$
11. $3x + 8y - z - 8 = 0$
12. $3x - 6y - 5z + 6 = 0$
13. $61x - 20y + 28z + 52 = 0$
14. $11x + 13y - 3z + 23 = 0$
15. $7x + 2y - z + 5 = 0$
16. $3x + 8y + 7z - 41 = 0$
17. $9x - 4y + 11z + 10 = 0$
18. $7x + 10y - 8z + 31 = 0$
19. $37x - 2y + 12z - 46 = 0$
20. $4x - 13y + 7z - 4 = 0$
21. $5x + 4y - 11z + 18 = 0$
22. $z + 3 = 0$
23. $4x - 7y + 3z + 28 = 0$
24. $5x - y + 2z - 3 = 0$
25. $13x + 2y - 3z + 41 = 0$
26. $4x + 3y - 20z + 8 = 0$
27. $7x - 3y + 20z - 41 = 0$
28. $y - 2 = 0$
29. $2y + z + 3 = 0$
30. $14x + y + 11z - 8 = 0$

Exercise set 2.

1. $d = \frac{4\sqrt{66}}{11}$
2. $d = \frac{27\sqrt{34}}{34}$
3. $d = \frac{11\sqrt{3}}{3}$
4. $d = \frac{\sqrt{26}}{26}$
5. $d = \frac{2\sqrt{14}}{7}$
6. $d = \frac{4\sqrt{6}}{3}$
7. $d = \frac{21\sqrt{22}}{44}$
8. $d = \frac{\sqrt{22}}{22}$
9. $d = \frac{33\sqrt{53}}{53}$
10. $d = \frac{\sqrt{6}}{2}$
11. $d = 2\sqrt{6}$
12. $d = \frac{15\sqrt{38}}{38}$
13. $d = \frac{40\sqrt{61}}{61}$
14. $d = \frac{12\sqrt{38}}{19}$
15. $d = \frac{25\sqrt{11}}{22}$
16. $d = \frac{23\sqrt{42}}{21}$
17. $d = \frac{25\sqrt{22}}{22}$
18. $d = 0$
19. $d = \frac{23\sqrt{42}}{42}$
20. $d = \frac{26\sqrt{3}}{9}$
21. $d = \frac{17\sqrt{6}}{6}$
22. $d = \frac{21\sqrt{22}}{22}$
23. $d = \frac{13\sqrt{10}}{10}$
24. $d = \frac{5\sqrt{11}}{11}$
25. $d = \frac{3\sqrt{65}}{13}$
26. $d = 2\sqrt{2}$
27. $d = \frac{16\sqrt{61}}{61}$
28. $d = \frac{29\sqrt{43}}{43}$
29. $d = \frac{33\sqrt{19}}{38}$
30. $d = \frac{10\sqrt{26}}{13}$

Exercise set 3.

1. $\cos\varphi = \frac{9\sqrt{95}}{95}$
2. $\cos\varphi = \frac{9}{13}$
3. $\cos\varphi = \frac{5\sqrt{3}}{9}$
4. $\cos\varphi = \frac{\sqrt{14}}{14}$
5. perpendicular (90°)
6. $t = \frac{7}{3}$: $\left(\frac{17}{3}; \frac{4}{3}; -\frac{7}{3}\right)$
7. $x - 2y + 4z - 16 = 0$
8. $3x - y + 2z + 1 = 0$
9. $\frac{x}{6} + \frac{y}{-4} + \frac{z}{2} = 1$
10. $d = 4$

3.3. Expansion of a vector of a linear space in a basis. Subspaces of a linear space

Exercise set 1.

1. $\vec{x} = -2\vec{e}_2 + \vec{e}_3$
2. $\vec{x} = \vec{e}_1 - 4\vec{e}_2 + \vec{e}_3$
3. $\vec{x} = 3\vec{e}_1 - 3\vec{e}_2 + 2\vec{e}_3$
4. $\vec{x} = 2\vec{e}_1 - 2\vec{e}_2 + \vec{e}_3$
5. $\vec{x} = -3\vec{e}_1 + 2\vec{e}_2 + \vec{e}_3$
6. $\vec{x} = -2\vec{e}_1 + 3\vec{e}_2 + 4\vec{e}_3$
7. $\vec{x} = 3\vec{e}_1 + 2\vec{e}_2 + 3\vec{e}_3$
8. $\vec{x} = -\vec{e}_2 - \vec{e}_3$
9. $\vec{x} = 3\vec{e}_1 - \vec{e}_2 + 2\vec{e}_3$
10. $\vec{x} = -2\vec{e}_1 - \vec{e}_2 - 2\vec{e}_3$
11. $\vec{x} = \vec{e}_1 + 4\vec{e}_2 - 4\vec{e}_3$
12. $\vec{x} = 3\vec{e}_2$
13. $\vec{x} = 2\vec{e}_1 - 2\vec{e}_2 - 3\vec{e}_3$
14. $\vec{x} = -3\vec{e}_1$
15. $\vec{x} = -3\vec{e}_1 - 2\vec{e}_2 - 4\vec{e}_3$
16. $\vec{x} = 2\vec{e}_2 - \vec{e}_3$
17. $\vec{x} = -2\vec{e}_1 - 3\vec{e}_2 + 4\vec{e}_3$
18. $\vec{x} = 2\vec{e}_1 + 4\vec{e}_2 - 2\vec{e}_3$
19. $\vec{x} = -3\vec{e}_1 + 2\vec{e}_2 + 3\vec{e}_3$
20. $\vec{x} = 2\vec{e}_1 + 4\vec{e}_3$
21. $\vec{x} = 3\vec{e}_1 + 4\vec{e}_2 - 2\vec{e}_3$
22. $\vec{x} = 3\vec{e}_1 + 3\vec{e}_2 + 4\vec{e}_3$
23. $\vec{x} = 2\vec{e}_1 + 3\vec{e}_2 - 3\vec{e}_3$
24. $\vec{x} = 2\vec{e}_1 - 3\vec{e}_2 + 4\vec{e}_3$
25. $\vec{x} = -3\vec{e}_1 + \vec{e}_2$
26. $\vec{x} = -3\vec{e}_1 + 3\vec{e}_2 + 4\vec{e}_3$
27. $\vec{x} = -2\vec{e}_2 + 4\vec{e}_3$
28. $\vec{x} = -\vec{e}_1 - \vec{e}_2 - 3\vec{e}_3$
29. $\vec{x} = -\vec{e}_1 + 4\vec{e}_2 + \vec{e}_3$
30. $\vec{x} = -4\vec{e}_2 + \vec{e}_3$

Exercise set 2.

1. rank = 2; linearly dependent, $\dim L = 2$
2. rank = 3; linearly independent, $\dim L = 3$
3. rank = 2; linearly dependent, $\dim L = 2$
4. rank = 3; linearly independent, $\dim L = 3$
5. rank = 2; linearly dependent, $\dim L = 2$
6. rank = 3; linearly independent, $\dim L = 3$
7. rank = 2; linearly dependent, $\dim L = 2$
8. rank = 3; linearly independent, $\dim L = 3$
9. rank = 2; linearly dependent, $\dim L = 2$
10. rank = 3; linearly independent, $\dim L = 3$
11. rank = 1; linearly dependent, $\dim L = 1$
12. rank = 3; linearly independent, $\dim L = 3$
13. rank = 2; linearly dependent, $\dim L = 2$
14. rank = 3; linearly independent, $\dim L = 3$
15. rank = 2; linearly dependent, $\dim L = 2$
16. rank = 3; linearly independent, $\dim L = 3$
17. rank = 2; linearly dependent, $\dim L = 2$
18. rank = 3; linearly independent, $\dim L = 3$
19. rank = 2; linearly dependent, $\dim L = 2$
20. rank = 3; linearly independent, $\dim L = 3$
21. rank = 2; linearly dependent, $\dim L = 2$
22. rank = 3; linearly independent, $\dim L = 3$
23. rank = 2; linearly dependent, $\dim L = 2$
24. rank = 3; linearly independent, $\dim L = 3$

25. rank = 2; linearly dependent, $\dim L = 2$
26. rank = 3; linearly independent, $\dim L = 3$
27. rank = 2; linearly dependent, $\dim L = 2$
28. rank = 1; linearly dependent, $\dim L = 1$
29. rank = 2; linearly dependent, $\dim L = 2$
30. rank = 3; linearly independent, $\dim L = 3$

Exercise set 3.

1. basis $(1;-1;0)$, $(1;0;-1)$; $\dim L = 2$
2. no: $(0;0) \notin M$
3. yes: $\vec{x} = 2\vec{a} + \vec{b}$
4. no (the system is inconsistent)
5. $p = 6 + 5(x-1) + (x-1)^2$
6. 4; $E_{11}, E_{12}, E_{21}, E_{22}$
7. $\vec{x} = 3\vec{e}_1 + 2\vec{e}_2$
8. dependent: $\vec{a} + \vec{b} + \vec{c} = \vec{0} \pmod 2$
9. basis $(-2;1;0)$, $(1;0;1)$; dim = 2
10. $2^2 = 4$ codewords: 0000, 1011, 0110, 1101

3.4. Definition of a Euclidean space. The norm of an element of a Euclidean space

Exercise set 1.

1. $(x,y) = 5$, $\|x\| = 3$, $\|y\| = \sqrt{34}$
2. $(x,y) = 2$, $\|x\| = \sqrt{31}$, $\|y\| = \sqrt{55}$
3. $(x,y) = -23$, $\|x\| = \sqrt{35}$, $\|y\| = \sqrt{46}$
4. $(x,y) = 1$, $\|x\| = \sqrt{51}$, $\|y\| = \sqrt{91}$
5. $(x,y) = -10$, $\|x\| = 7$, $\|y\| = \sqrt{6}$
6. $(x,y) = -13$, $\|x\| = 3\sqrt{2}$, $\|y\| = \sqrt{39}$
7. $(x,y) = -5$, $\|x\| = \sqrt{42}$, $\|y\| = \sqrt{67}$
8. $(x,y) = -7$, $\|x\| = \sqrt{17}$, $\|y\| = \sqrt{11}$
9. $(x,y) = -20$, $\|x\| = \sqrt{33}$, $\|y\| = \sqrt{37}$
10. $(x,y) = -26$, $\|x\| = 7$, $\|y\| = \sqrt{61}$
11. $(x,y) = 4$, $\|x\| = \sqrt{39}$, $\|y\| = \sqrt{59}$
12. $(x,y) = 33$, $\|x\| = \sqrt{67}$, $\|y\| = \sqrt{38}$
13. $(x,y) = -10$, $\|x\| = \sqrt{70}$, $\|y\| = \sqrt{41}$
14. $(x,y) = 15$, $\|x\| = \sqrt{46}$, $\|y\| = \sqrt{91}$
15. $(x,y) = 32$, $\|x\| = 2\sqrt{15}$, $\|y\| = \sqrt{30}$
16. $(x,y) = 49$, $\|x\| = \sqrt{42}$, $\|y\| = \sqrt{66}$
17. $(x,y) = -7$, $\|x\| = 2\sqrt{7}$, $\|y\| = \sqrt{13}$
18. $(x,y) = -23$, $\|x\| = 2\sqrt{19}$, $\|y\| = \sqrt{31}$
19. $(x,y) = -23$, $\|x\| = \sqrt{30}$, $\|y\| = \sqrt{58}$
20. $(x,y) = -8$, $\|x\| = \sqrt{22}$, $\|y\| = \sqrt{23}$
21. $(x,y) = 4$, $\|x\| = 2\sqrt{5}$, $\|y\| = \sqrt{14}$
22. $(x,y) = -8$, $\|x\| = \sqrt{39}$, $\|y\| = \sqrt{55}$
23. $(x,y) = -10$, $\|x\| = \sqrt{22}$, $\|y\| = \sqrt{37}$
24. $(x,y) = 32$, $\|x\| = \sqrt{42}$, $\|y\| = \sqrt{33}$
25. $(x,y) = -37$, $\|x\| = \sqrt{38}$, $\|y\| = \sqrt{91}$
26. $(x,y) = 31$, $\|x\| = 3\sqrt{5}$, $\|y\| = \sqrt{38}$
27. $(x,y) = 2$, $\|x\| = \sqrt{34}$, $\|y\| = \sqrt{19}$
28. $(x,y) = -6$, $\|x\| = 4$, $\|y\| = \sqrt{79}$
29. $(x,y) = 11$, $\|x\| = \sqrt{33}$, $\|y\| = \sqrt{55}$
30. $(x,y) = 45$, $\|x\| = 2\sqrt{19}$, $\|y\| = \sqrt{47}$

Exercise set 2.

1. $\cos\varphi = \frac{3\sqrt{705}}{470}$, $\rho(x,y) = \sqrt{89}$
2. $\cos\varphi = \frac{10\sqrt{462}}{231}$, $\rho(x,y) = \sqrt{14}$
3. $\cos\varphi = \frac{4\sqrt{1342}}{671}$, $\rho(x,y) = \sqrt{67}$
4. $\cos\varphi = -\frac{37\sqrt{3009}}{3009}$, $\rho(x,y) = 2\sqrt{46}$
5. $\cos\varphi = -\frac{13\sqrt{37}}{111}$, $\rho(x,y) = 5\sqrt{5}$
6. $\cos\varphi = \frac{\sqrt{295}}{177}$, $\rho(x,y) = \sqrt{94}$
7. $\cos\varphi = -\frac{2\sqrt{286}}{143}$, $\rho(x,y) = \sqrt{129}$
8. $\cos\varphi = \frac{9\sqrt{703}}{703}$, $\rho(x,y) = \sqrt{38}$
9. $\cos\varphi = -\frac{4}{15}$, $\rho(x,y) = \sqrt{58}$
10. $\cos\varphi = -\frac{17\sqrt{1426}}{713}$, $\rho(x,y) = \sqrt{145}$
11. $\cos\varphi = \frac{\sqrt{2193}}{51}$, $\rho(x,y) = 2\sqrt{2}$
12. $\cos\varphi = \frac{5\sqrt{14}}{21}$, $\rho(x,y) = \sqrt{10}$
13. $\cos\varphi = -\frac{\sqrt{1155}}{42}$, $\rho(x,y) = \sqrt{249}$
14. $\cos\varphi = \frac{2\sqrt{87}}{29}$, $\rho(x,y) = 2\sqrt{10}$
15. $\cos\varphi = \frac{25\sqrt{871}}{1742}$, $\rho(x,y) = \sqrt{69}$
16. $\cos\varphi = 0$, $\rho(x,y) = \sqrt{78}$
17. $\cos\varphi = \frac{\sqrt{14}}{63}$, $\rho(x,y) = \sqrt{65}$
18. $\cos\varphi = -\frac{\sqrt{462}}{42}$, $\rho(x,y) = \sqrt{138}$
19. $\cos\varphi = \frac{9\sqrt{35}}{70}$, $\rho(x,y) = \sqrt{19}$
20. $\cos\varphi = -\frac{17\sqrt{122}}{366}$, $\rho(x,y) = \sqrt{113}$
21. $\cos\varphi = 0$, $\rho(x,y) = 2\sqrt{17}$
22. $\cos\varphi = \frac{\sqrt{2}}{2}$, $\rho(x,y) = \sqrt{21}$
23. $\cos\varphi = \frac{\sqrt{273}}{78}$, $\rho(x,y) = 3\sqrt{6}$
24. $\cos\varphi = \frac{2\sqrt{2}}{21}$, $\rho(x,y) = \sqrt{69}$
25. $\cos\varphi = -\frac{6}{17}$, $\rho(x,y) = \sqrt{138}$
26. $\cos\varphi = \frac{\sqrt{15}}{30}$, $\rho(x,y) = \sqrt{82}$
27. $\cos\varphi = \frac{25\sqrt{1419}}{1419}$, $\rho(x,y) = \sqrt{26}$
28. $\cos\varphi = -\frac{7\sqrt{2}}{30}$, $\rho(x,y) = \sqrt{57}$
29. $\cos\varphi = \frac{4\sqrt{22}}{55}$, $\rho(x,y) = \sqrt{62}$
30. $\cos\varphi = \frac{11\sqrt{133}}{399}$, $\rho(x,y) = 2\sqrt{15}$

Exercise set 3.

1. yes: $(x,y) = 0$
2. $\frac{1}{5}(1;-2;2;4)$
3. $|8| \le \sqrt{14}\sqrt{21} \approx 17.15$
4. $\|x+y\| = \sqrt{290} \le 5 + 13 = 18$
5. $\lambda = -\frac{1}{2}$
6. $(x,y) = 0$, $\|x\| = \sqrt{14}$, $\|y\| = \sqrt{21}$
7. $(f,g) = \frac{1}{2}$, $\|f\| = \frac{1}{\sqrt{3}}$
8. $d = 2$
9. $\cos(u,v) = \frac{7}{\sqrt{14}\sqrt{6}} = \frac{7}{2\sqrt{21}}$
10. $\|x+y\|^2 = 10 = 2 + 8$, since $(x,y) = 0$

3.5. Constructing an orthonormal basis in a Euclidean space

Exercise set 1.

1. $\vec{b}_1 = (1; 0; 3)$, $\vec{b}_2 = (-\frac{3}{5}; -1; \frac{1}{5})$
2. $\vec{b}_1 = (-2; 3; 2)$, $\vec{b}_2 = (\frac{40}{17}; \frac{8}{17}; \frac{28}{17})$
3. $\vec{b}_1 = (-1; -3; -3)$, $\vec{b}_2 = (-\frac{45}{19}; -\frac{21}{19}; \frac{36}{19})$
4. $\vec{b}_1 = (2; 0; -1)$, $\vec{b}_2 = (\frac{9}{5}; 1; \frac{18}{5})$
5. $\vec{b}_1 = (-1; 2; 3)$, $\vec{b}_2 = (\frac{15}{7}; \frac{12}{7}; -\frac{3}{7})$
6. $\vec{b}_1 = (-2; 1; -3)$, $\vec{b}_2 = (-\frac{16}{7}; \frac{22}{7}; \frac{18}{7})$
7. $\vec{b}_1 = (-2; -2; 2)$, $\vec{b}_2 = (\frac{4}{3}; \frac{4}{3}; \frac{8}{3})$
8. $\vec{b}_1 = (2; -2; -1)$, $\vec{b}_2 = (-\frac{1}{3}; -\frac{5}{3}; \frac{8}{3})$
9. $\vec{b}_1 = (2; -3; -1)$, $\vec{b}_2 = (-\frac{8}{7}; -\frac{9}{7}; \frac{11}{7})$
10. $\vec{b}_1 = (3; 0; 2)$, $\vec{b}_2 = (-\frac{8}{13}; -2; \frac{12}{13})$
11. $\vec{b}_1 = (3; 0; 3)$, $\vec{b}_2 = (2; -1; -2)$
12. $\vec{b}_1 = (0; -1; 1)$, $\vec{b}_2 = (-3; \frac{5}{2}; \frac{5}{2})$
13. $\vec{b}_1 = (-1; -3; 0)$, $\vec{b}_2 = (-\frac{6}{5}; \frac{2}{5}; 3)$
14. $\vec{b}_1 = (-3; 3; 0)$, $\vec{b}_2 = (-\frac{5}{2}; -\frac{5}{2}; 1)$
15. $\vec{b}_1 = (-2; -2; 1)$, $\vec{b}_2 = (\frac{5}{9}; -\frac{13}{9}; -\frac{16}{9})$
16. $\vec{b}_1 = (-3; 3; -1)$, $\vec{b}_2 = (-\frac{5}{19}; \frac{5}{19}; \frac{30}{19})$
17. $\vec{b}_1 = (3; 1; 1)$, $\vec{b}_2 = (1; -2; -1)$
18. $\vec{b}_1 = (2; -1; -1)$, $\vec{b}_2 = (-\frac{4}{3}; -\frac{11}{6}; -\frac{5}{6})$
19. $\vec{b}_1 = (-3; -3; 3)$, $\vec{b}_2 = (\frac{1}{3}; -\frac{5}{3}; -\frac{4}{3})$
20. $\vec{b}_1 = (2; -1; 0)$, $\vec{b}_2 = (-\frac{4}{5}; -\frac{8}{5}; 2)$
21. $\vec{b}_1 = (-2; -1; -1)$, $\vec{b}_2 = (-\frac{7}{3}; \frac{7}{3}; \frac{7}{3})$
22. $\vec{b}_1 = (0; -3; 1)$, $\vec{b}_2 = (1; \frac{7}{10}; \frac{21}{10})$
23. $\vec{b}_1 = (-2; 0; 0)$, $\vec{b}_2 = (0; 1; 3)$
24. $\vec{b}_1 = (0; -2; -2)$, $\vec{b}_2 = (-1; -2; 2)$
25. $\vec{b}_1 = (-3; -2; -3)$, $\vec{b}_2 = (\frac{21}{11}; \frac{3}{11}; -\frac{23}{11})$
26. $\vec{b}_1 = (3; -1; 3)$, $\vec{b}_2 = (-\frac{1}{19}; -\frac{63}{19}; -\frac{20}{19})$
27. $\vec{b}_1 = (0; -3; 2)$, $\vec{b}_2 = (3; \frac{14}{13}; \frac{21}{13})$
28. $\vec{b}_1 = (-1; 2; 0)$, $\vec{b}_2 = (-\frac{6}{5}; -\frac{3}{5}; 1)$
29. $\vec{b}_1 = (-3; 1; 2)$, $\vec{b}_2 = (-\frac{5}{7}; -\frac{17}{7}; \frac{1}{7})$
30. $\vec{b}_1 = (-1; 1; -2)$, $\vec{b}_2 = (\frac{10}{3}; -\frac{4}{3}; -\frac{7}{3})$

Exercise set 2.

1. $\vec{b}_1 = (-2; -1; -1)$, $\vec{b}_2 = (0; \frac{1}{2}; -\frac{1}{2})$, $\vec{b}_3 = (-1; 1; 1)$
2. $\vec{b}_1 = (-2; 2; 0)$, $\vec{b}_2 = (\frac{3}{2}; \frac{3}{2}; 2)$, $\vec{b}_3 = (\frac{6}{17}; \frac{6}{17}; -\frac{9}{17})$
3. $\vec{b}_1 = (2; 0; -1)$, $\vec{b}_2 = (-\frac{1}{5}; 0; -\frac{2}{5})$, $\vec{b}_3 = (0; -1; 0)$
4. $\vec{b}_1 = (0; -2; 2)$, $\vec{b}_2 = (1; 0; 0)$, $\vec{b}_3 = (0; -\frac{3}{2}; -\frac{3}{2})$
5. $\vec{b}_1 = (2; 1; 2)$, $\vec{b}_2 = (\frac{4}{3}; \frac{2}{3}; -\frac{5}{3})$, $\vec{b}_3 = (-\frac{3}{5}; \frac{6}{5}; 0)$
6. $\vec{b}_1 = (2; -2; -1)$, $\vec{b}_2 = (\frac{1}{9}; \frac{8}{9}; -\frac{14}{9})$, $\vec{b}_3 = (\frac{24}{29}; \frac{18}{29}; \frac{12}{29})$
7. $\vec{b}_1 = (0; -2; 2)$, $\vec{b}_2 = (0; -1; -1)$, $\vec{b}_3 = (-1; 0; 0)$
8. $\vec{b}_1 = (-1; -2; 2)$, $\vec{b}_2 = (-\frac{22}{9}; \frac{1}{9}; -\frac{10}{9})$, $\vec{b}_3 = (\frac{38}{65}; -\frac{114}{65}; -\frac{19}{13})$
9. $\vec{b}_1 = (2; 0; 2)$, $\vec{b}_2 = (-\frac{1}{2}; 0; \frac{1}{2})$, $\vec{b}_3 = (0; 2; 0)$
10. $\vec{b}_1 = (-2; -2; 0)$, $\vec{b}_2 = (2; -2; 0)$, $\vec{b}_3 = (0; 0; -2)$
11. $\vec{b}_1 = (0; -1; 1)$, $\vec{b}_2 = (-1; -\frac{1}{2}; -\frac{1}{2})$, $\vec{b}_3 = (\frac{2}{3}; -\frac{2}{3}; -\frac{2}{3})$
12. $\vec{b}_1 = (-1; 0; 0)$, $\vec{b}_2 = (0; -1; -1)$, $\vec{b}_3 = (0; 1; -1)$
13. $\vec{b}_1 = (2; 0; 1)$, $\vec{b}_2 = (-\frac{2}{5}; -2; \frac{4}{5})$, $\vec{b}_3 = (-\frac{2}{3}; \frac{2}{3}; \frac{4}{3})$

14. $\vec{b}_1 = (2;\ 2;\ 2)$, $\vec{b}_2 = (1;\ -2;\ 1)$, $\vec{b}_3 = (-\frac{1}{2};\ 0;\ \frac{1}{2})$
15. $\vec{b}_1 = (1;\ -2;\ -1)$, $\vec{b}_2 = (\frac{1}{2};\ 1;\ -\frac{3}{2})$, $\vec{b}_3 = (\frac{4}{21};\ \frac{1}{21};\ \frac{2}{21})$
16. $\vec{b}_1 = (1;\ 2;\ -1)$, $\vec{b}_2 = (\frac{4}{3};\ -\frac{1}{3};\ \frac{2}{3})$, $\vec{b}_3 = (-\frac{1}{14};\ \frac{1}{7};\ \frac{3}{14})$
17. $\vec{b}_1 = (-2;\ 2;\ -1)$, $\vec{b}_2 = (\frac{13}{9};\ \frac{5}{9};\ -\frac{16}{9})$, $\vec{b}_3 = (-\frac{33}{50};\ -\frac{11}{10};\ -\frac{22}{25})$
18. $\vec{b}_1 = (1;\ 2;\ -2)$, $\vec{b}_2 = (\frac{2}{9};\ -\frac{5}{9};\ -\frac{4}{9})$, $\vec{b}_3 = (\frac{6}{5};\ 0;\ \frac{3}{5})$
19. $\vec{b}_1 = (-1;\ 0;\ 0)$, $\vec{b}_2 = (0;\ 1;\ -2)$, $\vec{b}_3 = (0;\ \frac{4}{5};\ \frac{2}{5})$
20. $\vec{b}_1 = (1;\ 2;\ 1)$, $\vec{b}_2 = (1;\ 0;\ -1)$, $\vec{b}_3 = (\frac{2}{3};\ -\frac{2}{3};\ \frac{2}{3})$
21. $\vec{b}_1 = (-1;\ 0;\ 1)$, $\vec{b}_2 = (-\frac{3}{2};\ -1;\ -\frac{3}{2})$, $\vec{b}_3 = (\frac{3}{11};\ -\frac{9}{11};\ \frac{3}{11})$
22. $\vec{b}_1 = (0;\ -1;\ 1)$, $\vec{b}_2 = (1;\ -\frac{1}{2};\ -\frac{1}{2})$, $\vec{b}_3 = (-\frac{2}{3};\ -\frac{2}{3};\ -\frac{2}{3})$
23. $\vec{b}_1 = (1;\ 1;\ 1)$, $\vec{b}_2 = (-\frac{1}{3};\ -\frac{1}{3};\ \frac{2}{3})$, $\vec{b}_3 = (-\frac{1}{2};\ \frac{1}{2};\ 0)$
24. $\vec{b}_1 = (-1;\ -2;\ 1)$, $\vec{b}_2 = (\frac{1}{3};\ -\frac{4}{3};\ -\frac{7}{3})$, $\vec{b}_3 = (-\frac{15}{11};\ \frac{5}{11};\ -\frac{5}{11})$
25. $\vec{b}_1 = (0;\ 2;\ 2)$, $\vec{b}_2 = (2;\ 0;\ 0)$, $\vec{b}_3 = (0;\ -\frac{1}{2};\ \frac{1}{2})$
26. $\vec{b}_1 = (-2;\ 0;\ -2)$, $\vec{b}_2 = (0;\ 2;\ 0)$, $\vec{b}_3 = (1;\ 0;\ -1)$
27. $\vec{b}_1 = (0;\ 1;\ -2)$, $\vec{b}_2 = (2;\ -\frac{8}{5};\ -\frac{4}{5})$, $\vec{b}_3 = (\frac{8}{9};\ \frac{8}{9};\ \frac{4}{9})$
28. $\vec{b}_1 = (0;\ 0;\ -1)$, $\vec{b}_2 = (1;\ 1;\ 0)$, $\vec{b}_3 = (\frac{1}{2};\ -\frac{1}{2};\ 0)$
29. $\vec{b}_1 = (-2;\ -1;\ 2)$, $\vec{b}_2 = (\frac{2}{3};\ \frac{4}{3};\ \frac{4}{3})$, $\vec{b}_3 = (\frac{4}{3};\ -\frac{4}{3};\ \frac{2}{3})$
30. $\vec{b}_1 = (-2;\ 1;\ 2)$, $\vec{b}_2 = (\frac{7}{9};\ -\frac{8}{9};\ \frac{11}{9})$, $\vec{b}_3 = (-\frac{18}{13};\ -\frac{24}{13};\ -\frac{6}{13})$

Exercise set 3.

1. $\mathrm{pr} = (\frac{27}{11};\ -\frac{27}{11};\ -\frac{18}{11})$, $\vec{h} = (-\frac{27}{11};\ -\frac{17}{11};\ -\frac{15}{11})$
2. $\mathrm{pr} = (\frac{3}{2};\ 1;\ -\frac{1}{2})$, $\vec{h} = (\frac{3}{2};\ -2;\ \frac{1}{2})$
3. $\mathrm{pr} = (2;\ 0;\ 2)$, $\vec{h} = (-2;\ 1;\ 2)$
4. $\mathrm{pr} = (-\frac{3}{11};\ -\frac{1}{11};\ \frac{1}{11})$, $\vec{h} = (\frac{25}{11};\ -\frac{43}{11};\ \frac{32}{11})$
5. $\mathrm{pr} = (\frac{5}{2};\ 0;\ -\frac{5}{2})$, $\vec{h} = (-\frac{3}{2};\ 3;\ -\frac{3}{2})$
6. $\pm\frac{1}{\sqrt{11}}(3;\ -1;\ 1)$
7. $(\vec{e}_i, \vec{e}_j) = \delta_{ij}$, so the basis is orthonormal
8. $\vec{x} = 3\vec{e}_1 + 3\vec{e}_2 + 0 \cdot \vec{e}_3$
9. $Q^T Q = E$, $Q^{-1} = Q^T = Q$
10. $\frac{1}{\sqrt{2}}(1;0;1)$, $\frac{1}{\sqrt{6}}(-1;2;1)$

Chapter 4. Linear Operators

4.1. Operations on linear operators. The space of linear operators. The matrix of a linear operator under a change of basis

Exercise set 1.

1. $\begin{pmatrix} -5 & -9 \\ -4 & 0 \end{pmatrix}$; $\begin{pmatrix} 8 & 4 \\ -2 & -4 \end{pmatrix}$; $\begin{pmatrix} -2 & 3 \\ 4 & 6 \end{pmatrix}$
2. $\begin{pmatrix} 0 & -5 \\ -6 & -4 \end{pmatrix}$; $\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$; $\begin{pmatrix} 0 & 1 \\ 0 & 5 \end{pmatrix}$
3. $\begin{pmatrix} -6 & -4 \\ 3 & -1 \end{pmatrix}$; $\begin{pmatrix} -7 & -12 \\ -1 & 9 \end{pmatrix}$; $\begin{pmatrix} -6 & 3 \\ 9 & 8 \end{pmatrix}$
4. $\begin{pmatrix} -5 & 1 \\ 1 & -4 \end{pmatrix}$; $\begin{pmatrix} 0 & 2 \\ 4 & -6 \end{pmatrix}$; $\begin{pmatrix} 0 & 4 \\ 2 & -6 \end{pmatrix}$
5. $\begin{pmatrix} 7 & -1 \\ 5 & 2 \end{pmatrix}$; $\begin{pmatrix} -9 & 3 \\ -6 & 6 \end{pmatrix}$; $\begin{pmatrix} 0 & -6 \\ -6 & -3 \end{pmatrix}$
6. $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$; $\begin{pmatrix} -2 & 0 \\ -1 & -3 \end{pmatrix}$; $\begin{pmatrix} -2 & 0 \\ 1 & -3 \end{pmatrix}$
7. $\begin{pmatrix} -1 & 3 \\ 3 & 1 \end{pmatrix}$; $\begin{pmatrix} 1 & 0 \\ 7 & -4 \end{pmatrix}$; $\begin{pmatrix} -2 & 2 \\ 3 & -1 \end{pmatrix}$
8. $\begin{pmatrix} 0 & -7 \\ 0 & 7 \end{pmatrix}$; $\begin{pmatrix} -3 & 4 \\ 0 & 2 \end{pmatrix}$; $\begin{pmatrix} -3 & -3 \\ 0 & 2 \end{pmatrix}$
9. $\begin{pmatrix} 1 & -4 \\ -5 & -2 \end{pmatrix}$; $\begin{pmatrix} 0 & 6 \\ -10 & -4 \end{pmatrix}$; $\begin{pmatrix} 2 & -8 \\ 9 & -6 \end{pmatrix}$
10. $\begin{pmatrix} -6 & 5 \\ -4 & 3 \end{pmatrix}$; $\begin{pmatrix} -2 & 1 \\ -2 & 1 \end{pmatrix}$; $\begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$
11. $\begin{pmatrix} 6 & 2 \\ 5 & 4 \end{pmatrix}$; $\begin{pmatrix} 7 & 8 \\ 15 & 12 \end{pmatrix}$; $\begin{pmatrix} 18 & 6 \\ 9 & 1 \end{pmatrix}$
12. $\begin{pmatrix} -9 & 0 \\ -11 & -6 \end{pmatrix}$; $\begin{pmatrix} 6 & 0 \\ 4 & 0 \end{pmatrix}$; $\begin{pmatrix} 6 & 0 \\ 13 & 0 \end{pmatrix}$
13. $\begin{pmatrix} 3 & 5 \\ -1 & 4 \end{pmatrix}$; $\begin{pmatrix} -9 & -3 \\ -3 & -8 \end{pmatrix}$; $\begin{pmatrix} -11 & -1 \\ -3 & -6 \end{pmatrix}$
14. $\begin{pmatrix} 6 & -8 \\ -4 & 12 \end{pmatrix}$; $\begin{pmatrix} 1 & -6 \\ -4 & 3 \end{pmatrix}$; $\begin{pmatrix} -3 & -8 \\ 0 & 7 \end{pmatrix}$
15. $\begin{pmatrix} 6 & -3 \\ -2 & 0 \end{pmatrix}$; $\begin{pmatrix} 15 & 0 \\ 0 & 0 \end{pmatrix}$; $\begin{pmatrix} 9 & -9 \\ -6 & 6 \end{pmatrix}$
16. $\begin{pmatrix} 0 & -5 & -4 \\ -2 & 4 & 6 \\ -1 & -6 & -3 \end{pmatrix}$; $\begin{pmatrix} 1 & 1 & -2 \\ 2 & 2 & -2 \\ -1 & -2 & 0 \end{pmatrix}$; $\begin{pmatrix} -1 & -1 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 3 \end{pmatrix}$
17. $\begin{pmatrix} 1 & 6 & -4 \\ 1 & -4 & -5 \\ 2 & 1 & 8 \end{pmatrix}$; $\begin{pmatrix} 0 & -5 & 7 \\ -2 & -10 & -6 \\ 3 & -1 & 5 \end{pmatrix}$; $\begin{pmatrix} -7 & 9 & -1 \\ 10 & 7 & 5 \\ -3 & -11 & -5 \end{pmatrix}$
18. $\begin{pmatrix} 2 & 1 & 0 \\ -1 & -3 & -1 \\ -3 & 2 & 3 \end{pmatrix}$; $\begin{pmatrix} 12 & -3 & 6 \\ 5 & -4 & -7 \\ 3 & -2 & -2 \end{pmatrix}$; $\begin{pmatrix} 3 & 1 & 6 \\ -3 & 0 & 9 \\ 11 & 3 & 3 \end{pmatrix}$
19. $\begin{pmatrix} 9 & 9 & -8 \\ -3 & 8 & 4 \\ 3 & -4 & 3 \end{pmatrix}$; $\begin{pmatrix} 2 & 22 & -12 \\ -1 & -1 & 11 \\ -9 & 3 & -6 \end{pmatrix}$; $\begin{pmatrix} 9 & 15 & 9 \\ -7 & 3 & 5 \\ 5 & 2 & -17 \end{pmatrix}$
20. $\begin{pmatrix} 2 & 2 & 1 \\ 3 & -4 & 5 \\ 7 & 4 & 5 \end{pmatrix}$; $\begin{pmatrix} 13 & 3 & 5 \\ 3 & 1 & 2 \\ -17 & -3 & -6 \end{pmatrix}$; $\begin{pmatrix} 9 & 6 & -7 \\ 2 & 2 & 3 \\ 8 & 6 & -3 \end{pmatrix}$
21. $\begin{pmatrix} 4 & -1 & 5 \\ 5 & 4 & -9 \\ -1 & 0 & 12 \end{pmatrix}$; $\begin{pmatrix} 1 & -2 & 2 \\ 0 & -4 & 10 \\ -1 & 0 & 13 \end{pmatrix}$; $\begin{pmatrix} 5 & -1 & 5 \\ -1 & -5 & -10 \\ 5 & 1 & 10 \end{pmatrix}$
22. $\begin{pmatrix} -6 & -5 & 5 \\ 6 & 5 & -3 \\ -11 & -10 & 7 \end{pmatrix}$; $\begin{pmatrix} 8 & 11 & -9 \\ 7 & 7 & -9 \\ 6 & 8 & -10 \end{pmatrix}$; $\begin{pmatrix} -1 & -1 & 3 \\ -6 & 0 & -8 \\ 3 & -3 & 6 \end{pmatrix}$
23. $\begin{pmatrix} -7 & 9 & -6 \\ 10 & 3 & 9 \\ -2 & -5 & 6 \end{pmatrix}$; $\begin{pmatrix} 8 & -8 & 21 \\ 7 & 2 & 6 \\ 4 & 8 & -9 \end{pmatrix}$; $\begin{pmatrix} 1 & -3 & 3 \\ 0 & 12 & 0 \\ 2 & -6 & -12 \end{pmatrix}$
24. $\begin{pmatrix} -5 & -8 & -1 \\ -3 & 1 & -2 \\ 2 & 1 & -3 \end{pmatrix}$; $\begin{pmatrix} -8 & -9 & -7 \\ 4 & 9 & 8 \\ 0 & 0 & 11 \end{pmatrix}$; $\begin{pmatrix} 6 & 1 & 5 \\ -9 & -4 & 3 \\ 3 & 5 & 10 \end{pmatrix}$
25. $\begin{pmatrix} -1 & -1 & 4 \\ -4 & 5 & -3 \\ 0 & -2 & 2 \end{pmatrix}$; $\begin{pmatrix} -3 & -6 & -5 \\ 6 & -10 & 8 \\ -12 & 0 & 0 \end{pmatrix}$; $\begin{pmatrix} -1 & 13 & 0 \\ 5 & -5 & 12 \\ 5 & -2 & -7 \end{pmatrix}$
26. $\begin{pmatrix} -1 & -2 & 3 \\ 5 & -2 & -3 \\ -4 & -4 & 1 \end{pmatrix}$; $\begin{pmatrix} 12 & 2 & 1 \\ 1 & -7 & -5 \\ 6 & 2 & 4 \end{pmatrix}$; $\begin{pmatrix} 4 & -2 & 2 \\ -1 & 5 & -9 \\ 4 & -8 & 0 \end{pmatrix}$
27. $\begin{pmatrix} -4 & 1 & -1 \\ 2 & 4 & -3 \\ 2 & -4 & 1 \end{pmatrix}$; $\begin{pmatrix} 1 & 5 & -3 \\ 0 & 9 & -3 \\ -5 & -2 & 2 \end{pmatrix}$; $\begin{pmatrix} 4 & 0 & 4 \\ -3 & 4 & -10 \\ 0 & -4 & 4 \end{pmatrix}$
28. $\begin{pmatrix} -3 & 0 & 1 \\ -2 & 4 & 4 \\ -1 & -1 & -1 \end{pmatrix}$; $\begin{pmatrix} -3 & -11 & -4 \\ -2 & -8 & -4 \\ -12 & -7 & 9 \end{pmatrix}$; $\begin{pmatrix} -8 & 10 & 9 \\ 10 & 5 & -6 \\ -6 & -14 & 1 \end{pmatrix}$

29. $\begin{pmatrix} -1 & 2 & -5 \\ 3 & -2 & 5 \\ 0 & 0 & 0 \end{pmatrix}$; $\begin{pmatrix} -8 & -13 & 4 \\ 9 & 12 & -4 \\ -2 & 5 & -2 \end{pmatrix}$; $\begin{pmatrix} 8 & 8 & 3 \\ -13 & -9 & -9 \\ 5 & 3 & 3 \end{pmatrix}$

30. $\begin{pmatrix} -6 & -6 & 3 \\ 6 & 3 & 0 \\ 3 & 4 & 3 \end{pmatrix}$; $\begin{pmatrix} 4 & 6 & 8 \\ 1 & 2 & 5 \\ 0 & 0 & -2 \end{pmatrix}$; $\begin{pmatrix} 4 & 6 & -10 \\ -5 & -8 & 14 \\ -6 & -6 & 8 \end{pmatrix}$

Exercise set 2.

1. $A' = \begin{pmatrix} -1 & 9 \\ 0 & 4 \end{pmatrix}$
2. $A' = \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$
3. $A' = \begin{pmatrix} -3 & -2 \\ 2 & 3 \end{pmatrix}$
4. $A' = \begin{pmatrix} -3 & 4 \\ -6 & 4 \end{pmatrix}$
5. $A' = \begin{pmatrix} 0 & -1 \\ 3 & -2 \end{pmatrix}$
6. $A' = \begin{pmatrix} 5 & -4 \\ 3 & -3 \end{pmatrix}$
7. $A' = \begin{pmatrix} 6 & 3 \\ -5 & 1 \end{pmatrix}$
8. $A' = \begin{pmatrix} 12 & -10 \\ 22 & -18 \end{pmatrix}$
9. $A' = \begin{pmatrix} 2 & -4 \\ 0 & -3 \end{pmatrix}$
10. $A' = \begin{pmatrix} -2 & -4 \\ -2 & 2 \end{pmatrix}$
11. $A' = \begin{pmatrix} 1 & -9 \\ 1 & 6 \end{pmatrix}$
12. $A' = \begin{pmatrix} -8 & 4 \\ -6 & 5 \end{pmatrix}$
13. $A' = \begin{pmatrix} -6 & 29 \\ -2 & 9 \end{pmatrix}$
14. $A' = \begin{pmatrix} -9 & -7 \\ 11 & 9 \end{pmatrix}$
15. $A' = \begin{pmatrix} -5 & 2 \\ -6 & 6 \end{pmatrix}$
16. $A' = \begin{pmatrix} -4 & 2 \\ -6 & 4 \end{pmatrix}$
17. $A' = \begin{pmatrix} 2 & -2 \\ 2 & 4 \end{pmatrix}$
18. $A' = \begin{pmatrix} 1 & 0 \\ -2 & 0 \end{pmatrix}$
19. $A' = \begin{pmatrix} -11 & 8 \\ -8 & 7 \end{pmatrix}$
20. $A' = \begin{pmatrix} 7 & -4 \\ 4 & 0 \end{pmatrix}$
21. $A' = \begin{pmatrix} -3 & -7 \\ 2 & 2 \end{pmatrix}$
22. $A' = \begin{pmatrix} -13 & 7 \\ -12 & 6 \end{pmatrix}$
23. $A' = \begin{pmatrix} 2 & -12 \\ 0 & -2 \end{pmatrix}$
24. $A' = \begin{pmatrix} 10 & 15 \\ -3 & -5 \end{pmatrix}$
25. $A' = \begin{pmatrix} 4 & -2 \\ 14 & -5 \end{pmatrix}$
26. $A' = \begin{pmatrix} -3 & 1 \\ 5 & -1 \end{pmatrix}$
27. $A' = \begin{pmatrix} 3 & 4 \\ 0 & -2 \end{pmatrix}$
28. $A' = \begin{pmatrix} -4 & -3 \\ -3 & -4 \end{pmatrix}$
29. $A' = \begin{pmatrix} -21 & -10 \\ 34 & 16 \end{pmatrix}$
30. $A' = \begin{pmatrix} 1 & 6 \\ 1 & 3 \end{pmatrix}$

Exercise set 3.

1. yes; $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$
2. no: $\mathcal{A}(\vec{0}) \neq \vec{0}$
3. $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; $(-1; 3)$
4. $\operatorname{diag}(1, 1, 0)$
5. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
6. $(-1; -3; 8)$
7. $\ker = \{t(-2; 1)\}$, $\operatorname{Im} = \{t(1; 2)\}$
8. $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$
9. $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$
10. $\det K = 11$, $\gcd(11, 26) = 1$, hence invertible

4.2. Eigenvalues and eigenvectors of a linear operator

Exercise set 1.

1. $\lambda = 4$: $(1; 0)$; $\lambda = 5$: $(1; 1)$
2. $\lambda = 0$: $(1; 0)$; $\lambda = 2$: $(1; -1)$
3. $\lambda = -4$: $(1; 0)$; $\lambda = 0$: $(1; -1)$
4. $\lambda = -3$: $(1; -1)$; $\lambda = 3$: $(1; -2)$
5. $\lambda = 0$: $(1; -2)$; $\lambda = 5$: $(1; -1)$
6. $\lambda = 0$: $(1; 0)$; $\lambda = 4$: $(2; 1)$
7. $\lambda = 1$: $(1; -1)$; $\lambda = 3$: $(2; -1)$
8. $\lambda = -2$: $(0; 1)$; $\lambda = 4$: $(1; 1)$
9. $\lambda = 2$: $(1; -2)$; $\lambda = 4$: $(0; 1)$
10. $\lambda = -3$: $(2; -1)$; $\lambda = -2$: $(1; 0)$
11. $\lambda = 0$: $(0; 1)$; $\lambda = 1$: $(1; 2)$
12. $\lambda = 3$: $(1; -1)$; $\lambda = 4$: $(2; -1)$
13. $\lambda = -2$: $(0; 1)$; $\lambda = 1$: $(1; -1)$
14. $\lambda = -2$: $(2; 1)$; $\lambda = 0$: $(1; 0)$
15. $\lambda = 2$: $(0; 1)$; $\lambda = 5$: $(1; 1)$
16. $\lambda = -4$: $(1; 0)$; $\lambda = -2$: $(2; -1)$
17. $\lambda = -1$: $(1; 0)$; $\lambda = 5$: $(1; -1)$
18. $\lambda = 0$: $(2; -1)$; $\lambda = 3$: $(1; 0)$
19. $\lambda = -2$: $(0; 1)$; $\lambda = 5$: $(1; -1)$
20. $\lambda = -1$: $(1; -2)$; $\lambda = 5$: $(0; 1)$
21. $\lambda = 1$: $(1; 2)$; $\lambda = 3$: $(0; 1)$
22. $\lambda = 1$: $(2; 1)$; $\lambda = 2$: $(1; 0)$
23. $\lambda = -3$: $(1; -2)$; $\lambda = 0$: $(1; -1)$
24. $\lambda = 1$: $(2; -1)$; $\lambda = 5$: $(1; 0)$
25. $\lambda = -3$: $(1; 0)$; $\lambda = 1$: $(2; 1)$
26. $\lambda = -2$: $(1; 0)$; $\lambda = -1$: $(1; 1)$
27. $\lambda = -1$: $(1; 1)$; $\lambda = 1$: $(2; 1)$
28. $\lambda = -1$: $(1; 0)$; $\lambda = 1$: $(1; 1)$
29. $\lambda = 0$: $(0; 1)$; $\lambda = 5$: $(1; 2)$
30. $\lambda = -4$: $(1; 0)$; $\lambda = 0$: $(1; -1)$

Exercise set 2.

1. $\lambda = -2$: $(1; -1; -1)$; $\lambda = -1$: $(0; 1; 0)$; $\lambda = 4$: $(1; 2; -2)$
2. $\lambda = -2$: $(1; 0; 0)$; $\lambda = 2$: $(0; 2; -1)$; $\lambda = 3$: $(1; 1; -1)$
3. $\lambda = -2$: $(0; 1; -1)$; $\lambda = 3$: $(1; 0; -1)$; $\lambda = 4$: $(0; 0; 1)$
4. $\lambda = -2$: $(0; 1; 0)$; $\lambda = 0$: $(1; -2; -1)$; $\lambda = 4$: $(0; 1; 1)$
5. $\lambda = -2$: $(0; 1; -2)$; $\lambda = -1$: $(0; 0; 1)$; $\lambda = 1$: $(1; 2; -1)$
6. $\lambda = -3$: $(0; 1; 1)$; $\lambda = -1$: $(1; -1; 0)$; $\lambda = 1$: $(1; -1; 1)$
7. $\lambda = -2$: $(2; -1; 2)$; $\lambda = -1$: $(1; -2; 2)$; $\lambda = 0$: $(0; 1; -1)$
8. $\lambda = -2$: $(1; 0; 2)$; $\lambda = -1$: $(0; 1; 0)$; $\lambda = 3$: $(1; 0; 1)$
9. $\lambda = -2$: $(1; 0; 0)$; $\lambda = 1$: $(0; 1; -1)$; $\lambda = 4$: $(2; -1; 0)$
10. $\lambda = -3$: $(2; 1; 1)$; $\lambda = -2$: $(1; 0; 0)$; $\lambda = 3$: $(2; 1; 0)$
11. $\lambda = 0$: $(1; 0; 0)$; $\lambda = 1$: $(1; 0; 1)$; $\lambda = 4$: $(2; -1; -1)$
12. $\lambda = 1$: $(1; -2; 0)$; $\lambda = 2$: $(0; 1; -1)$; $\lambda = 3$: $(1; -2; 1)$
13. $\lambda = -3$: $(1; 0; 0)$; $\lambda = -2$: $(2; 2; 1)$; $\lambda = 2$: $(1; 1; 0)$
14. $\lambda = -3$: $(2; 1; -2)$; $\lambda = 2$: $(0; 0; 1)$; $\lambda = 3$: $(1; 1; 2)$
15. $\lambda = -3$: $(1; 1; -1)$; $\lambda = -2$: $(1; 2; -1)$; $\lambda = 3$: $(1; 1; 0)$
16. $\lambda = 0$: $(1; -1; -2)$; $\lambda = 1$: $(0; 1; 0)$; $\lambda = 4$: $(1; 1; -1)$

17. $\lambda = -2$: $(1; 1; 1)$; $\lambda = -1$: $(2; 0; -1)$; $\lambda = 0$: $(2; 1; 0)$
18. $\lambda = -2$: $(1; -1; 1)$; $\lambda = 2$: $(0; 1; -2)$; $\lambda = 4$: $(0; 0; 1)$
19. $\lambda = -3$: $(1; -2; 2)$; $\lambda = -2$: $(0; 1; -1)$; $\lambda = 2$: $(1; 1; 0)$
20. $\lambda = -3$: $(0; 1; -1)$; $\lambda = 0$: $(1; 0; 1)$; $\lambda = 4$: $(2; 1; 0)$
21. $\lambda = -2$: $(1; 0; 0)$; $\lambda = 2$: $(1; -1; -1)$; $\lambda = 4$: $(0; 1; 2)$
22. $\lambda = 1$: $(1; -2; 2)$; $\lambda = 2$: $(1; -1; 1)$; $\lambda = 3$: $(1; -1; 0)$
23. $\lambda = -2$: $(1; 1; -2)$; $\lambda = -1$: $(1; 0; -1)$; $\lambda = 4$: $(1; 0; -2)$
24. $\lambda = 1$: $(1; 0; -1)$; $\lambda = 3$: $(0; 0; 1)$; $\lambda = 4$: $(2; -1; -1)$
25. $\lambda = 2$: $(1; 0; 0)$; $\lambda = 3$: $(1; -2; -1)$; $\lambda = 4$: $(2; 1; 1)$
26. $\lambda = 1$: $(0; 1; 1)$; $\lambda = 2$: $(1; 0; 1)$; $\lambda = 4$: $(1; -1; 1)$
27. $\lambda = -3$: $(1; 1; -1)$; $\lambda = -2$: $(1; 0; 0)$; $\lambda = 2$: $(1; -1; 2)$
28. $\lambda = -3$: $(1; 0; -1)$; $\lambda = 0$: $(1; 1; -1)$; $\lambda = 4$: $(1; -1; 0)$
29. $\lambda = -1$: $(2; -1; 0)$; $\lambda = 0$: $(1; 2; 2)$; $\lambda = 1$: $(1; 1; 1)$
30. $\lambda = -3$: $(1; 0; 0)$; $\lambda = -2$: $(1; 0; 1)$; $\lambda = 1$: $(0; 1; -2)$

Exercise set 3.

1. $-\lambda^3 + \lambda^2 + 7\lambda + 1$
2. $-\lambda^3 - 4\lambda^2 + 4\lambda + 13$
3. $-\lambda^3 + 2\lambda^2 + 17\lambda - 33$
4. $-\lambda^3 + 4\lambda^2 - 4\lambda - 9$
5. $-\lambda^3 - \lambda^2 + 8\lambda$
6. $-\lambda^3 - \lambda^2 + 3\lambda - 1$
7. A^2: $1; 4; 9$; A^{-1}: $1; \frac{1}{2}; -\frac{1}{3}$; $A + 2E$: $3; 4; -1$
8. $2; -3; 1$
9. $x = \frac{1}{3}(2; 1)$
10. $\lambda = 1$: $(1; -1)$, $\lambda = 3$: $(1; 1)$; $1 \cdot 1 + (-1) \cdot 1 = 0$

4.3. Linear operators with a basis of eigenvectors

Exercise set 1.

1. $T = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix}$
2. $T = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}$, $D = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix}$
3. $T = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$
4. $T = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & -2 \end{pmatrix}$
5. $T = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 0 \end{pmatrix}$
6. $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 3 \end{pmatrix}$
7. $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$
8. $T = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$
9. $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$

10. $T = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 1 \end{pmatrix}$
11. $T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 0 \end{pmatrix}$
12. $T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$
13. $T = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & -1 \end{pmatrix}$
14. $T = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix}$
15. $T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$
16. $T = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}$, $D = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$
17. $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix}$
18. $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & -2 \end{pmatrix}$
19. $T = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -4 & 0 \\ 0 & -3 \end{pmatrix}$
20. $T = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 4 \end{pmatrix}$
21. $T = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$, $D = \begin{pmatrix} -4 & 0 \\ 0 & 0 \end{pmatrix}$
22. $T = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$
23. $T = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$
24. $T = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$
25. $T = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$
26. $T = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & -2 \end{pmatrix}$
27. $T = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$
28. $T = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$
29. $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} -3 & 0 \\ 0 & -1 \end{pmatrix}$
30. $T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} -4 & 0 \\ 0 & -1 \end{pmatrix}$

Exercise set 2.

1. λ: -2, 1, 2; diagonalizable
2. λ: -3, 1, 4; diagonalizable
3. λ: 0 (double), 2; not diagonalizable
4. λ: -3, 0, 1; diagonalizable
5. λ: 2 (double), 3; diagonalizable
6. λ: -3, 3 (double); not diagonalizable
7. λ: -3, -1, 4; diagonalizable
8. λ: -1, 4 (double); diagonalizable
9. λ: -2 (double), -1; not diagonalizable
10. λ: 0, 3 (double); diagonalizable
11. λ: -2, 0, 2; diagonalizable
12. λ: -3 (double), -2; not diagonalizable
13. λ: -3, -2, 2; diagonalizable
14. λ: -2, -1, 3; diagonalizable
15. λ: 0 (double), 1; not diagonalizable
16. λ: -1 (double), 4; diagonalizable
17. λ: -1 (double), 0; diagonalizable
18. λ: -3, -2 (double); not diagonalizable
19. λ: -2, -1, 0; diagonalizable
20. λ: -3 (double), -1; diagonalizable

21. λ: -2, 2 (double); not diagonalizable
22. λ: -3, -2, 3; diagonalizable
23. λ: -3, 0, 3; diagonalizable
24. λ: -3 (double), -2; not diagonalizable
25. λ: -2, -1, 4; diagonalizable
26. λ: -3, 0, 3; diagonalizable
27. λ: -2 (double), -1; not diagonalizable
28. λ: 0 (double), 1; diagonalizable
29. λ: -1, 0, 2; diagonalizable
30. λ: 3 (triple); not diagonalizable

Exercise set 3.

1. $A^4 = \left(\begin{smallmatrix} -14 & 30 \\ -15 & 31 \end{smallmatrix}\right)$
2. $A^3 = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 8 \end{smallmatrix}\right)$
3. $A^4 = \left(\begin{smallmatrix} 16 & 0 \\ -15 & 1 \end{smallmatrix}\right)$
4. $A^4 = \left(\begin{smallmatrix} 81 & -160 \\ 0 & 1 \end{smallmatrix}\right)$
5. $A^4 = \left(\begin{smallmatrix} 16 & 0 \\ 0 & 1 \end{smallmatrix}\right)$
6. $A^5 = \left(\begin{smallmatrix} 487 & -244 \\ 488 & -245 \end{smallmatrix}\right)$
7. $\lambda_{1,2} = \frac{1\pm\sqrt{5}}{2}$; $F^n = \left(\begin{smallmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{smallmatrix}\right)$
8. $\lambda = 2$ is a double eigenvalue, but the eigenspace is spanned by the single vector $(1; 0)$
9. $Q = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$, $D = \operatorname{diag}(2, 4)$
10. $\lambda = 2; 5$: $2 + 5 = 7 = \operatorname{tr} A$, $2 \cdot 5 = 10 = \det A$

Chapter 5. Quadratic Forms, Second-Order Curves and Iterative Methods

5.1. Quadratic forms and their reduction to canonical form

Exercise set 1.

1. $\left(\begin{smallmatrix} -2 & 0 & 2 \\ 0 & 2 & 2 \\ 2 & 2 & 0 \end{smallmatrix}\right)$
2. $\left(\begin{smallmatrix} 3 & 3 & -2 \\ 3 & 3 & 3 \\ -2 & 3 & 4 \end{smallmatrix}\right)$
3. $\left(\begin{smallmatrix} -2 & 1 & 1 \\ 1 & -1 & 3 \\ 1 & 3 & -4 \end{smallmatrix}\right)$
4. $\left(\begin{smallmatrix} -4 & -1 & 1 \\ -1 & 2 & -3 \\ 1 & -3 & -2 \end{smallmatrix}\right)$
5. $\left(\begin{smallmatrix} -1 & -2 & 2 \\ -2 & -4 & 0 \\ 2 & 0 & 2 \end{smallmatrix}\right)$
6. $\left(\begin{smallmatrix} 3 & -3 & -1 \\ -3 & 4 & 0 \\ -1 & 0 & -3 \end{smallmatrix}\right)$
7. $\left(\begin{smallmatrix} 1 & -3 & -2 \\ -3 & -3 & -3 \\ -2 & -3 & -4 \end{smallmatrix}\right)$

8. $\begin{pmatrix} 3 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}$
9. $\begin{pmatrix} -2 & -2 & 2 \\ -2 & -4 & 0 \\ 2 & 0 & -1 \end{pmatrix}$
10. $\begin{pmatrix} -3 & 3 & -2 \\ 3 & -1 & 2 \\ -2 & 2 & -3 \end{pmatrix}$
11. $\begin{pmatrix} -3 & 1 & 0 \\ 1 & 4 & 3 \\ 0 & 3 & 2 \end{pmatrix}$
12. $\begin{pmatrix} -1 & -1 & -2 \\ -1 & 4 & 0 \\ -2 & 0 & 0 \end{pmatrix}$
13. $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & -1 & 3 \end{pmatrix}$
14. $\begin{pmatrix} 2 & -1 & -3 \\ -1 & -3 & -2 \\ -3 & -2 & 3 \end{pmatrix}$
15. $\begin{pmatrix} -4 & 0 & -3 \\ 0 & 4 & 1 \\ -3 & 1 & 4 \end{pmatrix}$
16. $\begin{pmatrix} -1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -3 \end{pmatrix}$
17. $\begin{pmatrix} -2 & 2 & -2 \\ 2 & 1 & 3 \\ -2 & 3 & 1 \end{pmatrix}$
18. $\begin{pmatrix} 1 & 3 & 1 \\ 3 & 0 & -2 \\ 1 & -2 & -3 \end{pmatrix}$
19. $\begin{pmatrix} -2 & -3 & 3 \\ -3 & -2 & 3 \\ 3 & 3 & 2 \end{pmatrix}$
20. $\begin{pmatrix} 1 & -2 & 2 \\ -2 & -4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$
21. $\begin{pmatrix} -1 & 0 & -2 \\ 0 & -4 & -1 \\ -2 & -1 & 3 \end{pmatrix}$
22. $\begin{pmatrix} 4 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & -2 \end{pmatrix}$
23. $\begin{pmatrix} 4 & -3 & 0 \\ -3 & 2 & 3 \\ 0 & 3 & 3 \end{pmatrix}$
24. $\begin{pmatrix} -4 & 3 & -1 \\ 3 & -4 & 3 \\ -1 & 3 & 0 \end{pmatrix}$
25. $\begin{pmatrix} 0 & -3 & 0 \\ -3 & -1 & -2 \\ 0 & -2 & 2 \end{pmatrix}$
26. $\begin{pmatrix} -3 & 3 & -1 \\ 3 & -2 & -2 \\ -1 & -2 & 3 \end{pmatrix}$
27. $\begin{pmatrix} -4 & -1 & -2 \\ -1 & 4 & 2 \\ -2 & 2 & 3 \end{pmatrix}$
28. $\begin{pmatrix} -4 & -2 & 3 \\ -2 & 1 & 3 \\ 3 & 3 & -2 \end{pmatrix}$
29. $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 2 & 0 \\ 3 & 0 & 3 \end{pmatrix}$
30. $\begin{pmatrix} 0 & -2 & 1 \\ -2 & 0 & -2 \\ 1 & -2 & -4 \end{pmatrix}$

Exercise set 2.

1. $y_1^2 - 4y_2^2$; indefinite
2. $-5y_1^2 + 5y_2^2$; indefinite
3. $4y_1^2 + 6y_2^2$; positive definite
4. $-4y_1^2 + y_2^2$; indefinite
5. $4y_1^2 - 4y_2^2$; indefinite
6. $5y_1^2 + y_2^2$; positive definite
7. $y_1^2 - 4y_2^2$; indefinite
8. $2y_1^2 - 2y_2^2$; indefinite
9. $y_1^2 + 6y_2^2$; positive definite
10. $6y_1^2 + y_2^2$; positive definite
11. $3y_1^2 - 2y_2^2$; indefinite
12. $-5y_1^2 + 5y_2^2$; indefinite
13. $-4y_1^2 + y_2^2$; indefinite
14. $-3y_1^2 + 3y_2^2$; indefinite
15. $-3y_1^2 - 5y_2^2$; negative definite
16. $-2y_1^2 + 3y_2^2$; indefinite
17. $5y_1^2 - 5y_2^2$; indefinite
18. $6y_1^2 - 4y_2^2$; indefinite
19. $3y_1^2 - y_2^2$; indefinite
20. $5y_1^2 - 5y_2^2$; indefinite
21. $-3y_1^2 + 2y_2^2$; indefinite
22. $4y_1^2 + 6y_2^2$; positive definite
23. $-4y_1^2 + 6y_2^2$; indefinite
24. $-3y_1^2 - y_2^2$; negative definite
25. $-4y_1^2 + 4y_2^2$; indefinite
26. $y_1^2 + 5y_2^2$; positive definite
27. $2y_1^2 - 2y_2^2$; indefinite
28. $2y_1^2 - 3y_2^2$; indefinite
29. $-3y_1^2 + 3y_2^2$; indefinite
30. $5y_1^2 + y_2^2$; positive definite

Exercise set 3.

1. $\Delta_1 = 1,\ \Delta_2 = 4,\ \Delta_3 = -9$; not positive definite
2. $\Delta_1 = 2,\ \Delta_2 = -2,\ \Delta_3 = -18$; not positive definite
3. $\Delta_1 = 5,\ \Delta_2 = 11,\ \Delta_3 = 30$; positive definite
4. $\Delta_1 = 4,\ \Delta_2 = 12,\ \Delta_3 = 32$; positive definite
5. $\Delta_1 = 5,\ \Delta_2 = 4,\ \Delta_3 = -5$; not positive definite
6. $\Delta_1 = 3,\ \Delta_2 = 5,\ \Delta_3 = 12$; positive definite
7. $\Delta_1 = 3,\ \Delta_2 = 2,\ \Delta_3 = -6$; not positive definite
8. $\Delta_1 = 4,\ \Delta_2 = 8,\ \Delta_3 = 22$; positive definite
9. $\Delta_1 = 3,\ \Delta_2 = 2,\ \Delta_3 = 7$; positive definite
10. $\Delta_1 = 1,\ \Delta_2 = 4,\ \Delta_3 = -13$; not positive definite

5.2. Reducing a second-order equation to canonical form

Exercise set 1.

1. ellipse; $O'(-2;-1)$, $\frac{X^2}{4} + \frac{Y^2}{16} = 1$
2. hyperbola; $O'(-2;-3)$, $\frac{X^2}{16} - \frac{Y^2}{1} = 1$
3. parabola; $O'(-1;0)$, $Y^2 = -4X$
4. ellipse; $O'(-2;4)$, $\frac{X^2}{25} + \frac{Y^2}{1} = 1$
5. hyperbola; $O'(-4;0)$, $\frac{X^2}{16} - \frac{Y^2}{16} = 1$
6. parabola; $O'(-1;-2)$, $Y^2 = 6X$

7. ellipse; $O'(1;0)$, $\frac{X^2}{9}+\frac{Y^2}{16}=1$
8. hyperbola; $O'(0;4)$, $\frac{X^2}{9}-\frac{Y^2}{9}=1$
9. parabola; $O'(0;-2)$, $Y^2=6X$
10. ellipse; $O'(4;-4)$, $\frac{X^2}{25}+\frac{Y^2}{9}=1$
11. hyperbola; $O'(4;0)$, $\frac{X^2}{25}-\frac{Y^2}{16}=1$
12. parabola; $O'(-2;0)$, $Y^2=-4X$
13. ellipse; $O'(-4;0)$, $\frac{X^2}{25}+\frac{Y^2}{16}=1$
14. hyperbola; $O'(-3;-3)$, $\frac{X^2}{25}-\frac{Y^2}{9}=1$
15. parabola; $O'(-3;2)$, $Y^2=4X$
16. ellipse; $O'(4;-1)$, $\frac{X^2}{25}+\frac{Y^2}{1}=1$
17. hyperbola; $O'(-3;3)$, $\frac{X^2}{9}-\frac{Y^2}{4}=1$
18. parabola; $O'(-4;4)$, $Y^2=4X$
19. ellipse; $O'(1;1)$, $\frac{X^2}{9}+\frac{Y^2}{16}=1$
20. hyperbola; $O'(-4;-3)$, $\frac{X^2}{9}-\frac{Y^2}{4}=1$
21. parabola; $O'(-3;-4)$, $Y^2=2X$
22. ellipse; $O'(-2;3)$, $\frac{X^2}{4}+\frac{Y^2}{1}=1$
23. hyperbola; $O'(3;4)$, $\frac{X^2}{9}-\frac{Y^2}{9}=1$
24. parabola; $O'(3;3)$, $Y^2=2X$
25. ellipse; $O'(4;1)$, $\frac{X^2}{4}+\frac{Y^2}{9}=1$
26. hyperbola; $O'(-3;-2)$, $\frac{X^2}{25}-\frac{Y^2}{9}=1$
27. parabola; $O'(-2;-4)$, $Y^2=6X$
28. ellipse; $O'(-3;-2)$, $\frac{X^2}{4}+\frac{Y^2}{9}=1$
29. hyperbola; $O'(-3;3)$, $\frac{X^2}{9}-\frac{Y^2}{16}=1$
30. parabola; $O'(1;-3)$, $Y^2=4X$

Exercise set 2.

1. $5x'^2+y'^2=20$; ellipse
2. $6x'^2+y'^2=8$; ellipse
3. $5x'^2-5y'^2=10$; hyperbola
4. $-x'^2-5y'^2=6$; empty set (imaginary ellipse)
5. $6x'^2+y'^2=20$; ellipse
6. $2x'^2-3y'^2=6$; hyperbola
7. $x'^2-4y'^2=20$; hyperbola
8. $3x'^2-2y'^2=6$; hyperbola
9. $3x'^2-3y'^2=12$; hyperbola
10. $5x'^2-5y'^2=4$; hyperbola
11. $4x'^2-4y'^2=6$; hyperbola
12. $3x'^2-2y'^2=8$; hyperbola
13. $5x'^2-y'^2=6$; hyperbola
14. $-2x'^2-4y'^2=6$; empty set (imaginary ellipse)
15. $3x'^2-y'^2=10$; hyperbola
16. $-x'^2-5y'^2=6$; empty set (imaginary ellipse)
17. $-3x'^2-5y'^2=12$; empty set (imaginary ellipse)
18. $3x'^2-y'^2=12$; hyperbola
19. $6x'^2+y'^2=12$; ellipse
20. $x'^2-y'^2=6$; hyperbola
21. $x'^2-3y'^2=8$; hyperbola
22. $-x'^2-5y'^2=6$; empty set (imaginary ellipse)
23. $2x'^2-3y'^2=6$; hyperbola
24. $6x'^2-4y'^2=8$; hyperbola
25. $2x'^2-4y'^2=6$; hyperbola
26. $6x'^2-4y'^2=12$; hyperbola
27. $5x'^2-5y'^2=4$; hyperbola
28. $3x'^2+y'^2=4$; ellipse
29. $x'^2-4y'^2=12$; hyperbola
30. $5x'^2-y'^2=4$; hyperbola

Exercise set 3.

1. $\delta = 36 > 0$, $\Delta = -1296 \neq 0$, $I\Delta < 0$ – ellipse
2. $O'(1;0)$
3. $\operatorname{tg} 2\alpha = \sqrt{3}$, $\alpha = 30°$
4. $C(3;-2)$, $R = 5$
5. $F(\pm 4;0)$, $e = \frac{4}{5}$
6. $y = \pm\frac{3}{4}x$, $e = \frac{5}{4}$
7. $F(2;0)$, $x = -2$
8. a pair of intersecting lines $y = \pm x$
9. $y^2 \equiv 8 + 4 + 3 \equiv 1$: $y = 1$ or $y = 6$
10. $\delta = 0$, $\Delta = -16 \neq 0$ – parabola

5.3. The iterative method for the approximate solution of systems of linear algebraic equations

Exercise set 1.

1. $x^{(1)} = (-1.875; 1.6667; -0.5)$, $x^{(2)} = (-2.2708; 1.0972; -0.0312)$; exact: $(-2; 1; 0)$
2. $x^{(1)} = (1.7273; -1.3333; 0.1111)$, $x^{(2)} = (2.0606; -1.9091; 0.835)$; exact: $(2; -2; 1)$
3. $x^{(1)} = (1; 2.5; -2)$, $x^{(2)} = (2.1667; 1.875; -1.8125)$; exact: $(2; 2; -2)$
4. $x^{(1)} = (-0.125; -0.2222; 3.2222)$, $x^{(2)} = (0.7639; -0.9244; 3.2886)$; exact: $(1; -1; 3)$
5. $x^{(1)} = (-2.2222; -0.4; 0.9)$, $x^{(2)} = (-1.9667; -1.0244; 0.7978)$; exact: $(-2; -1; 1)$
6. $x^{(1)} = (-2.8889; 3.2222; 0.5)$, $x^{(2)} = (-2.1173; 2.3704; 2.1111)$; exact: $(-2; 3; 2)$
7. $x^{(1)} = (1.4; -2.1667; 3.75)$, $x^{(2)} = (0.9667; -2.0083; 2.8583)$; exact: $(1; -2; 3)$
8. $x^{(1)} = (-0.125; 2.9; 0)$, $x^{(2)} = (0.2375; 2.8625; -0.9458)$; exact: $(0; 3; -1)$
9. $x^{(1)} = (1.875; 0.2; -4.5)$, $x^{(2)} = (3.5625; -1.075; -3.5625)$; exact: $(3; -1; -3)$
10. $x^{(1)} = (2.4286; -0.1667; 3)$, $x^{(2)} = (3.3333; 0.4286; 2.1905)$; exact: $(3; 0; 2)$
11. $x^{(1)} = (-2.75; 0.8571; 1.4)$, $x^{(2)} = (-3.2464; 0.2571; 2.1214)$; exact: $(-3; 0; 2)$
12. $x^{(1)} = (0.5; 2.25; -3.6667)$, $x^{(2)} = (-0.1111; 3.0417; -3.2222)$; exact: $(0; 3; -3)$
13. $x^{(1)} = (-0.3333; 2.3333; 2)$, $x^{(2)} = (0; 3; 2.0833)$; exact: $(0; 3; 2)$
14. $x^{(1)} = (-3.2222; -0.7143; 0.7273)$, $x^{(2)} = (-3.0649; -1.0707; 1.1183)$; exact: $(-3; -1; 1)$
15. $x^{(1)} = (-2.3333; -1.7; 0.1818)$, $x^{(2)} = (-2.2727; -0.9818; 1.2818)$; exact: $(-2; -1; 1)$

16. $x^{(1)} = (3.2222; -0.6667; -2.6667)$, $x^{(2)} = (2.2593; -2.3333; -3.6296)$; exact: $(2; -2; -3)$
17. $x^{(1)} = (-2.1667; -2.5; -3)$, $x^{(2)} = (-0.75; -3.4167; -2.1667)$; exact: $(-1; -3; -2)$
18. $x^{(1)} = (-0.3333; 1.125; 3.1429)$, $x^{(2)} = (-1.2817; -0.0536; 2.7738)$; exact: $(-1; 0; 3)$
19. $x^{(1)} = (-3.9; -0.7143; 2.5714)$, $x^{(2)} = (-2.9143; -1.0816; 1.6612)$; exact: $(-3; -1; 2)$
20. $x^{(1)} = (-0.125; 0.3333; -2.875)$, $x^{(2)} = (0.9531; 0.375; -2.9427)$; exact: $(1; 0; -3)$
21. $x^{(1)} = (-1.1667; -3.5; 1.75)$, $x^{(2)} = (-1.1667; -3.0625; 2.0417)$; exact: $(-1; -3; 2)$
22. $x^{(1)} = (-3; 0.5; 0.8571)$, $x^{(2)} = (-3; -0.2857; 0.1429)$; exact: $(-3; 0; 0)$
23. $x^{(1)} = (0.25; 0.2727; 1)$, $x^{(2)} = (0.0682; 0.0682; 0.9369)$; exact: $(0; 0; 1)$
24. $x^{(1)} = (-2.6; 3; -0.5)$, $x^{(2)} = (-3.3; 2.4143; 0.92)$; exact: $(-3; 3; 1)$
25. $x^{(1)} = (0.6364; 2.75; -0.8)$, $x^{(2)} = (1.6045; 2.7886; -1.9)$; exact: $(2; 3; -2)$
26. $x^{(1)} = (-3.0909; -1.25; 3)$, $x^{(2)} = (-2.2045; -2; 3.526)$; exact: $(-2; -2; 3)$
27. $x^{(1)} = (3.3333; -2.1429; -3)$, $x^{(2)} = (2.6905; -2.0952; -1.9286)$; exact: $(3; -2; -2)$
28. $x^{(1)} = (-0.25; 1.4444; 0.7143)$, $x^{(2)} = (-0.7897; 1.6865; 1.0556)$; exact: $(-1; 2; 1)$
29. $x^{(1)} = (2.3333; 2.6667; -3.3)$, $x^{(2)} = (2.8556; 3.2481; -3.1333)$; exact: $(3; 3; -3)$
30. $x^{(1)} = (-3; 1.75; -1.3333)$, $x^{(2)} = (-2.9074; 1; -0.75)$; exact: $(-3; 1; -1)$

Exercise set 2.

1. $\|B\|_\infty = \frac{5}{8}$; the method converges
2. $\|B\|_\infty = \frac{3}{5}$; the method converges
3. $\|B\|_\infty = 2$; the sufficient condition fails
4. $\|B\|_\infty = \frac{4}{7}$; the method converges
5. $\|B\|_\infty = \frac{6}{11}$; the method converges
6. $\|B\|_\infty = 3$; the sufficient condition fails
7. $\|B\|_\infty = \frac{6}{11}$; the method converges
8. $\|B\|_\infty = \frac{5}{9}$; the method converges
9. $\|B\|_\infty = \frac{7}{2}$; the sufficient condition fails
10. $\|B\|_\infty = \frac{1}{2}$; the method converges
11. $\|B\|_\infty = \frac{1}{2}$; the method converges
12. $\|B\|_\infty = \frac{8}{3}$; the sufficient condition fails
13. $\|B\|_\infty = \frac{4}{7}$; the method converges
14. $\|B\|_\infty = \frac{2}{3}$; the method converges
15. $\|B\|_\infty = 7$; the sufficient condition fails

16. $\|B\|_\infty = \frac{4}{9}$; the method converges
17. $\|B\|_\infty = \frac{4}{7}$; the method converges
18. $\|B\|_\infty = 6$; the sufficient condition fails
19. $\|B\|_\infty = \frac{4}{7}$; the method converges
20. $\|B\|_\infty = \frac{1}{2}$; the method converges
21. $\|B\|_\infty = \frac{7}{2}$; the sufficient condition fails
22. $\|B\|_\infty = \frac{2}{3}$; the method converges
23. $\|B\|_\infty = \frac{3}{5}$; the method converges
24. $\|B\|_\infty = 6$; the sufficient condition fails
25. $\|B\|_\infty = \frac{4}{9}$; the method converges
26. $\|B\|_\infty = \frac{1}{2}$; the method converges
27. $\|B\|_\infty = 7$; the sufficient condition fails
28. $\|B\|_\infty = \frac{2}{5}$; the method converges
29. $\|B\|_\infty = \frac{2}{5}$; the method converges
30. $\|B\|_\infty = 4$; the sufficient condition fails

Exercise set 3.

1. $x^{(3)} = (2.2109;\ 3.1875)$; exact $(2;\ 3)$
2. $x^{(3)} = (2.9438;\ 1.18)$; exact $(3;\ 1)$
3. $x^{(3)} = (2.1125;\ -2.12)$; exact $(2;\ -2)$
4. $x^{(3)} = (-4.0514;\ -3.1469)$; exact $(-4;\ -3)$
5. $x^{(3)} = (3.6694;\ -2.3829)$; exact $(4;\ -3)$
6. $x^{(3)} = (-1.0062;\ 2.9969)$; exact $(-1;\ 3)$
7. $x^{(3)} = (-1.9986;\ -0.9973)$; exact $(-2;\ -1)$
8. $x^{(3)} = (-0.0816;\ 2)$; exact $(0;\ 2)$
9. $x^{(3)} = (3.94;\ -0.775)$; exact $(4;\ -1)$
10. $x^{(3)} = (3.7333;\ 4.48)$; exact $(4;\ 4)$

5.4. The Seidel method for the approximate solution of systems of linear algebraic equations

Exercise set 1.

1. $x^{(1)} = (3.8571;\ 2.8095;\ 1.6667)$, $x^{(2)} = (3.1429;\ 2.1349;\ 1.9444)$; exact: $(3;\ 2;\ 2)$
2. $x^{(1)} = (-2;\ -3;\ 1.375)$, $x^{(2)} = (-0.9625;\ -3.075;\ 0.9859)$; exact: $(-1;\ -3;\ 1)$
3. $x^{(1)} = (-1.5;\ 0.5625;\ 1.9531)$, $x^{(2)} = (-0.9427;\ 1.0156;\ 1.9967)$; exact: $(-1;\ 1;\ 2)$
4. $x^{(1)} = (1;\ 2.25;\ -2.1667)$, $x^{(2)} = (1.1818;\ 3.0398;\ -2.0316)$; exact: $(1;\ 3;\ -2)$
5. $x^{(1)} = (-0.8182;\ -1.8455;\ 3.3327)$, $x^{(2)} = (0.2911;\ -0.8461;\ 2.911)$; exact: $(0;\ -1;\ 3)$
6. $x^{(1)} = (1;\ 0.8333;\ 3.375)$, $x^{(2)} = (-0.0324;\ -0.0733;\ 2.9878)$; exact: $(0;\ 0;\ 3)$
7. $x^{(1)} = (-1.75;\ -0.3056;\ -1.7396)$, $x^{(2)} = (-1.2387;\ -0.8626;\ -1.9485)$; exact: $(-1;\ -1;\ -2)$

8. $x^{(1)} = (-0.5455; -0.5124; 1.991)$, $x^{(2)} = (0.0907; 0.0231; 2.0184)$; exact: $(0; 0; 2)$
9. $x^{(1)} = (-0.8; -1.16; -1.64)$, $x^{(2)} = (-0.24; -1.82; -1.9229)$; exact: $(0; -2; -2)$
10. $x^{(1)} = (1.3333; -0.7273; -3.0606)$, $x^{(2)} = (2.596; -1.9063; -2.9341)$; exact: $(3; -2; -3)$
11. $x^{(1)} = (0.7778; 2.4333; 1.5767)$, $x^{(2)} = (1.7641; 2.8869; 1.9179)$; exact: $(2; 3; 2)$
12. $x^{(1)} = (-3; -1.25; 1.9375)$, $x^{(2)} = (-2.9297; -1.0342; 2.009)$; exact: $(-3; -1; 2)$
13. $x^{(1)} = (-2.125; -1.9688; 2.9609)$, $x^{(2)} = (-2.001; -1.9998; 2.9997)$; exact: $(-2; -2; 3)$
14. $x^{(1)} = (-0.1; -1.475; -2.8031)$, $x^{(2)} = (1.0359; -1.0582; -2.9972)$; exact: $(1; -1; -3)$
15. $x^{(1)} = (-0.3333; 2.9333; -0.0917)$, $x^{(2)} = (0.013; 3.0209; 0.0059)$; exact: $(0; 3; 0)$
16. $x^{(1)} = (-1; -1.25; -2.775)$, $x^{(2)} = (-1; -1.9438; -2.9831)$; exact: $(-1; -2; -3)$
17. $x^{(1)} = (-2.25; 3.3889; 0.9325)$, $x^{(2)} = (-0.8626; 2.977; 1.013)$; exact: $(-1; 3; 1)$
18. $x^{(1)} = (-2.7143; 1.5143; 2.0816)$, $x^{(2)} = (-3.0117; 0.979; 1.9967)$; exact: $(-3; 1; 2)$
19. $x^{(1)} = (3.375; -0.9062; 0.1205)$, $x^{(2)} = (3.0502; -1.0176; 0.0118)$; exact: $(3; -1; 0)$
20. $x^{(1)} = (0.75; 0.8333; 0.9028)$, $x^{(2)} = (0.9844; 0.9838; 0.992)$; exact: $(1; 1; 1)$
21. $x^{(1)} = (-1.1429; 1.7551; -1.7398)$, $x^{(2)} = (-1.8907; 1.9688; -1.9668)$; exact: $(-2; 2; -2)$
22. $x^{(1)} = (0.7143; -2.3333; 1.1476)$, $x^{(2)} = (-0.1163; -1.9508; 0.9749)$; exact: $(0; -2; 1)$
23. $x^{(1)} = (-1.125; 0.0179; -2.8357)$, $x^{(2)} = (-2.1839; 0.9793; -3.0572)$; exact: $(-2; 1; -3)$
24. $x^{(1)} = (-2.5714; 1.25; -1.1825)$, $x^{(2)} = (-1.8764; 0.9544; -0.9624)$; exact: $(-2; 1; -1)$
25. $x^{(1)} = (0.3333; 1.0952; 0.0736)$, $x^{(2)} = (-0.0072; 1.019; -0.0054)$; exact: $(0; 1; 0)$
26. $x^{(1)} = (2; 2.1667; 3)$, $x^{(2)} = (2.8611; 2.9537; 3)$; exact: $(3; 3; 3)$
27. $x^{(1)} = (-2; -1.6667; -3)$, $x^{(2)} = (-2.2222; -0.9259; -3.0741)$; exact: $(-2; -1; -3)$

28. $x^{(1)} = (-2.7778; 3.5972; -2.2817)$, $x^{(2)} = (-1.8322; 2.9086; -1.9499)$; exact: $(-2; 3; -2)$
29. $x^{(1)} = (-0.75; -3.375; -0.9531)$, $x^{(2)} = (0.0938; -3.0176; -0.9817)$; exact: $(0; -3; -1)$
30. $x^{(1)} = (-3.3333; 1.3; 1.0381)$, $x^{(2)} = (-2.9; 0.9624; 1.0035)$; exact: $(-3; 1; 1)$

Exercise set 2.

1. $x^{(3)} = (-0.9975; 1.001)$; exact $(-1; 1)$
2. $x^{(3)} = (3; -1)$; exact $(3; -1)$
3. $x^{(3)} = (-0.9974; 3.9996)$; exact $(-1; 4)$
4. $x^{(3)} = (1.0394; -2.0197)$; exact $(1; -2)$
5. $x^{(3)} = (-2; -1)$; exact $(-2; -1)$
6. $x^{(3)} = (3; 0)$; exact $(3; 0)$
7. $x^{(3)} = (0.875; -4.0625)$; exact $(1; -4)$
8. $x^{(3)} = (3; 1)$; exact $(3; 1)$
9. $x^{(3)} = (0.036; -4.0135)$; exact $(0; -4)$
10. $x^{(3)} = (2; 2)$; exact $(2; 2)$
11. $x^{(3)} = (-4; 0)$; exact $(-4; 0)$
12. $x^{(3)} = (-1.94; -3.024)$; exact $(-2; -3)$
13. $x^{(3)} = (-0.838; -3.081)$; exact $(-1; -3)$
14. $x^{(3)} = (-1.0098; 2.0025)$; exact $(-1; 2)$
15. $x^{(3)} = (0; 1)$; exact $(0; 1)$
16. $x^{(3)} = (-2.0208; 0.9844)$; exact $(-2; 1)$
17. $x^{(3)} = (-2; -3)$; exact $(-2; -3)$
18. $x^{(3)} = (-2; 0)$; exact $(-2; 0)$
19. $x^{(3)} = (2.9985; 2.0002)$; exact $(3; 2)$
20. $x^{(3)} = (-4; 0)$; exact $(-4; 0)$
21. $x^{(3)} = (-2; -1)$; exact $(-2; -1)$
22. $x^{(3)} = (-1.9927; -3.9984)$; exact $(-2; -4)$
23. $x^{(3)} = (2.9089; -3.0342)$; exact $(3; -3)$
24. $x^{(3)} = (0.9999; -4)$; exact $(1; -4)$
25. $x^{(3)} = (-2.689; -3.8134)$; exact $(-3; -4)$
26. $x^{(3)} = (1.0086; -3.0037)$; exact $(1; -3)$
27. $x^{(3)} = (-4; -3)$; exact $(-4; -3)$
28. $x^{(3)} = (-4; 0)$; exact $(-4; 0)$
29. $x^{(3)} = (1.0703; 3.9736)$; exact $(1; 4)$
30. $x^{(3)} = (-1; -1)$; exact $(-1; -1)$

Exercise set 3.

1. Jacobi: 0.0366; Seidel: 0.0048
2. Jacobi: 0.0508; Seidel: 0.001
3. Jacobi: 0.0248; Seidel: 0.0001
4. Jacobi: 0.1332; Seidel: 0.0017
5. Jacobi: 0.0508; Seidel: 0.0063
6. Jacobi: 0.0317; Seidel: 0.0027
7. Jacobi: 0.0402; Seidel: 0.0015
8. Jacobi: 0.048; Seidel: 0.0039
9. Jacobi: 0.0918; Seidel: 0.0098
10. Jacobi: 0.0069; Seidel: 0.0007

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